# Optimal Transport for Machine Learners

*Compact teaching notes*

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## 1 Optimal Matching between Point Clouds

### 1.1 Monge Problem for Discrete Points

This compact subsection starts from finite matching before relaxing to transport plans.

**Matching Problem.** Given a cost matrix $(\mathrm{C}_{i,j})_{i\in[\![n]\!],j\in[\![m]\!]}$ and assuming $n=m$, the optimal assignment problem aims to find a bijection $\sigma$ within the set $\mathrm{Perm}(n)$ of permutations of $n$ elements that solves

$$\min_{\sigma\in\mathrm{Perm}(n)} \frac{1}{n}\sum_{i=1}^{n} \mathrm{C}_{i,\sigma(i)}. \tag{1.1}$$

**1-D Case.**

**Proposition 1.1** (Monotone matching on the line)**.** *Assume that the points $(x_i)_i$ and $(y_j)_j$ are pairwise distinct. If the cost is of the form $\mathrm{C}_{i,j}=h(x_i-y_j)$, where $h:\mathbb{R}\to\mathbb{R}^+$ is strictly convex (for example, $\mathrm{C}_{i,j}=|x_i-y_j|^p$ for $p>1$), then any optimal $\sigma$ defines a strictly increasing map $x_i\mapsto y_{\sigma(i)}$ (and thus is unique), i.e., $\forall (i,i'),\quad (x_i-x_{i'})(y_{\sigma(i)}-y_{\sigma(i')})>0$.*

*Proof.* Indeed, if this property is violated, i.e., there exists $(i,i')$ such that $(x_i-x_{i'})(y_{\sigma(i)}-y_{\sigma(i')})<0$, then one can define a permutation $\tilde\sigma$ by swapping the match, i.e., $\tilde\sigma(i)=\sigma(i')$ and $\tilde\sigma(i')=\sigma(i)$, yielding a strictly better cost, as proved in the following fact.

Let $h:\mathbb{R}\to\mathbb{R}$ be strictly convex and let $x<x'$ and $y<y'$. Then $h(x-y)+h(x'-y')<h(x-y')+h(x'-y)$. We set the gap $d:=y'-y>0$ and define for every $s\in\mathbb{R}$ $D(s):=\frac{h(s)-h(s-d)}{d}$ and $\Delta:=h(x-y')+h(x'-y)-h(x-y)-h(x'-y')=d(D(x'-y)-D(x-y))$. Because $h$ is strictly convex, $D$ is strictly increasing. Since $x-y<x'-y$, monotonicity yields $D(x-y)<D(x'-y)$, that is $\Delta>0$. □

$x_{\sigma_X(1)}\leqslant x_{\sigma_X(2)}\leqslant\dots\quad$ and $\quad y_{\sigma_Y(1)}\leqslant y_{\sigma_Y(2)}\leqslant\dots$

**Optimal transport on the circle.** $d_{\mathbb{S}^1}(x,y):=\min_{k\in\mathbb{Z}}|x-y+k|,\qquad c_p(x,y):=d_{\mathbb{S}^1}(x,y)^p,\qquad p>1.$

**Proposition 1.2** (Discrete circle transport by a cut)**.** *Let $x_1,\dots,x_n$ and $y_1,\dots,y_n$ be two families of distinct points on $\mathbb{S}^1$, with equal weights. Let $x_{(1)},\dots,x_{(n)}$ and $y_{(1)},\dots,y_{(n)}$ denote any fixed cyclic orderings, with the convention $y_{(k+n)}=y_{(k)}$. For the cost $c_p$, $p>1$, an optimal assignment is one of the cyclic shifts $x_{(k)}\longmapsto y_{(k+s)},\qquad k\in[\![n]\!],\quad s\in\{0,\dots,n-1\}$, and is found by minimizing*

$$\sum_{k=1}^{n} d_{\mathbb{S}^1}\big(x_{(k)},y_{(k+s)}\big)^p$$

*over the $n$ possible shifts. Equivalently, for an optimal shift one can choose a cut $\theta\in\mathbb{S}^1\setminus(\{x_i\}_i\cup\{y_j\}_j)$ so that, after lifting all points to $(\theta,\theta+1)$ and sorting them, the optimal matching is the equal-rank monotone matching on this interval.*

*Proof.* Call two matched pairs cyclically inverted if the circular order of their source endpoints is opposite to the circular order of their target endpoints. Among optimal assignments, choose one with the smallest number of such inversions. The elementary exchange step is the circular analogue of the line argument in Proposition 1.1: if two matched pairs are inverted, then cutting the circle in a gap which does not meet the four endpoints and choosing integer lifts realizes the four geodesic distances involved in the exchange as ordinary distances between two ordered source lifts and two oppositely ordered target lifts. The one-dimensional Monge inequality for the strictly convex function $r\mapsto|r|^p$ then shows that swapping the two targets cannot increase the cost, and decreases it unless the four endpoints are in a degenerate tie configuration.

Thus an optimal assignment can be chosen with no cyclic inversion. A bijection between two finite cyclically ordered sets with no cyclic inversion is a rotation of the order, hence a cyclic shift. This shift specifies how the two cyclic orderings should be opened; after this cut, the rotation becomes an ordinary linear order and the matching is the equal-rank monotone assignment on the unfolded interval. Conversely, each cut gives one such cyclic shift, so minimizing over the finitely many shifts gives an optimal discrete circle assignment. Repeated points or ties are obtained by the same argument after an arbitrarily small perturbation and a limiting passage. This is the discrete fast circle-Monge construction. □

**Rational weights.**

**Proposition 1.3** (Rational weights as duplicated uniform matching)**.** *Let $\alpha=\sum_{i=1}^n\frac{k_i}{N}\delta_{x_i},\qquad \nu=\sum_{j=1}^m\frac{\ell_j}{N}\delta_{y_j},\qquad \sum_i k_i=\sum_j \ell_j=N$, with $k_i,\ell_j\in\mathbb{N}$. The discrete Kantorovich problem between $(\alpha,\nu)$ is equivalent to the uniform assignment problem obtained by replacing each $x_i$ by $k_i$ identical copies and each $y_j$ by $\ell_j$ identical copies. More precisely, after multiplying transport masses by $N$, optimal couplings correspond to optimal integer count matrices $(n_{ij})$ with row sums $k_i$ and column sums $\ell_j$, and these count matrices are exactly the collapsed form of assignments between the duplicated clouds.*

*Proof.* Any assignment between the duplicated source and target clouds defines integers $n_{ij}$ counting how many copied particles of type $x_i$ are matched to copied particles of type $y_j$. These counts satisfy $\sum_j n_{ij}=k_i$ and $\sum_i n_{ij}=\ell_j$, and the associated coupling $P_{ij}=n_{ij}/N$ has marginals $k_i/N$ and $\ell_j/N$. The assignment cost is $\frac{1}{N}\sum_{i,j}n_{ij}c(x_i,y_j)=\sum_{i,j}P_{ij}c(x_i,y_j)$. Conversely, any nonnegative integer count matrix with those row and column sums can be realized by matching the corresponding duplicated particles. Finally, the transportation constraint matrix is totally unimodular, so the linear problem with integer supplies and demands has an optimal integer count matrix. Thus the optimum of the rational-weight Kantorovich problem is the same as the optimum of the duplicated uniform assignment problem. □

**2D case.**

**Proposition 1.4** (Non-crossing optimal matchings)**.** *In dimension 2, for $c(x,y)=\|x-y\|$, if $\sigma$ is an optimal assignment, then segments $[x_i,y_{\sigma(i)}]$ cannot cross.*

*Proof.* If two segments $[x_i, y_{\sigma(i)}]$ and $[x_j, y_{\sigma(j)}]$ cross at an interior point $z$, then the triangle inequality gives $\|x_i - y_{\sigma(j)}\| + \|x_j - y_{\sigma(i)}\| < \|x_i - y_{\sigma(i)}\| + \|x_j - y_{\sigma(j)}\|$. The assignment obtained by swapping $(\sigma(i), \sigma(j))$ therefore has a strictly smaller cost, which contradicts optimality. □

$C_n = \frac{1}{n+1}\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi} n^{3/2}}$.

### 1.2 Matching Algorithms

This compact subsection records the algorithmic viewpoint on finite assignments.

**Algorithmic viewpoint.** Assignment algorithms combine primal matchings with dual certificates.

**Hungarian primal-dual method.** $u_i + v_j \leqslant C_{i,j} \qquad \forall i, j.$

$$\delta = \min_{i \in S,\ j \notin T} (C_{i,j} - u_i - v_j), \qquad u_i \leftarrow u_i + \delta\ (i \in S), \qquad v_j \leftarrow v_j - \delta\ (j \in T).$$

**Proposition 1.5** (Correctness and complexity of the Hungarian primal-dual method)**.** *Assume the Hungarian method terminates with a perfect matching $\sigma$ contained in the equality graph $E(u,v) = \{(i,j)\ ;\ u_i + v_j = C_{i,j}\}$, where $(u,v)$ is dual feasible, i.e. $u_i + v_j \leqslant C_{i,j}$ for all $(i,j)$. Then $\sigma$ is an optimal assignment. Moreover, the usual Hungarian updates terminate after finitely many augmentations. With maintained slacks, the method uses $O(n^3)$ arithmetic operations.*

*Proof.* For any permutation $\tau$, dual feasibility gives $\sum_i C_{i,\tau(i)} \geqslant \sum_i (u_i + v_{\tau(i)}) = \sum_i u_i + \sum_j v_j$. This is the weak duality lower bound. If $\sigma$ is contained in the equality graph, then $\sum_i C_{i,\sigma(i)} = \sum_i u_i + \sum_j v_j$, so the primal cost of $\sigma$ reaches the dual lower bound and is optimal.

At each successful augmentation, the matching cardinality increases by one, so there are at most $n$ augmentations. During one augmentation phase, the algorithm grows an alternating tree in the equality graph. If no augmenting path is available, the dual update uses the smallest slack of an edge leaving the current tree. For edges inside the tree, adding $\delta$ to source labels and subtracting $\delta$ from target labels preserves tightness; for edges from $S$ to $T^c$, the definition of $\delta$ preserves feasibility and makes at least one new edge tight; all other inequalities are unchanged or become looser. Thus the reachable sets strictly grow between two failed augmentation attempts, and they can grow at most $n$ times within one phase. If the current slacks $\min_{i \in S}(C_{i,j} - u_i - v_j)$ are updated when a source enters $S$, each tree expansion costs $O(n)$. A phase therefore costs $O(n^2)$, and the $n$ phases give $O(n^3)$ operations. Hence the method reaches a perfect optimal matching. □

## 2 Monge Problem between Measures

### 2.1 Measures

This compact subsection fixes the measure notation used throughout the handout.

**Histograms.**

**Definition 2.1** (Probability simplex)**.** The probability simplex of length $n$ is $\Sigma_n \coloneqq \left\{ \mathrm{a} \in \mathbb{R}^n_+\ ;\ \sum_{i=1}^n \mathrm{a}_i = 1 \right\}$. Its elements are also called probability vectors or histograms.

**Discrete measure, empirical measure.**

**Definition 2.2** (Discrete measure)**.** A discrete measure with weights $\mathrm{a}$ and locations $x_1, \dots, x_n \in \mathcal{X}$ is

$$\alpha = \sum_{i=1}^n \mathrm{a}_i \delta_{x_i}, \tag{2.1}$$

where $\delta_x$ is the Dirac mass at position $x$. It is a probability measure if $\mathrm{a} \in \Sigma_n$, and a positive measure if all weights $\mathrm{a}_i$ are nonnegative.

**General measures.** A Dirac measure $\delta_x$ is then defined as $\delta_x(A) = 1$ if $x \in A$ and 0 otherwise, and this extends by linearity for discrete measures of the form (2.1) as

$\alpha(A) = \sum_{x_i \in A} \mathrm{a}_i$ We denote $\mathcal{M}_+(\mathcal{X})$ the subset of all positive measures on $\mathcal{X}$, i.e. $\alpha(A) \geqslant 0$ (and $\alpha(\mathcal{X}) < +\infty$ for the measure to be finite). The set of probability measures is denoted $\mathcal{M}^1_+(\mathcal{X})$, which means that any $\alpha \in \mathcal{M}^1_+(\mathcal{X})$ is positive, and that $\alpha(\mathcal{X}) = 1$.

**Polish metric spaces.**

**Definition 2.3** (Polish metric space)**.** A metric space $(\mathcal{X}, d)$ is Polish if it is complete and separable: every Cauchy sequence converges in $\mathcal{X}$, and $\mathcal{X}$ contains a countable dense subset. More generally, a topological space is called Polish if its topology can be induced by some complete separable metric.

$\mathcal{S} = C([0,1]; \mathbb{R}^d), \qquad d_\infty(\gamma, \eta) \coloneqq \sup_{t \in [0,1]} \|\gamma(t) - \eta(t)\|$

**Definition 2.4** (Support of a measure)**.** The support $\mathrm{supp}(\alpha)$ of a Borel measure $\alpha$ on a metric space $\mathcal{X}$ is the smallest closed set of full $\alpha$-mass. Equivalently, $x \in \mathrm{supp}(\alpha)$ if every open ball centered at $x$ has positive $\alpha$-mass.

**Radon measures.** Using Lebesgue integration, a Borel measure can be used to compute the integral of measurable functions (i.e. such that level sets $\{x\ ;\ f(x) < t\}$ are Borel sets), and we denote this pairing as

$\langle f, \alpha \rangle \coloneqq \int f(x) \mathrm{d}\alpha(x)$. $\int_{\mathcal{X}} f(x) \mathrm{d}\alpha(x) = \sum_{i=1}^n \mathrm{a}_i f(x_i)$.

**Relative densities.**

**Definition 2.5** (Relative density)**.** If $\alpha$ is absolutely continuous with respect to a reference measure $\lambda$, its relative density is the Radon–Nikodym derivative $\rho_\alpha \coloneqq \frac{\mathrm{d}\alpha}{\mathrm{d}\lambda}, \qquad \mathrm{d}\alpha(x) = \rho_\alpha(x) \mathrm{d}\lambda(x)$. Equivalently, for all $h \in \mathcal{C}(\mathcal{X})$, $\int_{\mathcal{X}} h(x) \mathrm{d}\alpha(x) = \int_{\mathcal{X}} h(x) \rho_\alpha(x) \mathrm{d}\lambda(x)$.

**Total variation norm.**

**Definition 2.6** (Total variation)**.** For a finite signed Radon measure $\alpha$ on a compact space $\mathcal{X}$, $\|\alpha\|_{\mathrm{TV}} := \sup_{f\in\mathcal{C}(\mathcal{X})} \{\langle f, \alpha\rangle \,;\, \|f\|_\infty \leqslant 1\}$.

$|\alpha|(A) := \sup_{A=\cup_i B_i} \sum_i |\alpha(B_i)|$, where the supremum is over finite or countable measurable partitions of $A$. Thus, if $\alpha = \sum_i \mathrm{a}_i\delta_{x_i}$ with distinct atoms, $|\alpha| = \sum_i |\mathrm{a}_i|\delta_{x_i}$; if $\mathrm{d}\alpha(x) = \rho(x)\mathrm{d}\lambda(x)$, then $\mathrm{d}|\alpha|(x) = |\rho(x)|\mathrm{d}\lambda(x)$.

**Proposition 2.7** (Dual and measure definitions of total variation)**.** *For a finite signed Radon measure $\alpha$ on a compact space $\mathcal{X}$, $\|\alpha\|_{\mathrm{TV}} = |\alpha|(\mathcal{X})$.*

*Proof.* The inequality $\|\alpha\|_{\mathrm{TV}} \leqslant |\alpha|(\mathcal{X})$ follows from $\left|\int f\,\mathrm{d}\alpha\right| \leqslant \int |f|\,\mathrm{d}|\alpha| \leqslant \|f\|_\infty|\alpha|(\mathcal{X})$. For the reverse inequality, write the Jordan decomposition $\alpha = \alpha^+ - \alpha^-$, so that $|\alpha| = \alpha^+ + \alpha^-$. The measurable sign $s = \frac{\mathrm{d}\alpha}{\mathrm{d}|\alpha|}$ takes values in $\{-1, 1\}$ outside a $|\alpha|$-null set and satisfies $\mathrm{d}\alpha = s\,\mathrm{d}|\alpha|$. By regularity of Radon measures on compact spaces, $s$ can be approximated in $L^1(|\alpha|)$ by continuous functions $f_k$ with $\|f_k\|_\infty \leqslant 1$. Hence $\int f_k\,\mathrm{d}\alpha \to \int s\,\mathrm{d}\alpha = |\alpha|(\mathcal{X})$, which proves the equality. The final statement is the Riesz–Markov–Kakutani representation theorem with this norm. □

For two absolutely continuous measures $\mathrm{d}\alpha = \rho_\alpha\mathrm{d}\lambda$ and $\mathrm{d}\beta = \rho_\beta\mathrm{d}\lambda$, this gives the concrete formula $\|\alpha - \beta\|_{\mathrm{TV}} = \int_{\mathcal{X}} |\rho_\alpha(x) - \rho_\beta(x)|\,\mathrm{d}\lambda(x)$. $\alpha = \sum_{i=1}^n \tilde{a}_i\delta_{x_i}$, $\beta = \sum_{j=1}^m \tilde{b}_j\delta_{y_j}$, $a_k := \sum_{i:x_i=z_k} \tilde{a}_i$, $b_k := \sum_{j:y_j=z_k} \tilde{b}_j$. Then $\alpha = \sum_{k=1}^r a_k\delta_{z_k}$ and $\beta = \sum_{k=1}^r b_k\delta_{z_k}$. This convention merges masses located at the same point before comparing the two measures.
$\|\alpha - \beta\|_{\mathrm{TV}} = \sum_{k=1}^r |a_k - b_k|$.

**Probabilistic interpretation.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be an abstract probability space. If $X : \Omega \to \mathcal{X}$ is measurable, its law is $\alpha = X_\sharp\mathbb{P}$, i.e. $\alpha(A) = \mathbb{P}(\{\omega \in \Omega \,;\, X(\omega) \in A\})$ for Borel sets $A \subset \mathcal{X}$. $\int_{\mathcal{X}} f(x)\,\mathrm{d}\alpha(x) = \mathbb{E}[f(X)]$.

## 2.2 Push-Forward

For some continuous map $T : \mathcal{X} \to \mathcal{Y}$, we define the push-forward operator $T_\sharp : \mathcal{M}(\mathcal{X}) \to \mathcal{M}(\mathcal{Y})$.
For a Dirac mass, one has $T_\sharp\delta_x = \delta_{T(x)}$, and this formula is extended to arbitrary measures by linearity. In some sense, moving from $T$ to $T_\sharp$ is a way to linearize any map at the price of moving from a (possibly) finite-dimensional space $\mathcal{X}$ to the infinite-dimensional space $\mathcal{M}(\mathcal{X})$, and this idea is central to many convex relaxation methods, most notably Lasserre's relaxation.
For discrete measures (2.1), the push-forward operation consists simply in moving the positions of all the points in the support of the measure
$T_\sharp\alpha := \sum_i \mathrm{a}_i\delta_{T(x_i)}$.

**Definition 2.8** (Push-forward)**.** For $T : \mathcal{X} \to \mathcal{Y}$, the push-forward measure $\beta = T_\sharp\alpha \in \mathcal{M}(\mathcal{Y})$ of some $\alpha \in \mathcal{M}(\mathcal{X})$ satisfies

$$\forall\, h \in \mathcal{C}(\mathcal{Y}), \quad \int_{\mathcal{Y}} h(y)\mathrm{d}\beta(y) = \int_{\mathcal{X}} h(T(x))\mathrm{d}\alpha(x). \tag{2.2}$$

Equivalently, for any measurable set $B \subset \mathcal{Y}$, one has

$$\beta(B) = \alpha(\{x \in \mathcal{X} \,;\, T(x) \in B\}). \tag{2.3}$$

Note that $T_\sharp$ preserves positivity and total mass, so that if $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ then $T_\sharp\alpha \in \mathcal{M}_+^1(\mathcal{Y})$.

**Proposition 2.9** (Push-forward formula for densities)**.** *Let $\alpha$ and $\beta$ have densities $\rho_\alpha$ and $\rho_\beta$ on open subsets of $\mathbb{R}^d$. Assume that $T$ is a $C^1$ diffeomorphism and that $\beta = T_\sharp\alpha$. Then, for all $x$,*

$$\rho_\alpha(x) = |\det(T'(x))|\rho_\beta(T(x)). \tag{2.4}$$

*Equivalently, for $y = T(x)$, $\rho_\beta(y) = \rho_\alpha(T^{-1}y)\,\frac{1}{|\det(T'(T^{-1}y))|}$.*

*Proof.* Explicitly doing the change of variable $y = T(x)$, so that $\mathrm{d}y = |\det(T'(x))|\mathrm{d}x$ in formula (2.2) for measures with densities $(\rho_\alpha, \rho_\beta)$ on $\mathbb{R}^d$, one has for all $h \in \mathcal{C}(\mathcal{Y})$

$$\begin{aligned}\int_{\mathcal{Y}} h(y)\rho_\beta(y)\mathrm{d}y &= \int_{\mathcal{Y}} h(y)\mathrm{d}\beta(y) = \int_{\mathcal{X}} h(T(x))\mathrm{d}\alpha(x) = \int_{\mathcal{X}} h(T(x))\rho_\alpha(x)\mathrm{d}x\\ &= \int_{\mathcal{Y}} h(y)\rho_\alpha(T^{-1}y)\frac{\mathrm{d}y}{|\det(T'(T^{-1}y))|},\end{aligned}$$

which shows that $\rho_\beta(y) = \rho_\alpha(T^{-1}y)\frac{1}{|\det(T'(T^{-1}y))|}$. Since $T$ is a diffeomorphism, one obtains equivalently $\rho_\alpha(x) = |\det(T'(x))|\rho_\beta(T(x))$ where $T'(x) \in \mathbb{R}^{d\times d}$ is the Jacobian matrix of $T$ (the matrix formed by taking the gradient of each coordinate of $T$).
This implies, denoting $y = T(x)$ $|\det(T'(x))| = \frac{\rho_\alpha(x)}{\rho_\beta(y)}$. □

## 2.3 Monge's Formulation

This compact subsection records Monge transport before the Kantorovich relaxation.

**Monge problem.** Given $\alpha \in \mathcal{M}_+^1(\mathcal{X})$, $\beta \in \mathcal{M}_+^1(\mathcal{Y})$ and a cost $c : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}_+$, the Monge problem is

$$\mathcal{M}_c(\alpha, \beta) := \inf_{T:\mathcal{X}\to\mathcal{Y}} \left\{ \int_{\mathcal{X}} c(x, T(x))\mathrm{d}\alpha(x) \,;\, T_\sharp\alpha = \beta \right\}. \tag{2.5}$$

**Proposition 2.10** (Empirical Monge maps and matchings)**.** *Assume that the source atoms $x_1, \dots, x_n$ are distinct and that $\alpha = \frac{1}{n}\sum_{i=1}^n \delta_{x_i}, \qquad \beta = \frac{1}{n}\sum_{j=1}^n \delta_{y_j}$. If $T_\sharp\alpha = \beta$, then for each distinct target value $z$ in the support of $\beta$, exactly $n\beta(\{z\})$ source atoms are mapped to $z$. In particular, if the $y_j$ are distinct, then there exists a permutation $\sigma \in \mathrm{Perm}(n)$ such that $T(x_i) = y_{\sigma(i)}$ for all $i$. Conversely, every such assignment of source atoms to target atoms with the correct masses defines a feasible Monge map on the support of $\alpha$, and in the distinct-target case $\int_{\mathcal{X}} c(x, T(x))\mathrm{d}\alpha(x) = \frac{1}{n}\sum_{i=1}^n c(x_i, y_{\sigma(i)})$. If source locations are repeated, they should first be merged into atoms with larger masses; such atoms cannot be split by a Monge map.*

*Proof.* Since $T_\sharp\alpha = \beta$, all images $T(x_i)$ must belong to the support of $\beta$; otherwise the push-forward would give positive mass to a point outside that support. For any target atom $z$, $\beta(\{z\}) = \alpha(T^{-1}(\{z\})) = \frac{1}{n}\#\{i \,;\, T(x_i) = z\}$. This proves the counting statement. If all target atoms have mass $1/n$, each target receives exactly one source atom, which is a permutation. The converse and the cost identity follow by direct substitution. ☐

**Proposition 2.11** (Existence of transport maps from atomless sources)**.** *Let $\alpha$ and $\beta$ be Borel probability measures on Polish spaces, and assume that $\alpha$ is atomless. Then there exists a measurable map $T$ such that $T_\sharp\alpha = \beta$.*

*Proof.* A standard measure-isomorphism theorem identifies the atomless probability space generated by $\alpha$ with Lebesgue measure on $[0,1]$, modulo null sets. It is therefore enough to construct a map from $[0,1]$ to the target law $\beta$. Choose a Borel isomorphism $i$ from the support of $\beta$ onto a Borel subset of $[0,1]$, set $\nu = i_\sharp\beta$, and use the generalized inverse of the cumulative distribution function of $\nu$. This map sends Lebesgue measure on $[0,1]$ to $\nu$ and takes values in $i(\operatorname{supp}\beta)$ almost surely. Composing with $i^{-1}$ and then with the source isomorphism gives a measurable transport map from $\alpha$ to $\beta$. ☐

**Monge distance.** When $\mathcal{X} = \mathcal{Y}$ and $c(x,y) = d(x,y)^p$ for a metric $d$, set
$\mathcal{E}_\alpha(T) \coloneqq \int_{\mathcal{X}} d(x, T(x))^p \mathrm{d}\alpha(x)$.

$$\tilde{\mathcal{W}}_p(\alpha,\beta)^p \coloneqq \inf_{T:\mathcal{X}\to\mathcal{X}} \left\{\mathcal{E}_\alpha(T) \,;\, T_\sharp\alpha = \beta\right\}. \tag{2.6}$$

If the constraint set is empty, then $\tilde{\mathcal{W}}_p(\alpha,\beta) = +\infty$.

**Proposition 2.12** (Directed Monge distance)**.** *Assume that $\mathcal{X} = \mathcal{Y}$ is a metric space. The quantity $\tilde{\mathcal{W}}_p$ is nonnegative, vanishes only on the diagonal, and satisfies the triangle inequality. It is therefore a directed extended distance: it need not be symmetric and may take the value $+\infty$.*

*Proof.* Nonnegativity is immediate. If $\tilde{\mathcal{W}}_p(\alpha,\beta) = 0$, choose feasible maps $T_k$ with $\int d(x, T_k(x))^p \mathrm{d}\alpha(x) \to 0$. For every bounded 1-Lipschitz function $h$,

$$\left|\int h\mathrm{d}\beta - \int h\mathrm{d}\alpha\right| = \left|\int h(T_k(x)) - h(x)\mathrm{d}\alpha(x)\right| \leqslant \left(\int d(x, T_k(x))^p \mathrm{d}\alpha(x)\right)^{1/p} \longrightarrow 0.$$

Since bounded Lipschitz functions separate probability measures on metric spaces, $\alpha = \beta$.
If $\tilde{\mathcal{W}}_p(\alpha,\beta) = +\infty$ while both $\tilde{\mathcal{W}}_p(\alpha,\gamma)$ and $\tilde{\mathcal{W}}_p(\gamma,\beta)$ were finite, there would be maps $S_\sharp\alpha = \gamma$ and $T_\sharp\gamma = \beta$, hence $(T\circ S)_\sharp\alpha = \beta$, a contradiction. Thus the inequality is automatic whenever the left-hand side is infinite. In the finite case, fix $\varepsilon > 0$ and choose $\varepsilon$-minimizers $S_\sharp\alpha = \gamma$ and $T_\sharp\gamma = \beta$ such that $\mathcal{E}_\alpha(S)^{1/p} \leqslant \tilde{\mathcal{W}}_p(\alpha,\gamma) + \varepsilon, \qquad \mathcal{E}_\gamma(T)^{1/p} \leqslant \tilde{\mathcal{W}}_p(\gamma,\beta) + \varepsilon$. The composed map is feasible from $\alpha$ to $\beta$, and Minkowski's inequality gives

$$\begin{aligned}
\tilde{\mathcal{W}}_p(\alpha,\beta) &\leqslant \left(\int d(x, T(S(x)))^p \mathrm{d}\alpha(x)\right)^{1/p} \\
&\leqslant \left(\int d(x, S(x))^p \mathrm{d}\alpha(x)\right)^{1/p} + \left(\int d(S(x), T(S(x)))^p \mathrm{d}\alpha(x)\right)^{1/p} \\
&= \mathcal{E}_\alpha(S)^{1/p} + \mathcal{E}_\gamma(T)^{1/p} \leqslant \tilde{\mathcal{W}}_p(\alpha,\gamma) + \tilde{\mathcal{W}}_p(\gamma,\beta) + 2\varepsilon.
\end{aligned}$$

Letting $\varepsilon \to 0$ gives the result. ☐

## 2.4 Existence and Uniqueness of the Monge Map

This compact subsection summarizes the main existence and structure results for Monge maps.

**Brenier's theorem.**

**Theorem 2.13** (Brenier)**.** *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$ have finite second moments, and assume that $\alpha$ is absolutely continuous with respect to Lebesgue measure. For the quadratic cost $c(x,y) = \|x-y\|^2$, there exists a convex function $\varphi : \mathbb{R}^d \to \mathbb{R}\cup\{+\infty\}$ such that $T = \nabla\varphi, \qquad T_\sharp\alpha = \beta$, and $T$ is the unique optimal Monge map $\alpha$-almost everywhere. The optimal Kantorovich plan is $(\mathrm{Id}, T)_\sharp\alpha$.*

*Proof.* Kantorovich duality for the quadratic cost gives optimal potentials $(f,g)$ with equality $f(x) + g(y) = \|x-y\|^2$ on the support of any optimal plan. After subtracting the harmless quadratic terms, this equality is equivalent to the Fenchel equality $\varphi(x) + \varphi^*(y) = \langle x, y\rangle$ for a convex function $\varphi$. Hence the support of every optimal plan lies in the graph of the subdifferential $\partial\varphi$. Since $\alpha$ has a density and convex functions are differentiable Lebesgue-almost everywhere, $\partial\varphi(x)$ is a singleton for $\alpha$-almost every $x$. The plan is therefore concentrated on the graph of $T = \nabla\varphi$, which proves that the relaxed optimizer is induced by a Monge map. Any two optimal plans are concentrated on the same single-valued graph $\alpha$-almost everywhere, which gives uniqueness of the map. This is the standard route behind Brenier's polar factorization theorem; related existence and uniqueness results extend to more general strictly convex costs. ☐

$\langle\nabla\varphi(x) - \nabla\varphi(x'),\, x - x'\rangle \geqslant 0$.

**Definition 2.14** (Measures not charging hypersurfaces)**.** A Borel measure $\alpha$ on $\mathbb{R}^d$ does not charge hypersurfaces if $\alpha(S) = 0$ for every countably $(d-1)$-rectifiable set $S$, i.e. every set that can be covered, up to an $\mathcal{H}^{d-1}$-null set, by countably many $C^1$ hypersurfaces.

**Radial measures.** Radial symmetry gives a useful higher-dimensional case where the Brenier map reduces to a one-dimensional monotone rearrangement. A measure $\alpha$ on $\mathbb{R}^d$ is radial if $Q_\sharp\alpha = \alpha$ for every orthogonal map $Q$.

**Proposition 2.15** (Optimal transport between radial measures)**.** *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$ be radial probability measures with finite second moments, and assume that $\alpha$ is absolutely continuous with respect to Lebesgue measure. Define their radial distribution functions $F_\alpha(r) \coloneqq \alpha(\{x : \|x\| \leqslant r\})$, $\quad F_\beta(r) \coloneqq \beta(\{y : \|y\| \leqslant r\})$, $\quad r \geqslant 0$, and let $F_\beta^{-1}(u) \coloneqq \inf\{r \geqslant 0 : F_\beta(r) \geqslant u\}$ be the generalized inverse. Set $\tau(r) \coloneqq F_\beta^{-1}(F_\alpha(r))$. Then the quadratic Brenier map from $\alpha$ to $\beta$ is the radial map $T(0) = 0$, $\quad T(x) = \frac{\tau(\|x\|)}{\|x\|}x \quad$ for $x \neq 0$. Moreover, if $\alpha_R = (x \mapsto \|x\|)_\sharp\alpha$ is the law of the source radius, then $\mathcal{W}_2^2(\alpha, \beta) = \int_0^\infty |r - \tau(r)|^2 \, \mathrm{d}\alpha_R(r)$.*

*Proof.* Let $\alpha_R = (\|\cdot\|)_\sharp\alpha$ and $\beta_R = (\|\cdot\|)_\sharp\beta$ be the radial laws. Since $\alpha$ is absolutely continuous, $\alpha_R$ has no atoms. The map $\tau = F_\beta^{-1} \circ F_\alpha$ is therefore the monotone one-dimensional transport from $\alpha_R$ to $\beta_R$. If $X \sim \alpha$ and $R = \|X\|$, then $\tau(R)$ has law $\beta_R$. Moreover, by radiality, the conditional direction $X/\|X\|$ given $R = r$ is uniform on the sphere for $\alpha_R$-a.e. $r > 0$. Thus $T$ changes only the radius, from $R$ to $\tau(R)$, and keeps the uniform angular distribution. It follows that $T_\sharp\alpha$ is radial with radial law $\beta_R$, hence $T_\sharp\alpha = \beta$.
The function $\tau$ is nondecreasing and nonnegative. Define $\psi(r) \coloneqq \int_0^r \tau(s)\mathrm{d}s$ on $\mathbb{R}_+$. Then $\psi$ is convex and nondecreasing, so $\varphi(x) \coloneqq \psi(\|x\|)$ is convex on $\mathbb{R}^d$. Since $\alpha$ is absolutely continuous, it does not charge the origin nor the radii where the monotone function $\tau$ is discontinuous. Thus $\nabla\varphi(x) = \frac{\tau(\|x\|)}{\|x\|}x = T(x) \qquad$ for $\alpha$-a.e. $x$. The map $T$ is therefore a gradient of a convex potential and pushes $\alpha$ to $\beta$. Brenier's theorem identifies it as the unique quadratic optimal Monge map. The displayed cost formula follows by substituting $T(x)$ in $\int \|x - T(x)\|^2 \mathrm{d}\alpha(x)$. □

If $\alpha$ is not absolutely continuous, the same radial idea is still useful but has to be interpreted at the level of couplings of the radial variables: atoms on spheres may have to be split, so a Monge map need not exist.

**Polar factorization.** Brenier's theorem does more than solve one transport problem: it provides a canonical way to extract the "monotone part" of an arbitrary map. Suppose one starts from a square-integrable deformation $u : \Omega \to \mathbb{R}^d$, for instance a velocity snapshot, an image deformation or a generative map.

**Proposition 2.16** (Polar factorization)**.** *Let $\Omega \subset \mathbb{R}^d$ be endowed with normalized Lebesgue measure $\lambda$, and let $u \in L^2(\Omega; \mathbb{R}^d)$. Assume that the law $\nu = u_\sharp\lambda$ has finite second moment. Then there exist a measure-preserving map $s : \Omega \to \Omega$ and a convex function $\varphi$ such that $u = \nabla\varphi \circ s \qquad \lambda$-a.e. The Brenier factor $\nabla\varphi$ is unique $\lambda$-almost everywhere.*

*Proof.* By Brenier's theorem there is a unique gradient of a convex function $T = \nabla\varphi$ transporting $\lambda$ to $\nu$. The maps $u$ and $T$ have the same image law. The rearrangement theorem for non-atomic probability spaces gives a measure-preserving map $s$ such that $u = T \circ s$; equivalently, $s$ chooses, with the correct conditional probabilities, preimages of $u(x)$ under $T$. Uniqueness of the Brenier factor follows from Theorem 2.13. □

The name is meant to echo the usual polar decomposition of matrices. This analogy becomes literal for linear maps under the Gaussian reference measure.
$A = SO, \qquad O = S^\dagger A,$

**Displacement interpolation.**

**Definition 2.17** (Monge and McCann displacement interpolation)**.** If $T_\sharp\alpha = \beta$, the Monge interpolation generated by $T$ is the curve $T_t(x) \coloneqq (1-t)x + tT(x), \qquad \alpha_t \coloneqq (T_t)_\sharp\alpha, \qquad t \in [0,1]$. For the quadratic cost, when $T$ is the optimal Brenier map, this curve is called McCann's displacement interpolation.

**Proposition 2.18** (Directed Monge displacement geodesics)**.** *Let $p \geqslant 1$, let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$ have finite $p$-th moments, and let $T$ be an optimal map for $\tilde{\mathcal{W}}_p(\alpha, \beta)$ in (2.6). Set $\alpha_t = (T_t)_\sharp\alpha$ with $T_t = (1-t)\mathrm{Id} + tT$. Assume that, for every $t < 1$, $T_t$ is one-to-one on a full $\alpha$-measure Borel set, so that $T_t^{-1}$ is defined $\alpha_t$-almost everywhere. Then, for $0 \leqslant s \leqslant t \leqslant 1$, $\tilde{\mathcal{W}}_p(\alpha_s, \alpha_t) = (t-s)\tilde{\mathcal{W}}_p(\alpha, \beta)$. Thus $t \mapsto \alpha_t$ is an oriented constant-speed geodesic for the directed Monge distance. In particular, for $p = 2$, this applies to the Brenier map under the hypotheses of Theorem 2.13.*

*Proof.* The case $s = t$ is trivial, so assume $s < t$. Since $s < 1$, the inverse $T_s^{-1}$ is defined $\alpha_s$-almost everywhere along the transported particles. Define $S_{s,t} \coloneqq T_t \circ T_s^{-1} \qquad \alpha_s$-a.e. Then $(S_{s,t})_\sharp\alpha_s = \alpha_t$, and, using the optimality of $T$,

$$\begin{aligned}\tilde{\mathcal{W}}_p(\alpha_s, \alpha_t)^p &\leqslant \int \|S_{s,t}(z) - z\|^p \mathrm{d}\alpha_s(z) \\ &= \int \|T_t(x) - T_s(x)\|^p \mathrm{d}\alpha(x) = (t-s)^p \tilde{\mathcal{W}}_p(\alpha, \beta)^p.\end{aligned}$$

The same particle construction gives $\tilde{\mathcal{W}}_p(\alpha, \alpha_s) \leqslant s\tilde{\mathcal{W}}_p(\alpha, \beta)$. If $t < 1$, the map $T \circ T_t^{-1}$ sends $\alpha_t$ to $\beta$ and gives $\tilde{\mathcal{W}}_p(\alpha_t, \beta) \leqslant (1-t)\tilde{\mathcal{W}}_p(\alpha, \beta)$; if $t = 1$, this latter distance is zero. Using the triangle inequality from Proposition 2.12,

$$\tilde{\mathcal{W}}_p(\alpha, \beta) \leqslant \tilde{\mathcal{W}}_p(\alpha, \alpha_s) + \tilde{\mathcal{W}}_p(\alpha_s, \alpha_t) + \tilde{\mathcal{W}}_p(\alpha_t, \beta) \leqslant s\tilde{\mathcal{W}}_p(\alpha, \beta) + \tilde{\mathcal{W}}_p(\alpha_s, \alpha_t) + (1-t)\tilde{\mathcal{W}}_p(\alpha, \beta).$$

Hence $\tilde{\mathcal{W}}_p(\alpha_s, \alpha_t) \geqslant (t-s)\tilde{\mathcal{W}}_p(\alpha, \beta)$, which proves equality. For a Brenier map $T = \nabla\varphi$, the map $T_t$ is the gradient of $x \mapsto (1-t)\frac{\|x\|^2}{2} + t\varphi(x)$, which is $(1-t)$-strongly convex for every $t < 1$. On the full-measure set where $\varphi$ is differentiable, this gives $\langle T_t(x) - T_t(y), x - y\rangle \geqslant (1-t)\|x - y\|^2$, so $T_t$ is injective there. □

**Regularity and the Monge–Ampère equation.**

**Proposition 2.19** (Caffarelli regularity)**.** *Let $\Omega, \Lambda \subset \mathbb{R}^d$ be bounded uniformly convex domains with $C^2$ boundaries. Let $\alpha = \rho(x)\mathrm{d}x$ be supported on $\Omega$ and $\beta = \eta(y)\mathrm{d}y$ be supported on $\Lambda$, with $0 < m \leqslant \rho, \eta \leqslant M < +\infty$. If $\rho, \eta \in C^\alpha$ for some $\alpha \in (0,1)$, then the Brenier potential $\varphi$ transporting $\alpha$ to $\beta$ is strictly convex and satisfies $\varphi \in C^{2,\alpha}_{\mathrm{loc}}(\Omega)$; in particular $\nabla\varphi$ is locally Hölder. Under the corresponding boundary compatibility and smoothness assumptions, the regularity holds up to the boundary.*

*Proof.* The potential solves the Monge–Ampère equation $\det(\nabla^2\varphi(x)) = \frac{\rho(x)}{\eta(\nabla\varphi(x))}$ in the Alexandrov sense, with second boundary condition $\nabla\varphi(\Omega) = \Lambda$. The density bounds and convexity of the domains give strict convexity and localization of sections. Caffarelli's interior theory then yields $C^{2,\alpha}_{\mathrm{loc}}$ estimates for $\varphi$; the boundary statement follows from the boundary regularity theory under uniform convexity and compatibility assumptions. □

For smooth densities, the change-of-variables formula (2.4) gives the Monge–Ampère equation

$$\det(\nabla^2\varphi(x))\rho_\beta(\nabla\varphi(x)) = \rho_\alpha(x). \tag{2.7}$$

**Proposition 2.20** (Linearization of the Monge–Ampère equation)**.** *Let $\rho_\varepsilon = \rho_0 + \varepsilon r + o(\varepsilon)$ be a smooth perturbation of a positive reference density $\rho_0$ on a smooth bounded domain, with $\int r = 0$. If $T_\varepsilon(x) = x + \varepsilon\nabla u(x) + o(\varepsilon)$ transports $\rho_0$ to $\rho_\varepsilon$, then, to first order, $-\nabla\cdot(\rho_0\nabla u) = r$. In particular, when $\rho_0$ is constant, the linearized equation is $-\Delta u = r/\rho_0$.*

*Proof.* The change-of-variables equation for $T_\varepsilon$ is $\rho_0(x) = \rho_\varepsilon(T_\varepsilon(x))\det(\nabla T_\varepsilon(x))$. Expanding $\rho_\varepsilon(x+\varepsilon\nabla u) = \rho_0(x) + \varepsilon r(x) + \varepsilon\langle\nabla\rho_0,\, \nabla u\rangle + o(\varepsilon)$ and $\det(I+\varepsilon\nabla^2 u) = 1 + \varepsilon\Delta u + o(\varepsilon)$ gives

$$\rho_0 = \rho_0 + \varepsilon\big(r + \langle\nabla\rho_0,\, \nabla u\rangle + \rho_0\Delta u\big) + o(\varepsilon) = \rho_0 + \varepsilon\big(r + \nabla\cdot(\rho_0\nabla u)\big) + o(\varepsilon).$$

The first-order term must vanish. □

### 2.5 Beyond the Quadratic Euclidean Cost

This compact subsection records how Monge theory changes beyond the quadratic cost.

**$W_p$ costs..** $\nabla f(x) = \nabla_x c(x, T(x))$. $T(x) = x - \|\nabla f(x)\|^{q-2}\nabla f(x)$, $\quad \frac{1}{p} + \frac{1}{q} = 1.$

**Squared geodesic distance.** $T(x) = \exp_x(-\nabla\varphi(x))$,

**Twist condition.**

**Definition 2.21** (Twist condition)**.** Let $X, Y \subset \mathbb{R}^d$ and let $c \in C^1(X\times Y)$. The cost $c$ satisfies the twist condition from $X$ to $Y$ if, for every fixed $x \in X$, the map $y \in Y \longmapsto \nabla_x c(x,y) \in \mathbb{R}^d$ is injective. The dual twist condition is the analogous injectivity of $x \mapsto \nabla_y c(x,y)$ for every fixed $y$.

**Proposition 2.22** (Twist prevents splitting)**.** *Assume that $c \in C^1(X\times Y)$ satisfies the twist condition from $X$ to $Y$. Suppose that optimal pairs for the cost $c$ admit a source potential $f$ with the following first-order certificate: $f$ is differentiable on a full $\alpha$-measure subset of $X$, and whenever $x$ belongs to this differentiability set and $y$ can be paired optimally with $x$, one has $\nabla f(x) = \nabla_x c(x,y)$. Then, for $\alpha$-almost every source point $x$, there is at most one target point $y$ that can be paired optimally with $x$.*

*Proof.* Let $(f,g)$ be optimal Kantorovich potentials and let $\pi$ be an optimal plan. On $\mathrm{supp}(\pi)$ one has $f(x)+g(y) = c(x,y)$, while dual feasibility gives $f(z)+g(y) \leqslant c(z,y)$ for all $z$. Thus, for each contact pair $(x,y)$, the function $z \mapsto c(z,y) - g(y)$ touches $f$ from above at $x$. At a point where $f$ is differentiable, this gives $\nabla f(x) = \nabla_x c(x,y)$. If the cost is twisted, this equation determines at most one $y$. Hence no optimal plan can split the mass of such an $x$ between several target points. Since differentiability holds on a full $\alpha$-measure set, the plan is concentrated on the graph of a measurable map there. □

The quadratic cost satisfies twist since $\nabla_x\|x-y\|^2 = 2(x-y)$; the bilinear cost $c(x,y) = -\langle x,\, y\rangle$ satisfies twist since $\nabla_x c(x,y) = -y$; more generally $c(x,y) = h(x-y)$ is twisted when $h$ is smooth strictly convex and $\nabla h$ is injective. On a Riemannian manifold, the squared geodesic cost is twisted locally away from the cut locus.

**Ma–Trudinger–Wang curvature.**

**Definition 2.23** (Weak MTW condition)**.** Assume that $c \in C^4(X\times Y)$, that $X, Y \subset \mathbb{R}^d$, that $c$ is twisted, and that the mixed Hessian $\nabla^2_{xy}c(x,y)$ is invertible. For $(x,p)$ in the image of the change of variables $(x,y) \mapsto (x, \nabla_x c(x,y))$, let $Y(x,p)$ be defined by $\nabla_x c(x, Y(x,p)) = p$, and set $A_{ij}(x,p) \coloneqq \partial^2_{x_i x_j}c(x, Y(x,p))$. The cost satisfies the weak MTW condition if $\sum_{i,j,k,l}\partial^2_{p_k p_l}A_{ij}(x,p)\,\xi_i\xi_j\eta_k\eta_l \geqslant 0 \quad$ whenever $\langle\xi,\, \eta\rangle = 0$. A strong MTW condition asks for a positive lower bound, proportional to $\|\xi\|^2\|\eta\|^2$, on the same orthogonal directions.

**Proposition 2.24** (MTW condition and regularity)**.** *Assume that $c$ is smooth, twisted and non-degenerate, that the source and target domains satisfy the usual $c$-convexity assumptions, and that the source and target densities are positive and bounded above and below. Under the weak MTW condition, optimal maps are continuous for smooth positive data. Under the corresponding strong MTW condition and higher smoothness of the data and domains, one obtains higher regularity estimates for the optimal map.*

*Proof.* This is the regularity theorem of Ma–Trudinger–Wang and Loeper, stated here in its structural form. The $c$-convex potential solves a generated Jacobian equation. Twist and non-degeneracy turn the potential into a single-valued map, while the weak MTW inequality controls the convexity of contact sets of $c$-convex functions. This prevents the contact set from tearing apart and yields continuity. Strong MTW gives a quantitative curvature lower bound; combined with smooth densities, boundary geometry and elliptic estimates for the generated Jacobian equation, it yields the usual higher regularity. Loeper also proved that nonnegative MTW curvature is essentially necessary for continuity for arbitrary smooth positive data. □

### 2.6 One-Dimensional Transport and Quantiles

This compact subsection collects the explicit one-dimensional formulas.

**Cumulative and quantile functions.**

**Definition 2.25** (Cumulative and quantile functions)**.** For $\alpha \in \mathcal{M}^1_+(\mathbb{R})$, its cumulative distribution function is

$$\mathcal{C}_\alpha(x) \coloneqq \alpha((-\infty, x]). \tag{2.8}$$

Its generalized inverse, or quantile function, is

$$\mathcal{C}_\alpha^{-1}(r) \coloneqq \inf\{x \in \mathbb{R}\,;\, \mathcal{C}_\alpha(x) \geqslant r\}, \qquad r \in (0,1). \tag{2.9}$$

**Proposition 2.26** (Quantile push-forward)**.** *One has $(\mathcal{C}_\alpha^{-1})_\sharp \mathrm{Leb}_{[0,1]} = \alpha$. If $\alpha$ has no atoms, then $(\mathcal{C}_\alpha)_\sharp \alpha = \mathrm{Leb}_{[0,1]}$.*

*Proof.* For simplicity, assume first that $\alpha$ has a strictly positive density, so that $\mathcal{C}_\alpha$ is strictly increasing and continuous. Denote $\gamma \coloneqq (\mathcal{C}_\alpha^{-1})_\sharp \mathrm{Leb}_{[0,1]}$. It is enough to prove that $\mathcal{C}_\gamma = \mathcal{C}_\alpha$. For every $x$,

$$\mathcal{C}_\gamma(x) = \int_0^1 \mathbf{1}_{(-\infty,x]}(\mathcal{C}_\alpha^{-1}(z))\mathrm{d}z = \int_0^1 \mathbf{1}_{[0,\mathcal{C}_\alpha(x)]}(z)\mathrm{d}z = \mathcal{C}_\alpha(x),$$

where we used $\mathcal{C}_\alpha^{-1}(z) \leqslant x$ if and only if $z \leqslant \mathcal{C}_\alpha(x)$. General measures follow from the same argument with generalized inverses and right-continuity of the cumulative distribution function. If $\alpha$ has no atoms, the probability integral transform gives $(\mathcal{C}_\alpha)_\sharp \alpha = \mathrm{Leb}_{[0,1]}$. ☐

### 1-D Monge solutions.

**Theorem 2.27** (One-dimensional Monge solution)**.** *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R})$, assume that $\alpha$ has no atoms, and let $h : \mathbb{R} \to \mathbb{R}$ be convex. Consider the cost $c(x, y) = h(x - y)$. Write $q_\alpha \coloneqq \mathcal{C}_\alpha^{-1}$ and $q_\beta \coloneqq \mathcal{C}_\beta^{-1}$, and assume that $h(q_\alpha - q_\beta) \in L^1(0, 1)$. Then*

$$T = q_\beta \circ \mathcal{C}_\alpha = \mathcal{C}_\beta^{-1} \circ \mathcal{C}_\alpha \tag{2.10}$$

*satisfies $T_\sharp\alpha = \beta$ and minimizes $\min_{S_\sharp\alpha=\beta} \int h(x - S(x))\mathrm{d}\alpha(x)$. In particular, $h(t) = |t|^p$ gives the usual one-dimensional Monge map for every $p \geqslant 1$ whenever the source has no atoms.*

*Proof.* By Proposition 2.26, atomlessness of $\alpha$ gives $(\mathcal{C}_\alpha)_\sharp\alpha = \mathrm{Leb}_{[0,1]}$, while $(q_\beta)_\sharp \mathrm{Leb}_{[0,1]} = \beta$. Hence $T_\sharp\alpha = \beta$.
Let $S$ be any admissible map and set $\pi_S = (\mathrm{Id}, S)_\sharp\alpha$. The one-dimensional uncrossing theorem for relaxed couplings, proved later as Theorem 3.20, shows that the quantile coupling $\pi^\star = (q_\alpha, q_\beta)_\sharp \mathrm{Leb}_{[0,1]}$ is optimal among all couplings. Since $\alpha$ has no atoms, $(\mathrm{Id}, T)_\sharp\alpha = \pi^\star$, and therefore $\int h(x - T(x))\mathrm{d}\alpha(x) = \int h(x - y)\mathrm{d}\pi^\star(x, y) \leqslant \int h(x - y)\mathrm{d}\pi_S(x, y) = \int h(x - S(x))\mathrm{d}\alpha(x)$. This proves optimality of $T$ among Monge maps. ☐

For discrete measures, one cannot directly apply the map formula when the source has atoms, but if the measures are uniform on the same number of Dirac masses, then it is exactly the sorting formula of Proposition 1.1.

**Proposition 2.28** (One-dimensional Wasserstein formulas)**.** *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R})$ have finite $p$-th moments. For every $p \geqslant 1$,*

$$\mathcal{W}_p(\alpha, \beta)^p = \int_0^1 \left|\mathcal{C}_\alpha^{-1}(r) - \mathcal{C}_\beta^{-1}(r)\right|^p \mathrm{d}r = \|\mathcal{C}_\alpha^{-1} - \mathcal{C}_\beta^{-1}\|_{L^p([0,1])}^p. \tag{2.11}$$

*For $p = 1$, this is equivalently*

$$\mathcal{W}_1(\alpha, \beta) = \|\mathcal{C}_\alpha - \mathcal{C}_\beta\|_{L^1(\mathbb{R})} = \int_\mathbb{R} |\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(x)|\,\mathrm{d}x \tag{2.12}$$

$$= \int_\mathbb{R} \left|\int_{-\infty}^x \mathrm{d}(\alpha - \beta)\right| \mathrm{d}x. \tag{2.13}$$

*Proof.* The first formula follows from Theorem 3.20: the optimal coupling is obtained by taking the same quantile level $r$ for both measures. For $p = 1$, use the layer-cake identity. If $q_\alpha$ and $q_\beta$ are the two quantile functions, then $\int_0^1 |q_\alpha(r) - q_\beta(r)|\mathrm{d}r = \int_\mathbb{R} \lambda\big(\{r : q_\alpha(r) \leqslant x < q_\beta(r)\} \cup \{r : q_\beta(r) \leqslant x < q_\alpha(r)\}\big)\mathrm{d}x$. The measure of the displayed set is exactly $|\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(x)|$ for almost every $x$. ☐

**Proposition 2.29** (Bobkov–Ledoux cumulative formula)**.** *Let $\alpha, \beta \in \mathcal{M}_+^1([a, b])$ and write $s_+ \coloneqq \max(s, 0)$. Then*

$$\mathcal{W}_2^2(\alpha, \beta) = 2 \int_a^b \int_x^b \Big[\big(\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(y)\big)_+ + \big(\mathcal{C}_\beta(x) - \mathcal{C}_\alpha(y)\big)_+\Big]\mathrm{d}y\,\mathrm{d}x. \tag{2.14}$$

*Proof.* For any $u, v \in [a, b]$, one has $|u - v|^2 = 2 \int_a^b \int_x^b \Big[\mathbf{1}_{\{u \leqslant x \leqslant y < v\}} + \mathbf{1}_{\{v \leqslant x \leqslant y < u\}}\Big]\mathrm{d}y\,\mathrm{d}x$, because, when $u < v$, the integration domain is the triangle $u \leqslant x \leqslant y < v$, whose area is $(v - u)^2/2$, and the case $v < u$ is symmetric. Apply this identity with $u = q_\alpha(r)$ and $v = q_\beta(r)$, where $q_\alpha = \mathcal{C}_\alpha^{-1}$ and $q_\beta = \mathcal{C}_\beta^{-1}$, and integrate with respect to $r \in (0, 1)$. Fubini's theorem applies since the integrand is bounded. For $x \leqslant y$, the generalized inverse identities give, up to endpoint sets of zero Lebesgue measure, $\lambda\{r\,;\, q_\alpha(r) \leqslant x \text{ and } q_\beta(r) > y\} = \big(\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(y)\big)_+$, and the same argument with $\alpha$ and $\beta$ exchanged gives the second positive part. ☐

$\mathcal{C}_{\alpha_t}^{-1}(r) = (1 - t)\mathcal{C}_\alpha^{-1}(r) + t\mathcal{C}_\beta^{-1}(r), \qquad r \in (0, 1).$

### OT on trees.

**Proposition 2.30** (Cumulative formula on a tree)**.** *Let $\mathsf{T} = (V, E, \ell)$ be a finite weighted tree, where each edge $e$ has length $\ell_e > 0$, and let $d_\mathsf{T}$ be its geodesic distance. Choose a root $o$. For an edge $e = (u, v)$ oriented away from $o$, let $V_e$ be the set of vertices in the connected component containing $v$ after removing $e$. If $\alpha, \beta \in \mathcal{P}(V)$, then*

$$\mathcal{W}_{1,d_\mathsf{T}}(\alpha, \beta) = \min_{\pi\in\mathcal{U}(\alpha,\beta)} \sum_{x,y\in V} d_\mathsf{T}(x, y)\pi_{xy} = \sum_{e\in E} \ell_e\, |\alpha(V_e) - \beta(V_e)|\,. \tag{2.15}$$

*The value is computable in $O(|V|)$ arithmetic operations once the signed masses $\alpha - \beta$ are stored on the vertices, by one postorder traversal of the rooted tree.*

*Proof.* Let $\sigma = \alpha - \beta$. Removing an edge $e$ splits the tree into $V_e$ and its complement. For any coupling $\pi$, the net amount of mass crossing $e$ from $V_e$ to $V \setminus V_e$ minus the amount crossing in the reverse direction is fixed and equals $\sigma(V_e)$. Hence the total amount of transported mass whose path uses $e$ is at least $|\sigma(V_e)|$. Since every unit of mass crossing $e$ pays the length $\ell_e$, summing over edges gives $\sum_{x,y} d_{\mathsf{T}}(x,y)\pi_{xy} = \sum_{e\in E} \ell_e \sum_{x,y:\, e\in[x,y]} \pi_{xy} \geqslant \sum_{e\in E} \ell_e|\sigma(V_e)|$, where $[x,y]$ is the unique path between $x$ and $y$. Process the rooted tree from the leaves to the root. At a vertex $v \neq o$, compute the total signed mass in the subtree below $v$, namely $F_e = \sigma(V_e)$ for the edge $e$ joining $v$ to its parent. If $F_e > 0$, send $F_e$ units of surplus from this subtree to the parent side; if $F_e < 0$, import $-F_e$ units from the parent side into the subtree. After this operation the subtree has zero net imbalance and can be collapsed into its parent. Continuing upward balances all subtrees because $\sigma(V) = 0$. Decomposing these edge fluxes into source-to-target paths yields a coupling whose traffic across each edge is exactly $|F_e|$, and therefore whose cost is the right-hand side of (2.15). The same postorder pass computes all $F_e$, proving the complexity claim. □

**Triangular rearrangements.**

**Proposition 2.31** (Knothe–Rosenblatt triangular rearrangement)**.** *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$. Assume, for simplicity, that the first marginal of $\alpha$ and the one-dimensional conditional laws of $\alpha$ used below are atomless, and that regular conditional distributions are fixed. There is a triangular map $T(x_1,\dots,x_d) = (T_1(x_1), T_2(x_1,x_2),\dots,T_d(x_1,\dots,x_d))$ such that $T_\sharp\alpha = \beta$ and, for each $k$, the function $x_k \mapsto T_k(x_1,\dots,x_k)$ is nondecreasing for $\alpha$-almost every value of $(x_1,\dots,x_{k-1})$.*

*Proof.* The construction is recursive. For $k=1$, let $T_1$ be the monotone rearrangement between the first marginal of $\alpha$ and the first marginal of $\beta$. Suppose that $T_1,\dots,T_{k-1}$ have already been constructed. Write $x_{<k} = (x_1,\dots,x_{k-1})$ and $T_{<k} = (T_1,\dots,T_{k-1})$. By induction, $(T_{<k})_\sharp\alpha_{<k} = \beta_{<k}$, where $\alpha_{<k}$ and $\beta_{<k}$ are the first $(k-1)$-coordinate marginals. Let $\alpha^k_{x_{<k}}$ and $\beta^k_{y_{<k}}$ be regular conditional laws of the $k$-th coordinate given the previous coordinates. Define $T_k(x_{<k},\cdot)$ as the one-dimensional monotone rearrangement from $\alpha^k_{x_{<k}}$ to $\beta^k_{T_{<k}(x_{<k})}$.
The map $T_k(x_{<k},\cdot)$ is nondecreasing by the one-dimensional rearrangement theorem. The chain rule for disintegrations then shows that, after step $k$, the first $k$ coordinates of $T_\sharp\alpha$ have the same law as the first $k$ coordinates of $\beta$. At $k=d$ this gives $T_\sharp\alpha = \beta$. Target atoms are handled by generalized quantiles. If a source conditional law has atoms that must be split, the same recursive construction defines a triangular Markov kernel rather than a deterministic map. □

**Proposition 2.32** (Anisotropic Brenier maps converge to Knothe–Rosenblatt)**.** *Let $\alpha, \beta$ be compactly supported probability measures on $\mathbb{R}^d$ with densities bounded above and below on their rectangular supports, and assume that the conditional laws entering Proposition 2.31 are atomless. For $\varepsilon > 0$, set $c_\varepsilon(x,y) := \sum_{k=1}^d \varepsilon^{k-1}|x_k - y_k|^2$. Let $T_\varepsilon$ be the Monge map from $\alpha$ to $\beta$ for the cost $c_\varepsilon$, and let $T_{\mathrm{KR}}$ be the triangular Knothe–Rosenblatt rearrangement with the coordinate order used above. Then $T_\varepsilon \longrightarrow T_{\mathrm{KR}}$ in $L^2(\alpha;\mathbb{R}^d)$ as $\varepsilon \to 0$.*

*Proof.* Let $\pi_\varepsilon = (\mathrm{Id}, T_\varepsilon)_\sharp\alpha$. Since the supports are compact, a subsequence converges weakly to some coupling $\pi$ between $\alpha$ and $\beta$. The optimality of $\pi_\varepsilon$ for $F_\varepsilon(\gamma) = \sum_{k=1}^d \varepsilon^{k-1}\int |x_k - y_k|^2\, \mathrm{d}\gamma(x,y)$ first implies, by letting $\varepsilon \to 0$, that $\pi$ minimizes the one-dimensional quadratic cost in the first coordinate among all couplings. Since the first marginal of $\alpha$ is atomless, the one-dimensional minimizer is the monotone rearrangement, so $y_1 = T_1(x_1)$ under $\pi$.
Now restrict attention to couplings satisfying this first-coordinate constraint. Subtract the common minimal value of the first-coordinate cost, divide the optimality inequality by $\varepsilon$, and let $\varepsilon \to 0$. The limit coupling must minimize the second-coordinate quadratic cost among all couplings that already realize the first monotone rearrangement. Disintegrating with respect to $(x_1,y_1)$ reduces this constrained problem to the one-dimensional monotone rearrangement between the conditional laws of $x_2$ and $y_2$. Hence $y_2 = T_2(x_1,x_2)$ under $\pi$.
Repeating the same subtraction-and-rescaling argument gives, for every $k$, the conditional monotone rearrangement $y_k = T_k(x_1,\dots,x_k)$. Thus every weak limit of $(\pi_\varepsilon)_\varepsilon$ is concentrated on the graph of $T_{\mathrm{KR}}$. This graph coupling is unique, so the whole family $\pi_\varepsilon$ converges weakly to $(\mathrm{Id}, T_{\mathrm{KR}})_\sharp\alpha$.
Finally, let $X \sim \alpha$. The graph couplings are the laws of $(X, T_\varepsilon(X))$, and they converge weakly to the law of $(X, T_{\mathrm{KR}}(X))$. To turn this into convergence of maps, fix $\zeta > 0$. By Lusin's theorem, there is a compact set $K$ with $\alpha(K) > 1-\zeta$ on which $T_{\mathrm{KR}}$ is continuous. On $K$, the set $\{(x,y)\,;\, x \in K,\ \|y - T_{\mathrm{KR}}(x)\| \geqslant \delta\}$ is closed and has zero mass under the limiting graph coupling. Portmanteau's theorem gives

$$\limsup_{\varepsilon\to 0} \alpha(\{x\,;\, x\in K,\ \|T_\varepsilon(x) - T_{\mathrm{KR}}(x)\| \geqslant \delta\}) = 0.$$

Adding the complement of $K$ and letting $\zeta \to 0$ proves convergence in $\alpha$-probability. Since all maps take values in a common compact set, this convergence is uniformly integrable and hence holds in $L^2(\alpha)$. □

## 2.7 Gaussian Measures and the Bures Metric

This compact subsection records the closed-form Gaussian geometry.

**One-dimensional Gaussians.** Let $\alpha = \mathcal{N}(m_\alpha, \sigma_\alpha^2)$ and $\beta = \mathcal{N}(m_\beta, \sigma_\beta^2)$ be two nondegenerate Gaussians on $\mathbb{R}$. Then $T(x) = \frac{\sigma_\beta}{\sigma_\alpha}(x - m_\alpha) + m_\beta$ satisfies $T_\sharp\alpha = \beta$. It is the derivative of the convex function $\varphi(x) = \frac{\sigma_\beta}{2\sigma_\alpha}(x-m_\alpha)^2 + m_\beta x$,

$$\mathcal{W}_2(\alpha,\beta)^2 = \int_{\mathbb{R}} \left(\frac{\sigma_\beta}{\sigma_\alpha}(x-m_\alpha) + m_\beta - x\right)^2 \mathrm{d}\alpha(x) = (m_\alpha - m_\beta)^2 + (\sigma_\alpha - \sigma_\beta)^2.$$

The formula extends by continuity to Dirac masses, although the affine Monge map itself only pushes a Dirac source to another Dirac. Thus the OT geometry of one-dimensional Gaussians is the Euclidean geometry of the half-plane $(m,\sigma) \in \mathbb{R}\times\mathbb{R}_+$.

**Multivariate Gaussians.**

$$\alpha = \mathcal{N}(\mathbf{m}_\alpha, \mathbf{\Sigma}_\alpha), \qquad \beta = \mathcal{N}(\mathbf{m}_\beta, \mathbf{\Sigma}_\beta), \qquad T(x) = \mathbf{m}_\beta + A(x - \mathbf{m}_\alpha), \tag{2.16}$$

**Proposition 2.33** (Affine push-forward of Gaussians)**.** *One has $T_\sharp\alpha = \beta$ if and only if*

$$A\mathbf{\Sigma}_\alpha A^\top = \mathbf{\Sigma}_\beta. \tag{2.17}$$

*Proof.* An affine function maps a Gaussian to a Gaussian, so the law of $T(X)$ is determined by its mean and covariance. If $X \sim \alpha$ and $Y = T(X)$, then $\mathbb{E}(Y) = \mathbf{m}_\beta + A\mathbb{E}(X - \mathbf{m}_\alpha) = \mathbf{m}_\beta$, and $\mathbb{E}((Y - \mathbf{m}_\beta)(Y - \mathbf{m}_\beta)^\top) = A\mathbb{E}((X - \mathbf{m}_\alpha)(X - \mathbf{m}_\alpha)^\top)A^\top = A\boldsymbol{\Sigma}_\alpha A^\top$. Thus $A\boldsymbol{\Sigma}_\alpha A^\top = \boldsymbol{\Sigma}_\beta$ is necessary and sufficient for $Y$ to have the same mean and covariance as $\beta$. $\square$

**Definition 2.34** (Bures metric)**.** For positive semidefinite covariance matrices $\Sigma$ and $\Lambda$, the Bures metric is

$$\mathcal{B}(\Sigma, \Lambda)^2 := \operatorname{tr}\left(\Sigma + \Lambda - 2(\Sigma^{1/2}\Lambda\Sigma^{1/2})^{1/2}\right). \tag{2.18}$$

**Proposition 2.35** (Gaussian $\mathcal{W}_2$ formula and Bures covariance term)**.** *Assume that $\boldsymbol{\Sigma}_\alpha$ and $\boldsymbol{\Sigma}_\beta$ are positive definite. The unique symmetric positive-definite solution of $A\boldsymbol{\Sigma}_\alpha A = \boldsymbol{\Sigma}_\beta$ is*

$$A = \boldsymbol{\Sigma}_\alpha^{-1/2}\left(\boldsymbol{\Sigma}_\alpha^{1/2}\boldsymbol{\Sigma}_\beta\boldsymbol{\Sigma}_\alpha^{1/2}\right)^{1/2}\boldsymbol{\Sigma}_\alpha^{-1/2}. \tag{2.19}$$

*The affine map $T(x) = \mathbf{m}_\beta + A(x - \mathbf{m}_\alpha)$ is the optimal quadratic-cost transport from $\mathcal{N}(\mathbf{m}_\alpha, \boldsymbol{\Sigma}_\alpha)$ to $\mathcal{N}(\mathbf{m}_\beta, \boldsymbol{\Sigma}_\beta)$, and*

$$\mathcal{W}_2(\alpha, \beta)^2 = \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \mathcal{B}(\boldsymbol{\Sigma}_\alpha, \boldsymbol{\Sigma}_\beta)^2, \tag{2.20}$$

*where $\mathcal{B}$ is the Bures metric of Definition 2.34.*

*Proof.* Multiplying $A\boldsymbol{\Sigma}_\alpha A = \boldsymbol{\Sigma}_\beta$ on the left and right by $\boldsymbol{\Sigma}_\alpha^{1/2}$ gives $(\boldsymbol{\Sigma}_\alpha^{1/2}A\boldsymbol{\Sigma}_\alpha^{1/2})^2 = \boldsymbol{\Sigma}_\alpha^{1/2}\boldsymbol{\Sigma}_\beta\boldsymbol{\Sigma}_\alpha^{1/2}$. Since $A$ is symmetric positive, $\boldsymbol{\Sigma}_\alpha^{1/2}A\boldsymbol{\Sigma}_\alpha^{1/2}$ is symmetric positive and is therefore the unique positive square root of the right-hand side. Conversely, the matrix in (2.19) is symmetric positive and satisfies the covariance equation.
By Proposition 2.33, this affine map pushes $\alpha$ to $\beta$. It is the gradient of a convex quadratic potential, so Brenier's theorem implies optimality. If $X \sim \alpha$, then

$$\begin{aligned}
\mathbb{E}\|X - T(X)\|^2 &= \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \mathbb{E}\|(I - A)(X - \mathbf{m}_\alpha)\|^2 \\
&= \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \operatorname{tr}((I - A)\boldsymbol{\Sigma}_\alpha(I - A)^\top) \\
&= \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \operatorname{tr}(\boldsymbol{\Sigma}_\alpha) + \operatorname{tr}(A\boldsymbol{\Sigma}_\alpha A) - 2\operatorname{tr}(A\boldsymbol{\Sigma}_\alpha) \\
&= \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \operatorname{tr}(\boldsymbol{\Sigma}_\alpha) + \operatorname{tr}(\boldsymbol{\Sigma}_\beta) - 2\operatorname{tr}((\boldsymbol{\Sigma}_\alpha^{1/2}\boldsymbol{\Sigma}_\beta\boldsymbol{\Sigma}_\alpha^{1/2})^{1/2}),
\end{aligned}$$

which is the desired expression. The formula for singular covariance matrices follows by adding $\eta I$ and letting $\eta \downarrow 0$. $\square$

The same formula can be read without first restricting to Gaussian measures. Let $\Phi_2$ keep only raw second moments.
**Proposition 2.36** (Bures metric as a second-moment quotient)**.** *For $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$, set its raw second-moment matrix to be $\Phi_2(\alpha) := \int_{\mathbb{R}^d} uu^\top \mathrm{d}\alpha(u) \in \mathbb{S}_+^d$. For $\Sigma, \Lambda \in \mathbb{S}_+^d$, $\min_{\Phi_2(\alpha)=\Sigma,\ \Phi_2(\nu)=\Lambda} \mathcal{W}_2(\alpha, \nu)^2 = \mathcal{B}(\Sigma, \Lambda)^2$. In other words, the Bures metric is the Wasserstein quotient distance induced on positive semidefinite matrices by identifying all probability measures with the same raw second-moment matrix.*

*Proof.* Assume first that $\Sigma$ and $\Lambda$ are positive definite. Fix two admissible laws $\alpha, \nu$ and an arbitrary coupling $\pi$ between them, and write $K := \int_{\mathbb{R}^d \times \mathbb{R}^d} uv^\top \mathrm{d}\pi(u, v)$. Then $\int \|u - v\|^2 \mathrm{d}\pi(u, v) = \operatorname{tr}(\Sigma) + \operatorname{tr}(\Lambda) - 2\operatorname{tr}(K)$. The block second-moment matrix

$$\int \begin{pmatrix} u \\ v \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}^\top \mathrm{d}\pi(u, v) = \begin{pmatrix} \Sigma & K \\ K^\top & \Lambda \end{pmatrix}$$

is positive semidefinite. The Schur complement gives $R := \Sigma^{-1/2}K\Lambda^{-1/2}, \qquad RR^\top \preceq I$, so $K = \Sigma^{1/2}R\Lambda^{1/2}$ with $\|R\|_{\mathrm{op}} \leqslant 1$. By duality between the nuclear and operator norms,

$$\operatorname{tr}(K) \leqslant \sup_{\|R\|_{\mathrm{op}} \leqslant 1} \operatorname{tr}(\Sigma^{1/2}R\Lambda^{1/2}) = \|\Lambda^{1/2}\Sigma^{1/2}\|_* = \operatorname{tr}\left((\Sigma^{1/2}\Lambda\Sigma^{1/2})^{1/2}\right).$$

For singular matrices, fix $\eta > 0$. Adding $\eta I$ to the two diagonal blocks preserves block positivity, so the same Schur-complement argument gives $\operatorname{tr}(K) \leqslant \operatorname{tr}\left(((\Sigma + \eta I)^{1/2}(\Lambda + \eta I)(\Sigma + \eta I)^{1/2})^{1/2}\right)$. Letting $\eta \downarrow 0$ and using continuity of the matrix square root gives the same trace bound with $\Sigma, \Lambda$. Hence every admissible coupling has cost at least $\mathcal{B}(\Sigma, \Lambda)^2$.
Conversely, choose the centered Gaussian laws $\mathcal{N}(0, \Sigma)$ and $\mathcal{N}(0, \Lambda)$. They satisfy the two second-moment constraints, and Proposition 2.35 gives $\mathcal{W}_2(\mathcal{N}(0, \Sigma), \mathcal{N}(0, \Lambda))^2 = \mathcal{B}(\Sigma, \Lambda)^2$, again by continuity in the singular case. This admissible pair attains the lower bound, proving the claim. $\square$

$$\Sigma = \begin{pmatrix} a & c \\ c & b \end{pmatrix}, \qquad t = \frac{a + b}{\sqrt{2}}, \qquad u = \frac{a - b}{\sqrt{2}}, \qquad v = \sqrt{2}\,c,$$

$a = \frac{t+u}{\sqrt{2}}, \qquad b = \frac{t-u}{\sqrt{2}}, \qquad c = \frac{v}{\sqrt{2}}$. $t^2 - u^2 - v^2 = 2(ab - c^2) = 2\det(\Sigma)$. $t \geqslant \sqrt{u^2 + v^2}$. Thus the cone of $2 \times 2$ covariance matrices is the Lorentz, or ice-cream, cone. The Bures distance is not the ambient Euclidean distance in these coordinates: on the positive definite interior it is the geodesic distance of a smooth nonlinear geometry, whose geodesics are the covariance parts of Gaussian optimal transport rather than the straight chords inherited from $\mathbb{R}^3$.
**Proposition 2.37** (Metric and convexity properties of the Bures term)**.** *The function $\mathcal{B}$ is a distance on positive semidefinite covariance matrices. Moreover, $\mathcal{B}^2$ is jointly convex: for $t \in [0, 1]$, $\mathcal{B}^2((1 - t)\Sigma_0 + t\Sigma_1, (1 - t)\Lambda_0 + t\Lambda_1) \leqslant (1 - t)\mathcal{B}^2(\Sigma_0, \Lambda_0) + t\mathcal{B}^2(\Sigma_1, \Lambda_1)$.*

*Proof.* The key identity is the Procrustes representation $\mathcal{B}^2(\Sigma,\Lambda)=\min_{Q^\top Q=I}\|\Sigma^{1/2}-\Lambda^{1/2}Q\|_F^2$. Indeed, expanding the square gives $\operatorname{tr}\Sigma+\operatorname{tr}\Lambda-2\max_{Q^\top Q=I}\operatorname{tr}(\Sigma^{1/2}Q^\top\Lambda^{1/2})$, and the orthogonal Procrustes formula identifies the maximum with $\operatorname{tr}((\Sigma^{1/2}\Lambda\Sigma^{1/2})^{1/2})$. Symmetry, positivity and separation follow immediately from this representation. The triangle inequality follows by choosing two almost optimal orthogonal matrices $Q_1, Q_2$ and writing $\|\Sigma^{1/2}-\Gamma^{1/2}Q_2Q_1\|_F \leqslant \|\Sigma^{1/2}-\Lambda^{1/2}Q_1\|_F+\|\Lambda^{1/2}-\Gamma^{1/2}Q_2\|_F$. Letting the two choices become optimal proves the metric property.
For convexity, use the equivalent factor formulation $\mathcal{B}^2(\Sigma,\Lambda)=\min_{UU^\top=\Sigma,\,VV^\top=\Lambda}\|U-V\|_F^2$. Choose nearly optimal factors $(U_0,V_0)$ and $(U_1,V_1)$ for the two pairs, and define block factors $U_t=[\sqrt{1-t}\,U_0,\sqrt{t}\,U_1]$, $\quad V_t=[\sqrt{1-t}\,V_0,\sqrt{t}\,V_1]$. Then $U_tU_t^\top=(1-t)\Sigma_0+t\Sigma_1$ and $V_tV_t^\top=(1-t)\Lambda_0+t\Lambda_1$, while $\|U_t-V_t\|_F^2=(1-t)\|U_0-V_0\|_F^2+t\|U_1-V_1\|_F^2$. Taking the infimum over the initial factors proves joint convexity. □

## 3 Kantorovich Relaxation

### 3.1 Discrete Relaxation

The discrete relaxation replaces deterministic assignments by mass-splitting matrices.

**Couplings and product plans.** $\alpha=\sum_i \mathrm{a}_i\delta_{x_i},\qquad \beta=\sum_j \mathrm{b}_j\delta_{y_j}$.

**Definition 3.1** (Discrete couplings and mass conservation)**.** Admissible couplings are only constrained to satisfy conservation of mass:

$$\mathrm{U}(\mathrm{a},\mathrm{b}) := \left\{\mathrm{P}\in\mathbb{R}_+^{n\times m}\ ;\ \mathrm{P}\mathbb{1}_m=\mathrm{a}\quad\text{and}\quad \mathrm{P}^\top\mathbb{1}_n=\mathrm{b}\right\}. \tag{3.1}$$

Equivalently,

$$\mathrm{P}\mathbb{1}_m=\left(\sum_j \mathrm{P}_{i,j}\right)_i\in\mathbb{R}^n,\qquad \mathrm{P}^\top\mathbb{1}_n=\left(\sum_i \mathrm{P}_{i,j}\right)_j\in\mathbb{R}^m.$$

**Definition 3.2** (Discrete product coupling)**.** Given weights $\mathrm{a}\in\Sigma_n$ and $\mathrm{b}\in\Sigma_m$, the discrete product, or trivial, coupling is $(\mathrm{a}\otimes\mathrm{b})_{i,j}:=\mathrm{a}_i\mathrm{b}_j$. It belongs to $\mathrm{U}(\mathrm{a},\mathrm{b})$ and corresponds to choosing the source and target labels independently.

**Proposition 3.3** (Discrete product optimality is degenerate)**.** *Assume that the zero-mass rows and columns have been removed, so that* $\mathrm{a}_i>0$ *and* $\mathrm{b}_j>0$*, and let* $\mathrm{C}$ *be a finite cost matrix. The product plan* $\mathrm{a}\otimes\mathrm{b}$ *minimizes* $\mathrm{P}\mapsto\langle\mathrm{C},\mathrm{P}\rangle$ *over* $\mathrm{U}(\mathrm{a},\mathrm{b})$ *if and only if every coupling* $\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})$ *minimizes it.*

*Proof.* The reverse implication is immediate. Assume conversely that $\mathrm{a}\otimes\mathrm{b}$ is optimal and let $\mathrm{Q}\in\mathrm{U}(\mathrm{a},\mathrm{b})$ be arbitrary. Since all entries of $\mathrm{a}\otimes\mathrm{b}$ are positive, there exists $t>0$ small enough that $\mathrm{R}:=(1+t)(\mathrm{a}\otimes\mathrm{b})-t\mathrm{Q}$ has nonnegative entries. Its row and column sums are $\mathrm{R}\mathbb{1}_m=(1+t)\mathrm{a}-t\mathrm{a}=\mathrm{a}$, $\quad \mathrm{R}^\top\mathbb{1}_n=(1+t)\mathrm{b}-t\mathrm{b}=\mathrm{b}$, so $\mathrm{R}\in\mathrm{U}(\mathrm{a},\mathrm{b})$. Moreover, $\mathrm{a}\otimes\mathrm{b}=\frac{1}{1+t}\mathrm{R}+\frac{t}{1+t}\mathrm{Q}$. By optimality of $\mathrm{a}\otimes\mathrm{b}$, both $\langle\mathrm{C},\mathrm{R}\rangle$ and $\langle\mathrm{C},\mathrm{Q}\rangle$ are at least $\langle\mathrm{C},\mathrm{a}\otimes\mathrm{b}\rangle$. Taking the scalar product of the convex-combination identity with $\mathrm{C}$ gives $\langle\mathrm{C},\mathrm{a}\otimes\mathrm{b}\rangle=\frac{1}{1+t}\langle\mathrm{C},\mathrm{R}\rangle+\frac{t}{1+t}\langle\mathrm{C},\mathrm{Q}\rangle$. A convex average of two numbers not smaller than $\langle\mathrm{C},\mathrm{a}\otimes\mathrm{b}\rangle$ can equal $\langle\mathrm{C},\mathrm{a}\otimes\mathrm{b}\rangle$ only if both numbers are equal to it. Hence $\langle\mathrm{C},\mathrm{Q}\rangle=\langle\mathrm{C},\mathrm{a}\otimes\mathrm{b}\rangle$, and $\mathrm{Q}$ is optimal. Since $\mathrm{Q}$ was arbitrary, all couplings are optimal. □

Thus the product plan is mainly a feasibility witness. Except in the degenerate situation where the linear cost is constant on the whole transportation polytope, it is not expected to solve optimal transport.

**Linear-programming structure.**

$$\mathrm{L}_{\mathrm{C}}(\mathrm{a},\mathrm{b}) := \min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})}\langle\mathrm{C},\mathrm{P}\rangle := \min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})}\sum_{i,j}\mathrm{C}_{i,j}\mathrm{P}_{i,j}. \tag{3.2}$$

**Proposition 3.4** (Sparse optimal plans)**.** *Assume* $\mathrm{a}_i>0$, $\mathrm{b}_j>0$ *and* $\sum_i\mathrm{a}_i=\sum_j\mathrm{b}_j=1$*. The linear program* (3.2) *admits an optimal coupling with at most* $n+m-1$ *nonzero entries.*

*Proof.* The transportation polytope is compact, so a linear objective attains its minimum at an extreme point. The row and column marginal equations have rank $n+m-1$: the only linear redundancy is that both sets of constraints impose the same total mass. The support-graph argument below is the combinatorial form of this basic-feasible-variable count.
Let $\mathrm{P}$ be an extreme point and let $E=\{(i,j):\mathrm{P}_{ij}>0\}$ be its support graph on the bipartite vertex set $\{1,\dots,n\}\cup\{1,\dots,m\}$. If this graph contains a cycle, orient the cycle and put alternating signs $+1,-1$ on its edges, obtaining a nonzero matrix $H$ supported on $E$ with zero row and column sums. For sufficiently small $t>0$, both $\mathrm{P}+tH$ and $\mathrm{P}-tH$ are nonnegative couplings, and $\mathrm{P}$ is their midpoint, contradicting extremality. Thus the support graph is a forest. Since a forest on $n+m$ vertices has at most $n+m-1$ edges, the claim follows. □

**Proposition 3.5** (North-west corner feasible plan)**.** *Let* $\mathrm{a}\in\mathbb{R}_+^n$ *and* $\mathrm{b}\in\mathbb{R}_+^m$ *have the same positive total mass. Starting from* $(i,j)=(1,1)$ *with residual masses* $r_i=\mathrm{a}_i$ *and* $s_j=\mathrm{b}_j$*, skip zero residuals, set* $\mathrm{P}_{ij}=\min(r_i,s_j)$*, subtract this value from both residuals, and advance every index whose residual has become zero. Repeat until all residual masses are exhausted.*

*Proof.* All assignments are nonnegative. At each step, the mass placed in entry $(i,j)$ is subtracted from exactly one current row residual and one current column residual, so no row or column can receive more mass than prescribed. Conversely, an index is advanced only when its residual has been fully filled. When the algorithm stops, the total assigned mass is $\sum_i a_i=\sum_j b_j$, hence all row and column sums are exactly a and b.
Each positive assignment exhausts at least one current row or one current column. Before the final assignment, at most $n-1$ row advances and $m-1$ column advances can occur without terminating the construction. Hence the number of positive entries is at most $(n-1)+(m-1)+1=n+m-1$. For acyclicity, view the positive support as a bipartite graph. Once a row or column index is advanced, it never appears again, so each new positive edge either starts a new component or attaches at least one new vertex to the component currently being swept. No edge is ever added between two old vertices of the same component, so no cycle can be created. □

### One-dimensional cases.

**Proposition 3.6** (One-dimensional weighted sweep)**.** *Let $x_1 \leqslant \cdots \leqslant x_n$ and $y_1 \leqslant \cdots \leqslant y_m$ be points on the line, and let $c(x,y) = h(x-y)$ with $h$ convex. The north-west corner plan between the sorted weighted atoms is optimal for (3.2).*

*Proof.* The north-west plan is monotone: if $i < i'$ and $j > j'$, it cannot put positive mass on both $(i,j)$ and $(i',j')$, because the sweep exhausts rows and columns in increasing order. Conversely, any feasible plan with a crossing pair of positive entries can be improved by moving a small mass $\eta$ from $(i,j)$ and $(i',j')$ to $(i,j')$ and $(i',j)$. The two marginals are unchanged, and convexity of $h$ gives $h(x_i - y_j) + h(x_{i'} - y_{j'}) \geqslant h(x_i - y_{j'}) + h(x_{i'} - y_j)$ for $i < i'$ and $j' < j$, with strict inequality for strictly convex $h$ and distinct points. Repeating this uncrossing procedure until no crossing remains yields a monotone optimal plan. There is only one monotone feasible plan with the prescribed sorted marginals, namely the sweep plan: it pairs the leftmost remaining source mass with the leftmost remaining target mass at every step. Sorting costs $O(n\log n + m\log m)$ and the sweep uses at most $n+m-1$ assignments. □

### Permutation matrices as couplings.

**Definition 3.7** (Permutation matrices)**.** For a permutation $\sigma \in \mathrm{Perm}(n)$, its permutation matrix $P_\sigma$ is

$$(P_\sigma)_{i,j} = \begin{cases} 1 & \text{if} \quad j = \sigma(i) \\ 0 & \text{otherwise.} \end{cases}$$

The set of all permutation matrices is $\mathcal{P}_n^{\mathrm{perm}} := \{P_\sigma \; ; \; \sigma \in \mathrm{Perm}(n)\}$.

$\langle \mathrm{C},\, P_\sigma/n\rangle = \frac{1}{n}\sum_{i=1}^n C_{i,\sigma(i)}$.

**Definition 3.8** (Birkhoff polytope)**.** The Birkhoff polytope is the convex set of bistochastic matrices $\mathcal{B}_n := \left\{P \in \mathbb{R}_+^{n\times n} \; ; \; P\mathbb{1}_n = \mathbb{1}_n \quad \text{and} \quad P^\top \mathbb{1}_n = \mathbb{1}_n\right\}$.

Then $\mathrm{U}(\mathbb{1}_n/n, \mathbb{1}_n/n) = \mathcal{B}_n/n$, and permutation couplings are included in this convex relaxation. More precisely, $\mathcal{P}_n^{\mathrm{perm}} = \mathcal{B}_n \cap \{0,1\}^{n\times n} \subset \mathcal{B}_n$, $\min_{P\in\mathcal{B}_n} \langle \mathrm{C},\, P\rangle \leqslant \min_{P\in\mathcal{P}_n^{\mathrm{perm}}} \langle \mathrm{C},\, P\rangle$.

**Definition 3.9** (Extreme points)**.** For a compact convex set $\mathcal{C}$ in a finite-dimensional vector space, $\mathrm{Extr}(\mathcal{C}) := \{x \in \mathcal{C} \; ; \; x = (y+z)/2,\; y,z \in \mathcal{C} \Rightarrow y = z = x\}$.

**Proposition 3.10** (Existence of extreme points)**.** *If $\mathcal{C}$ is a non-empty compact convex subset of a finite-dimensional vector space, then* $\mathrm{Extr}(\mathcal{C})$ *is non-empty.*

*Proof.* Among all non-empty faces of $\mathcal{C}$, choose one of minimal affine dimension. If this face contained two distinct points, maximizing a linear functional that is not constant on the face would produce a non-empty proper exposed subface, contradicting minimality. Hence the minimal face is a singleton, and its point is extreme. □

**Proposition 3.11** (Linear programs have extreme minimizers)**.** *Let $\mathcal{C}$ be non-empty and compact. For every linear form $\ell$,* $\mathrm{Extr}(\mathcal{C}) \cap \mathrm{argmin}_{x\in\mathcal{C}}\, \ell(x) \neq \emptyset$.

*Proof.* The set $S = \mathrm{argmin}_{x\in\mathcal{C}}\, \ell(x)$ is non-empty, compact and convex. By Proposition 3.10, it has an extreme point $x$. If $x = (y+z)/2$ with $y, z \in \mathcal{C}$, then by linearity and optimality of $x$, both $y$ and $z$ also minimize $\ell$ on $\mathcal{C}$, hence $y, z \in S$. Since $x$ is extreme in $S$, $y = z = x$. Thus $x$ is extreme in $\mathcal{C}$. □

**Theorem 3.12** (Birkhoff–von Neumann)**.** *The extreme points of $\mathcal{B}_n$ are exactly the permutation matrices.*

*Proof.* We first prove that permutation matrices are extreme. Let $P_\sigma \in \mathcal{P}_n^{\mathrm{perm}}$ and assume that $P_\sigma = \frac{Q+R}{2}$ with $Q, R \in \mathcal{B}_n$. Every bistochastic matrix has entries in $[0,1]$. Since the only extreme points of $[0,1]$ are 0 and 1, each entry of $P_\sigma$ fixes the corresponding entries of $Q$ and $R$: if $(P_\sigma)_{ij} = 0$, then $Q_{ij} = R_{ij} = 0$, while if $(P_\sigma)_{ij} = 1$, then $Q_{ij} = R_{ij} = 1$. Hence $Q = R = P_\sigma$, so $P_\sigma$ is extreme.
Pick $P \in \mathcal{B}_n \setminus \mathcal{P}_n^{\mathrm{perm}}$. Since an integral bistochastic matrix is necessarily a permutation matrix, $P$ has at least one fractional entry. We shall split $P = \frac{Q+R}{2}$ with $Q, R \in \mathcal{B}_n$ and $Q \neq R$, proving that $P$ is not extreme.
Associate with $P$ the bipartite graph whose left vertices are the rows, whose right vertices are the columns, and whose edges are the fractional entries $0 < P_{ij} < 1$. An entry equal to 1 uses the whole mass of its row and column, so it is isolated in the positive support and does not appear in this fractional graph. If a left vertex $i$ is incident to a fractional edge $(i, j_1)$, then it must be incident to at least one other fractional edge. Indeed, the row sum is one; after the contribution $P_{i,j_1} \in (0,1)$, a positive amount $1 - P_{i,j_1}$ remains in the same row, and it cannot be carried by an entry equal to 1. The same argument applies to right vertices, using the column sums. Thus every non-isolated vertex of the fractional graph has degree at least two.
Starting from any fractional edge, one may therefore walk through adjacent fractional edges without immediately backtracking and without getting stuck. Since the graph is finite, some vertex is eventually visited twice; the portion of the walk between the two visits contains a cycle. Choose a shortest such cycle and write it in alternating form $(i_1, j_1, i_2, j_2, \ldots, i_p, j_p)$, $\quad i_{p+1} = i_1$, where both $(i_s, j_s)$ and $(i_{s+1}, j_s)$ are fractional for every $s$. The minimality of the cycle implies that the vertices $i_s$ are all distinct and that the vertices $j_s$ are all distinct. In particular, $0 < P_{i_s,j_s} < 1 \quad \text{and} \quad 0 < P_{i_{s+1},j_s} < 1$. Define $\varepsilon := \min_{1\leqslant s\leqslant p}\{P_{i_s,j_s},\, P_{i_{s+1},j_s},\, 1 - P_{i_s,j_s},\, 1 - P_{i_{s+1},j_s}\}$. All these numbers are positive, so $\varepsilon > 0$. Split the cycle edges into the two alternating families $A := \{(i_s, j_s)\}_{s=1}^p$, $\quad B := \{(i_{s+1}, j_s)\}_{s=1}^p$.

$$Q_{ij} := \begin{cases} P_{ij}, & (i,j) \notin A \cup B, \\ P_{ij} + \varepsilon/2, & (i,j) \in A, \\ P_{ij} - \varepsilon/2, & (i,j) \in B, \end{cases} \qquad R_{ij} := \begin{cases} P_{ij}, & (i,j) \notin A \cup B, \\ P_{ij} - \varepsilon/2, & (i,j) \in A, \\ P_{ij} + \varepsilon/2, & (i,j) \in B. \end{cases}$$

By the definition of $\varepsilon$, all modified entries stay in $[0,1]$, so $Q$ and $R$ are nonnegative. Each row vertex $i_s$ of the cycle is incident to exactly one edge of $A$ and one edge of $B$; the $+\varepsilon/2$ and $-\varepsilon/2$ perturbations therefore cancel in that row. The same cancellation holds in each column vertex $j_s$, and all other rows and columns are unchanged. $Q\mathbb{1}_n = R\mathbb{1}_n = \mathbb{1}_n$, $\quad Q^\top\mathbb{1}_n = R^\top\mathbb{1}_n = \mathbb{1}_n$, so $Q, R \in \mathcal{B}_n$. Finally, $Q \neq R$ because $\varepsilon > 0$ and the cycle is non-empty, while by construction $P = (Q+R)/2$. Thus $P$ is not extreme. □

**Corollary 3.13** (Kantorovich for matching)**.** *If $m = n$ and $\mathrm{a} = \mathrm{b} = \mathbb{1}_n/n$, then the discrete Kantorovich problem (3.2) admits an optimal solution of the form $P_\sigma/n$. The associated permutation $\sigma$ solves the assignment problem of Section 1.1.*

*Proof.* The feasible set is $\mathcal{B}_n/n$. By Proposition 3.11, the linear objective has an optimal extreme point. Since scaling preserves extreme points and Theorem 3.12 identifies the extreme points of $\mathcal{B}_n$, this optimizer is $P_\sigma/n$ for some permutation $\sigma$. Its cost is exactly $n^{-1}\sum_i C_{i,\sigma(i)}$, so $\sigma$ is an optimal assignment. □

## 3.2 Linear-Programming Algorithms

This compact subsection locates discrete OT inside network and transportation linear programs.

### Transportation simplex and network simplex.

### Interior-point methods.

$$\mathrm{P}_\varepsilon := \underset{\substack{\mathrm{P}\mathbb{1}_m=\mathrm{a},\ \mathrm{P}^\top\mathbb{1}_n=\mathrm{b} \\ \mathrm{P}_{ij}>0}}{\operatorname{argmin}} \quad \langle \mathrm{C}, \mathrm{P}\rangle - \varepsilon \sum_{i,j} \log \mathrm{P}_{ij}, \tag{3.3}$$

where $\varepsilon > 0$ is decreased along the algorithm. The barrier is singular at the boundary, so each iterate stays strictly inside the transportation polytope; as $\varepsilon \downarrow 0$, the central path approaches the set of LP minimizers.

## 3.3 Relaxation for Arbitrary Measures

This compact subsection passes from finite transport matrices to general couplings.

### Continuous couplings.

**Definition 3.14** (Marginals of a joint measure)**.** Let $\pi \in \mathcal{M}_+^1(\mathcal{X}\times\mathcal{Y})$ and let $P_\mathcal{X}(x,y) = x$, $P_\mathcal{Y}(x,y) = y$ be the coordinate projections. The marginals of $\pi$ are $\pi_1 := (P_\mathcal{X})_\sharp\pi \in \mathcal{M}_+^1(\mathcal{X})$, $\quad \pi_2 := (P_\mathcal{Y})_\sharp\pi \in \mathcal{M}_+^1(\mathcal{Y})$. Equivalently, for all bounded continuous test functions $f$ on $\mathcal{X}$ and $g$ on $\mathcal{Y}$, $\int_{\mathcal{X}\times\mathcal{Y}} f(x)\mathrm{d}\pi(x,y) = \int_\mathcal{X} f\mathrm{d}\pi_1$, $\quad \int_{\mathcal{X}\times\mathcal{Y}} g(y)\mathrm{d}\pi(x,y) = \int_\mathcal{Y} g\mathrm{d}\pi_2$.

$\int_\mathcal{Y} \mathrm{d}\pi(x,y) = \mathrm{d}\alpha(x)$, $\quad \int_\mathcal{X} \mathrm{d}\pi(x,y) = \mathrm{d}\beta(y)$, which is made rigorous by Definition 3.14, or equivalently by the identities $\pi(A\times\mathcal{Y}) = \alpha(A)$ and $\pi(\mathcal{X}\times B) = \beta(B)$ for measurable sets $A \subset \mathcal{X}$ and $B \subset \mathcal{Y}$.

**Definition 3.15** (Couplings)**.** Given $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ and $\beta \in \mathcal{M}_+^1(\mathcal{Y})$, the set of couplings between $\alpha$ and $\beta$ is

$$\mathcal{U}(\alpha,\beta) := \left\{\pi \in \mathcal{M}_+^1(\mathcal{X}\times\mathcal{Y})\ ;\ \pi_1 = \alpha \quad \text{and} \quad \pi_2 = \beta\right\}. \tag{3.4}$$

This is the continuous analogue of the transportation polytope (3.1).

**Definition 3.16** (Tensor product and trivial coupling)**.** Given $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ and $\beta \in \mathcal{M}_+^1(\mathcal{Y})$, the tensor product coupling $\alpha\otimes\beta$ is the probability measure on $\mathcal{X}\times\mathcal{Y}$ defined by

$$\int_{\mathcal{X}\times\mathcal{Y}} h(x,y)\mathrm{d}(\alpha\otimes\beta)(x,y) = \int_\mathcal{X}\left(\int_\mathcal{Y} h(x,y)\mathrm{d}\beta(y)\right)\mathrm{d}\alpha(x)$$

for every bounded measurable $h$. It is also called the trivial coupling because it makes the two coordinates independent.

$$\int_{\mathcal{X}\times\mathcal{Y}} f(x)\mathrm{d}(\alpha\otimes\beta)(x,y) = \left(\int_\mathcal{X} f(x)\mathrm{d}\alpha(x)\right)\left(\int_\mathcal{Y}\mathrm{d}\beta(y)\right) = \int f\mathrm{d}\alpha,$$

**Proposition 3.17** (Product optimality is degenerate)**.** *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces and that $c \in \mathcal{C}(\mathcal{X}\times\mathcal{Y})$. $\alpha\otimes\beta \in \underset{\pi\in\mathcal{U}(\alpha,\beta)}{\operatorname{argmin}} \int c\,\mathrm{d}\pi$, every coupling in $\mathcal{U}(\alpha,\beta)$ is optimal. They are also equivalent to the additive decomposition of the cost on the product support, $c(x,y) = u(x) + v(y)$.*

*Proof.* If every coupling is optimal, then $\alpha\otimes\beta$ is optimal. Conversely, assume that $\alpha\otimes\beta$ is optimal. We first show that, for every $x_0, x_1 \in \operatorname{supp}(\alpha)$ and $y_0, y_1 \in \operatorname{supp}(\beta)$, $c(x_0,y_0) + c(x_1,y_1) = c(x_0,y_1) + c(x_1,y_0)$. Indeed, if this equality failed, after exchanging $y_0$ and $y_1$ if necessary one would have a strict inequality $c(x_0,y_0) + c(x_1,y_1) > c(x_0,y_1) + c(x_1,y_0)$. By continuity, the strict inequality persists with a uniform margin on small neighborhoods $U_0, U_1$ of $x_0, x_1$ and $V_0, V_1$ of $y_0, y_1$, chosen disjoint within each pair. Since the four points lie in the supports, these neighborhoods have positive marginal mass. Denote by $\alpha_i$ and $\beta_i$ the normalized restrictions of $\alpha$ to $U_i$ and of $\beta$ to $V_i$, and choose $0 < \lambda \leqslant \min\{\alpha(U_0)\beta(V_0), \alpha(U_1)\beta(V_1)\}$. The exchanged measure $\tilde\pi = \alpha\otimes\beta - \lambda\,\alpha_0\otimes\beta_0 - \lambda\,\alpha_1\otimes\beta_1 + \lambda\,\alpha_0\otimes\beta_1 + \lambda\,\alpha_1\otimes\beta_0$ is nonnegative and has the same two marginals as $\alpha\otimes\beta$. The uniform strict inequality on the neighborhoods implies that $\int c\,\mathrm{d}\tilde\pi < \int c\,\mathrm{d}(\alpha\otimes\beta)$, contradicting optimality.
Fixing any $x_\star \in \operatorname{supp}(\alpha)$ and $y_\star \in \operatorname{supp}(\beta)$, the equality of cross differences gives, for all $(x,y) \in \operatorname{supp}(\alpha)\times\operatorname{supp}(\beta)$, $c(x,y) = c(x,y_\star) + c(x_\star,y) - c(x_\star,y_\star)$. Thus $c = u + v$ on the product support. Every coupling is concentrated on this product support, so for any $\pi \in \mathcal{U}(\alpha,\beta)$, $\int c\,\mathrm{d}\pi = \int u\,\mathrm{d}\alpha + \int v\,\mathrm{d}\beta$, which depends only on the marginals. Hence all couplings are optimal. □

The tensor product is therefore a trivial feasible coupling, not a typical optimizer. Product optimality means that the cost cannot distinguish between dependences once the marginals are fixed.
If there exists a map $T : \mathcal{X} \to \mathcal{Y}$ such that $T_\sharp\alpha = \beta$, then the Monge map induces the graph coupling $\pi = (\mathrm{Id}, T)_\sharp\alpha \in \mathcal{U}(\alpha,\beta)$, characterized by
$\int_{\mathcal{X}\times\mathcal{Y}} h(x,y)\mathrm{d}\pi(x,y) = \int_\mathcal{X} h(x,T(x))\mathrm{d}\alpha(x)$. Applying this identity to $h(x,y) = f(x)$ or $h(x,y) = g(y)$ gives respectively $\pi_1 = \alpha$ and $\pi_2 = \beta$. Thus graph couplings are precisely the Kantorovich representation of deterministic Monge maps.

### Continuous Kantorovich problem.

$$\mathcal{L}_c(\alpha,\beta) := \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\mathrm{d}\pi(x,y). \tag{3.5}$$

**Proposition 3.18** (Existence on compact spaces)**.** *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces and that $c \in \mathcal{C}(\mathcal{X}\times\mathcal{Y})$. Then the Kantorovich problem (3.5) admits at least one minimizer.*

*Proof.* The constraint set is non-empty because it contains the product coupling $\alpha \otimes \beta$. It is closed for weak convergence of measures because the marginal constraints are preserved under weak convergence. Since $\Omega := \mathcal{X} \times \mathcal{Y}$ is compact, Proposition 12.5, applied to $\Omega$, gives compactness of $\mathcal{P}(\Omega)$ for the Wasserstein topology. On compact spaces this topology is the weak topology by Proposition 3.37; therefore $\mathcal{U}(\alpha, \beta)$ is compact. Finally, the functional $\pi \mapsto \int c\mathrm{d}\pi$ is weakly continuous because $c$ is continuous and bounded. The minimum is thus attained. □

$\mathcal{P}_p(\mathcal{X}) := \left\{\alpha \in \mathcal{M}_+^1(\mathcal{X})\ ;\ \int d(x, x_0)^p \mathrm{d}\alpha(x) < +\infty\right\},$

**Monge–Kantorovich equivalence.**

**Corollary 3.19** (Monge–Kantorovich equivalence under Brenier)**.** *Assume that $\alpha$ is absolutely continuous with respect to Lebesgue measure and that $c(x, y) = \|x - y\|^2$. If $T$ is the Brenier map solving Monge's problem, then $\pi = (\mathrm{Id}, T)_\sharp\alpha$ is the unique optimal coupling solving the Kantorovich problem. In particular, Monge and Kantorovich costs are the same.*

*Proof.* The proof of Brenier's theorem shows that the support of any optimal Kantorovich plan is contained in the subdifferential $\partial\varphi$ of a convex function $\varphi$. When $\alpha$ has a density, $\varphi$ is differentiable $\alpha$-almost everywhere, so $\partial\varphi(x) = \{\nabla\varphi(x)\}$ for $\alpha$-almost every $x$. Thus every optimal coupling is concentrated on the graph of $T = \nabla\varphi$ and must equal $(\mathrm{Id}, T)_\sharp\alpha$. The graph coupling is feasible and optimal, and the two formulations have the same value. □

**Kantorovich solution in 1-D.**

**Theorem 3.20** (One-dimensional Kantorovich solution)**.** *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R})$ and let $h : \mathbb{R} \to \mathbb{R}$ be convex. Consider the cost $c(x, y) = h(x - y)$. Write $q_\alpha := \mathcal{C}_\alpha^{-1}$ and $q_\beta := \mathcal{C}_\beta^{-1}$, and assume that $h(q_\alpha - q_\beta) \in L^1(0, 1)$. Then the quantile coupling $\pi^\star = (q_\alpha, q_\beta)_\sharp \mathrm{Leb}_{[0,1]} = (\mathcal{C}_\alpha^{-1}, \mathcal{C}_\beta^{-1})_\sharp \mathrm{Leb}_{[0,1]}$ minimizes $\min\limits_{\pi\in\mathcal{U}(\alpha,\beta)} \int h(x - y)\mathrm{d}\pi(x, y)$. In particular, $h(t) = |t|^p$ gives the usual one-dimensional optimal coupling for every $p \geqslant 1$.*

*Proof.* The push-forward statement $\pi^\star \in \mathcal{U}(\alpha, \beta)$ follows from Proposition 2.26.
The key point is the one-dimensional uncrossing inequality. If $x < x'$ and $y > y'$, set $a = x - y$, $\delta = x' - x > 0$ and $\eta = y - y' > 0$. Convexity of $h$ implies that increments are monotone, hence $h(a + \eta) - h(a) \leqslant h(a + \delta + \eta) - h(a + \delta)$, which is exactly $h(x - y) + h(x' - y') \geqslant h(x - y') + h(x' - y)$. Thus removing a crossing never increases the cost. For a finite transport matrix on two ordered grids, if $i < i'$ and $j > j'$ carry crossed masses $P_{ij}$ and $P_{i'j'}$, one may move $\theta = \min(P_{ij}, P_{i'j'})$ units of mass from the crossed entries $(i, j)$, $(i', j')$ to the uncrossed entries $(i, j')$, $(i', j)$. The row and column sums are unchanged, and the preceding inequality shows that the cost does not increase. Repeating this elementary uncrossing step yields an ordered plan; on an ordered uniform quantile grid, this is the diagonal plan.
For general measures, apply the same argument in quantile coordinates. Let $\pi \in \mathcal{U}(\alpha, \beta)$. Let $U_\alpha$ and $U_\beta$ be uniform random variables on $(0, 1)$, and denote by $K_\alpha(x, \mathrm{d}r)$ and $K_\beta(y, \mathrm{d}s)$ regular conditional laws of $U_\alpha$ given $q_\alpha(U_\alpha) = x$ and of $U_\beta$ given $q_\beta(U_\beta) = y$. Given $(X, Y) \sim \pi$, draw conditionally $R \sim K_\alpha(X, \cdot)$ and $S \sim K_\beta(Y, \cdot)$. The law $\gamma$ of $(R, S)$ has uniform marginals and satisfies $\pi = (q_\alpha, q_\beta)_\sharp\gamma$. Approximate $\gamma$ by transport matrices on uniform partitions of $(0, 1)$. The finite uncrossing argument lowers each such matrix to the diagonal quantile plan. Passing to the limit, using truncation if the convex cost is unbounded, gives $\int_0^1 h(q_\alpha(r) - q_\beta(r))\mathrm{d}r \leqslant \int h(x - y)\mathrm{d}\pi(x, y)$ for every $\pi \in \mathcal{U}(\alpha, \beta)$. □

### 3.4 $c$-Cyclical Monotonicity

This compact subsection records the support condition characterizing optimal couplings.

**Support and $c$-cyclical monotonicity.**

**Definition 3.21** (Support)**.** For a Radon measure $\pi$ on $\mathcal{X}\times\mathcal{Y}$, $\mathrm{supp}(\pi) := \{(x, y)\ ;\ \pi(U \times V) > 0 \text{ for every open } U \ni x,\ V \ni y\}$.

**Definition 3.22** ($c$-cyclical monotonicity)**.** A set $\Gamma \subset \mathcal{X} \times \mathcal{Y}$ is $c$-cyclically monotone if, for every $k \geqslant 2$, every finite family $(x_i, y_i)_{i=1}^k \subset \Gamma$ and every permutation $\sigma$ of $\{1, \dots, k\}$, $\sum_{i=1}^k c(x_i, y_i) \leqslant \sum_{i=1}^k c(x_i, y_{\sigma(i)})$.

$\sum_{i=1}^k c(x_i, y_i) \leqslant \sum_{i=1}^k c(x_i, y_{i+1}), \qquad y_{k+1} = y_1.$

**Optimal matching to optimal transport.** Let the marginals be uniform on $n$ points, $\alpha = \frac{1}{n}\sum_{i=1}^n \delta_{x_i}$ and $\beta = \frac{1}{n}\sum_{i=1}^n \delta_{y_i}$. By Corollary 3.13, there exists an optimal plan induced by a permutation. Its support $\Gamma = \{(x_i, y_{\sigma(i)})\}_i$ is $c$-cyclically monotone: otherwise exchanging the finitely many targets along a violating cycle would lower the matching cost.

**Theorem 3.23** (Optimal plans are $c$-cyclically monotone)**.** *Assume $c$ is continuous. For any optimal plan $\pi$ solving the Kantorovich problem* (3.5)*,* $\mathrm{supp}(\pi)$ *is $c$-cyclically monotone.*

*Proof.* Suppose that $\mathrm{supp}(\pi)$ is not $c$-cyclically monotone. Then there exist points $(x_i, y_i)_{i=1}^k$ in the support and a permutation $\sigma$ such that $\sum_i c(x_i, y_i) > \sum_i c(x_i, y_{\sigma(i)})$. By continuity of $c$, after shrinking neighborhoods $U_i \ni x_i$ and $V_i \ni y_i$, the same strict inequality holds uniformly for every choice of points in these neighborhoods: $\sum_i c(u_i, v_i) > \sum_i c(u_i, \tilde{v}_{\sigma(i)}) \quad (u_i \in U_i,\ v_i \in V_i,\ \tilde{v}_{\sigma(i)} \in V_{\sigma(i)})$. Choose the sets so that $\pi(U_i \times V_i) > 0$. Because there are only finitely many rectangles, one can choose $\lambda > 0$ small enough that the scaled restrictions $\pi_i = \lambda\frac{\pi|_{U_i\times V_i}}{\pi(U_i\times V_i)}$ have common mass $\lambda$ and satisfy $\sum_i \pi_i \leqslant \pi$. Let $\alpha_i = (P_\mathcal{X})_\sharp\pi_i$ and $\beta_i = (P_\mathcal{Y})_\sharp\pi_i$. Define $\tilde{\pi} = \pi - \sum_i \pi_i + \sum_i \frac{\alpha_i\otimes\beta_{\sigma(i)}}{\lambda}$. The removed and reinserted first marginals are both $\sum_i \alpha_i$, and the removed and reinserted second marginals are both $\sum_i \beta_i$ because $\sigma$ is a permutation. Hence $\tilde{\pi} \in \mathcal{U}(\alpha, \beta)$. Integrating the uniform strict inequality against the product probability $\otimes_i(\pi_i/\lambda)$ shows that the reinserted crossed terms have strictly smaller cost than the removed diagonal terms. This contradicts the optimality of $\pi$. □

**Monotonicity.** Assume the optimal plan is induced by a measurable map $T : \mathcal{X} \to \mathcal{Y}$, i.e. $\pi = (\mathrm{Id}, T)_\sharp\alpha$. For any $k$ points $x_1, \dots, x_k$ in the domain, cyclical monotonicity reads
$\sum_{i=1}^k c(x_i, T(x_i)) \leqslant \sum_{i=1}^k c(x_i, T(x_{i+1})), \qquad x_{k+1} = x_1$. For $c(x, y) = \frac{1}{2}\|x - y\|^2$, taking $k = 2$ gives, for any $x, y$,
$\langle T(x) - T(y), x - y\rangle \geqslant 0,$

**One dimension.** $|x - T(x)|^p + |y - T(y)|^p \leqslant |x - T(y)|^p + |y - T(x)|^p$, which is equivalent to $T(x) \leqslant T(y)$ whenever $x < y$. Thus $T$ must be nondecreasing, recovering the classical monotone rearrangement.

### 3.5 Metric Properties: Wasserstein Distances

This compact subsection records the metric consequences of the gluing lemma.

**OT defines a distance.**

**Lemma 3.24** (Discrete gluing lemma)**.** *Given* $(\mathrm{a},\mathrm{b},\mathrm{c}) \in \Sigma_n \times \Sigma_p \times \Sigma_m$*, let* $\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})$ *and* $\mathrm{Q} \in \mathrm{U}(\mathrm{b},\mathrm{c})$*. Then there exists a 3-D tensor coupling* $\mathrm{S} \in \mathbb{R}_+^{n\times p\times m}$ *such that the 2-D marginals satisfy* $\sum_k \mathrm{S}_{i,j,k} = \mathrm{P}_{i,j}$ *and* $\sum_i \mathrm{S}_{i,j,k} = \mathrm{Q}_{j,k}$. $\mathrm{R}_{i,k} := \sum_j \mathrm{S}_{i,j,k}$*, belongs to* $\mathrm{U}(\mathrm{a},\mathrm{c})$*. For the canonical construction below, this glued coupling is the twisted matrix product* $\mathrm{R} = \mathrm{P}\,\mathrm{diag}(1/\mathrm{b})\mathrm{Q}$, $\mathrm{R}_{i,k} = \sum_{j:\mathrm{b}_j>0} \frac{\mathrm{P}_{i,j}\mathrm{Q}_{j,k}}{\mathrm{b}_j}$*. In the matrix notation,* $1/\mathrm{b}_j$ *is understood as 0 when* $\mathrm{b}_j = 0$*.*

*Proof.* One verifies that

$$\mathrm{S}_{i,j,k} = \begin{cases} \frac{\mathrm{P}_{i,j}\mathrm{Q}_{j,k}}{\mathrm{b}_j} & \text{if} \quad \mathrm{b}_j \neq 0 \\ 0 \text{ otherwise} \end{cases} \tag{3.6}$$

is acceptable. Indeed, if $\mathrm{b}_j \neq 0$ $\sum_k \mathrm{S}_{i,j,k} = \sum_k \frac{\mathrm{P}_{i,j}\mathrm{Q}_{j,k}}{\mathrm{b}_j} = \frac{\mathrm{P}_{i,j}}{\mathrm{b}_j}(\mathrm{Q}\mathbb{1}_m)_j = \frac{\mathrm{P}_{i,j}}{\mathrm{b}_j}\mathrm{b}_j$. If $\mathrm{b}_j = 0$, then necessarily $\mathrm{P}_{i,j} = 0$ and $\sum_k \mathrm{S}_{i,j,k} = 0 = \mathrm{P}_{i,j}$. The same computation gives the other prescribed marginal:

$$\sum_i \mathrm{S}_{i,j,k} = \begin{cases} \frac{\mathrm{Q}_{j,k}}{\mathrm{b}_j}\sum_i \mathrm{P}_{i,j} = \mathrm{Q}_{j,k} & \text{if} \quad \mathrm{b}_j > 0 \\ 0 = \mathrm{Q}_{j,k} \quad \text{if} \quad \mathrm{b}_j = 0. \end{cases}$$

Summing over $j$ then gives the displayed formula for R. Its row and column sums are $\sum_k \mathrm{R}_{i,k} = \sum_j \mathrm{P}_{i,j} = \mathrm{a}_i$, $\sum_i \mathrm{R}_{i,k} = \sum_j \mathrm{Q}_{j,k} = \mathrm{c}_k$, so $\mathrm{R} \in \mathrm{U}(\mathrm{a},\mathrm{c})$. □

When the cost matrix is the $p$th power of a distance matrix, the discrete Kantorovich value becomes a metric on histograms.

**Definition 3.25** (Discrete Wasserstein distance)**.** Let $\mathrm{D} \in \mathbb{R}_+^{n\times n}$ be a distance matrix on $[\![n]\!]$ and let $p \geqslant 1$. The discrete $p$-Wasserstein distance between histograms $\mathrm{a},\mathrm{b} \in \Sigma_n$ is

$$\mathrm{W}_p(\mathrm{a},\mathrm{b}) := \mathrm{L}_{\mathrm{D}^p}(\mathrm{a},\mathrm{b})^{1/p}. \tag{3.7}$$

It depends on the chosen ground distance D.

**Proposition 3.26** (Metric property of discrete Wasserstein distance)**.** *For every distance matrix* D *on* $[\![n]\!]$*, Definition 3.25 defines a distance on* $\Sigma_n$*:* $\mathrm{W}_p$ *is symmetric, positive,* $\mathrm{W}_p(\mathrm{a},\mathrm{b}) = 0$ *if and only if* $\mathrm{a} = \mathrm{b}$*, and it satisfies the triangle inequality* $\forall\, \mathrm{a},\mathrm{b},\mathrm{c} \in \Sigma_n$, $\mathrm{W}_p(\mathrm{a},\mathrm{c}) \leqslant \mathrm{W}_p(\mathrm{a},\mathrm{b}) + \mathrm{W}_p(\mathrm{b},\mathrm{c})$.

*Proof.* For symmetry, since $\mathrm{D}^p$ is symmetric, we use the fact that if $\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})$ is optimal for $\mathrm{W}_p(\mathrm{a},\mathrm{b})$, then $\mathrm{P}^\top \in \mathrm{U}(\mathrm{b},\mathrm{a})$ is optimal for $\mathrm{W}_p(\mathrm{b},\mathrm{a})$. For definiteness, since $\mathrm{C} = \mathrm{D}^p$ has a null diagonal, $\mathrm{W}_p(\mathrm{a},\mathrm{b}) = 0$ is achieved by the diagonal coupling $\mathrm{P}^\star = \mathrm{diag}(\mathrm{a}) = \mathrm{diag}(\mathrm{b})$ when $\mathrm{a} = \mathrm{b}$; by positivity of all off-diagonal elements of $\mathrm{D}^p$, $\mathrm{W}_p(\mathrm{a},\mathrm{b}) > 0$ whenever $\mathrm{a} \neq \mathrm{b}$ because any admissible coupling then has a nonzero element outside the diagonal.
To prove the triangle inequality in this discrete setting, we consider $\mathrm{a},\mathrm{b},\mathrm{c} \in \Sigma_n$, and let P and Q be two optimal solutions of the transport problems between a and b, and b and c respectively.
We use the gluing Lemma 3.24 which defines $\mathrm{S} \in \mathbb{R}_+^{n^3}$ with marginals $\sum_k \mathrm{S}_{\cdot,\cdot,k} = \mathrm{P}$ and $\sum_i \mathrm{S}_{i,\cdot,\cdot} = \mathrm{Q}$. We define $\mathrm{R} = \sum_j \mathrm{S}_{\cdot,j,\cdot}$, which is an element of $\mathrm{U}(\mathrm{a},\mathrm{c})$.

$$\begin{tikzcd} a \arrow[r, "P"] \arrow[rr, bend right, "R"'] & b \arrow[r, "Q"] & c \end{tikzcd}$$

Note that if one assumes $\mathrm{b} > 0$ then $\mathrm{R} = \mathrm{P}\,\mathrm{diag}(1/\mathrm{b})\mathrm{Q}$.
The triangle inequality follows from

$$\begin{aligned}
\mathrm{W}_p(\mathrm{a},\mathrm{c}) &= \Big(\min_{\tilde{\mathrm{R}}\in\mathrm{U}(\mathrm{a},\mathrm{c})} \langle \tilde{\mathrm{R}},\, \mathrm{D}^p\rangle\Big)^{1/p} \leqslant \langle \mathrm{R},\, \mathrm{D}^p\rangle^{1/p} \\
&= \Big(\sum_{i,k} \mathrm{D}^p_{ik} \sum_j \mathrm{S}_{i,j,k}\Big)^{1/p} \leqslant \Big(\sum_{i,j,k} \Big(\mathrm{D}_{ij} + \mathrm{D}_{j,k}\Big)^p \mathrm{S}_{i,j,k}\Big)^{1/p} \\
&\leqslant \Big(\sum_{i,j,k} \mathrm{D}^p_{ij}\mathrm{S}_{i,j,k}\Big)^{1/p} + \Big(\sum_{i,j,k} \mathrm{D}^p_{j,k}\mathrm{S}_{i,j,k}\Big)^{1/p} \\
&= \Big(\sum_{i,j} \mathrm{D}^p_{i,j} \sum_k \mathrm{S}_{i,j,k}\Big)^{1/p} + \Big(\sum_{j,k} \mathrm{D}^p_{j,k} \sum_i \mathrm{S}_{i,j,k}\Big)^{1/p} \\
&= \Big(\sum_{i,j} \mathrm{D}^p_{i,j}\mathrm{P}_{i,j}\Big)^{1/p} + \Big(\sum_{j,k} \mathrm{D}^p_{j,k}\mathrm{Q}_{j,k}\Big)^{1/p} = \mathrm{W}_p(\mathrm{a},\mathrm{b}) + \mathrm{W}_p(\mathrm{b},\mathrm{c}).
\end{aligned}$$

The first inequality follows from the feasibility of R, the second is the usual triangle inequality for elements in D, and the third comes from Minkowski's inequality. □

**Continuous gluing.**

**Lemma 3.27** (Gluing lemma)**.** *Let* $(\alpha,\beta,\gamma) \in \mathcal{M}_+^1(\mathcal{X}) \times \mathcal{M}_+^1(\mathcal{Y}) \times \mathcal{M}_+^1(\mathcal{Z})$ *where* $(\mathcal{X},\mathcal{Y},\mathcal{Z})$ *are Polish spaces in the sense of Definition 2.3. Given* $\pi \in \mathcal{U}(\alpha,\beta)$ *and* $\xi \in \mathcal{U}(\beta,\gamma)$*, then there exists a tensor coupling measure* $\sigma \in \mathcal{M}_+(\mathcal{X}\times\mathcal{Y}\times\mathcal{Z})$ *such that* $(P_{\mathcal{X},\mathcal{Y}})_\sharp\sigma = \pi$ *and* $(P_{\mathcal{Y},\mathcal{Z}})_\sharp\sigma = \xi$ *where we denoted the projector* $P_{\mathcal{X},\mathcal{Y}}(x,y,z) = (x,y)$ *and* $P_{\mathcal{Y},\mathcal{Z}}(x,y,z) = (y,z)$*.*

*Proof.* The proof of this fundamental result is involved since it requires using the disintegration of measure (which corresponds to conditional probabilities).
The disintegration of measures is applicable because the spaces are Polish.
We disintegrate $\pi$ and $\xi$ against $\beta$ to obtain two families $(\pi_y)_{y\in\mathcal{Y}}$ and $(\xi_y)_{y\in\mathcal{Y}}$ of probability distributions on $\mathcal{X}$ and $\mathcal{Z}$. These families are defined by the fact that $\forall\, h \in \mathcal{C}(\mathcal{X}\times\mathcal{Y})$, $\quad \int_{\mathcal{Y}} \left( \int_{\mathcal{X}} h(x,y)\mathrm{d}\pi_y(x) \right)\mathrm{d}\beta(y) = \int h(x,y)\mathrm{d}\pi(x,y)$. and similarly for $\xi$.
When $\beta = \sum_j \mathrm{b}_j\delta_{y_j}$ and $\pi = \sum_{i,j} \mathrm{P}_{i,j}\delta_{(x_i,y_j)}$, then this conditional distribution is defined on the support of $\beta$ as $\pi_{y_j} = \sum_i \frac{\mathrm{P}_{i,j}}{\mathrm{b}_j}\delta_{x_i}$ (and similarly for $\xi$).
The glued measure is then defined by the conditional-product formula $\forall\, g \in \mathcal{C}(\mathcal{X}\times\mathcal{Y}\times\mathcal{Z})$, $\int g(x,y,z)\mathrm{d}\sigma(x,y,z) = \int g(x,y,z)\mathrm{d}\pi_y(x)\mathrm{d}\xi_y(z)\mathrm{d}\beta(y)$. For discrete measures, this matches the definition (3.6), since $\sigma = \sum_{i,j,k} \mathrm{S}_{i,j,k}\delta_{x_i,y_j,z_k}$ where $\mathrm{S}_{i,j,k} = \frac{\mathrm{P}_{i,j}}{\mathrm{b}_j}\frac{\mathrm{Q}_{j,k}}{\mathrm{b}_j}\mathrm{b}_j$. $\square$

**Definition 3.28** (Wasserstein distance)**.** Let $(\mathcal{X},d)$ be a metric space and $p \geqslant 1$. For $\alpha,\beta \in \mathcal{P}_p(\mathcal{X})$, the $p$-Wasserstein distance is

$$\mathcal{W}_p(\alpha,\beta) := \mathcal{L}_{d^p}(\alpha,\beta)^{1/p} = \left( \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{X}} d(x,y)^p \mathrm{d}\pi(x,y) \right)^{1/p}. \tag{3.8}$$

It depends on the ground distance $d$.

**Proposition 3.29** (Metric property of the Wasserstein distance)**.** *Definition 3.28 defines a distance: $\mathcal{W}_p$ is symmetric, positive, $\mathcal{W}_p(\alpha,\beta) = 0$ if and only if $\alpha = \beta$, and it satisfies the triangle inequality $\forall\,(\alpha,\beta,\gamma) \in \mathcal{P}_p(\mathcal{X})^3$, $\quad \mathcal{W}_p(\alpha,\gamma) \leqslant \mathcal{W}_p(\alpha,\beta) + \mathcal{W}_p(\beta,\gamma)$.*

*Proof.* The symmetry follows from the fact that since $d$ is symmetric, if $\pi(x,y)$ is optimal for $\mathcal{L}_{d^p}(\alpha,\beta)$, then $\pi(y,x) \in \mathcal{U}(\beta,\alpha)$ is optimal for $\mathcal{L}_{d^p}(\beta,\alpha)$.
If $\mathcal{L}_{d^p}(\alpha,\beta) = 0$, then necessarily an optimal coupling $\pi$ is supported on the diagonal $\Delta := \{(x,x)\}_x \subset \mathcal{X}^2$.
We denote $\lambda(x)$ the corresponding measure on the diagonal, i.e. such that $\int h(x,y)\mathrm{d}\pi(x,y) = \int h(x,x)\mathrm{d}\lambda(x)$.
Then since $\pi \in \mathcal{U}(\alpha,\beta)$ necessarily $\lambda = \alpha$ and $\lambda = \beta$ so that $\alpha = \beta$.
For the triangle inequality, we consider optimal couplings $\pi \in \mathcal{U}(\alpha,\beta)$ and $\xi \in \mathcal{U}(\beta,\gamma)$ and we glue them according to the Lemma 3.27.
We define the composition of the two couplings $(\pi,\xi)$ as $\rho := (P_{\mathcal{X},\mathcal{Z}})_\sharp\sigma$.
Note that if $\pi$ and $\xi$ are couplings induced by two Monge maps $T_{\mathcal{X}}(x)$ and $T_{\mathcal{Y}}(y)$, then $\rho$ is itself induced by the Monge map $T_{\mathcal{Y}} \circ T_{\mathcal{X}}$, so that this notion of composition of coupling generalizes the composition of maps.
The triangular inequality follows from

$$\begin{aligned}
\mathcal{W}_p(\alpha,\gamma) &\leqslant \left( \int_{\mathcal{X}\times\mathcal{Z}} d(x,z)^p \mathrm{d}\rho(x,z) \right)^{1/p} = \left( \int_{\mathcal{X}\times\mathcal{Y}\times\mathcal{Z}} d(x,z)^p \mathrm{d}\sigma(x,y,z) \right)^{1/p} \\
&\leqslant \left( \int_{\mathcal{X}\times\mathcal{Y}\times\mathcal{Z}} (d(x,y)+d(y,z))^p \mathrm{d}\sigma(x,y,z) \right)^{1/p} \\
&\leqslant \left( \int_{\mathcal{X}\times\mathcal{Y}\times\mathcal{Z}} d(x,y)^p \mathrm{d}\sigma(x,y,z) \right)^{1/p} + \left( \int_{\mathcal{X}\times\mathcal{Y}\times\mathcal{Z}} d(y,z)^p \mathrm{d}\sigma(x,y,z) \right)^{1/p} \\
&= \left( \int_{\mathcal{X}\times\mathcal{Y}} d(x,y)^p \mathrm{d}\pi(x,y) \right)^{1/p} + \left( \int_{\mathcal{Y}\times\mathcal{Z}} d(y,z)^p \mathrm{d}\xi(y,z) \right)^{1/p} = \mathcal{W}_p(\alpha,\beta) + \mathcal{W}_p(\beta,\gamma).
\end{aligned}$$

$\square$

**Interpolation induced by an optimal plan.**

**Definition 3.30** ($\mathcal{W}_2$ geodesic induced by an optimal plan)**.** Let $\alpha_0,\alpha_1 \in \mathcal{P}_2(\mathbb{R}^d)$, and let $\pi^\star \in \mathcal{U}(\alpha_0,\alpha_1)$ be optimal for $\mathcal{W}_2^2(\alpha_0,\alpha_1)$. For $t \in [0,1]$, define $e_t(x,y) := (1-t)x + ty$, $\qquad \alpha_t := (e_t)_\sharp\pi^\star$. The curve $(\alpha_t)_{t\in[0,1]}$ is the displacement, or McCann, $\mathcal{W}_2$ geodesic induced by $\pi^\star$.

**Proposition 3.31** (Optimal-plan interpolation is a $\mathcal{W}_2$ geodesic)**.** *Let $(\alpha_t)_{t\in[0,1]}$ be defined by Definition 3.30. Then, for every $0 \leqslant s \leqslant t \leqslant 1$, $\mathcal{W}_2(\alpha_s,\alpha_t) = (t-s)\,\mathcal{W}_2(\alpha_0,\alpha_1)$. Thus $t \mapsto \alpha_t$ is a constant-speed geodesic for the metric $\mathcal{W}_2$.*

*Proof.* Push the optimal plan $\pi^\star$ forward by $(e_s,e_t)$. $\int \|z-z'\|^2\mathrm{d}\gamma_{s,t}(z,z') = \int \|e_t(x,y)-e_s(x,y)\|^2\mathrm{d}\pi^\star(x,y) = (t-s)^2\,\mathcal{W}_2^2(\alpha_0,\alpha_1)$. Hence $\mathcal{W}_2(\alpha_s,\alpha_t) \leqslant (t-s)\,\mathcal{W}_2(\alpha_0,\alpha_1)$. Applying this upper bound to the three pairs $(0,s)$, $(s,t)$ and $(t,1)$, and using the triangle inequality of Proposition 3.29, gives $\mathcal{W}_2(\alpha_0,\alpha_1) \leqslant \mathcal{W}_2(\alpha_0,\alpha_s) + \mathcal{W}_2(\alpha_s,\alpha_t) + \mathcal{W}_2(\alpha_t,\alpha_1) \leqslant \mathcal{W}_2(\alpha_0,\alpha_1)$. All inequalities are therefore equalities, in particular the middle segment has the claimed length. $\square$

**General geodesic spaces.** For Dirac masses in Euclidean space, the $\mathcal{W}_2$ geodesic from $\delta_x$ to $\delta_y$ is $t \mapsto \delta_{(1-t)x+ty}$. The same idea extends to any geodesic metric space $(\mathcal{X},d)$, meaning that each pair of points can be joined by a constant-speed metric geodesic. For each pair $(x,y)$, one replaces the Euclidean segment by a curve $\gamma^{x,y} : [0,1] \to \mathcal{X}$ such that $\gamma_0^{x,y} = x$, $\gamma_1^{x,y} = y$, and $d(\gamma_s^{x,y},\gamma_t^{x,y}) = |t-s|d(x,y)$.

**Comparison with Monge.**

**Proposition 3.32** (Kantorovich as the plan-space relaxation of Monge)**.** *Let $(\mathcal{X},d)$ be a compact metric space, let $p \geqslant 1$, and let $\alpha,\beta \in \mathcal{P}(\mathcal{X})$ with $\alpha$ atomless. Then $\mathcal{W}_p(\alpha,\beta)^p = \inf_{T_\sharp\alpha=\beta}\int_{\mathcal{X}} d(x,T(x))^p\mathrm{d}\alpha(x)$. The infimum on the Monge side need not be attained.*

*Proof.* Let $\pi \in \mathcal{U}(\alpha,\beta)$. Choose finite Borel partitions $(A_i)_i$ and $(B_j)_j$ of $\mathcal{X}$ whose cells have diameter at most $\varepsilon$. Set $m_{ij} = \pi(A_i\times B_j)$. Since $\alpha$ is atomless and $\sum_j m_{ij} = \alpha(A_i)$, each $A_i$ can be partitioned into Borel sets $A_{ij}$ with $\alpha(A_{ij}) = m_{ij}$. If $m_{ij} > 0$, then $\beta(B_j) > 0$, and Proposition 2.11 gives a measurable map $T_{ij} : A_{ij} \to B_j$, $\qquad (T_{ij})_\sharp\frac{\alpha|_{A_{ij}}}{m_{ij}} = \frac{\beta|_{B_j}}{\beta(B_j)}$. Define $T$ by $T = T_{ij}$ on $A_{ij}$, ignoring null cells. Then $T_\sharp\alpha = \beta$, because the contributions landing in $B_j$ sum to $\beta|_{B_j}$, and the graph coupling $(\mathrm{Id},T)_\sharp\alpha$ has the same masses $m_{ij}$ as $\pi$ on all rectangles $A_i\times B_j$.

Let $h \in \mathcal{C}(\mathcal{X} \times \mathcal{X})$. By uniform continuity on the compact product, the oscillation of $h$ on every rectangle $A_i \times B_j$ goes to zero as $\varepsilon \to 0$. Since $\pi$ and $(\mathrm{Id}, T)_\sharp\alpha$ have the same mass on each rectangle, their integrals against $h$ differ by at most this maximal oscillation. Taking partitions with mesh tending to zero therefore gives graph couplings $(\mathrm{Id}, T_k)_\sharp\alpha$ converging weakly to $\pi$. Applying this to the continuous function $(x, y) \mapsto d(x, y)^p$ gives the convergence of costs.
Define

$$\mathcal{G}(\alpha, \beta) := \{(\mathrm{Id}, T)_\sharp\alpha \,;\, T_\sharp\alpha = \beta\} \subset \mathcal{U}(\alpha, \beta), \qquad F_p(\pi) := \int_{\mathcal{X}\times\mathcal{X}} d(x, y)^p \mathrm{d}\pi(x, y),$$

and let $G$ be the Monge graph functional which equals $F_p$ on $\mathcal{G}(\alpha, \beta)$ and $+\infty$ otherwise. The function $F_p$ is weakly continuous on $\mathcal{U}(\alpha, \beta)$ and satisfies $F_p \leqslant G$. If $H$ is any weakly lower-semicontinuous function with $H \leqslant G$, then for the graph approximation above, $H(\pi) \leqslant \liminf_k H((\mathrm{Id}, T_k)_\sharp\alpha) \leqslant \lim_k F_p((\mathrm{Id}, T_k)_\sharp\alpha) = F_p(\pi)$. Thus $F_p$ is the largest weakly lower-semicontinuous minorant of $G$, i.e. the lower-semicontinuous envelope. Minimizing this envelope over all couplings gives the Kantorovich value, and the density argument gives the same infimum when restricting to graph couplings. □

**Corollary 3.33** (Lower-semicontinuous envelope of the Monge $p$-cost)**.** *Assume, in addition, that atomless probability measures are dense in $\mathcal{P}(\mathcal{X})$ for the $\mathcal{W}_p$ topology. Then $(\alpha, \beta) \mapsto \mathcal{W}_p(\alpha, \beta)^p$ is the lower-semicontinuous envelope, on $\mathcal{P}(\mathcal{X}) \times \mathcal{P}(\mathcal{X})$ for the product $\mathcal{W}_p$ topology, of the extended Monge cost*

$$\tilde{\mathcal{W}}_p(\alpha, \beta)^p = \inf_{\substack{T:\mathcal{X}\to\mathcal{X}\\ T_\sharp\alpha=\beta}} \int_{\mathcal{X}} d(x, T(x))^p \mathrm{d}\alpha(x),$$

*with the convention $\tilde{\mathcal{W}}_p(\alpha, \beta)^p = +\infty$ if no admissible map exists.*

*Proof.* For every admissible map $T$, the graph plan $(\mathrm{Id}, T)_\sharp\alpha$ is a coupling, hence $\mathcal{W}_p(\alpha, \beta)^p \leqslant \tilde{\mathcal{W}}_p(\alpha, \beta)^p$. Since $\mathcal{W}_p^p$ is continuous for the product $\mathcal{W}_p$ topology, it is a lower-semicontinuous minorant of the extended Monge cost.
Conversely, let $H$ be any lower-semicontinuous minorant of the extended Monge cost. Fix $(\alpha, \beta)$ and choose atomless $\alpha_k \to \alpha$ in $\mathcal{W}_p$. By Proposition 3.32, $\tilde{\mathcal{W}}_p(\alpha_k, \beta)^p = \mathcal{W}_p(\alpha_k, \beta)^p$. Therefore $H(\alpha, \beta) \leqslant \liminf_k H(\alpha_k, \beta) \leqslant \lim_k \mathcal{W}_p(\alpha_k, \beta)^p = \mathcal{W}_p(\alpha, \beta)^p$. Thus no larger lower-semicontinuous minorant exists. □

### 3.6 Metric Properties: Topology and Applications

This compact subsection recalls how Wasserstein distances metrize weak convergence.

**Topological role.** Wasserstein distances refine weak convergence by adding moment control.

**Convergence in law topology.** On a bounded metric space, all $\mathcal{W}_p$ distances define the same topology, although they are not equivalent as distances.

**Proposition 3.34** (Equivalence of Wasserstein distances on compact spaces)**.** *One has for $p \leqslant q$ $\mathcal{W}_p(\alpha, \beta) \leqslant \mathcal{W}_q(\alpha, \beta) \leqslant \mathrm{diam}(\mathcal{X})^{\frac{q-p}{q}} \mathcal{W}_p(\alpha, \beta)^{\frac{p}{q}}$ where $\mathrm{diam}(\mathcal{X}) := \sup_{x,y} d(x, y)$.*

*Proof.* The left inequality follows from Jensen inequality, $\varphi(\int c(x, y)\mathrm{d}\pi(x, y)) \leqslant \int \varphi(c(x, y))\mathrm{d}\pi(x, y)$, applied to any probability distribution $\pi$ and to the convex function $\varphi(r) = r^{q/p}$ with $c(x, y) = d(x, y)^p$, so that one gets $\left(\int d(x, y)^p \mathrm{d}\pi(x, y)\right)^{\frac{q}{p}} \leqslant \int d(x, y)^q \mathrm{d}\pi(x, y)$. The right inequality follows from $d(x, y)^q \leqslant \mathrm{diam}(\mathcal{X})^{q-p} d(x, y)^p$. □

**Definition 3.35** (Weak* topology)**.** $(\alpha_k)_k$ converges weakly* to $\alpha$ in $\mathcal{M}_+^1(\mathcal{X})$ (denoted $\alpha_k \rightharpoonup \alpha$) if and only if for any bounded continuous function $f \in \mathcal{C}_b(\mathcal{X})$, $\int_{\mathcal{X}} f \mathrm{d}\alpha_k \to \int_{\mathcal{X}} f \mathrm{d}\alpha$. On compact spaces, $\mathcal{C}_b(\mathcal{X}) = \mathcal{C}(\mathcal{X})$, which is why the boundedness is often left implicit there.

In terms of random vectors, if $X_n \sim \alpha_n$ and $X \sim \alpha$ (not necessarily defined on the same probability space), weak convergence corresponds to convergence in law of $X_n$ toward $X$.
$\|\alpha - \beta\|_{\mathrm{TV}} = |\alpha - \beta|(\mathcal{X})$,

**Proposition 3.36** (Total variation as Wasserstein for the discrete metric)**.** *Denoting $d$ the 0/1 distance such that $d(x, x) = 0$ and $d(x, y) = 1$ if $x \neq y$, then $\mathcal{W}_p(\alpha, \beta)^p = \frac{1}{2}\|\alpha - \beta\|_{\mathrm{TV}}$.*

*Proof.* For the sake of simplicity, we do the proof for discrete measures with weights $(\mathrm{a}, \mathrm{b})$ and without loss of generality assume they have the same support $(x_i)_i$ and we denote $\mathrm{D} := (d(x_i, x_j))_{i,j}$ which is 0 on the diagonal and one outside.
Also since $d^p = d$ we consider $p = 1$. We denote $\mathrm{c}_i = \min(\mathrm{a}_i, \mathrm{b}_i)$. By conservation of mass, for every $\mathrm{P} \in \mathrm{U}(\mathrm{a}, \mathrm{b})$, $\mathrm{P}_{i,i} \leqslant \mathrm{c}_i$, thus $\langle \mathrm{P}, \mathrm{D}\rangle = \sum_{i\neq j} \mathrm{P}_{i,j} = 1 - \sum_i \mathrm{P}_{i,i} \geqslant 1 - \sum_i \mathrm{c}_i$. We need to show that this bound is tight, namely to construct $\hat{P} \in \mathrm{U}(\mathrm{a}, \mathrm{b})$ such that $\mathrm{diag}(\hat{P}) = \mathrm{c}$. Let $\bar{\mathrm{a}} := \mathrm{a} - \mathrm{c} = (\mathrm{a} - \mathrm{b})_+ \geqslant 0$ and $\bar{\mathrm{b}} := \mathrm{b} - \mathrm{c} = (\mathrm{b} - \mathrm{a})_+ \geqslant 0$ If $\bar{\mathrm{a}} = \bar{\mathrm{b}} = 0$, then $\mathrm{a} = \mathrm{b}$ and the diagonal coupling is optimal. Otherwise, one has $\frac{\bar{\mathrm{a}}\otimes\bar{\mathrm{b}}}{\langle\bar{\mathrm{a}}, \mathbb{1}\rangle} \in \mathrm{U}(\bar{\mathrm{a}}, \bar{\mathrm{b}})$ and we remark that $\langle\bar{\mathrm{a}}, \mathbb{1}\rangle = \langle\bar{\mathrm{b}}, \mathbb{1}\rangle = 1 - \langle\mathrm{c}, \mathbb{1}\rangle$.
Thus denoting $\hat{\mathrm{P}} := \mathrm{diag}(\mathrm{c}) + \frac{\bar{\mathrm{a}}\otimes\bar{\mathrm{b}}}{\langle\bar{\mathrm{a}}, \mathbb{1}\rangle} \geqslant 0$ satisfies $\hat{\mathrm{P}}\mathbb{1} = \mathrm{c} + \bar{\mathrm{a}} = \mathrm{a}$ and $\hat{\mathrm{P}}^\top\mathbb{1} = \mathrm{c} + \bar{\mathrm{b}} = \mathrm{b}$ so that $\hat{P} \in \mathrm{U}(\mathrm{a}, \mathrm{b})$ is a coupling so that $\mathrm{diag}(\hat{P}) = \mathrm{diag}(\mathrm{c})$ since $\mathrm{diag}(\bar{\mathrm{a}} \otimes \bar{\mathrm{b}}) = 0$. We thus conclude that

$$\mathrm{W}_1(\mathrm{a}, \mathrm{b}) = \langle\mathrm{D}, \hat{\mathrm{P}}\rangle = \sum_{i,j} \frac{\bar{\mathrm{a}}_i\bar{\mathrm{b}}_j}{\langle\bar{\mathrm{a}}, \mathbb{1}\rangle} = \sum_i \bar{\mathrm{a}}_i = \sum_i \bar{\mathrm{b}}_i = \frac{1}{2}\sum_i(\bar{\mathrm{a}}_i + \bar{\mathrm{b}}_i) = \frac{1}{2}\|\mathrm{a} - \mathrm{b}\|_{\mathrm{TV}}.$$

□

$\|\delta_{x_n} - \delta_x\|_{\mathrm{TV}} = 2$ and $\mathcal{W}_p(\delta_{x_n}, \delta_x) = d(x_n, x)$.

**Proposition 3.37** (Wasserstein metrizes weak convergence on compact spaces)**.** *If $\mathcal{X}$ is compact, $\alpha_k \rightharpoonup \alpha$ if and only if $\mathcal{W}_p(\alpha_k, \alpha) \to 0$.*

*Proof.* For $p = 1$, this is the Kantorovich–Rubinstein metrization theorem: by duality, $\mathcal{W}_1$ is the supremum over 1-Lipschitz test functions, and on a compact metric space this class is compact modulo constants by Arzelà–Ascoli. Thus convergence in $\mathcal{W}_1$ is equivalent to weak convergence. Proposition 3.34 then shows that all Wasserstein distances $\mathcal{W}_p$ induce the same convergent sequences on compact spaces. Hence weak convergence is equivalent to convergence in $\mathcal{W}_p$ for every $p \geqslant 1$. $\square$

$\int d(x, x_0)^p \mathrm{d}\alpha_k(x) \longrightarrow \int d(x, x_0)^p \mathrm{d}\alpha(x)$.

**Proposition 3.38** (Comparison with total variation on discrete spaces). *One has*

$$\frac{d_{\min}}{2}\|\alpha - \beta\|_{\mathrm{TV}} \leqslant \mathcal{W}_1(\alpha, \beta) \leqslant \frac{d_{\max}}{2}\|\alpha - \beta\|_{\mathrm{TV}} \quad \text{where} \quad \begin{cases} d_{\min} := \inf_{x \neq y} d(x, y) \\ d_{\max} := \sup_{x,y} d(x, y) \end{cases}$$

*Proof.* We denote $d_0(x, y)$ the distance such that $d_0(x, x) = 0$ and $d_0(x, y) = 1$ for $x \neq y$. One has $d_{\min} d_0(x, y) \leqslant d(x, y) \leqslant d_{\max} d_0(x, y)$ so that integrating this against any $\pi \in \mathcal{U}(\alpha, \beta)$ and taking the minimum among those $\pi$ gives the result using Proposition 3.36. $\square$

This bound is sharp, as this can be observed by taking $\alpha = \delta_x$ and $\beta = \delta_y$, in which case the bound simply reads if $x \neq y$ $d_{\min} \leqslant d(x, y) \leqslant d_{\max}$.

### 3.7 Distributional Robustness and $\mathcal{W}_\infty$

This compact subsection records the robust-optimization uses of Wasserstein balls.

**DRO ambiguity sets.** Wasserstein distances are also used to define ambiguity sets around an empirical law. Given samples $z_i$ and $\hat{\alpha}_n = \frac{1}{n}\sum_i \delta_{z_i}$, a distributionally robust optimization (DRO) problem replaces the empirical risk $\frac{1}{n}\sum_i \ell_\theta(z_i)$ by $\sup_{\beta:\, \mathcal{W}_p(\beta, \hat{\alpha}_n) \leqslant \rho} \int \ell_\theta(z) \mathrm{d}\beta(z)$,

$$\sup_{\beta:\, \mathcal{W}_p(\beta, \hat{\alpha}_n)^p \leqslant \rho^p} \int \ell_\theta \mathrm{d}\beta = \inf_{\lambda \geqslant 0} \lambda \rho^p + \frac{1}{n}\sum_{i=1}^n \sup_z \{\ell_\theta(z) - \lambda d(z, z_i)^p\}. \tag{3.9}$$

Thus the robust risk is an empirical risk in which each sample is replaced by its worst penalized perturbation. For $p = 1$ and an $L_\theta$-Lipschitz loss, the Kantorovich–Rubinstein dual gives the transparent upper bound $\sup_{\beta:\, \mathcal{W}_1(\beta, \hat{\alpha}_n) \leqslant \rho} \int \ell_\theta \mathrm{d}\beta \leqslant \frac{1}{n}\sum_i \ell_\theta(z_i) + \rho L_\theta$,

**Proposition 3.39** (Convexity of transport costs). *For any nonnegative lower-semicontinuous cost $c$, the value $(\alpha, \beta) \mapsto \mathcal{L}_c(\alpha, \beta)$ is jointly convex. In particular, for a ground metric $d$ and $p \geqslant 1$, the map $(\alpha, \beta) \mapsto \mathcal{W}_p(\alpha, \beta)^p$ is jointly convex. The distance $\mathcal{W}_1$ is jointly convex, but $\mathcal{W}_p$ itself need not be convex for $p > 1$.*

*Proof.* Let $\pi_0 \in \mathcal{U}(\alpha_0, \beta_0)$ and $\pi_1 \in \mathcal{U}(\alpha_1, \beta_1)$ be $\eta$-optimal. Then $(1-t)\pi_0 + t\pi_1$ is a coupling between $(1-t)\alpha_0 + t\alpha_1$ and $(1-t)\beta_0 + t\beta_1$, and its cost is the corresponding convex combination of the two costs. Letting $\eta \to 0$ proves joint convexity of $\mathcal{L}_c$. Taking $c = d^p$ gives convexity of $\mathcal{W}_p^p$. For $p = 1$, this is convexity of $\mathcal{W}_1$ itself. For $p > 1$, the root can destroy convexity: on the real line, $F(t) := \mathcal{W}_p((1-t)\delta_0 + t\delta_1, \delta_0) = t^{1/p}$ satisfies $F(1/2) > (F(0) + F(1))/2$. $\square$

$\theta \mapsto \sup_{\beta:\, \mathcal{W}_p(\beta, \hat{\alpha}_n) \leqslant \rho} \int \ell_\theta \, \mathrm{d}\beta$

**$\mathcal{W}_\infty$ robustness.**

$$\mathcal{W}_\infty(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \operatorname*{ess\,sup}_{(x,y) \sim \pi} d(x, y) \tag{3.10}$$

is the limit of $\mathcal{W}_p(\alpha, \beta)$ as $p \to \infty$ on bounded spaces, not the limit of the convex programs defining $\mathcal{W}_p^p$. It minimizes the worst displacement rather than an average displacement, and the resulting optimization is no longer a linear convex program because of the essential-supremum objective.

**Proposition 3.40** ($\mathcal{W}_\infty$ robust envelope around an empirical law). *Let $(\mathcal{Z}, d)$ be a Polish metric space. Let $\hat{\alpha} = \sum_{i=1}^n a_i \delta_{z_i}$ with $a_i > 0$ and $\sum_i a_i = 1$, and assume that the closed balls $\overline{B}(z_i, \rho)$ are compact. For any real-valued upper-semicontinuous loss $\ell$,* $\sup_{\beta:\, \mathcal{W}_\infty(\beta, \hat{\alpha}) \leqslant \rho} \int \ell(z) \mathrm{d}\beta(z) = \sum_{i=1}^n a_i \sup_{z \in \overline{B}(z_i, \rho)} \ell(z)$.

*Proof.* If $\mathcal{W}_\infty(\beta, \hat{\alpha}) \leqslant \rho$, then, by symmetry, there are couplings $\pi_m \in \mathcal{U}(\hat{\alpha}, \beta)$ whose essential displacements are at most $\rho + 1/m$. Since the two marginals are fixed probability measures on a Polish space, $\mathcal{U}(\hat{\alpha}, \beta)$ is tight and closed, hence weakly compact by Prokhorov's theorem. After extraction, $\pi_m \rightharpoonup \pi \in \mathcal{U}(\hat{\alpha}, \beta)$. For every $\eta > 0$, the closed set $F_\eta = \{(x, z) : d(x, z) \leqslant \rho + \eta\}$ has $\pi_m(F_\eta) = 1$ for all sufficiently large $m$. Portmanteau's theorem gives $\pi(F_\eta) = 1$. Letting $\eta \downarrow 0$ along a countable sequence gives $\pi(\{(x, z) : d(x, z) \leqslant \rho\}) = 1$. Disintegrating $\pi$ with respect to the first marginal gives $\pi = \sum_i a_i \delta_{z_i} \otimes \nu_i$, where each $\nu_i$ is supported in the closed ball $\overline{B}(z_i, \rho)$ and $\beta = \sum_i a_i \nu_i$. Hence $\int \ell \, \mathrm{d}\beta = \sum_i a_i \int \ell \, \mathrm{d}\nu_i \leqslant \sum_i a_i \sup_{\overline{B}(z_i, \rho)} \ell$. The reverse inequality follows by choosing, for each $i$, a maximizer $z_i^\star \in \overline{B}(z_i, \rho)$ and setting $\beta = \sum_i a_i \delta_{z_i^\star}$. The coupling $\sum_i a_i \delta_{(z_i, z_i^\star)}$ has essential displacement at most $\rho$, so this $\beta$ is feasible and attains the displayed value. $\square$

## 4 Dual Problem

### 4.1 Discrete dual

The Kantorovich problem (3.2) is a linear program so that one can equivalently compute its value by solving a dual linear program.

**Definition 4.1** (Admissible potentials). For a discrete problem with marginal sizes $n, m$ and cost matrix $\mathrm{C} \in \mathbb{R}^{n \times m}$, a pair $(\mathrm{f}, \mathrm{g}) \in \mathbb{R}^n \times \mathbb{R}^m$ is admissible if it lies below the cost:

$$\mathrm{R}(\mathrm{a}, \mathrm{b}) := \{(\mathrm{f}, \mathrm{g}) \in \mathbb{R}^n \times \mathbb{R}^m \; ; \; \forall (i, j) \in [\![n]\!] \times [\![m]\!], \mathrm{f}_i + \mathrm{g}_j \leqslant \mathrm{C}_{i,j}\}. \tag{4.1}$$

Equivalently, $\mathrm{f} \oplus \mathrm{g} \leqslant \mathrm{C}$ entrywise. The notation suppresses the dependence on C, which is fixed in the surrounding problem.

**Proposition 4.2** (Discrete Kantorovich duality). *One has*

$$\mathrm{L}_{\mathrm{C}}(\mathrm{a}, \mathrm{b}) = \max_{(\mathrm{f}, \mathrm{g}) \in \mathrm{R}(\mathrm{a}, \mathrm{b})} \langle \mathrm{f}, \mathrm{a} \rangle + \langle \mathrm{g}, \mathrm{b} \rangle \tag{4.2}$$

*Proof.* For the sake of completeness, let us derive this dual problem using Lagrangian duality. The Lagrangian associated to (3.2) reads

$$\min_{\mathrm{P}\geqslant 0} \max_{(\mathrm{f},\mathrm{g})\in\mathbb{R}^n\times\mathbb{R}^m} \langle \mathrm{C}, \mathrm{P}\rangle + \langle \mathrm{a} - \mathrm{P}\mathbb{1}_m, \mathrm{f}\rangle + \langle \mathrm{b} - \mathrm{P}^\top \mathbb{1}_n, \mathrm{g}\rangle. \tag{4.3}$$

For a linear program, if the primal constraint set is non-empty, one can always exchange the min and the max and get the same value. We thus consider $\max_{(\mathrm{f},\mathrm{g})\in\mathbb{R}^n\times\mathbb{R}^m} \langle \mathrm{a}, \mathrm{f}\rangle + \langle \mathrm{b}, \mathrm{g}\rangle + \min_{\mathrm{P}\geqslant 0} \langle \mathrm{C} - \mathrm{f}\mathbb{1}_m^\top - \mathbb{1}_n \mathrm{g}^\top, \mathrm{P}\rangle$. We conclude by remarking that

$$\min_{\mathrm{P}\geqslant 0} \langle \mathrm{Q}, \mathrm{P}\rangle = \begin{cases} 0 & \text{if} \quad \mathrm{Q} \geqslant 0 \\ -\infty & \text{otherwise} \end{cases}$$

so that the constraint reads $\mathrm{C} - \mathrm{f}\mathbb{1}_m^\top - \mathbb{1}_n \mathrm{g}^\top = \mathrm{C} - \mathrm{f} \oplus \mathrm{g} \geqslant 0$. ☐

$$\mathrm{Supp}(\mathrm{P}) \subset \{(i,j) \in [\![n]\!] \times [\![m]\!] \; ; \; \mathrm{f}_i + \mathrm{g}_j = \mathrm{C}_{i,j}\}\,. \tag{4.4}$$

### 4.2 Auction Algorithm and Dual Prices

Consider the square assignment problem with costs $C_{i,j}$ and rewrite it as the profit maximization problem with $a_{i,j} = -C_{i,j}$. The auction algorithm keeps prices $p_j$ on the target points and a partial assignment.
$v_i = \max_j (a_{i,j} - p_j)$, $\quad j_i \in \operatorname{argmax}_j (a_{i,j} - p_j)$, $\quad w_i = \max_{j\neq j_i} (a_{i,j} - p_j)$.

$$p_{j_i} \leftarrow p_{j_i} + v_i - w_i + \varepsilon.$$

For fixed prices $p$, eliminating the bidder utilities $u_i$ in the dual minimization gives the convex objective
$D(p) = \sum_j p_j + \sum_i \max_j (a_{i,j} - p_j)$,

**Definition 4.3** ($\varepsilon$-complementary slackness)**.** An assignment $\sigma$ and prices $p$ satisfy $\varepsilon$-complementary slackness if, for every source $i$, $a_{i,\sigma(i)} - p_{\sigma(i)} \geqslant \max_j (a_{i,j} - p_j) - \varepsilon$.

**Proposition 4.4** (Auction optimality certificate)**.** *If a complete assignment $\sigma$ satisfies $\varepsilon$-complementary slackness, then it is $n\varepsilon$-optimal for the profit maximization problem, or equivalently $n\varepsilon$-optimal for the original cost minimization problem. If all costs are integers and $\varepsilon < 1/n$, then $\sigma$ is optimal.*

*Proof.* Let $\tau$ be any assignment. By $\varepsilon$-complementary slackness, $a_{i,\tau(i)} - p_{\tau(i)} \leqslant \max_j (a_{i,j} - p_j) \leqslant a_{i,\sigma(i)} - p_{\sigma(i)} + \varepsilon$. Summing over $i$ cancels prices, because both $\sigma$ and $\tau$ are permutations: $\sum_i a_{i,\tau(i)} \leqslant \sum_i a_{i,\sigma(i)} + n\varepsilon$. Thus no assignment has profit more than $n\varepsilon$ above that of $\sigma$. Since $a = -C$, the same statement says that the cost of $\sigma$ is at most $n\varepsilon$ above the minimum cost. If the costs are integers, all assignment costs are integers; a gap strictly smaller than one therefore forces the gap to be zero. ☐

### 4.3 General formulation

**Proposition 4.5** (Kantorovich duality)**.** *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces and that $c \in \mathcal{C}(\mathcal{X} \times \mathcal{Y})$. Then*

$$\mathcal{L}_c(\alpha,\beta) = \max_{(f,g)\in\mathcal{R}(c)} \int_{\mathcal{X}} f(x)\mathrm{d}\alpha(x) + \int_{\mathcal{Y}} g(y)\mathrm{d}\beta(y), \tag{4.5}$$

*where the set of admissible dual potentials is*

$$\mathcal{R}(c) := \{(f,g) \in \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y}) \; ; \; \forall (x,y), f(x) + g(y) \leqslant c(x,y)\}\,. \tag{4.6}$$

*Here, $(f,g)$ is a pair of continuous functions, often called "Kantorovich potentials". The same formula extends under the usual lower-semicontinuity and integrability assumptions, replacing maxima by suprema when dual optimizers need not exist.*

*Proof.* Weak duality is immediate: if $f(x)+g(y) \leqslant c(x,y)$ and $\pi \in \mathcal{U}(\alpha,\beta)$, then $\int f\mathrm{d}\alpha + \int g\mathrm{d}\beta = \int (f(x)+g(y))\mathrm{d}\pi(x,y) \leqslant \int c\mathrm{d}\pi$. Taking the supremum over admissible potentials and the infimum over couplings gives "$\leqslant$".
For the reverse inequality, view the primal problem as a linear program over the locally convex space of Radon measures, paired with $\mathcal{C}(\mathcal{X} \times \mathcal{Y})$. The affine map $\pi \mapsto (\pi_1, \pi_2)$ is continuous for the weak topology, the feasible set is non-empty because it contains $\alpha \otimes \beta$, and the cost is continuous and bounded on compact sets. Since the set of probability measures on the compact product is weakly compact, the set of attainable cost-marginal triples is closed after adding the epigraph variable below. The separating-hyperplane theorem applied to the convex set of attainable triples $\left\{\left(\pi_1, \pi_2, \int c\mathrm{d}\pi + r\right) : \pi \geqslant 0, \; r \geqslant 0\right\}$ gives a continuous affine separator, hence functions $(f,g)$ and a scalar multiplier which can be normalized so that $f \oplus g \leqslant c$. The separating inequality then states that the supremum over such potentials is at least the primal value. This proves equality. The same argument is the infinite-dimensional analogue of the finite linear-programming proof in Proposition 4.2. ☐

$$\mathrm{Supp}(\pi) \subset \{(x,y) \in \mathcal{X} \times \mathcal{Y} \; ; \; f(x) + g(y) = c(x,y)\}\,. \tag{4.7}$$

For the one-dimensional quadratic cost, the continuous potentials can be read from the monotone map $T = F_\beta^{-1} \circ F_\alpha$: on the active graph, $f'(x) = 2(x - T(x))$ and $g = f^c$.
$s(x,y) := c(x,y) - f(x) - g(y)$. Dual feasibility is the statement $s \geqslant 0$, while complementary slackness says that an optimal coupling can only live where this slack vanishes.

### 4.4 $c$-transforms

This compact subsection collects the transform formulas behind continuous duality.

**Best-response potentials and the $c$-transform.** $\sup_{g\in\mathcal{C}(\mathcal{Y})} \left\{\int g\mathrm{d}\beta\,;\,\forall\,(x,y), g(y)\leqslant c(x,y)-f(x)\right\}$. $\forall\,y\in\mathcal{Y},\quad g(y)\leqslant f^c(y)$

**Definition 4.6** ($c$-transform)**.** For a function $f:\mathcal{X}\to\mathbb{R}\cup\{-\infty\}$, its $c$-transform is

$$\forall\,y\in\mathcal{Y},\quad f^c(y) \coloneqq \inf_{x\in\mathcal{X}} c(x,y)-f(x). \tag{4.8}$$

For a function $g:\mathcal{Y}\to\mathbb{R}\cup\{-\infty\}$, the $\bar{c}$-transform associated with $\bar{c}(y,x)=c(x,y)$ is $\forall\,x\in\mathcal{X},\quad g^{\bar{c}}(x)\coloneqq \inf_{y\in\mathcal{Y}} c(x,y)-g(y)$.

**Proposition 4.7** ($c$-transforms solve the semi-relaxed problems)**.** *For fixed $f$, the maximizers of the dual objective over all $g$ such that $f\oplus g\leqslant c$ are exactly the functions satisfying $g=f^c$ $\beta$-almost everywhere. Equivalently, $f^c$ gives the value of the one-marginal primal problem $\inf_{\pi:\,\pi_2=\beta}\int c(x,y)\mathrm{d}\pi(x,y)-\int f(x)\mathrm{d}\pi_1(x)=\int f^c(y)\mathrm{d}\beta(y)$. Symmetrically, for fixed $g$, the maximizers over $f$ are the functions satisfying $f=g^{\bar{c}}$ $\alpha$-almost everywhere.*

*Proof.* The constraint $f(x)+g(y)\leqslant c(x,y)$ for all $x$ is equivalent, for each fixed $y$, to $g(y)\leqslant \inf_x c(x,y)-f(x)=f^c(y)$. Since $\beta$ is nonnegative, the largest possible value of $\int g\mathrm{d}\beta$ is obtained by saturating this pointwise upper bound on the support of $\beta$. The proof for $f=g^{\bar{c}}$ is identical after exchanging the two marginals.
For the primal formula, disintegrate any feasible $\pi$ as $\pi(\mathrm{d}x,\mathrm{d}y)=\pi_y(\mathrm{d}x)\beta(\mathrm{d}y)$. Then

$$\int c\mathrm{d}\pi-\int f\mathrm{d}\pi_1=\int\left(\int (c(x,y)-f(x))\mathrm{d}\pi_y(x)\right)\mathrm{d}\beta(y)\geqslant\int f^c(y)\mathrm{d}\beta(y).$$

If minimizers admit a measurable selection, equality is obtained by choosing $\pi_y$ supported on minimizers of $x\mapsto c(x,y)-f(x)$. Otherwise one uses approximate measurable selections and lets the approximation error vanish. □

The map $(f,g)\mapsto(g^{\bar{c}},f^c)$ replaces dual potentials by better ones, in the sense that it preserves feasibility and improves the dual objective. Functions of the form $f^c$ and $g^{\bar{c}}$ are called $c$-concave and $\bar{c}$-concave functions.
**Proposition 4.8** (Lipschitz stability of $c$-transforms)**.** *If $c$ is $L$-Lipschitz with respect to its second variable, uniformly in the first one and for the metric $d_{\mathcal{Y}}$ on $\mathcal{Y}$, then $f^c$ is $L$-Lipschitz.*

*Proof.* For each $x$, set $F_x(y)=c(x,y)-f(x)$ and $F(y)=f^c(y)=\inf_x F_x(y)$. Since all the functions $F_x$ are $L$-Lipschitz,

$$|F(y)-F(y')|=\left|\inf_x F_x(y)-\inf_x F_x(y')\right|\leqslant\sup_x|F_x(y)-F_x(y')|\leqslant Ld_{\mathcal{Y}}(y,y').$$

□

**Euclidean case.** The Euclidean quadratic cost is the model case where $c$-transforms become ordinary convex conjugates after removing the quadratic terms. This is the algebraic bridge between Kantorovich duality and Brenier maps.
$\int\|x-y\|^2\mathrm{d}\pi(x,y)=\int\|x\|^2\mathrm{d}\alpha(x)+\int\|y\|^2\mathrm{d}\beta(y)-2\int\langle x,\,y\rangle\mathrm{d}\pi(x,y)$. For $c(x,y)=-\langle x,\,y\rangle$, one has
$f^c(y)=\inf_x-\langle x,\,y\rangle-f(x)=-(-f)^*(y),\qquad h^*(y)\coloneqq\sup_x\langle x,\,y\rangle-h(x)$. Thus $c$-concave functions are negatives of convex functions. In the one-dimensional bilinear model case, the hard double $c$-transform is therefore an operation of taking concave envelopes.

**The failure of alternate optimization.**

**Proposition 4.9** (Algebra of $c$-transforms)**.**

$$\text{(i) } f\leqslant f'\Rightarrow f^c\geqslant f'^c,\qquad\text{(ii) } f^{c\bar{c}}\geqslant f,\qquad\text{(iii) } g^{\bar{c}c}\geqslant g,\qquad\text{(iv) } f^{c\bar{c}c}=f^c.$$

*Proof.* The first inequality (i) follows from the definition of $c$-transforms (because of the $-$ sign). To prove (ii), expanding the definition of $f^{c\bar{c}}$ we have

$$\left(f^{c\bar{c}}\right)(x)=\min_y c(x,y)-f^c(y)=\min_y c(x,y)-\min_{x'}(c(x',y)-f(x')).$$

Now, since $-\min_{x'}\left(c(x',y)-f(x')\right)\geqslant-(c(x,y)-f(x))$, we recover $(f^{c\bar{c}})(x)\geqslant\min_y c(x,y)-c(x,y)+f(x)=f(x)$. The relation $g^{\bar{c}c}\geqslant g$ is obtained in the same way.
Now, to prove (iv), we first apply (ii) and then (i) with $f'=f^{c\bar{c}}$ to have $f^c\geqslant f^{c\bar{c}c}$. Then we apply (iii) to $g=f^c$ to obtain $f^c\leqslant f^{c\bar{c}c}$.

□

$$(f,g)\mapsto(f,f^c)\mapsto(f^{c\bar{c}},f^c)\mapsto(f^{c\bar{c}},f^{c\bar{c}c})=(f^{c\bar{c}},f^c)\ldots \tag{4.9}$$

For the bilinear cost $c(x,y)=-xy$ on a compact interval, the $c$-concave functions are ordinary concave functions and $f^{c\bar{c}}$ is the smallest concave majorant of $f$. In that model case, a hard transform removes non-concave oscillations in one closure step rather than producing a gradual ascent.

# 5 Semi-discrete and $\mathcal{W}_1$

## 5.1 Semi-dual

$\sup_{f,g\in\mathcal{C}(\mathcal{X})\times\mathcal{C}(\mathcal{Y})}\mathcal{E}(f,g)$ where $\mathcal{E}(f,g)$ is the dual objective, with value $-\infty$ when the feasibility constraint fails. One can optimize out $g$ exactly and obtain the following semi-dual problem

$$\sup_{f\in\mathcal{C}(\mathcal{X})}\tilde{\mathcal{E}}(f)\coloneqq\mathcal{E}(f,f^c)=\sup_g\mathcal{E}(f,g)=\int_{\mathcal{X}}f\mathrm{d}\alpha+\int_{\mathcal{Y}}f^c\mathrm{d}\beta. \tag{5.1}$$

## 5.2 Semi-discrete

This compact subsection records the Laguerre-cell formulation of semi-discrete OT.

**Discrete target and Laguerre cells.** One can adapt Definition 4.6 of the $\bar{c}$-transform to this setting by restricting the minimization to the finite support $\{y_j\}_{j=1}^m$ of $\beta$. Equivalently, one applies the same definition with the discrete target space $\mathcal{Y} = \{y_j\}_{j=1}^m$ and identifies a vector $\mathrm{g} \in \mathbb{R}^m$ with the function $g : \mathcal{Y} \to \mathbb{R}$ defined by $g(y_j) = \mathrm{g}_j$:

$$\forall \mathrm{g} \in \mathbb{R}^m,\ \forall x \in \mathcal{X}, \quad \mathrm{g}^{\bar{c}}(x) := \min_{j \in [\![m]\!]} c(x, y_j) - \mathrm{g}_j. \tag{5.2}$$

$$\mathcal{L}_c(\alpha, \beta) = \max_{\mathrm{g} \in \mathbb{R}^m} \mathcal{E}(\mathrm{g}) := \int_{\mathcal{X}} \mathrm{g}^{\bar{c}}(x) \mathrm{d}\alpha(x) + \sum_j \mathrm{g}_j \mathrm{b}_j. \tag{5.3}$$

**Definition 5.1** (Laguerre cells and power diagrams)**.** For sites $(y_j)_{j=1}^m$ and weights $\mathrm{g} \in \mathbb{R}^m$, the Laguerre cell associated with $y_j$ is

$$\mathbb{L}_j(\mathrm{g}) := \{x \in \mathcal{X} \ ; \ \forall j' \neq j, c(x, y_j) - \mathrm{g}_j \leqslant c(x, y_{j'}) - \mathrm{g}_{j'}\}. \tag{5.4}$$

The cells cover $\mathcal{X}$; after arbitrary tie-breaking on common boundaries, they induce a disjoint partition. When $c(x, y) = \|x - y\|^2$, this Laguerre decomposition is also called a power diagram. If g is constant, it reduces to the ordinary Voronoi diagram.

**Mass balance.**

$$\mathcal{E}(\mathrm{g}) = \sum_{j=1}^m \int_{\mathbb{L}_j(\mathrm{g})} (c(x, y_j) - \mathrm{g}_j)\, \mathrm{d}\alpha(x) + \langle \mathrm{g}, \mathrm{b} \rangle. \tag{5.5}$$

**Proposition 5.2** (Gradient of the semi-discrete dual)**.** *If $\alpha$ gives zero mass to the Laguerre cell boundaries, then $\mathcal{E}$ is differentiable at* g *and* $\forall j \in [\![m]\!], \quad \nabla \mathcal{E}(\mathrm{g})_j = \mathrm{b}_j - \int_{\mathbb{L}_j(\mathrm{g})} \mathrm{d}\alpha$.

*Proof.* For $\alpha$-almost every $x$, the minimizing index in $\min_j c(x, y_j) - \mathrm{g}_j$ is unique. If this index is $j(x)$, then the directional derivative in a direction $h \in \mathbb{R}^m$ is

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \min_j \big(c(x, y_j) - (\mathrm{g}_j + t h_j)\big) = -h_{j(x)}.$$

Dominated convergence gives $\mathrm{d}\mathcal{E}(\mathrm{g})[h] = -\sum_j h_j \int_{\mathbb{L}_j(\mathrm{g})} \mathrm{d}\alpha + \sum_j h_j \mathrm{b}_j$, which is the announced gradient formula. □

**Stochastic optimization.**

$$\mathcal{E}(\mathrm{g}) = \int_{\mathcal{X}} E(\mathrm{g}, x) \mathrm{d}\alpha(x) = \mathbb{E}_X(E(\mathrm{g}, X)), \qquad E(\mathrm{g}, x) := \mathrm{g}^{\bar{c}}(x) + \langle \mathrm{g}, \mathrm{b} \rangle. \tag{5.6}$$

$$\nabla_{\mathrm{g}} E(\mathrm{g}, x) = \big(\mathrm{b}_j - \mathbb{1}_{\mathbb{L}_j(\mathrm{g})}(x)\big)_{j=1}^m,$$

$$\mathrm{g}^{(\ell+1)} := \mathrm{g}^{(\ell)} + \tau_\ell \nabla_{\mathrm{g}} E(\mathrm{g}^{(\ell)}, x_\ell). \tag{5.7}$$

$j_\ell \in \operatorname{argmin}_j \big(c(x_\ell, y_j) - \mathrm{g}_j^{(\ell)}\big)$, $\mathrm{g}_j^{(\ell+1)} = \mathrm{g}_j^{(\ell)} + \tau_\ell \big(\mathrm{b}_j - \mathbb{1}_{\{j = j_\ell\}}\big)$.

$$\tau_\ell := \frac{\tau_0}{1 + \ell/\ell_0}, \tag{5.8}$$

where $\ell_0$ is a warmup scale. Under standard stochastic-approximation assumptions, one obtains the usual sublinear rate $\mathcal{E}(\mathrm{g}^\star) - \mathbb{E}\big(\mathcal{E}(\mathrm{g}^{(\ell)})\big) = O(\ell^{-1/2})$, where $\mathrm{g}^\star$ is a maximizer and the expectation is over the i.i.d. samples.

## 5.3 Optimal Quantization

$$\mathcal{Q}_m(\alpha) := \min_{Y = (y_j)_{j=1}^m,\ \mathrm{b} \in \Sigma_m} \mathcal{W}_p \left( \alpha, \sum_{j=1}^m \mathrm{b}_j \delta_{y_j} \right). \tag{5.9}$$

**Free masses and prescribed weights.**

**Proposition 5.3** (Quantization rate and curse of dimensionality)**.** *Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and assume $\alpha = \rho \mathrm{d}x$ on $\Omega$, with $0 < \rho_- \leqslant \rho \leqslant \rho_+ < +\infty$. Then, for fixed $p \geqslant 1$, there exist constants $0 < c \leqslant C < +\infty$ such that $c\, m^{-1/d} \leqslant \mathcal{Q}_m(\alpha) \leqslant C\, m^{-1/d}$.*

*Proof.* For the upper bound, partition $\Omega$ into $m$ cells of diameter at most $Cm^{-1/d}$, up to boundary effects, and put one codepoint in each nonempty cell. Sending each point to the codepoint in its cell gives a transport distance bounded by $Cm^{-1/d}$.
For the lower bound, fix any set $Y$ of $m$ codepoints and write $d_Y(x) = \min_j \|x - y_j\|$. Since the density is bounded above, the mass of the $t$-neighborhood of $Y$ is at most $Cmt^d$. Choosing $t_0 \simeq m^{-1/d}$ small enough gives $\alpha(\{d_Y > t\}) \geqslant c$ for $0 < t < t_0$. Hence $\int d_Y(x)^p \mathrm{d}\alpha(x) = \int_0^{+\infty} p t^{p-1} \alpha(\{d_Y > t\}) \mathrm{d}t \geqslant c t_0^p \simeq c m^{-p/d}$. Taking the $p$-th root and minimizing over $Y$ proves the lower bound. □

**Lloyd algorithm.**

**Proposition 5.4** (Free masses give Voronoi cells)**.** *For the cost $c(x, y) = d(x, y)^p$, fix distinct codepoints $Y = (y_j)_{j=1}^m$. Duplicate codepoints can be merged beforehand. Minimizing over the weights $\mathrm{b} \in \Sigma_m$ gives*

$$\min_{\mathrm{b} \in \Sigma_m} \mathcal{W}_p^p \left( \alpha, \sum_j \mathrm{b}_j \delta_{y_j} \right) = \int_{\mathcal{X}} \min_{1 \leqslant j \leqslant m} c(x, y_j) \mathrm{d}\alpha(x).$$

*An optimal coupling is induced by sending each $x$ to a nearest codepoint. The corresponding cells are the Voronoi cells $\mathbb{V}_j(Y) := \{x \ ; \ \forall j',\ c(x, y_j) \leqslant c(x, y_{j'})\}$, up to arbitrary tie-breaking on common boundaries.*

*Proof.* For any coupling between $\alpha$ and a measure supported on $Y$, the conditional destination of a point $x$ belongs to $Y$, hence its conditional cost is at least $\min_j c(x, y_j)$. Integrating gives the lower bound. Conversely, choose a measurable nearest-codepoint map $T_Y(x) \in \operatorname{argmin}_j c(x, y_j)$, breaking ties measurably, and set $\mathrm{b}_j = \alpha(T_Y^{-1}(y_j))$. Then $(T_Y)_\sharp\alpha = \sum_j \mathrm{b}_j \delta_{y_j}$ and the induced transport reaches the displayed lower bound. ☐

$\mathcal{Q}_m(\alpha)^p = \min_Y \mathcal{F}(Y), \qquad \mathcal{F}(Y) := \int_{\mathcal{X}} \min_{1 \leqslant j \leqslant m} c(x, y_j) \mathrm{d}\alpha(x)$. $y_j \in \operatorname{argmin}_y \int_{\mathbb{V}_j(Y)} c(x, y) \mathrm{d}\alpha(x)$. $y_j = \frac{\int_{\mathbb{V}_j(Y)} x \mathrm{d}\alpha(x)}{\int_{\mathbb{V}_j(Y)} \mathrm{d}\alpha}$.

**Continuous Lloyd flow.** $a_j(Y) = \alpha(\mathbb{V}_j(Y)), \qquad b_j(Y) = \frac{1}{a_j(Y)} \int_{\mathbb{V}_j(Y)} x \,\mathrm{d}\alpha(x)$ $y_j^{(k+1)} = y_j^{(k)} + \tau\big(b_j(Y^{(k)}) - y_j^{(k)}\big), \qquad 0 < \tau \leqslant 1$, $\dot{y}_j(t) = b_j(Y_t) - y_j(t)$. $\nabla_{y_j}\mathcal{F}(Y) = 2\int_{\mathbb{V}_j(Y)} (y_j - x) \mathrm{d}\alpha(x) = 2a_j(Y)(y_j - b_j(Y))$, $\dot{y}_j(t) = -\frac{1}{2a_j(Y_t)} \nabla_{y_j}\mathcal{F}(Y_t)$. $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}(Y_t) = -2\sum_j a_j(Y_t) \|b_j(Y_t) - y_j(t)\|^2 \leqslant 0$. If $\alpha_t = \sum_j w_j \delta_{y_j(t)}$ carries fixed positive weights, independent of the Voronoi masses $a_j(Y_t)$, Proposition 13.1 turns this labelled particle ODE into the weak continuity equation
$\partial_t \alpha_t + \operatorname{div}(v_{\alpha_t}\alpha_t) = 0, \qquad v_{\alpha_t}(y_j(t)) = b_j(Y_t) - y_j(t)$, in the sense of the measure evolutions of Section 13. The weights $w_j$ in this transport equation are auxiliary weights for the moving labelled particles; they are not the Voronoi masses used to define the quantization energy.
$\nu_{Y_t} = \sum_j a_j(Y_t)\delta_{y_j(t)}$, $\partial_t \nu_{Y_t} + \operatorname{div}(v_t \nu_{Y_t}) = \sum_j \dot{a}_j(Y_t)\delta_{y_j(t)}, \qquad v_t(y_j(t)) = \dot{y}_j(t)$.
**Mean-field limit and ultrafast diffusion.** $\mathcal{G}_\rho(f) := \int_\Omega \rho(x) f(x)^{-r} \mathrm{d}x, \qquad \int_\Omega f \,\mathrm{d}x = 1$,

$$\partial_t f = -r \operatorname{div}\left(f \nabla\left(\frac{\rho}{f^{r+1}}\right)\right),$$

$$\partial_t u = -\frac{r+1}{\omega} \operatorname{div}\big(\omega \nabla(u^{-r})\big),$$

**Quantization with fixed equal weights.** $\nu_Y := \frac{1}{m}\sum_{j=1}^m \delta_{y_j}, \qquad \mathcal{F}_{\mathrm{eq}}(Y) := \frac{1}{2}\mathcal{W}_2^2(\alpha, \nu_Y)$, $\bar{x}_j(Y) := m\int_{C_j(Y)} x \mathrm{d}\alpha(x)$. $\nabla_{y_j}\mathcal{F}_{\mathrm{eq}}(Y) = \frac{1}{m}\big(y_j - \bar{x}_j(Y)\big)$. The $\mathcal{W}_2$ metric on equal-weight empirical measures induces the particle metric $g_Y(U, V) = m^{-1}\sum_j \langle u_j, v_j\rangle$. Hence the associated $\mathcal{W}_2$ gradient flow is the coupled system
$\dot{y}_j(t) = \bar{x}_j(Y_t) - y_j(t), \qquad j = 1, \ldots, m$. Equivalently, $\nu_{Y_t}$ satisfies a continuity equation with velocity $v_t(y_j(t)) = \bar{x}_j(Y_t) - y_j(t)$. This is the so-called finite-particle $\mathcal{W}_2$ gradient-flow viewpoint developed more systematically in Chapter 14; the fixed-weight cells are Laguerre cells rather than the free-mass Voronoi cells used by Lloyd's method.
**Equal-weight quantization on the line.**
**Proposition 5.5** (One-dimensional equal-weight quantization)**.** *Let $\alpha \in \mathcal{M}_+^1(\mathbb{R})$ have finite second moment and quantile function $Q = \mathcal{C}_\alpha^{-1}$. For*

$$\mathcal{Q}_m^{\mathrm{eq}}(\alpha)^2 := \min_{y_1 \leqslant \cdots \leqslant y_m} \mathcal{W}_2^2\left(\alpha, \frac{1}{m}\sum_{i=1}^m \delta_{y_i}\right),$$

*set $I_i = ((i-1)/m, i/m]$. Then the sorted minimizer is unique and its ith atom is $y_i^\star = m\int_{I_i} Q(u)\mathrm{d}u$, and $\mathcal{Q}_m^{\mathrm{eq}}(\alpha)^2 = \sum_{i=1}^m \int_{I_i} |Q(u) - y_i^\star|^2 \,\mathrm{d}u$. If $Q \in C^1([0,1])$, then $m^2\mathcal{Q}_m^{\mathrm{eq}}(\alpha)^2 \longrightarrow \frac{1}{12}\int_0^1 |Q'(u)|^2 \,\mathrm{d}u$.*

*Proof.* After sorting the atoms, the quantile formula for $\mathcal{W}_2$ gives

$$\mathcal{W}_2^2\left(\alpha, \frac{1}{m}\sum_{i=1}^m \delta_{y_i}\right) = \sum_{i=1}^m \int_{I_i} |Q(u) - y_i|^2 \,\mathrm{d}u.$$

The minimization therefore decouples over the intervals $I_i$, and the best constant approximation of $Q$ on $I_i$ in $L^2$ is its average on $I_i$. This proves the formula for $y_i^\star$ and the energy.
Denote by $\bar{Q}_{I_i} = m\int_{I_i} Q(u)\mathrm{d}u$ the interval average. If $Q$ is $C^1$, set $h = 1/m$ and write $I_i = (a_i, a_i + h]$. Uniform Taylor expansion gives, for $v \in [0,1]$, $Q(a_i + hv) = Q(a_i) + hvQ'(a_i) + h\, r_i(v), \qquad \max_i \sup_{v \in [0,1]} |r_i(v)| \to 0$. Subtracting the average over $v \in [0,1]$ and integrating gives $\int_{I_i} \left|Q(u) - \bar{Q}_{I_i}\right|^2 \mathrm{d}u = \frac{h^3}{12}|Q'(a_i)|^2 + o(h^3)$, uniformly in $i$. Summing over $i$ yields a Riemann sum for $\frac{1}{12}\int_0^1 |Q'(u)|^2 \,\mathrm{d}u$. ☐

Thus the common deterministic rule $m^{-1}\sum_i \delta_{Q((i-1/2)/m)}$ should be read as a midpoint approximation of the optimal bin-average formula. It has the same leading squared error under the same smoothness assumptions.
**Proposition 5.6** (Quantile-process asymptotics for random placement)**.** *Let $\alpha \in \mathcal{M}_+^1(\mathbb{R})$ have quantile $Q \in C^1([0,1])$, and let $\widehat{\alpha}_m = m^{-1}\sum_{i=1}^m \delta_{X_i}$ with $X_i$ iid with law $\alpha$. If $B$ denotes the standard Brownian bridge on $[0,1]$, then $m\,\mathcal{W}_2^2(\alpha, \widehat{\alpha}_m) \xrightarrow{\text{law}} \int_0^1 B(u)^2 |Q'(u)|^2 \,\mathrm{d}u$, and, in expectation,*

$$m\,\mathbb{E}\big[\mathcal{W}_2^2(\alpha, \widehat{\alpha}_m)\big] \longrightarrow \int_0^1 u(1-u)\,|Q'(u)|^2 \,\mathrm{d}u.$$

*Proof.* Let $\widehat{Q}_m$ be the empirical quantile function. The one-dimensional formula gives $\mathcal{W}_2^2(\alpha, \widehat{\alpha}_m) = \int_0^1 \left|\widehat{Q}_m(u) - Q(u)\right|^2 \mathrm{d}u$. Write $X_i = Q(U_i)$ with $U_i$ iid uniform on $[0,1]$, and let $\widehat{U}_m^{-1}$ be the empirical quantile function of the $U_i$. Then $\widehat{Q}_m = Q \circ \widehat{U}_m^{-1}$. The classical uniform quantile-process theorem gives $\sqrt{m}\,(\widehat{U}_m^{-1} - \mathrm{Id}) \xrightarrow{\text{law}} -B$ in $L^2(0,1)$. Since $Q \in C^1([0,1])$, the map $u \mapsto Q(u)$ is Hadamard differentiable along these perturbations, and therefore $\sqrt{m}\,(\widehat{Q}_m - Q) \xrightarrow{\text{law}} -B\,Q'$ in $L^2(0,1)$. Since $B$ and $-B$ have the same law, the sign disappears in the squared $L^2$ norm. The continuous mapping theorem gives the distributional convergence. Standard second-moment bounds for the uniform quantile process, together with the boundedness of $Q'$, give uniform integrability, hence convergence of expectations. Since $\mathbb{E}[B(u)^2] = u(1-u)$, Fubini's theorem gives the displayed expectation limit. ☐

### 5.4 Wasserstein-1 norm

This compact subsection summarizes the dual and graph formulations of $\mathcal{W}_1$.

**$c$-transform for $\mathcal{W}_1$.** Assume that $d$ is a distance on $\mathcal{X} = \mathcal{Y}$ and take the ground cost $c(x, y) = d(x, y)$. We denote the Lipschitz constant of $f \in \mathcal{C}(\mathcal{X})$ by

**Definition 5.7** (Lipschitz constant)**.** For a function $f : \mathcal{X} \to \mathbb{R}$ on a metric space $(\mathcal{X}, d)$, its Lipschitz constant is

$$\mathrm{Lip}(f) := \sup \left\{ \frac{|f(x) - f(y)|}{d(x, y)} \; ; \; x \neq y \right\}. \tag{5.10}$$

The function is 1-Lipschitz when $\mathrm{Lip}(f) \leqslant 1$.

**Proposition 5.8** ($c$-transforms and 1-Lipschitz functions)**.** *Suppose $\mathcal{X} = \mathcal{Y}$ and $c(x, y) = d(x, y)$. Then there exists $g$ such that $f = g^c$ if and only if $\mathrm{Lip}(f) \leqslant 1$. Furthermore, if $\mathrm{Lip}(f) \leqslant 1$, then $f^c = -f$.*

*Proof.* First suppose $f = g^c$ for some $g$. For $x, y \in \mathcal{X}$,

$$|f(x) - f(y)| = \left| \inf_z [d(x, z) - g(z)] - \inf_z [d(y, z) - g(z)] \right| \leqslant \sup_z |d(x, z) - d(y, z)| \leqslant d(x, y),$$

where the last inequality is the reverse triangle inequality. Thus $\mathrm{Lip}(f) \leqslant 1$.
If $\mathrm{Lip}(f) \leqslant 1$, then $f(x) \leqslant f(y) + d(x, y)$, so $d(x, y) - f(x) \geqslant -f(y)$ for all $x$ and hence $f^c(y) \geqslant -f(y)$. Taking $x = y$ gives $f^c(y) \leqslant -f(y)$. Therefore $f^c = -f$. Applying the same property to $-f$ gives $(-f)^c = f$, so every 1-Lipschitz function is $c$-concave. □

$$\mathcal{W}_1(\alpha, \beta) = \max_f \left\{ \int_{\mathcal{X}} f \mathrm{d}(\alpha - \beta) \; ; \; \mathrm{Lip}(f) \leqslant 1 \right\} =: \|\alpha - \beta\|_{W_1}. \tag{5.11}$$

For a discrete signed measure $\alpha - \beta = \sum_k r_k \delta_{z_k}$ with $\sum_k r_k = 0$, (5.11) becomes the finite-dimensional linear program

$$\mathcal{W}_1(\alpha, \beta) = \max_{(f_k)_k} \left\{ \sum_k f_k r_k \; ; \; \forall\, k, \ell, \; |f_k - f_\ell| \leqslant d(z_k, z_\ell) \right\}. \tag{5.12}$$

This linear program can be solved by generic interior-point or first-order methods. If $N$ support points are involved, however, it still contains $O(N^2)$ Lipschitz constraints, mirroring the $O(nm)$ coupling variables of the original discrete Kantorovich LP; the dual formulation alone does not remove the all-pairs structure.
$\mathcal{W}_1(\alpha, \beta) = \max_{(f_k)_k} \left\{ \sum_k f_k r_k \; ; \; \forall\, k, \; |f_{k+1} - f_k| \leqslant z_{k+1} - z_k \right\}.$

**$\mathcal{W}_1$ on Euclidean spaces.**

$$\mathcal{W}_1(\alpha, \beta) = \sup_f \left\{ \int_{\mathbb{R}^d} f(\mathrm{d}\alpha - \mathrm{d}\beta) \; ; \; \|\nabla f\|_\infty \leqslant 1 \right\}. \tag{5.13}$$

$-\iota_{\|\cdot\|_{\mathbb{R}^d} \leqslant 1}(u) = \inf_v \langle u, \, v \rangle + \|v\|_{\mathbb{R}^d},$

$$\begin{aligned}
\mathcal{W}_1(\alpha, \beta) &= \sup_f \inf_{s(x) \in \mathbb{R}^d} \int_{\mathbb{R}^d} f \mathrm{d}\xi + \int \langle \nabla f(x), \, s(x) \rangle \mathrm{d}x + \int \|s(x)\|_{\mathbb{R}^d} \mathrm{d}x \\
&= \inf_{s(x) \in \mathbb{R}^d} \int \|s(x)\| \mathrm{d}x + \sup_f \int f(x)(\mathrm{d}\xi - \mathrm{div}(s)\mathrm{d}x)
\end{aligned}$$

$$\mathcal{W}_1(\alpha, \beta) = \inf_s \left\{ \int_{\mathbb{R}^d} \|s(x)\|_{\mathbb{R}^d} \mathrm{d}x \; ; \; \mathrm{div}(s) = \alpha - \beta \right\}, \tag{5.14}$$

**Graph distances and Beckmann flows.**

**Definition 5.9** (Graph geodesic distance)**.** Let $G = (V, E)$ be a connected finite graph with positive edge lengths $(\ell_e)_{e \in E}$. The graph geodesic distance between two vertices is $d_G(i, j) = \min_{\gamma : i \rightsquigarrow j} \sum_{e \in \gamma} \ell_e$. The minimum is over all paths $\gamma$ joining $i$ to $j$.

**Proposition 5.10** ($\mathcal{W}_1$ and Beckmann flow on a graph)**.** *Let $G = (V, E)$ be a connected finite graph with positive edge lengths $(\ell_e)_{e \in E}$ and graph geodesic distance $d_G$. For two probability vectors $\mathrm{a}, \mathrm{b}$ on $V$, set $r = \mathrm{a} - \mathrm{b}$ and orient each edge $e = (i, j)$. If $(\nabla_G f)_e = f_j - f_i$, $\quad \mathrm{div}_G = -\nabla_G^*$ are the finite-difference gradient and its negative adjoint, then a positive flow on the oriented edge $i \to j$ has positive divergence at $i$ and negative divergence at $j$. With this convention,*

$$\mathcal{W}_{1,G}(\mathrm{a}, \mathrm{b}) = \max_f \left\{ \sum_{i \in V} f_i r_i \; ; \; |f_i - f_j| \leqslant \ell_e \quad \forall e = (i, j) \right\} = \min_m \left\{ \sum_{e \in E} \ell_e |m_e| \; ; \; \mathrm{div}_G \, m = r \right\}.$$

*The vector $m_e$ is an oriented edge flow, and the constraint $\mathrm{div}_G \, m = r$ is conservation of mass at each vertex.*

*Proof.* The edge constraint $|f_i - f_j| \leqslant \ell_e$ implies, by summing along paths, that $|f_i - f_j| \leqslant d_G(i, j)$ for all vertices. Conversely, any 1-Lipschitz function for $d_G$ satisfies the edge constraints because each edge is a path of length $\ell_e$. The first equality is therefore the Kantorovich–Rubinstein formula on the metric space $(V, d_G)$.
For the second equality, write the graph Beckmann problem and dualize its equality constraint with a potential $f$: $\inf_m \sum_e \ell_e |m_e| + \sup_f \sum_i f_i (r_i - (\mathrm{div}_G \, m)_i)$. Using $\mathrm{div}_G = -\nabla_G^*$, the coupling term is $\sum_e m_e (\nabla_G f)_e$. The minimization over each scalar flow $m_e$ is finite exactly when $|(\nabla_G f)_e| \leqslant \ell_e$, and is then equal to zero. The dual problem is therefore precisely the graph Lipschitz dual above. Strong duality holds because this is a finite-dimensional linear program with a nonempty feasible set: connectedness and $\sum_i r_i = 0$ allow the signed surplus to be routed along paths. This proves the graph Beckmann formula. □

# 6 Divergences and Dual Norms

## 6.1 Dual norms (Integral Probability Metrics)

This compact subsection compares weak discrepancies through their discriminator classes.

**Test-function viewpoint.** Dual norms compare signed measures by restricting admissible test functions.

**Integral probability metrics.**

**Definition 6.1** (Dual norm and integral probability metric)**.** For a symmetric convex set $B$ of measurable functions, define on signed measures $\xi$

$$\|\xi\|_B := \sup_f \left\{ \int_{\mathcal{X}} f(x) \mathrm{d}\xi(x) \; ; \; f \in B \right\}. \tag{6.1}$$

When this quantity is applied to $\alpha - \beta$ for probability measures, it is often called an integral probability metric.

**Proposition 6.2** (Metrization by dual norms)**.** *Assume that $\mathcal{X}$ is compact, that $B = -B$, and that the measures considered are probability measures.*

*1. If every function in $\mathcal{C}(\mathcal{X})$ can be uniformly approximated by elements of* $\operatorname{Span}(B)$*, then $\|\alpha_n - \alpha\|_B \to 0$ implies $\alpha_n \rightharpoonup \alpha$.*

*2. If $B \subset \mathcal{C}(\mathcal{X})$ is compact for $\|\cdot\|_\infty$, then $\alpha_n \rightharpoonup \alpha$ implies $\|\alpha_n - \alpha\|_B \to 0$.*

*Proof.* For the first implication, $\|\alpha_n - \alpha\|_B \to 0$ and the symmetry of $B$ imply $|\langle f, \alpha_n - \alpha\rangle| \leqslant \|\alpha_n - \alpha\|_B$ for $f \in B$. By linearity, integrals converge for every $h \in \operatorname{Span}(B)$. Let $u \in \mathcal{C}(\mathcal{X})$ and choose $h \in \operatorname{Span}(B)$ with $\|u - h\|_\infty \leqslant \eta$. Since $\alpha_n$ and $\alpha$ are probabilities, $|\langle u, \alpha_n - \alpha\rangle| \leqslant |\langle h, \alpha_n - \alpha\rangle| + 2\eta$. Taking the limsup as $n \to \infty$ and then letting $\eta \to 0$ gives $\langle u, \alpha_n\rangle \to \langle u, \alpha\rangle$ for all $u \in \mathcal{C}(\mathcal{X})$, which is weak convergence.

For the second implication, assume $\alpha_n \rightharpoonup \alpha$ and choose a subsequence $(\alpha_{n_k})_k$ such that $\|\alpha_{n_k} - \alpha\|_B \to \limsup_n \|\alpha_n - \alpha\|_B$. Since $B$ is compact and the map $f \mapsto \langle f, \alpha_{n_k} - \alpha\rangle$ is continuous on $B$, the supremum is attained by some $f_{n_k} \in B$. Extracting a further subsequence if needed, $f_{n_k} \to f$ uniformly for some $f \in B$. Then $\langle f_{n_k}, \alpha_{n_k} - \alpha\rangle = \langle f, \alpha_{n_k} - \alpha\rangle + \langle f_{n_k} - f, \alpha_{n_k}\rangle - \langle f_{n_k} - f, \alpha\rangle$. The first term tends to zero by weak convergence and the last two by uniform convergence. Hence every limsup subsequence has limit zero, proving $\|\alpha_n - \alpha\|_B \to 0$. □

**Corollary 6.3** (Wasserstein metrizes weak convergence)**.** *On a compact metric space, $\mathcal{W}_p$ metrizes weak convergence on probability measures for every $p \geqslant 1$.*

*Proof.* For $p = 1$, take $B = \{f \; ; \; \operatorname{Lip}(f) \leqslant 1\}$. The span of $B$ contains all Lipschitz functions, and Lipschitz functions are dense in $\mathcal{C}(\mathcal{X})$ on compact metric spaces.

Conversely, constants do not change the pairing with $\alpha_n - \alpha$. Fix $x_0 \in \mathcal{X}$ and normalize potentials by $f(x_0) = 0$. The normalized unit Lipschitz ball is uniformly bounded by $\operatorname{diam}(\mathcal{X})$ and equicontinuous, hence compact in $\|\cdot\|_\infty$ by Arzelà–Ascoli. Proposition 6.2 gives $\mathcal{W}_1(\alpha_n, \alpha) \to 0$. Proposition 3.34 shows that all $\mathcal{W}_p$ distances induce the same topology on a compact space, so the result follows for every $p \geqslant 1$. □

## 6.2 Dual RKHS Norms and Maximum Mean Discrepancies

**Definition 6.4** (Positive and conditionally positive kernels)**.** A symmetric function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is positive definite if for every $n \geqslant 1$, every $x_1, \ldots, x_n \in \mathcal{X}$ and every $r \in \mathbb{R}^n$,

$$\sum_{i,j=1}^n r_i r_j k(x_i, x_j) \geqslant 0. \tag{6.2}$$

It is conditionally positive definite if the same inequality is required only for zero-sum vectors, $\langle r, \mathbb{1}_n\rangle = 0$.

**Definition 6.5** (Kernel norm and MMD)**.** Let $k$ be positive definite. More generally, let $k$ be conditionally positive definite and restrict attention to signed measures of total mass zero. For a signed measure $\xi$ with finite kernel energy, the associated norm is

$$\|\xi\|_k^2 := \iint_{\mathcal{X} \times \mathcal{X}} k(x, y) \mathrm{d}\xi(x) \mathrm{d}\xi(y). \tag{6.3}$$

For two probability measures, the maximum mean discrepancy associated with $k$ is $\operatorname{MMD}_k(\alpha, \beta) := \|\alpha - \beta\|_k$.

**Proposition 6.6** (Kernel norm as an RKHS dual norm)**.** *Let $\mathcal{H}$ be the RKHS with reproducing kernel $k$, and assume that the kernel mean embedding $m_\xi := \int k(x, \cdot) \mathrm{d}\xi(x)$ is well-defined for the signed measure $\xi$. Then $\|\xi\|_k = \sup_{\|h\|_{\mathcal{H}} \leqslant 1} \int h(x) \mathrm{d}\xi(x)$, so $\|\cdot\|_k$ is the dual norm in the sense of (6.1) associated with the RKHS unit ball.*

*Proof.* By the reproducing property,

$$\int h(x) \mathrm{d}\xi(x) = \left\langle h, \int k(x, \cdot) \mathrm{d}\xi(x) \right\rangle_{\mathcal{H}} = \langle h, m_\xi\rangle_{\mathcal{H}}.$$

Cauchy–Schwarz gives $\sup_{\|h\|_{\mathcal{H}} \leqslant 1} \int h \mathrm{d}\xi = \|m_\xi\|_{\mathcal{H}}$. Finally, $\|m_\xi\|_{\mathcal{H}}^2 = \iint k(x, y) \mathrm{d}\xi(x) \mathrm{d}\xi(y)$, which is exactly (6.3). □

**Proposition 6.7** (Universal kernels metrize weak convergence)**.** *Assume that $\mathcal{X}$ is compact and that the RKHS generated by the continuous kernel $k$ is dense in $\mathcal{C}(\mathcal{X})$ for the uniform norm. Then*

$$\operatorname{MMD}_k(\alpha_n, \alpha) \to 0 \iff \alpha_n \rightharpoonup \alpha$$

*for probability measures on $\mathcal{X}$.*

*Proof.* If $\mathrm{MMD}_k(\alpha_n, \alpha) \to 0$, then integrals of all RKHS functions converge. For any $h \in \mathcal{C}(\mathcal{X})$ and any $\eta > 0$, choose $g \in \mathcal{H}$ with $\|h - g\|_\infty \leqslant \eta$. Since $\alpha_n$ and $\alpha$ are probabilities,

$$\left|\int h \,\mathrm{d}(\alpha_n - \alpha)\right| \leqslant 2\eta + \left|\int g \,\mathrm{d}(\alpha_n - \alpha)\right|,$$

and the last term tends to zero. This proves weak convergence. Conversely, if $\alpha_n \rightharpoonup \alpha$, then $\alpha_n \otimes \alpha_n$, $\alpha_n \otimes \alpha$ and $\alpha \otimes \alpha$ converge weakly on the compact product space. Applying this to the continuous bounded function $k$ in the identity $\mathrm{MMD}_k(\alpha_n, \alpha)^2 = \iint k \,\mathrm{d}\alpha_n \mathrm{d}\alpha_n - 2\iint k \,\mathrm{d}\alpha_n \mathrm{d}\alpha + \iint k \,\mathrm{d}\alpha \mathrm{d}\alpha$ gives convergence to zero. ☐

In the special case where $\alpha$ is a discrete measure, one thus has the simple expression

$$\|\alpha\|_k^2 = \sum_{i=1}^n \sum_{i'=1}^n \mathrm{a}_i \mathrm{a}_{i'} \mathrm{k}_{i,i'} = \langle \mathrm{ka}, \mathrm{a} \rangle \quad \text{where} \quad \mathrm{k}_{i,i'} := k(x_i, x_{i'}).$$

$$\|\alpha - \beta\|_k^2 = \sum_{i,i'} \mathrm{a}_i \mathrm{a}_{i'} k(x_i, x_{i'}) + \sum_{j,j'} \mathrm{b}_j \mathrm{b}_{j'} k(y_j, y_{j'}) - 2 \sum_{i,j} \mathrm{a}_i \mathrm{b}_j k(x_i, y_j). \tag{6.4}$$

### 6.3 $\varphi$-divergences

This compact subsection records the density-ratio divergences used to contrast strong and weak topologies.

**Definition by density ratios.**

**Definition 6.8** (Entropy function)**.** A function $\varphi : \mathbb{R} \to \mathbb{R} \cup \{\infty\}$ is an entropy function if it is lower semicontinuous, convex, $\operatorname{dom}\varphi \subset [0, \infty[$, and satisfies the feasibility condition $\operatorname{dom}\varphi \cap (0, +\infty) \neq \emptyset$. The speed of growth of $\varphi$ at $\infty$ is described by

$$\varphi'_\infty = \lim_{x \to +\infty} \varphi(x)/x \in \mathbb{R} \cup \{\infty\}.$$

If $\varphi'_\infty = \infty$, then $\varphi$ grows faster than any linear function and $\varphi$ is said to be *superlinear.*

**Definition 6.9** ($\varphi$-Divergences)**.** Let $\varphi$ be an entropy function. For $\alpha, \beta \in \mathcal{M}(\mathcal{X})$, let $\frac{\mathrm{d}\alpha}{\mathrm{d}\beta}\beta + \alpha^\perp$ be the Lebesgue decomposition of $\alpha$ with respect to $\beta$: this means that $\alpha$ is uniquely decomposed as $\alpha^{\mathrm{ac}} + \alpha^\perp$, with $\alpha^{\mathrm{ac}} \ll \beta$, $\alpha^\perp \perp \beta$, and $\alpha^{\mathrm{ac}} = (\mathrm{d}\alpha/\mathrm{d}\beta)\beta$. The divergence $\mathcal{D}_\varphi$ is defined by

$$\mathcal{D}_\varphi(\alpha|\beta) := \int_{\mathcal{X}} \varphi\left(\frac{\mathrm{d}\alpha}{\mathrm{d}\beta}\right) \mathrm{d}\beta + \varphi'_\infty \alpha^\perp(\mathcal{X}) \tag{6.5}$$

if $\alpha, \beta$ are nonnegative and $\infty$ otherwise.

$$\alpha = \sum_i \mathrm{a}_i \delta_{x_i} \quad \text{and} \quad \beta = \sum_i \mathrm{b}_i \delta_{x_i} \tag{6.6}$$

$$\mathrm{D}_\varphi(\mathrm{a}|\mathrm{b}) = \sum_{i \in \mathrm{Supp}(\mathrm{b})} \varphi\left(\frac{\mathrm{a}_i}{\mathrm{b}_i}\right) \mathrm{b}_i + \varphi'_\infty \sum_{i \notin \mathrm{Supp}(\mathrm{b})} \mathrm{a}_i, \tag{6.7}$$

where $\mathrm{Supp}(\mathrm{b}) := \{i \in [\![n]\!] \,;\, \mathrm{b}_i \neq 0\}$.

**Proposition 6.10** (Basic properties of $\varphi$-divergences)**.** *If $\varphi$ is an entropy function, then $\mathcal{D}_\varphi$ is jointly 1-homogeneous, convex and weak-* lower semicontinuous in $(\alpha, \beta)$.*

*Proof.* One defines the associated perspective function

$$\forall\, (u, v) \in (\mathbb{R}_+)^2, \quad \psi(u, v) = \begin{cases} \varphi(u/v)v & \text{if} \quad v \neq 0 \\ u\varphi'_\infty & \text{if} \quad v = 0 \end{cases}$$

The joint 1-homogeneity follows from the definition of this perspective. $\mathrm{D}_\varphi(\mathrm{a}|\mathrm{b}) = \sum_i \psi(\mathrm{a}_i, \mathrm{b}_i)$, and it is enough to show that $\psi$ is convex on $(\mathbb{R}_+)^2$. We first prove this on $\mathbb{R}_+ \times \mathbb{R}_+^*$; the extension to $v = 0$ follows by lower semicontinuity of the recession value $u\varphi'_\infty$.
Indeed, for any $\lambda \in [0, 1]$, $\tau = 1 - \lambda$, set $\theta_1 = \frac{\tau v_1}{\tau v_1 + \lambda v_2}$, $\quad \theta_2 = \frac{\lambda v_2}{\tau v_1 + \lambda v_2}$. Then $\theta_1 + \theta_2 = 1$ and $\frac{\tau u_1 + \lambda u_2}{\tau v_1 + \lambda v_2} = \theta_1 \frac{u_1}{v_1} + \theta_2 \frac{u_2}{v_2}$. Convexity of $\varphi$ therefore gives

$$\varphi\left(\frac{\tau u_1 + \lambda u_2}{\tau v_1 + \lambda v_2}\right)(\tau v_1 + \lambda v_2) \leqslant \tau v_1 \varphi\left(\frac{u_1}{v_1}\right) + \lambda v_2 \varphi\left(\frac{u_2}{v_2}\right).$$

In the general measure case, weak-* lower semicontinuity is the standard lower-semicontinuity theorem for convex integral functionals with recession extension; in the discrete case it is immediate from the lower semicontinuity of $\psi$.

☐

**Proposition 6.11** (Non-negativity of $\varphi$-divergences)**.** *Assume that $\varphi$ is normalized by $\varphi(1) = 0$. For probability distributions $(\alpha, \beta) \in \mathcal{M}_+^1(\mathcal{X})$, one has $\mathcal{D}_\varphi(\alpha|\beta) \geqslant 0$. If $\varphi$ is strictly convex, then one has $\mathcal{D}_\varphi(\alpha|\beta) = 0$ if and only if $\alpha = \beta$.*

*Proof.* Let $m = \alpha + \beta$ and write $a = \frac{\mathrm{d}\alpha}{\mathrm{d}m}$ and $b = \frac{\mathrm{d}\beta}{\mathrm{d}m}$. Using the perspective function $\psi$ from the previous proof, $\mathcal{D}_\varphi(\alpha|\beta) = \int \psi(a, b)\mathrm{d}m$. Since $\alpha$ and $\beta$ are probabilities, $\int a\mathrm{d}m = \int b\mathrm{d}m = 1$. Jensen's inequality and $\psi(1, 1) = \varphi(1) = 0$ give $\mathcal{D}_\varphi(\alpha|\beta) \geqslant \psi\left(\int a\mathrm{d}m, \int b\mathrm{d}m\right) = 0$. If $\varphi$ is strictly convex, equality in Jensen forces $a = b$ $m$-almost everywhere, hence $\alpha = \beta$. In the general non-probability case, if $\varphi \geqslant 0$ then the divergence is positive by construction. ☐

**Classical examples and topology.**

**Main families of $\varphi$-divergences.** $\varphi_\gamma(s) = \frac{s^\gamma - \gamma s + \gamma - 1}{\gamma(\gamma-1)}$ $\quad (\gamma \neq 0, 1)$

$$\mathrm{JS}(\alpha,\beta)^2 = \frac{1}{2}\mathrm{KL}\left(\alpha\middle|\frac{\alpha+\beta}{2}\right) + \frac{1}{2}\mathrm{KL}\left(\beta\middle|\frac{\alpha+\beta}{2}\right),$$

**Variational dual formula.**

**Proposition 6.12** (Dual expression)**.** *A $\varphi$-divergence can be expressed using the Legendre transform $\varphi^{*,\geqslant 0}(s) := \sup_{t\in\mathbb{R}^+} st - \varphi(t)$ (notice that we restrict the function to the positive real) of $\varphi$ as*

$$\mathcal{D}_\varphi(\alpha|\beta) = \sup_{f:\mathcal{X}\to\mathbb{R}} \int_{\mathcal{X}} f(x)\mathrm{d}\alpha(x) - \int_{\mathcal{X}} \varphi^{*,\geqslant 0}(f(x))\mathrm{d}\beta(x). \tag{6.8}$$

*which equivalently reads that the Legendre transform of $\mathcal{D}_\varphi(\cdot|\beta)$ reads*

$$\forall\, f \in \mathcal{C}(\mathcal{X}), \quad \mathcal{D}^*_\varphi(f|\beta) = \int_{\mathcal{X}} \varphi^{*,\geqslant 0}(f(x))\mathrm{d}\beta(x). \tag{6.9}$$

*Proof.* We first consider the superlinear case $\varphi'_\infty = +\infty$, so that $\mathcal{D}_\varphi(\alpha|\beta) = +\infty$ if $\alpha$ does not have a density $\rho \geqslant 0$ with respect to $\beta$, $\mathrm{d}\alpha = \rho\mathrm{d}\beta$. Thus the Legendre-Fenchel transform of $\mathcal{D}_\varphi(\cdot|\beta)$ reads

$$\begin{aligned}\mathcal{D}^*_\varphi(f|\beta) &= \sup_{\rho\geqslant 0} \int_{\mathcal{X}} f(x)\rho(x)\mathrm{d}\beta(x) - \int_{\mathcal{X}} \varphi(\rho(x))\mathrm{d}\beta(x) \\ &= \int_{\mathcal{X}} \sup_{\rho(x)\geqslant 0} \left(f(x)\rho(x) - \varphi(\rho(x))\right)\mathrm{d}\beta(x) = \int_{\mathcal{X}} \varphi^{*,\geqslant 0}(f(x))\mathrm{d}\beta(x).\end{aligned}$$

Fenchel–Moreau then gives the displayed dual expression. For a general entropy, the same argument is applied to the perspective with its recession term; the singular part is exactly encoded by the effective domain of $\varphi^{*,\geqslant 0}$. ☐

### 6.4 GANs via Duality

This compact subsection connects dual discrepancies with adversarial generative modeling.

**Divergence-based adversarial losses.**

$$\min_\theta \mathcal{D}_\varphi(\alpha_\theta|\beta) = \min_\theta \sup_f \int_{\mathcal{X}} f(x)\mathrm{d}\alpha_\theta(x) - \mathcal{D}^*_\varphi(f|\beta) = \min_\theta \sup_f \int_{\mathcal{X}} f(g_\theta(z))\mathrm{d}\zeta(z) - \frac{1}{m}\sum_j \varphi^*(f(y_j)).$$

$\min_\theta \max_\xi \int_{\mathcal{Z}} f_\xi(g_\theta(z))\mathrm{d}\zeta(z) - \frac{1}{m}\sum_j \varphi^*(f_\xi(y_j))$. $\varphi_{\mathrm{JS}}(s) = s\log s - (s+1)\log\frac{s+1}{2}$, $\quad \varphi^*_{\mathrm{JS}}(u) = -\log(2-e^u)$, $\quad u < \log 2$,

**Dual norms and integral probability metrics.**

$$\min_\theta \|\alpha_\theta - \beta\|_B = \min_\theta \sup_{f\in B} \int_{\mathcal{X}} f(x)\mathrm{d}(\alpha_\theta - \beta)(x) = \min_\theta \sup_{f\in B} \int_{\mathcal{Z}} f(g_\theta(z))\mathrm{d}\zeta - \frac{1}{m}\sum_j f(y_j).$$

## 7 Entropic Regularization: Sinkhorn Algorithm

### 7.1 Entropic Regularization for Discrete Measures

**Definition 7.1** (Discrete Shannon–Boltzmann entropy)**.** For a nonnegative matrix $\mathrm{P}$, its Shannon–Boltzmann entropy is $\mathrm{H}(\mathrm{P}) := -\sum_{i,j} \mathrm{P}_{i,j}\log(\mathrm{P}_{i,j})$, with the convention $0\log(0) = 0$.

$$\mathrm{L}^\varepsilon_{\mathrm{C}}(\mathrm{a},\mathrm{b}) := \min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \langle \mathrm{P},\, \mathrm{C}\rangle - \varepsilon\mathrm{H}(\mathrm{P}). \tag{7.1}$$

**Proposition 7.2** (Existence and uniqueness of entropic OT)**.** *Assume that* $\mathrm{a}, \mathrm{b}$ *are probability histograms and that* $\mathrm{C}$ *is finite. For every $\varepsilon > 0$, problem (7.1) admits a unique minimizer. If all entries of* $\mathrm{a}$ *and* $\mathrm{b}$ *are positive, then this minimizer is positive on every entry.*

*Proof.* The transport polytope $\mathrm{U}(\mathrm{a},\mathrm{b})$ is non-empty and compact, and the objective is continuous on it with the convention $0\log 0 = 0$, so a minimizer exists. On the relative interior of the polytope, $-\partial^2\mathrm{H}(\mathrm{P}) = \mathrm{diag}(1/\mathrm{P}_{i,j})$ is positive definite on every non-zero feasible direction. Hence $-\mathrm{H}$ is strictly convex on the polytope and $\langle \mathrm{P},\, \mathrm{C}\rangle - \varepsilon\mathrm{H}(\mathrm{P})$ is strictly convex, which implies uniqueness.

If $\mathrm{a}_i, \mathrm{b}_j > 0$ and a minimizer had $\mathrm{P}_{i,j} = 0$, then for small $t > 0$ the perturbation $\mathrm{P}_t = (1-t)\mathrm{P} + t\,\mathrm{a}\otimes\mathrm{b}$ remains feasible. The directional derivative of the entropic part at $t = 0$ is $-\infty$ because the derivative of $r\log r$ at $0$ is $-\infty$ along a positive direction. Thus the objective decreases for small $t$, contradicting optimality. Therefore the minimizer is strictly positive. ☐

**Smoothing effect.**

> **Remark 7.3 (Entropy barriers versus generic LP barriers).** The entropy used in Sinkhorn is not the generic logarithmic barrier of interior-point linear programming. Its special feature is separability over the entries of P, which turns Bregman projections onto marginal constraints into diagonal matrix scalings.

### 7.2 Sinkhorn's Algorithm

**Proposition 7.4** (Scaling form of entropic OT)**.** *P is the unique solution to* (7.1) *if and only if there exists* $(\mathrm{u},\mathrm{v}) \in \mathbb{R}_+^n \times \mathbb{R}_+^m$ *such that*

$$\forall\,(i,j) \in [\![n]\!] \times [\![m]\!], \quad \mathrm{P}_{i,j} = \mathrm{u}_i \mathrm{K}_{i,j} \mathrm{v}_j \quad \textit{where} \quad \mathrm{K}_{i,j} := e^{-\frac{\mathrm{C}_{i,j}}{\varepsilon}}, \tag{7.2}$$

*and* $\mathrm{P} \in \mathcal{U}(\mathrm{a},\mathrm{b})$.

*Proof.* Without loss of generality, we assume $\mathrm{a}_i, \mathrm{b}_j > 0$ (otherwise, rows or columns with zero mass are fixed to zero and can be removed). By Proposition 7.2, the minimizer is strictly positive on the remaining support.
We can thus ignore the positivity constraint when introducing two dual variables $\mathrm{f} \in \mathbb{R}^n, \mathrm{g} \in \mathbb{R}^m$ for each marginal constraint so that the Lagrangian of (7.1) reads

$$\mathcal{E}(\mathrm{P},\mathrm{f},\mathrm{g}) = \langle \mathrm{P},\, \mathrm{C}\rangle + \varepsilon \sum_{i,j} \mathrm{P}_{i,j} \log(\mathrm{P}_{i,j}) + \langle \mathrm{f},\, \mathrm{a} - \mathrm{P}\mathbb{1}_m\rangle + \langle \mathrm{g},\, \mathrm{b} - \mathrm{P}^\top \mathbb{1}_n\rangle.$$

Considering first-order conditions (where we ignore the positivity constraint as explained above), we have

$$\frac{\partial \mathcal{E}(\mathrm{P},\mathrm{f},\mathrm{g})}{\partial \mathrm{P}_{i,j}} = \mathrm{C}_{i,j} + \varepsilon(\log(\mathrm{P}_{i,j}) + 1) - \mathrm{f}_i - \mathrm{g}_j = 0.$$

which results, in an optimal P coupling of the regularized problem, in the expression $\mathrm{P}_{i,j} = e^{\frac{\mathrm{f}_i + \mathrm{g}_j - \mathrm{C}_{i,j}}{\varepsilon} - 1}$ which can be rewritten in the form provided in the proposition using non-negative vectors $\mathrm{u}_i := e^{\mathrm{f}_i/\varepsilon - 1}$ and $\mathrm{v}_j := e^{\mathrm{g}_j/\varepsilon}$. ☐

$$\mathrm{diag}(\mathrm{u})\mathrm{K}\,\mathrm{diag}(\mathrm{v})\mathbb{1}_m = \mathrm{a}, \quad \text{and} \quad \mathrm{diag}(\mathrm{v})\mathrm{K}^\top \mathrm{diag}(\mathrm{u})\mathbb{1}_n = \mathrm{b}, \tag{7.3}$$

$$\mathrm{u} \odot (\mathrm{K}\mathrm{v}) = \mathrm{a} \quad \text{and} \quad \mathrm{v} \odot (\mathrm{K}^\top \mathrm{u}) = \mathrm{b} \tag{7.4}$$

where $\odot$ corresponds to the entry-wise multiplication of vectors.
An intuitive way to try to solve these equations is to solve them iteratively, by modifying first u so that it satisfies the left-hand side of Equation (7.4) and then v to satisfy its right-hand side. These two updates define Sinkhorn's algorithm

$$\mathrm{u}^{(\ell+1)} := \frac{\mathrm{a}}{\mathrm{K}\mathrm{v}^{(\ell)}} \quad \text{and} \quad \mathrm{v}^{(\ell+1)} := \frac{\mathrm{b}}{\mathrm{K}^\top \mathrm{u}^{(\ell+1)}}, \tag{7.5}$$

initialized with an arbitrary positive vector, for instance $\mathrm{v}^{(0)} = \mathbb{1}_m$. The division operator used above between two vectors is to be understood entry-wise.
A chief computational advantage of Sinkhorn's algorithm, besides its simplicity, is that the only expensive step is multiplication by the Gibbs kernel. Its complexity therefore scales like $Cnm$, where $C$ is the number of Sinkhorn iterations.

### 7.3 Reformulation using relative entropy

**Definition 7.5** (Discrete relative entropy)**.** For nonnegative matrices P, Q of the same size, the generalized relative entropy, or Kullback–Leibler divergence, is

$$\mathrm{KL}(\mathrm{P}|\mathrm{Q}) := \sum_{i,j} \mathrm{P}_{i,j} \log\left(\frac{\mathrm{P}_{i,j}}{\mathrm{Q}_{i,j}}\right) - \mathrm{P}_{i,j} + \mathrm{Q}_{i,j}. \tag{7.6}$$

The convention is $0\log(0) = 0$, and $\mathrm{KL}(\mathrm{P}|\mathrm{Q}) = +\infty$ if there exists $(i,j)$ such that $\mathrm{Q}_{i,j} = 0$ but $\mathrm{P}_{i,j} \neq 0$.

For the specific case of comparing probability distributions, where P and Q have the same total mass, this further simplifies to $\mathrm{KL}(\mathrm{P}|\mathrm{Q}) = \sum_{i,j} \mathrm{P}_{i,j} \log\left(\frac{\mathrm{P}_{i,j}}{\mathrm{Q}_{i,j}}\right)$. For the reference matrix $\mathrm{Q} = \mathbb{1}_{n\times m}$, one has
$-\mathrm{KL}(\mathrm{P}|\mathbb{1}_{n\times m}) = \mathrm{H}(\mathrm{P}) + \sum_{i,j} \mathrm{P}_{i,j} - n\,m$. On fixed-mass couplings the last two terms are constant, so KL regularization with reference $\mathbb{1}_{n\times m}$ is equivalent to subtracting Shannon–Boltzmann entropy. KL is a particular instance of both a $\varphi$-divergence (as defined in Section 6.3) and a Bregman divergence; up to standard affine rescalings, it is the canonical overlap between these two families.

**Proposition 7.6** (Relative entropy is distance-like)**.** *Let* $\mathrm{P},\mathrm{Q} \in \mathbb{R}_+^{n\times m}$ *have the same total mass and assume* $\mathrm{Q}_{i,j} > 0$ *on the support of* P*. Then* $\mathrm{KL}(\mathrm{P}|\mathrm{Q}) \geqslant 0$*, with equality if and only if* $\mathrm{P} = \mathrm{Q}$.

*Proof.* Write $\varphi(s) = s\log s - s + 1$. Convexity gives $\varphi(s) \geqslant \varphi(1) + \varphi'(1)(s-1) = 0$, and strict convexity gives equality only at $s = 1$. Hence $\mathrm{KL}(\mathrm{P}|\mathrm{Q}) = \sum_{i,j} \mathrm{Q}_{i,j}\varphi(\mathrm{P}_{i,j}/\mathrm{Q}_{i,j}) \geqslant 0$, with equality only when $\mathrm{P}_{i,j}/\mathrm{Q}_{i,j} = 1$ for all entries with $\mathrm{Q}_{i,j} > 0$. The support convention rules out positive P where $\mathrm{Q} = 0$, so equality is equivalent to $\mathrm{P} = \mathrm{Q}$. ☐

Equivalently, when P and Q have the same total mass, it reads
$\mathrm{KL}(\mathrm{P}|\mathrm{Q}) = \sum_{i,j} \varphi(\mathrm{P}_{i,j}/\mathrm{Q}_{i,j})\mathrm{Q}_{i,j}$. where $\varphi(s) = s\log(s)$. For any convex $\varphi$ such that $\varphi(1) = 0$, one has indeed by Jensen
$\sum_{i,j} \varphi(\mathrm{P}_{i,j}/\mathrm{Q}_{i,j})\mathrm{Q}_{i,j} \geqslant \varphi(\sum_{i,j} \mathrm{P}_{i,j}/\mathrm{Q}_{i,j}\mathrm{Q}_{i,j}) = \varphi(\sum_{i,j} \mathrm{P}_{i,j}) = \varphi(1) = 0$.

$$\min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \langle \mathrm{P},\, \mathrm{C}\rangle + \varepsilon \mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}). \tag{7.7}$$

For the balanced problem with fixed positive marginals, however, the choice of tensor-product reference does not affect the selected coupling: it only adds a constant to the objective, as shown in the following proposition. In particular, (7.7) and (7.1) have the same unique solution.

**Proposition 7.7** (Reference measure shift for KL)**.** *After removing zero-mass rows and columns, assume that* $\mathrm{a},\mathrm{a}' \in \Sigma_n$ *and* $\mathrm{b},\mathrm{b}' \in \Sigma_m$ *have positive entries. For every* $\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})$*, one has* $\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) = \mathrm{KL}(\mathrm{P}|\mathrm{a}'\otimes\mathrm{b}') - \mathrm{KL}(\mathrm{a}|\mathrm{a}') - \mathrm{KL}(\mathrm{b}|\mathrm{b}')$*. In particular,* (7.7) *and* (7.1) *have the same unique minimizer.*

*Proof.* Expanding the logarithm and using the marginal constraints gives

$$\begin{aligned}\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) &= \mathrm{KL}(\mathrm{P}|\mathrm{a}'\otimes\mathrm{b}') + \sum_i \mathrm{a}_i \log\frac{\mathrm{a}'_i}{\mathrm{a}_i} + \sum_j \mathrm{b}_j \log\frac{\mathrm{b}'_j}{\mathrm{b}_j} \\ &= \mathrm{KL}(\mathrm{P}|\mathrm{a}'\otimes\mathrm{b}') - \mathrm{KL}(\mathrm{a}|\mathrm{a}') - \mathrm{KL}(\mathrm{b}|\mathrm{b}').\end{aligned}$$

□

The tensor-product reference is nevertheless useful when supports vary, because it makes explicit which entries are allowed to vanish. It is also the normalization that passes cleanly to the continuous formulation below, where the reference measure is $\alpha\otimes\beta$ rather than an ambient Lebesgue measure.

**Proposition 7.8** (Convergence with $\varepsilon$)**.** *Assume, after removing zero-mass rows and columns, that* a *and* b *are positive and that* C *is finite. The unique solution* $\mathrm{P}_\varepsilon$ *of* (7.1) *converges to the optimal solution with maximal entropy within the set of all optimal solutions of the Kantorovich problem, namely*

$$\mathrm{P}_\varepsilon \xrightarrow{\varepsilon\to 0} \underset{\mathrm{P}}{\operatorname{argmin}} \ \{-\mathrm{H}(\mathrm{P})\ ;\ \mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b}), \langle \mathrm{P},\, \mathrm{C}\rangle = \mathrm{L}_{\mathrm{C}}(\mathrm{a},\mathrm{b})\} \tag{7.8}$$

*so that in particular*

$$\mathrm{L}^{\varepsilon}_{\mathrm{C}}(\mathrm{a},\mathrm{b}) \xrightarrow{\varepsilon\to 0} \mathrm{L}_{\mathrm{C}}(\mathrm{a},\mathrm{b}).$$

*Moreover,*

$$\mathrm{P}_\varepsilon \xrightarrow{\varepsilon\to\infty} \mathrm{a}\otimes\mathrm{b}. \tag{7.9}$$

*Proof.* **Case** $\varepsilon\to 0$**.** We denote $\mathrm{P}_\ell$ the solution of (7.1) for $\varepsilon=\varepsilon_\ell$.
Since $\mathrm{U}(\mathrm{a},\mathrm{b})$ is bounded, we can extract a sequence (that we do not relabel for the sake of simplicity) such that $\mathrm{P}_\ell\to\mathrm{P}^\star$. Since $\mathrm{U}(\mathrm{a},\mathrm{b})$ is closed, $\mathrm{P}^\star\in\mathrm{U}(\mathrm{a},\mathrm{b})$. Using the equivalent KL-normalized formulation (7.7), optimality of P and $\mathrm{P}_\ell$ for their respective optimization problems (for $\varepsilon=0$ and $\varepsilon=\varepsilon_\ell$) gives

$$0\leqslant \langle \mathrm{C},\, \mathrm{P}_\ell\rangle - \langle \mathrm{C},\, \mathrm{P}\rangle \leqslant \varepsilon_\ell(\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) - \mathrm{KL}(\mathrm{P}_\ell|\mathrm{a}\otimes\mathrm{b})). \tag{7.10}$$

Since KL is continuous, taking the limit $\ell\to+\infty$ in this expression shows that $\langle \mathrm{C},\, \mathrm{P}^\star\rangle = \langle \mathrm{C},\, \mathrm{P}\rangle$ so that $\mathrm{P}^\star$ is a feasible point of (7.8). Furthermore, dividing by $\varepsilon_\ell$ in (7.10) and taking the limit shows that $\mathrm{KL}(\mathrm{P}^\star|\mathrm{a}\otimes\mathrm{b})\leqslant\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b})$, which shows that $\mathrm{P}^\star$ is a solution of (7.8). Since the solution $\mathrm{P}_0^\star$ to this program is unique by strict convexity of $\mathrm{KL}(\cdot|\mathrm{a}\otimes\mathrm{b})$ on the optimal face, one has $\mathrm{P}^\star=\mathrm{P}_0^\star$, and the whole sequence is converging.
**Case** $\varepsilon\to+\infty$**.** Subtracting $\min_{i,j}\mathrm{C}_{i,j}$ from the cost changes every feasible objective by the same constant, so it does not change the minimizer. We can therefore assume $\mathrm{C}\geqslant 0$. Evaluating the energy at $\mathrm{a}\otimes\mathrm{b}$ (which belongs to the constraint set $\mathrm{U}(\mathrm{a},\mathrm{b})$), one has $\langle \mathrm{C},\, \mathrm{P}_\varepsilon\rangle + \varepsilon\mathrm{KL}(\mathrm{P}_\varepsilon|\mathrm{a}\otimes\mathrm{b}) \leqslant \langle \mathrm{C},\, \mathrm{a}\otimes\mathrm{b}\rangle + \varepsilon\times 0$ and since $\langle \mathrm{C},\, \mathrm{P}_\varepsilon\rangle\geqslant 0$, this leads to $\mathrm{KL}(\mathrm{P}_\varepsilon|\mathrm{a}\otimes\mathrm{b})\leqslant\varepsilon^{-1}\langle \mathrm{C},\, \mathrm{a}\otimes\mathrm{b}\rangle\leqslant\frac{\|\mathrm{C}\|_\infty}{\varepsilon}$ so that $\mathrm{KL}(\mathrm{P}_\varepsilon|\mathrm{a}\otimes\mathrm{b})\to 0$ and thus $\mathrm{P}_\varepsilon\to\mathrm{a}\otimes\mathrm{b}$ since KL is a valid divergence. □

## 7.4 General Formulation

This compact subsection records the continuous entropic formulation and its limiting regimes.

**Measure formulation.**

$$\mathcal{L}^{\varepsilon}_c(\alpha,\beta) \coloneqq \min_{\pi\in\mathcal{U}(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\mathrm{d}\pi(x,y) + \varepsilon\,\mathrm{KL}(\pi|\alpha\otimes\beta) \tag{7.11}$$

**Definition 7.9** (Relative entropy of measures)**.** For nonnegative measures $\pi$ and $\xi$ on $\mathcal{X}\times\mathcal{Y}$, the relative entropy is

$$\mathrm{KL}(\pi|\xi) \coloneqq \int_{\mathcal{X}\times\mathcal{Y}} \log\Big(\frac{\mathrm{d}\pi}{\mathrm{d}\xi}(x,y)\Big)\mathrm{d}\pi(x,y) + \int_{\mathcal{X}\times\mathcal{Y}} (\mathrm{d}\xi(x,y) - \mathrm{d}\pi(x,y)). \tag{7.12}$$

By convention, $\mathrm{KL}(\pi|\xi)=+\infty$ if $\pi$ is not absolutely continuous with respect to $\xi$.

For fixed balanced marginals, the specific product reference in the entropy only matters up to additive constants, exactly as in Proposition 7.7, provided the alternative reference marginals are mutually absolutely continuous with $\alpha$ and $\beta$. Its support and absolute-continuity structure are nevertheless essential: they determine which couplings have finite entropy.

**Probabilistic interpretation.** The KL term measures the dependence created by the coupling.

**Definition 7.10** (Mutual information)**.** If $(X,Y)\sim\pi$ have marginals $X\sim\alpha$ and $Y\sim\beta$, the mutual information of the pair is $\mathcal{I}(X,Y)\coloneqq\mathrm{KL}(\pi|\alpha\otimes\beta)$. It is nonnegative and vanishes if and only if $X$ and $Y$ are independent.

$\inf_{X\sim\alpha,\, Y\sim\beta}\mathbb{E}\big(c(X,Y)\big)+\varepsilon\mathcal{I}(X,Y)$.

**Convergence with** $\varepsilon$**.**

**Proposition 7.11** (Convergence with $\varepsilon$ for measures)**.** *Let* $\mathcal{X},\mathcal{Y}$ *be compact metric spaces, let* $c\in\mathcal{C}(\mathcal{X}\times\mathcal{Y})$*, and let* $\alpha\in\mathcal{P}(\mathcal{X})$*,* $\beta\in\mathcal{P}(\mathcal{Y})$*. Assume that finite-entropy couplings are dense in* $\mathcal{U}(\alpha,\beta)$ *for weak convergence with convergence of the cost integral. For* $\varepsilon>0$*, let* $\pi_\varepsilon$ *be a minimizer of* (7.11)*. Then* $\mathcal{L}^{\varepsilon}_c(\alpha,\beta)\longrightarrow\mathcal{L}_c(\alpha,\beta)$ *as* $\varepsilon\downarrow 0$*, and every weak cluster point of* $(\pi_\varepsilon)_\varepsilon$ *is an optimal plan for the unregularized Kantorovich problem. If the latter optimal plan is unique, then* $\pi_\varepsilon$ *converges weakly to it. In particular, for* $\mathcal{X}=\mathcal{Y}\subset\mathbb{R}^d$*,* $c(x,y)=\|x-y\|^2$ *and* $\alpha$ *absolutely continuous,* $\pi_\varepsilon\rightharpoonup(\mathrm{Id},T)_\sharp\alpha$*, where* $T$ *is the Brenier map. As* $\varepsilon\to+\infty$*,*

$$\pi_\varepsilon\longrightarrow\alpha\otimes\beta \quad \textit{in total variation,} \qquad \mathcal{L}^{\varepsilon}_c(\alpha,\beta)\longrightarrow\int_{\mathcal{X}\times\mathcal{Y}} c\,\mathrm{d}\alpha\mathrm{d}\beta.$$

*Proof.* Existence follows from compactness of $\mathcal{U}(\alpha,\beta)$ and lower semicontinuity of the relative entropy. For the limit $\varepsilon \downarrow 0$, take any sequence $\varepsilon_k \downarrow 0$ and a weak cluster point $\pi^\star$ of $\pi_{\varepsilon_k}$. Since $\mathrm{KL}(\pi|\alpha\otimes\beta) \geqslant 0$ on probability couplings, $\liminf_k \mathcal{L}_c^{\varepsilon_k}(\alpha,\beta) \geqslant \liminf_k \int c\,\mathrm{d}\pi_{\varepsilon_k} = \int c\,\mathrm{d}\pi^\star \geqslant \mathcal{L}_c(\alpha,\beta)$. Conversely, by the density assumption, for every $\delta > 0$ there exists a coupling $\pi^\delta$ with finite relative entropy and $\int c\,\mathrm{d}\pi^\delta \leqslant \mathcal{L}_c(\alpha,\beta)+\delta$. Testing (7.11) with $\pi^\delta$ gives $\limsup_k \mathcal{L}_c^{\varepsilon_k}(\alpha,\beta) \leqslant \int c\,\mathrm{d}\pi^\delta$. Letting $\delta \downarrow 0$ proves convergence of the values and forces $\int c\,\mathrm{d}\pi^\star = \mathcal{L}_c(\alpha,\beta)$. Uniqueness of the unregularized optimizer then identifies all cluster points. Brenier's theorem gives the stated quadratic consequence when $\alpha$ is absolutely continuous.
For the limit $\varepsilon \to +\infty$, subtract a constant from $c$ and assume $c \geqslant 0$. Since $\alpha\otimes\beta$ is admissible and has zero relative entropy with respect to itself, $\int c\,\mathrm{d}\pi_\varepsilon + \varepsilon\,\mathrm{KL}(\pi_\varepsilon|\alpha\otimes\beta) \leqslant \int c\,\mathrm{d}\alpha\mathrm{d}\beta$. Hence $\mathrm{KL}(\pi_\varepsilon|\alpha\otimes\beta) \leqslant \varepsilon^{-1}\int c\,\mathrm{d}\alpha\mathrm{d}\beta$. Pinsker's inequality gives total-variation convergence to $\alpha\otimes\beta$, and the convergence of the values follows by continuity of $c$. □

**Large-temperature expansion.** When $\varepsilon$ is large, the entropy dominates and the optimal plan is a small perturbation of the product coupling. The useful quantity is the component of the cost that cannot be absorbed into row and column potentials.
**Proposition 7.12** (Large-temperature expansion)**.** *Let $r = \alpha\otimes\beta$ and assume $c \in L^\infty(r)$. Define the product averages $\bar{c}_{\mathcal{X}}(x) := \int_{\mathcal{Y}} c(x,y)\,\mathrm{d}\beta(y)$, $\bar{c}_{\mathcal{Y}}(y) := \int_{\mathcal{X}} c(x,y)\,\mathrm{d}\alpha(x)$, $\bar{c} := \int_{\mathcal{X}\times\mathcal{Y}} c\,\mathrm{d}r$, and the centered interaction $c_0(x,y) := c(x,y)-\bar{c}_{\mathcal{X}}(x)-\bar{c}_{\mathcal{Y}}(y)+\bar{c}$. Assume that the large-temperature branch $p_\varepsilon = \mathrm{d}\pi_\varepsilon/\mathrm{d}r$, $f_\varepsilon, g_\varepsilon$ admits an expansion to third order in $\varepsilon^{-1}$ near 0 in $L^\infty(r)$, with gauge $\int g_\varepsilon\,\mathrm{d}\beta = 0$. Then $p_\varepsilon(x,y) = 1 - \frac{c_0(x,y)}{\varepsilon} + O(\varepsilon^{-2})$ in $L^2(r)$, and*

$$\mathcal{L}_c^\varepsilon(\alpha,\beta) = \bar{c} - \frac{1}{2\varepsilon}\int_{\mathcal{X}\times\mathcal{Y}} c_0(x,y)^2\,\mathrm{d}r(x,y) + \frac{1}{6\varepsilon^2}\int_{\mathcal{X}\times\mathcal{Y}} c_0(x,y)^3\,\mathrm{d}r(x,y) + O(\varepsilon^{-3}).$$

*Moreover, with $A(x) := \int_{\mathcal{Y}} c_0(x,y)^2\,\mathrm{d}\beta(y)$, $B(y) := \int_{\mathcal{X}} c_0(x,y)^2\,\mathrm{d}\alpha(x)$, $\sigma^2 := \int_{\mathcal{X}\times\mathcal{Y}} c_0^2\,\mathrm{d}r$, one has $f_\varepsilon(x) = \bar{c}_{\mathcal{X}}(x) - \frac{A(x)}{2\varepsilon} + O(\varepsilon^{-2})$, and $g_\varepsilon(y) = \bar{c}_{\mathcal{Y}}(y) - \bar{c} + \frac{\sigma^2 - B(y)}{2\varepsilon} + O(\varepsilon^{-2})$ in the same gauge.*

*Proof.* The optimality equation can be written $p_\varepsilon(x,y) = \exp\left(\frac{f_\varepsilon(x)+g_\varepsilon(y)-c(x,y)}{\varepsilon}\right)$, with the constraints $\int p_\varepsilon(x,y)\,\mathrm{d}\beta(y) = 1$ and $\int p_\varepsilon(x,y)\,\mathrm{d}\alpha(x) = 1$. Write $p_\varepsilon = 1 + \varepsilon^{-1}q + O(\varepsilon^{-2})$. The linearized marginal constraints impose $\int q(x,y)\,\mathrm{d}\beta(y) = 0$, $\int q(x,y)\,\mathrm{d}\alpha(x) = 0$. Expanding the objective gives

$$\int c\,p_\varepsilon\,\mathrm{d}r + \varepsilon\int p_\varepsilon\log(p_\varepsilon)\,\mathrm{d}r = \bar{c} + \frac{1}{\varepsilon}\left(\int c\,q\,\mathrm{d}r + \frac{1}{2}\int q^2\,\mathrm{d}r\right) + O(\varepsilon^{-2}).$$

Thus $q$ is the minimizer of the quadratic expression over zero-row and zero-column perturbations. Orthogonal projection in $L^2(r)$ gives $q = -c_0$, because $c_0$ has zero conditional means. This proves the first-order density and value expansions.
To identify the next displayed terms, write $p_\varepsilon = 1+\varepsilon^{-1}q+\varepsilon^{-2}s+O(\varepsilon^{-3})$ and $f_\varepsilon = f_0+\varepsilon^{-1}f_1+O(\varepsilon^{-2})$, $g_\varepsilon = g_0+\varepsilon^{-1}g_1+O(\varepsilon^{-2})$, where $f_0 = \bar{c}_{\mathcal{X}}$ and $g_0 = \bar{c}_{\mathcal{Y}} - \bar{c}$. Since $p_\varepsilon = \exp\left(-\frac{c_0}{\varepsilon} + \frac{f_1(x)+g_1(y)}{\varepsilon^2} + O(\varepsilon^{-3})\right)$, one has $s = f_1 + g_1 + c_0^2/2$. The second-order marginal constraints and the gauge $\int g_1\,\mathrm{d}\beta = 0$ give $f_1 = -A/2$ and $g_1 = (\sigma^2 - B)/2$. Finally, $p_\varepsilon\log p_\varepsilon = \varepsilon^{-1}q + \varepsilon^{-2}\left(s+\frac{1}{2}q^2\right) + \varepsilon^{-3}\left(qs - \frac{1}{6}q^3\right) + O(\varepsilon^{-4})$, and the zero-conditional-mean property of $c_0$ cancels all additive terms in $s$. The order-$\varepsilon^{-2}$ contribution to the value is therefore $\frac{1}{2}\int c_0^3\,\mathrm{d}r - \frac{1}{3}\int c_0^3\,\mathrm{d}r = \frac{1}{6}\int c_0^3\,\mathrm{d}r$. □

**Small-temperature expansion for smooth densities.**
**Proposition 7.13** (Small-temperature quadratic expansion)**.** *Let $\Omega \subset \mathbb{R}^d$ be a smooth compact domain and let $\alpha = \rho_0\,\mathrm{d}x$, $\beta = \rho_1\,\mathrm{d}x$ have smooth positive densities. Assume that the quadratic optimal map $T$ from $\alpha$ to $\beta$ is a smooth diffeomorphism and that the McCann interpolation $\alpha_t = \rho_t\,\mathrm{d}x = ((1-t)\mathrm{Id}+tT)_\sharp\alpha$ satisfies $\mathcal{I}_0(\alpha,\beta) := \int_0^1\int_\Omega \|\nabla\log\rho_t(x)\|^2\rho_t(x)\,\mathrm{d}x\,\mathrm{d}t < +\infty$. For $c(x,y) = \|x-y\|^2$, define $\mathrm{H}(\alpha) := \int_\Omega \rho_0\log\rho_0\,\mathrm{d}x$ and similarly for $\beta$. Then, as $\varepsilon \downarrow 0$,*

$$\mathcal{L}^\varepsilon_{\|\cdot\|^2}(\alpha,\beta) = \mathcal{W}_2^2(\alpha,\beta) - \frac{d\varepsilon}{2}\log(\pi\varepsilon) - \frac{\varepsilon}{2}\left(\mathrm{H}(\alpha)+\mathrm{H}(\beta)\right) + \frac{\varepsilon^2}{16}\mathcal{I}_0(\alpha,\beta) + o(\varepsilon^2).$$

*If $f_\varepsilon, g_\varepsilon$ are optimal Sinkhorn potentials and $f_0, g_0$ are Kantorovich potentials for the quadratic cost, normalized by $\int g_\varepsilon\,\mathrm{d}\beta = \int g_0\,\mathrm{d}\beta = 0$, then under the same smooth non-degeneracy assumptions, $f_\varepsilon + \frac{d\varepsilon}{2}\log(\pi\varepsilon) \longrightarrow f_0$, $g_\varepsilon \longrightarrow g_0$, locally uniformly. The order-$\varepsilon$ correction to the potentials is obtained by the Laplace prefactor in the soft $c$-transform equations and depends on the endpoint densities and on the Hessian of the quadratic contact phase.*

*Proof.* For the Brownian reference with heat kernel variance $T$, the dynamic Schrödinger formulation gives $T\mathcal{C}_T(\alpha,\beta) = \frac{1}{2}\mathcal{W}_2^2(\alpha,\beta) + \frac{T}{2}\left(\mathrm{H}(\alpha)+\mathrm{H}(\beta)\right) + \frac{T^2}{8}\mathcal{I}_0(\alpha,\beta) + o(T^2)$, under the stated regularity assumptions. Since $\mathcal{C}_T(\alpha,\beta) = \inf_{\pi\in\mathcal{U}(\alpha,\beta)}\mathrm{KL}(\pi|p_T(x,y)\,\mathrm{d}x\mathrm{d}y)$, $p_T(x,y) = (2\pi T)^{-d/2}e^{-\|x-y\|^2/(2T)}$, multiplying by $2T$ and setting $\varepsilon = 2T$ gives $\mathcal{L}^\varepsilon_{\|\cdot\|^2}(\alpha,\beta) = 2T\mathcal{C}_T(\alpha,\beta) - 2T\left(\mathrm{H}(\alpha)+\mathrm{H}(\beta)\right) - dT\log(2\pi T)$, which is exactly the displayed expansion because $2T = \varepsilon$. The potential statement follows by applying Laplace's method to the soft $c$-transform equations. In the chosen gauge the universal Gaussian prefactor is placed in $f_\varepsilon$; changing the additive gauge redistributes the same scalar shift between the two potentials. □

### 7.5 Dual of Sinkhorn

This compact subsection records the dual potentials and soft transforms for Sinkhorn.

**Discrete dual.**
**Proposition 7.14** (Dual of entropic OT)**.** *The optimal value of (7.7) is*

$$\min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \langle \mathrm{P}, \mathrm{C}\rangle + \varepsilon\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) = \max_{\mathrm{f}\in\mathbb{R}^n,\mathrm{g}\in\mathbb{R}^m} \langle \mathrm{f}, \mathrm{a}\rangle + \langle \mathrm{g}, \mathrm{b}\rangle - \varepsilon\sum_{i,j}\exp\left(\frac{\mathrm{f}_i + \mathrm{g}_j - \mathrm{C}_{i,j}}{\varepsilon}\right)\mathrm{a}_i\mathrm{b}_j + \varepsilon. \tag{7.13}$$

*The optimal $(\mathrm{f},\mathrm{g})$ are linked to scalings $(\mathrm{u},\mathrm{v})$ appearing in (7.2) through*

$$\mathrm{u}_i = \mathrm{a}_i e^{\mathrm{f}_i/\varepsilon} \quad \text{and} \quad \mathrm{v}_j = \mathrm{b}_j e^{\mathrm{g}_j/\varepsilon}. \tag{7.14}$$

*Proof.* We introduce Lagrange multipliers and consider $\min_{P \geqslant 0} \max_{f,g} \langle C, P\rangle + \varepsilon \mathrm{KL}(P|a \otimes b) + \langle a - P\mathbb{1}, f\rangle + \langle b - P^\top \mathbb{1}, g\rangle$. Finite-dimensional convex duality allows us to exchange the minimum over P with the maximum over (f, g), giving

$$\max_{f,g} \langle f, a\rangle + \langle g, b\rangle + \varepsilon \min_{P \geqslant 0} \left( \mathrm{KL}(P|a \otimes b) - \langle \frac{f \oplus g - C}{\varepsilon}, P\rangle \right) = \langle f, a\rangle + \langle g, b\rangle - \varepsilon \mathrm{KL}^* \left( \frac{f \oplus g - C}{\varepsilon} |a \otimes b \right).$$

One concludes by using (6.9) for $\varphi(r) = r\log(r) - r + 1$ $\mathrm{KL}^*(H|a \otimes b) = \sum_{i,j} \varphi^*(H_{i,j}) a_i b_j$. Indeed, the scalar maximization $\varphi^*(s) = \sup_{r \geqslant 0}\{rs - r\log r + r - 1\}$ has first-order condition $s - \log r = 0$, hence $r = e^s$ and $\varphi^*(s) = e^s - 1$. □

**Discrete soft $c$-transforms.**

$$a_i - e^{\frac{f_i}{\varepsilon}} a_i \sum_j \exp\left(\frac{g_j - C_{i,j}}{\varepsilon}\right) b_j = 0$$

$$f_i = -\varepsilon \log \sum_j \exp\left(\frac{g_j - C_{i,j}}{\varepsilon}\right) b_j.$$

**Definition 7.15** (Soft-min and discrete soft $c$-transform)**.** For $h \in \mathbb{R}^m$ and weights $b \in \Sigma_m$, the weighted soft-min at temperature $\varepsilon > 0$ is $\min_b^\varepsilon(h) := -\varepsilon \log \sum_j e^{-h_j/\varepsilon} b_j$. It converges to $\min_j h_j$ as $\varepsilon \to 0$. Given a cost matrix C, the discrete soft $c$-transforms are

$$f_i = \min\nolimits_b^\varepsilon(C_{i,\cdot} - g) \tag{7.15}$$

and

$$g_j = \min\nolimits_a^\varepsilon(C_{\cdot,j} - f). \tag{7.16}$$

$\min_b^\varepsilon(h - \mathrm{cst}) = \min_b^\varepsilon(h) - \mathrm{cst}$

**Continuous dual and soft-transforms.** For general, not necessarily discrete, measures $(\alpha, \beta)$, the KL-regularized problem (7.11) has the concave dual

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) = \sup_{f \in \mathcal{C}(\mathcal{X}),\, g \in \mathcal{C}(\mathcal{Y})} \mathcal{D}_\varepsilon(f, g), \tag{7.17}$$

$$\mathcal{D}_\varepsilon(f, g) := \int_{\mathcal{X}} f(x) \mathrm{d}\alpha(x) + \int_{\mathcal{Y}} g(y) \mathrm{d}\beta(y) - \varepsilon \int_{\mathcal{X} \times \mathcal{Y}} \left( e^{\frac{f(x)+g(y)-c(x,y)}{\varepsilon}} - 1 \right) \mathrm{d}\alpha(x) \mathrm{d}\beta(y). \tag{7.18}$$

**Definition 7.16** (Continuous soft $c$-transforms)**.** For $f \in \mathcal{C}(\mathcal{X})$ and $g \in \mathcal{C}(\mathcal{Y})$, define

$$f^{c,\varepsilon}(y) := -\varepsilon \log\left( \int_{\mathcal{X}} e^{\frac{f(x)-c(x,y)}{\varepsilon}} \mathrm{d}\alpha(x) \right), \qquad y \in \mathcal{Y}, \tag{7.19}$$

$$g^{\bar{c},\varepsilon}(x) := -\varepsilon \log\left( \int_{\mathcal{Y}} e^{\frac{g(y)-c(x,y)}{\varepsilon}} \mathrm{d}\beta(y) \right), \qquad x \in \mathcal{X}. \tag{7.20}$$

**Proposition 7.17** (Existence and uniqueness of entropic dual potentials)**.** *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact and that $c$ is continuous. The dual problem (7.17) has solutions, and the set of solutions is of the form $(f^\star + \lambda, g^\star - \lambda), \qquad \lambda \in \mathbb{R}$.*

*Proof.* Normalize potentials by imposing $\int f \mathrm{d}\alpha = 0$. Replacing any pair $(f, g)$ by the corresponding soft $c$-transforms does not decrease the dual objective, because each soft transform is the exact maximizer in one block variable. For transformed potentials, the oscillations are bounded by the oscillation of the cost: $\|f\|_V + \|g\|_V \leqslant 2(\sup c - \inf c)$. Moreover the modulus of continuity of the soft transforms is controlled by that of $c$; for instance $|g^{\bar{c},\varepsilon}(x) - g^{\bar{c},\varepsilon}(x')| \leqslant \sup_y |c(x, y) - c(x', y)|$. After normalization, maximizing sequences are therefore uniformly bounded and equicontinuous. Arzelà–Ascoli gives a uniformly converging subsequence, and continuity of (7.18) gives a maximizer.

For uniqueness, use strict convexity of $H \mapsto \int e^{H/\varepsilon} \mathrm{d}(\alpha \otimes \beta)$ on the image of $(f, g) \mapsto H = f \oplus g - c$, modulo constants. If two optimal pairs exist, their midpoint is also optimal; strict convexity forces the two functions $f \oplus g$ to agree $\alpha \otimes \beta$-almost everywhere. Since the potentials are continuous on compact supports, this implies that $f - f'$ is constant and $g - g'$ is the opposite constant. □

**Neural dual solvers.** $\Phi(x) + \Psi(y) \geqslant \langle x, y\rangle$. For the quadratic cost this is the same statement after subtracting the quadratic terms, using the convexity properties of soft transforms. One can therefore maximize the continuous dual (7.18) over parameterized convex potentials, estimating the integrals by stochastic samples.

### 7.6 Path-Space Schrödinger Problem

This compact subsection connects entropic OT with path-space Schrödinger bridges.

**Unregularized path-space transport.** Let $\Omega = C([0, 1]; \mathcal{X})$ be a path space and denote by
$e_t : \Omega \to \mathcal{X}, \qquad e_t(\omega) = \omega_t$ $(e_0)_\sharp M = \alpha, \qquad (e_1)_\sharp M = \beta$

$$\inf_{M \in \mathcal{P}(\Omega)} \left\{ \int_\Omega \mathcal{A}(\omega) \, \mathrm{d}M(\omega) \; ; \; (e_0)_\sharp M = \alpha, \; (e_1)_\sharp M = \beta \right\}. \tag{7.21}$$

For the quadratic Wasserstein geometry on $\mathbb{R}^d$, one takes

$$\mathcal{A}(\omega) = \begin{cases} \int_0^1 \|\dot{\omega}_t\|^2 \, \mathrm{d}t, & \text{if } \omega \text{ is absolutely continuous,} \\ +\infty, & \text{otherwise.} \end{cases}$$

$$c_{\mathcal{A}}(x, y) := \inf_{\omega \in \Omega} \{\mathcal{A}(\omega) \; ; \; e_0(\omega) = x, \; e_1(\omega) = y\}. \tag{7.22}$$

For the quadratic action above, the minimizing path is the straight segment $\omega_t = (1 - t)x + ty$, and $c_{\mathcal{A}}(x, y) = \|x - y\|^2$.

**Proposition 7.18** (Endpoint reduction of path-space transport)**.** *Assume that minimizing paths in (7.22) can be selected measurably, or more generally that the infimum can be approximated by measurable selections. Then (7.21) has the same value as the Kantorovich problem $\inf_{\pi\in\mathcal{U}(\alpha,\beta)}\int_{\mathcal{X}\times\mathcal{X}} c_{\mathcal{A}}(x,y)\,\mathrm{d}\pi(x,y)$. Moreover, if $\pi^\star$ is an optimal endpoint coupling and $\omega^{x,y}$ is an optimal path from $x$ to $y$, then $M^\star=\int_{\mathcal{X}\times\mathcal{X}}\delta_{\omega^{x,y}}\,\mathrm{d}\pi^\star(x,y)$ is an optimal path law.*

*Proof.* Let $M$ be any feasible path law and set $\pi=(e_0,e_1)_\sharp M$. Then $\pi\in\mathcal{U}(\alpha,\beta)$ and, by the definition of $c_{\mathcal{A}}$, $\int_\Omega \mathcal{A}(\omega)\,\mathrm{d}M(\omega)\geqslant\int_{\mathcal{X}\times\mathcal{X}} c_{\mathcal{A}}(x,y)\,\mathrm{d}\pi(x,y)$. This proves that the path-space value is at least the Kantorovich value. Conversely, given a coupling $\pi$ and a measurable selection $(x,y)\mapsto\omega^{x,y}$ with action arbitrarily close to $c_{\mathcal{A}}(x,y)$, the mixture of Dirac path laws $M=\int\delta_{\omega^{x,y}}\,\mathrm{d}\pi(x,y)$ has endpoints $\alpha$ and $\beta$ and action equal, up to the selected approximation error, to $\int c_{\mathcal{A}}\,\mathrm{d}\pi$. Optimizing over $\pi$ and letting the approximation error vanish proves equality. If exact minimizing paths are selected for an optimal $\pi^\star$, the displayed $M^\star$ is optimal. □

**Entropic path-space problem.** Let $\mathcal{R}^\varepsilon\in\mathcal{P}(\Omega)$ be a reference path law, for instance a Brownian or Langevin dynamics at noise level $\varepsilon$. Schrödinger's dynamic problem is the entropy projection

$$\mathrm{SB}_\varepsilon(\alpha,\beta):=\inf_{M\in\mathcal{P}(\Omega)}\left\{\varepsilon\,\mathrm{KL}(M|\mathcal{R}^\varepsilon)\;;\;(e_0)_\sharp M=\alpha,\;(e_1)_\sharp M=\beta\right\}. \tag{7.23}$$

**Stochastic-control interpretation.** For Brownian references, the entropy projection has a direct control meaning. To fix constants, suppose that $\mathcal{R}^\varepsilon$ is the law of
$\mathrm{d}X_t=\sqrt{\varepsilon}\,\mathrm{d}B_t,\qquad X_0\sim\alpha,\ \mathrm{d}X_t=u_t\,\mathrm{d}t+\sqrt{\varepsilon}\,\mathrm{d}B_t,\qquad X_0\sim\alpha.$

$$\mathrm{KL}(M|\mathcal{R}^\varepsilon)=\frac{1}{2\varepsilon}\mathbb{E}_M\int_0^1\|u_t\|^2\,\mathrm{d}t,\qquad \varepsilon\,\mathrm{KL}(M|\mathcal{R}^\varepsilon)=\frac{1}{2}\mathbb{E}_M\int_0^1\|u_t\|^2\,\mathrm{d}t.$$

If the reference initial law is not $\alpha$, an additional term $\varepsilon\,\mathrm{KL}(\alpha|\mathcal{R}^\varepsilon_0)$ appears, but it is fixed by the endpoint constraint. Hence, up to this fixed endpoint cost, the Schrödinger bridge can be read as
$\inf_u\left\{\frac{1}{2}\mathbb{E}\int_0^1\|u_t\|^2\,\mathrm{d}t\;;\;\mathrm{d}X_t=u_t\,\mathrm{d}t+\sqrt{\varepsilon}\,\mathrm{d}B_t,\quad X_0\sim\alpha,\ X_1\sim\beta\right\}$, $u_t^\star(x)=\varepsilon\nabla\log h_t(x)$.

**Viscous Benamou–Brenier formulations.** $\partial_t\rho_t+\mathrm{div}(\rho_t v_t)=\frac{\sigma}{2}\Delta\rho_t$, $\int_0^1\int\frac{1}{2}\|v_t(x)\|^2\rho_t(x)\mathrm{d}x\mathrm{d}t$. $u_t=v_t-\frac{\sigma}{2}\nabla\log\rho_t$, $\qquad\partial_t\rho_t+\mathrm{div}(\rho_t u_t)=0$.

$$\begin{aligned}\int_0^1\int\frac{1}{2}\|v_t\|^2\rho_t\,\mathrm{d}x\,\mathrm{d}t&=\int_0^1\int\left(\frac{1}{2}\|u_t\|^2+\frac{\sigma^2}{8}\|\nabla\log\rho_t\|^2\right)\rho_t\,\mathrm{d}x\,\mathrm{d}t\\&\quad+\frac{\sigma}{2}\left[\int\rho_1\log\rho_1\,\mathrm{d}x-\int\rho_0\log\rho_0\,\mathrm{d}x\right].\end{aligned}$$

$$\int_0^1\int\left(\frac{1}{2}\|u_t(x)\|^2+\frac{\sigma^2}{8}\|\nabla\log\rho_t(x)\|^2\right)\rho_t(x)\mathrm{d}x\mathrm{d}t.$$

Thus the Schrödinger bridge is a least-action interpolation with both transport kinetic energy and a Fisher-information penalty. If one instead writes the viscous equation with diffusion coefficient $\sigma\Delta\rho_t$, the same formula is obtained after replacing $\sigma$ above by $2\sigma$, so the Fisher coefficient becomes $\sigma^2/2$.

$$\mathcal{R}^\varepsilon(\mathrm{d}\omega)=\int\mathcal{R}^{\varepsilon,x,y}(\mathrm{d}\omega)\,\mathcal{R}^\varepsilon_{01}(\mathrm{d}x,\mathrm{d}y),\qquad\mathcal{R}^\varepsilon_{01}:=(e_0,e_1)_\sharp\mathcal{R}^\varepsilon,$$

where $\mathcal{R}^{\varepsilon,x,y}$ is the reference bridge conditioned on endpoints $(x,y)$. Similarly, for any feasible $M$, write
$M(\mathrm{d}\omega)=\int M^{x,y}(\mathrm{d}\omega)\,\pi(\mathrm{d}x,\mathrm{d}y),\qquad\pi:=(e_0,e_1)_\sharp M.$

**Proposition 7.19** (Endpoint reduction of the Schrödinger problem)**.** *Assume that the regular conditional laws above exist and that the relative-entropy chain rule applies, with value $+\infty$ when absolute continuity fails. Then*

$$\mathrm{SB}_\varepsilon(\alpha,\beta)=\inf_{\pi\in\mathcal{U}(\alpha,\beta)}\varepsilon\,\mathrm{KL}(\pi|\mathcal{R}^\varepsilon_{01}). \tag{7.24}$$

*For a fixed endpoint coupling $\pi$ with finite $\mathrm{KL}(\pi|\mathcal{R}^\varepsilon_{01})$, the minimizing path law is the mixture of reference bridges*

$$M^\pi=\int\mathcal{R}^{\varepsilon,x,y}\,\mathrm{d}\pi(x,y). \tag{7.25}$$

*Proof.* If $M$ has finite relative entropy with respect to $\mathcal{R}^\varepsilon$, then $\pi=(e_0,e_1)_\sharp M$ is necessarily absolutely continuous with respect to $\mathcal{R}^\varepsilon_{01}$. The chain rule for relative entropy gives

$$\mathrm{KL}(M|\mathcal{R}^\varepsilon)=\mathrm{KL}(\pi|\mathcal{R}^\varepsilon_{01})+\int_{\mathcal{X}\times\mathcal{X}}\mathrm{KL}(M^{x,y}|\mathcal{R}^{\varepsilon,x,y})\,\mathrm{d}\pi(x,y). \tag{7.26}$$

The second term is nonnegative and vanishes exactly when $M^{x,y}=\mathcal{R}^{\varepsilon,x,y}$ for $\pi$-almost every $(x,y)$. Thus, once an endpoint coupling $\pi$ with finite $\mathrm{KL}(\pi|\mathcal{R}^\varepsilon_{01})$ is fixed, the best path law is (7.25), and the remaining minimization is precisely (7.24). If no such endpoint coupling exists, both sides are $+\infty$. □

**Brownian bridges and Sinkhorn couplings.** For $\mathcal{X} = \mathbb{R}^d$, take $\mathcal{R}^\varepsilon$ to be a Brownian reference dynamics, up to the conventional scaling of $\varepsilon$. Its endpoint law has a heat-kernel density of the form

$$p_\varepsilon(x, y) \propto \exp\left(-\frac{\|x-y\|^2}{\varepsilon}\right)$$

$$\mathcal{R}^\varepsilon_{01}(\mathrm{d}x, \mathrm{d}y) \propto \exp\left(-\frac{c(x,y)}{\varepsilon}\right) \alpha(\mathrm{d}x)\beta(\mathrm{d}y).$$

This includes the usual heat-kernel reference after rewriting it with respect to $\alpha \otimes \beta$, whenever the one-time endpoint densities are fixed and mutually absolutely continuous. The additional one-body density factors only add constants under the marginal constraints.
$\varepsilon\,\mathrm{KL}(\pi|\mathcal{R}^\varepsilon_{01}) = \int c(x, y)\,\mathrm{d}\pi(x, y) + \varepsilon\,\mathrm{KL}(\pi|\alpha \otimes \beta) + \text{constant}$, where the constant does not depend on $\pi$. Hence (7.24) is exactly the continuous Sinkhorn problem (7.11), up to an additive constant in the value.
$M^\star_\varepsilon = \int \mathcal{R}^{\varepsilon,x,y}\,\mathrm{d}\pi^\star_\varepsilon(x, y)$.

### 7.7 Marginal-Dependent Problems

Let $\mathcal{F}$ and $\mathcal{G}$ be proper convex lower semicontinuous functionals on finite nonnegative measures on $\mathcal{X}$ and $\mathcal{Y}$. The unregularized marginal-dependent transport problem is

$$\inf_{\pi \in \mathcal{M}_+(\mathcal{X}\times\mathcal{Y})} \int c(x, y)\mathrm{d}\pi(x, y) + \mathcal{F}(\pi_1) + \mathcal{G}(\pi_2), \tag{7.27}$$

where $\pi_1 = (\mathrm{p}_\mathcal{X})_\sharp\pi$ and $\pi_2 = (\mathrm{p}_\mathcal{Y})_\sharp\pi$ are the two marginals of $\pi$. Entropic regularization turns this problem into the scaling-friendly form

$$\inf_{\pi \in \mathcal{M}_+(\mathcal{X}\times\mathcal{Y})} \int c(x, y)\mathrm{d}\pi(x, y) + \mathcal{F}(\pi_1) + \mathcal{G}(\pi_2) + \varepsilon\,\mathrm{KL}(\pi|\alpha \otimes \beta), \tag{7.28}$$

where $\alpha$ and $\beta$ are reference measures. In this subsection, when the total mass of $\pi$ is not fixed, the KL term is understood in the generalized sense associated with $\varphi(s) = s\log s - s + 1$.
$\mathrm{KL}(\pi|\lambda) = \int \big(\rho\log\rho - \rho + 1\big)\mathrm{d}\lambda$

$$\inf_{\mathrm{P}\in\mathbb{R}^{n\times m}_+} \langle \mathrm{C},\, \mathrm{P}\rangle + \mathsf{F}(\mathrm{P}\mathbb{1}_m) + \mathsf{G}(\mathrm{P}^\top\mathbb{1}_n) + \varepsilon\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}). \tag{7.29}$$

$\inf_{\mathrm{P}\geqslant 0} \mathsf{F}(\mathrm{P}\mathbb{1}_m) + \mathsf{G}(\mathrm{P}^\top\mathbb{1}_n) + \varepsilon\mathrm{KL}(\mathrm{P}|\mathrm{K})$.

**Proposition 7.20** (Dual and scaling for marginal penalties)**.** *Assume $a_i, b_j > 0$, $\varepsilon > 0$, and a Fenchel qualification condition, for instance the existence of a matrix* $\mathrm{P} > 0$ *such that* $\mathrm{P}\mathbb{1}_m$ *and* $\mathrm{P}^\top\mathbb{1}_n$ *belong to the relative interiors of* $\mathrm{dom}(\mathsf{F})$ *and* $\mathrm{dom}(\mathsf{G})$*. The Fenchel dual of* (7.29) *is*

$$\sup_{\mathrm{f}\in\mathbb{R}^n, \mathrm{g}\in\mathbb{R}^m} -\mathsf{F}^*(-\mathrm{f}) - \mathsf{G}^*(-\mathrm{g}) - \varepsilon\sum_{i,j} a_ib_j\left[\exp\left(\frac{\mathrm{f}_i + \mathrm{g}_j - \mathrm{C}_{ij}}{\varepsilon}\right) - 1\right]. \tag{7.30}$$

*Equivalently, up to the additive constant $\varepsilon\sum_{i,j} a_ib_j$, the last term is*

$$-\varepsilon\sum_{i,j} \mathrm{K}_{ij}\exp\left(\frac{\mathrm{f}_i + \mathrm{g}_j}{\varepsilon}\right).$$

*If $\mathrm{u} = e^{\mathrm{f}/\varepsilon}$ and $\mathrm{v} = e^{\mathrm{g}/\varepsilon}$, exact block ascent in the two dual variables is the generalized Sinkhorn cycle*

$$\begin{aligned} r &\leftarrow \mathrm{prox}^{\mathrm{KL}}_{\mathsf{F}/\varepsilon}(\mathrm{K}\mathrm{v}), & \mathrm{u} &\leftarrow r \oslash (\mathrm{K}\mathrm{v}),\\ s &\leftarrow \mathrm{prox}^{\mathrm{KL}}_{\mathsf{G}/\varepsilon}(\mathrm{K}^\top\mathrm{u}), & \mathrm{v} &\leftarrow s \oslash (\mathrm{K}^\top\mathrm{u}),\\ \mathrm{P} &= \mathrm{diag}(\mathrm{u})\mathrm{K}\,\mathrm{diag}(\mathrm{v}), \end{aligned} \tag{7.31}$$

*where divisions are entrywise and*

$$\mathrm{prox}^{\mathrm{KL}}_{\mathsf{h}}(z) \coloneqq \operatorname*{argmin}_{r\in\mathbb{R}^d_+} \mathsf{h}(r) + \mathrm{KL}(r|z). \tag{7.32}$$

*In particular,* $\mathrm{prox}^{\mathrm{KL}}_{\mathsf{F}/\varepsilon}(z)$ *is the minimizer of* $\mathsf{F}(r) + \varepsilon\mathrm{KL}(r|z)$*.*

*Proof.* Introduce independent variables $r$ and $s$ for the two marginals and use dual variables $\mathrm{f}, \mathrm{g}$ for the constraints $r = \mathrm{P}\mathbb{1}_m$ and $s = \mathrm{P}^\top\mathbb{1}_n$. The Lagrangian contains $\mathsf{F}(r) + \langle \mathrm{f},\, r\rangle + \mathsf{G}(s) + \langle \mathrm{g},\, s\rangle + \langle \mathrm{C},\, \mathrm{P}\rangle + \varepsilon\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) - \langle \mathrm{f}\oplus\mathrm{g},\, \mathrm{P}\rangle$. Minimizing over $r$ and $s$ gives $-\mathsf{F}^*(-\mathrm{f})$ and $-\mathsf{G}^*(-\mathrm{g})$. For each scalar entry, the convex conjugate of $p \mapsto C_{ij}p + \varepsilon(p\log(p/(a_ib_j)) - p + a_ib_j)$ is $q \mapsto \varepsilon a_ib_j(e^{(q - C_{ij})/\varepsilon} - 1)$. Minimizing over $\mathrm{P}$ with $q = \mathrm{f}_i + \mathrm{g}_j$ gives (7.30).
For the scaling form, fix $\mathrm{v} > 0$, set $\widetilde{\mathrm{K}} = \mathrm{K}\,\mathrm{diag}(\mathrm{v})$ and $z = \widetilde{\mathrm{K}}\mathbb{1}_m = \mathrm{K}\mathrm{v}$. Updating the row scaling means looking for $\mathrm{P} = \mathrm{diag}(\mathrm{u})\widetilde{\mathrm{K}}$, so its row marginal is $r = \mathrm{P}\mathbb{1}_m = \mathrm{u}\odot z$. The row-wise chain rule gives $\mathrm{KL}(\mathrm{diag}(\mathrm{u})\widetilde{\mathrm{K}}|\widetilde{\mathrm{K}}) = \sum_i\big(r_i\log(r_i/z_i) - r_i + z_i\big) = \mathrm{KL}(r|z)$. Thus exact optimization of this block is equivalent to minimizing $\mathsf{F}(r) + \varepsilon\mathrm{KL}(r|z)$ over $r \geqslant 0$, hence $r = \mathrm{prox}^{\mathrm{KL}}_{\mathsf{F}/\varepsilon}(z)$ and $\mathrm{u} = r \oslash z$. The column update is identical with $w = \mathrm{K}^\top\mathrm{u}$. □

When $\mathsf{F} = \iota_{\{\mathrm{a}\}}$ and $\mathsf{G} = \iota_{\{\mathrm{b}\}}$, the first two dual terms are $\langle \mathrm{f},\, \mathrm{a}\rangle + \langle \mathrm{g},\, \mathrm{b}\rangle$, and one recovers the usual entropic dual (7.18). The classical Sinkhorn update is the special case in which the KL proximal maps return the prescribed marginals a and b.

- **Hard marginal constraint.** If $\mathsf{F} = \iota_{\{\mathrm{a}\}}$, then $\mathrm{prox}^{\mathrm{KL}}_{\mathsf{F}/\varepsilon}(z) = \mathrm{a}$.
- **KL marginal relaxation.** If $\mathsf{F}(r) = \tau\mathrm{KL}(r|\mathrm{a})$ with $\tau > 0$, then, coordinatewise, $\mathrm{prox}^{\mathrm{KL}}_{\mathsf{F}/\varepsilon}(z) = z^{\varepsilon/(\tau+\varepsilon)} \odot \mathrm{a}^{\tau/(\tau+\varepsilon)}$, and the scaling update becomes $\mathrm{u} \leftarrow (\mathrm{a}\oslash z)^{\tau/(\tau+\varepsilon)}$, the damped scaling of unbalanced Sinkhorn.

- **Pointwise bounds.** If $\mathsf{F} = \iota_{\{\ell \leqslant r \leqslant u\}}$, then the proximal map is the coordinatewise clipping $\operatorname{prox}^{\mathrm{KL}}_{\mathsf{F}/\varepsilon}(z)_i = \min\{u_i, \max\{\ell_i, z_i\}\}$.
- **Total-variation marginal relaxation.** If $\mathsf{F}(r) = \tau\|r - \mathrm{a}\|_1$ and $\lambda = \tau/\varepsilon$, then the proximal map is coordinatewise

$$\operatorname{prox}^{\mathrm{KL}}_{\mathsf{F}/\varepsilon}(z)_i = \begin{cases} z_i e^{\lambda}, & z_i < a_i e^{-\lambda}, \\ a_i, & a_i e^{-\lambda} \leqslant z_i \leqslant a_i e^{\lambda}, \\ z_i e^{-\lambda}, & z_i > a_i e^{\lambda}. \end{cases}$$

- **Fixed total mass.** If $\mathsf{F} = \iota_{\{\langle r, \mathbb{1}\rangle = m\}}$, then $\operatorname{prox}^{\mathrm{KL}}_{\mathsf{F}/\varepsilon}(z) = mz/\langle z, \mathbb{1}\rangle$.

### 7.8 Heat Kernels and Hopf–Cole Transforms

This compact subsection records heat-kernel and Hopf–Cole viewpoints on soft transforms.

**Geodesics in heat.**

$$h_t(x, y) = (4\pi t)^{-d/2} \exp\left(-\frac{\|x - y\|^2}{4t}\right).$$

Up to normalization, the Sinkhorn Gibbs kernel for the quadratic cost is therefore a Gaussian heat kernel with heat time $t = \varepsilon/4$. A neighboring but distinct construction uses the resolvent of the positive Laplacian.
$-4t \log h_t(x, y) \longrightarrow d_M(x, y)^2 \qquad (t \downarrow 0)$.

$$e^{-tL} = \lim_{q\to\infty} \left(I + \frac{t}{q} L\right)^{-q}.$$

Thus a single inverse shifted Laplacian $(I + \tau L)^{-1}$ gives a Laplace/Yukawa smoothing, while repeated resolvent steps approximate the Gaussian heat kernel. On a triangulated surface or image grid, $L$ is replaced by a discrete Laplacian $L_h$, and one applies sparse linear solves rather than dense Gaussian matrices.

**Soft Hopf–Lax and Hopf–Cole.** $c(x, y) = \frac{1}{2}\|x - y\|^2$,

$$(-h)^{\bar{c}}(x) = \inf_y \left\{ h(y) + \frac{1}{2}\|x - y\|^2 \right\},$$

where the sign only reflects the convention of Definition 4.6. In the present entropic setting, the parameter of interest is instead the temperature $\varepsilon$.

$$(-h)^{\bar{c},\varepsilon}(x) = -\varepsilon \log \int \exp\left(-\frac{h(y) + \|x - y\|^2/2}{\varepsilon}\right) \mathrm{d}y. \tag{7.33}$$

$$G_\varepsilon(z) \coloneqq (2\pi\varepsilon)^{-d/2} \exp\left(-\frac{\|z\|^2}{2\varepsilon}\right)$$

$$(-h)^{\bar{c},\varepsilon}(x) = -\varepsilon \log \left(G_\varepsilon * e^{-h/\varepsilon}\right)(x) - \frac{\varepsilon d}{2} \log(2\pi\varepsilon). \tag{7.34}$$

Thus a soft quadratic $c$-transform is a Gaussian convolution followed by a logarithm, up to an explicit additive constant independent of $x$. This is the bridge between soft-minimum operators, heat kernels and entropic transport potentials.

**Proposition 7.21** (Soft quadratic $c$-transform and Legendre approximation)**.** *Let $f : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ be such that the integrals below are finite, and introduce the quadratic shift $\mathsf{S}f(y) \coloneqq f(y) - \frac{1}{2}\|y\|^2$. For $\varepsilon > 0$, define the soft conjugate by applying the soft $\bar{c}$-transform to the shifted function:*

$$f^{*,\varepsilon}(p) \coloneqq \frac{1}{2}\|p\|^2 - \big(-\mathsf{S}f\big)^{\bar{c},\varepsilon}(p). \tag{7.35}$$

*Then*

$$f^{*,\varepsilon}(p) = \varepsilon \log \int_{\mathbb{R}^d} \exp\left(\frac{\langle p, y\rangle - f(y)}{\varepsilon}\right) \mathrm{d}y, \tag{7.36}$$

*and equivalently*

$$f^{*,\varepsilon}(p) = \frac{1}{2}\|p\|^2 + \varepsilon \log \left(G_\varepsilon * e^{-\mathsf{S}f/\varepsilon}\right)(p) + \frac{\varepsilon d}{2} \log(2\pi\varepsilon). \tag{7.37}$$

*If, for instance, $f$ is proper, lower semicontinuous and superlinear, then $f^{*,\varepsilon}(p) \to f^*(p)$ for every $p$ as $\varepsilon \to 0$.*

*Proof.* The hard identity follows by completing the square:

$$\inf_y \left\{ \mathsf{S}f(y) + \frac{1}{2}\|p - y\|^2 \right\} = \frac{1}{2}\|p\|^2 - \sup_y\{\langle p, y\rangle - f(y)\} = \frac{1}{2}\|p\|^2 - f^*(p).$$

This explains the shift: it turns the Legendre–Fenchel transform into a quadratic $\bar{c}$-transform. Replacing the hard transform by its soft version gives (7.35). Expanding the square in (7.33) with the function $\mathsf{S}f$ gives

$$\exp\left(-\frac{\mathsf{S}f(y) + \|p - y\|^2/2}{\varepsilon}\right) = \exp\left(-\frac{\|p\|^2}{2\varepsilon}\right) \exp\left(\frac{\langle p, y\rangle - f(y)}{\varepsilon}\right),$$

which proves (7.36). Formula (7.37) is the Gaussian-convolution identity (7.34) applied to $\mathsf{S}f$. Under the stated superlinearity assumption, the functions $y \mapsto \langle p, y\rangle - f(y)$ have compact upper level sets, so Laplace's principle applied to (7.36) gives the convergence to $f^*(p)$. ∎

$\partial_s \varphi_s + \frac{1}{2}\|\nabla\varphi_s\|^2 = \frac{\varepsilon}{2}\Delta\varphi_s$, $\partial_s v_s + (v_s \cdot \nabla)v_s = \frac{\varepsilon}{2}\Delta v_s$.

### 7.9 Other Convex Regularizers

Let $\varphi$ be an entropy function in the sense of Definition 6.8, and recall the $\varphi$-divergence $\mathcal{D}_\varphi$ from Definition 6.9. For $\varepsilon > 0$, define the $\varphi$-regularized transport value

$$\mathcal{L}^{\varepsilon}_{c,\varphi}(\alpha, \beta) \coloneqq \min_{\pi \in \mathcal{U}(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\mathrm{d}\pi(x,y) + \varepsilon \mathcal{D}_\varphi(\pi|\alpha\otimes\beta). \tag{7.38}$$

**Proposition 7.22** (Dual and density law for $\varphi$-regularized OT)**.** *Under the usual Fenchel–Rockafellar qualification assumptions, for instance compact spaces, continuous $c$, and finite value in (7.38), one has*

$$\mathcal{L}^{\varepsilon}_{c,\varphi}(\alpha, \beta) = \sup_{f\in\mathcal{C}(\mathcal{X}),\, g\in\mathcal{C}(\mathcal{Y})} \int f\mathrm{d}\alpha + \int g\mathrm{d}\beta - \varepsilon \int \varphi^{*,\geqslant 0}\left(\frac{f(x)+g(y)-c(x,y)}{\varepsilon}\right)\mathrm{d}\alpha(x)\mathrm{d}\beta(y). \tag{7.39}$$

*If an optimal plan has density $r^\star = \frac{\mathrm{d}\pi^\star}{\mathrm{d}(\alpha\otimes\beta)}$ and optimal potentials $(f^\star, g^\star)$, then $\frac{f^\star(x)+g^\star(y)-c(x,y)}{\varepsilon} \in \partial\varphi(r^\star(x,y))$ $\quad \alpha\otimes\beta$-a.e. In the smooth interior this reads*

$$r^\star(x,y) = (\varphi')^{-1}\left(\frac{f^\star(x)+g^\star(y)-c(x,y)}{\varepsilon}\right).$$

*Proof.* Introduce dual variables $(f, g)$ for the two marginal constraints. For fixed $(f, g)$, the minimization over $\pi$ gives

$$\int f\mathrm{d}\alpha + \int g\mathrm{d}\beta + \inf_{\pi\in\mathcal{M}_+(\mathcal{X}\times\mathcal{Y})}\left\{\int \big(c-(f\oplus g)\big)\mathrm{d}\pi + \varepsilon\mathcal{D}_\varphi(\pi|\alpha\otimes\beta)\right\}.$$

Using the Legendre formula (6.9) for the convex functional $\mathcal{D}_\varphi(\cdot|\alpha\otimes\beta)$, the infimum equals

$$-\varepsilon\int \varphi^{*,\geqslant 0}\left(\frac{f(x)+g(y)-c(x,y)}{\varepsilon}\right)\mathrm{d}\alpha(x)\mathrm{d}\beta(y),$$

which gives (7.39). Equality in the Fenchel inequality is equivalent to the subgradient inclusion $\frac{f^\star\oplus g^\star - c}{\varepsilon} \in \partial\varphi(r^\star)$, and inversion of $\varphi'$ gives the density law when $\varphi$ is differentiable and the optimizer is in the interior of its domain. □

For the KL entropy $\varphi(r) = r\log r - r + 1$, one has $\varphi^{*,\geqslant 0}(s) = e^s - 1$. Taking this parameter to be the Sinkhorn temperature $\varepsilon$ in (7.39) recovers exactly the continuous Sinkhorn dual (7.18). Other choices replace the exponential law by another scalar transfer function:

$$\begin{aligned}
&\varphi(r) = r\log r - r + 1 && \Rightarrow && r^\star = e^s, \\
&\varphi(r) = r - \log r - 1 && \Rightarrow && r^\star = (1-s)^{-1} \quad (s<1), \qquad s \coloneqq \frac{f^\star\oplus g^\star - c}{\varepsilon}. \\
&\varphi(r) = \tfrac{1}{2}(r-1)^2 && \Rightarrow && r^\star = (1+s)_+,
\end{aligned}$$

**Generalized soft $c$-transforms.**

$$g^{\bar{c},\varepsilon,\varphi}(x) \in \operatorname*{argmin}_{u\in\mathbb{R}} \left\{\varepsilon\int_{\mathcal{Y}} \varphi^{*,\geqslant 0}\left(\frac{u+g(y)-c(x,y)}{\varepsilon}\right)\mathrm{d}\beta(y) - u\right\}. \tag{7.40}$$

$$\int_{\mathcal{Y}} (\varphi^{*,\geqslant 0})'\left(\frac{u+g(y)-c(x,y)}{\varepsilon}\right)\mathrm{d}\beta(y) = 1.$$

This equation says that the conditional density induced by the current dual score has unit $\beta$-mass, hence it is exactly the row-normalization step in dual variables. The update $f^{c,\varepsilon,\varphi}$ on $\mathcal{Y}$ is defined symmetrically by exchanging $(x, \alpha, f)$ and $(y, \beta, g)$.

**Bregman vs. $\varphi$-divergence regularization.**

**Definition 7.23** (Measure Bregman divergence)**.** If $\Phi$ is a differentiable convex functional on a convex class of nonnegative measures and $\xi$ is a reference measure in its domain, the measure Bregman divergence generated by $\Phi$ is

$$B_\Phi(\pi|\xi) \coloneqq \Phi(\pi) - \Phi(\xi) - \int \delta\Phi(\xi)\mathrm{d}(\pi-\xi), \tag{7.41}$$

where $\delta\Phi(\xi)$ is the first variation and the formula is understood whenever the right-hand side is well-defined.

**Proposition 7.24** (Dual comparison: Bregman vs. density-ratio penalties)**.** *Fix the marginals $\alpha, \beta$ and set $\xi \coloneqq \alpha\otimes\beta$. Let $\Phi$ be a convex Gateaux-differentiable functional on nonnegative measures on $\mathcal{X}\times\mathcal{Y}$. Its convex conjugate is, for a continuous test function $u$,*

$$\Phi^*(u) \coloneqq \sup_{\pi\geqslant 0}\left\{\int u\,\mathrm{d}\pi - \Phi(\pi)\right\}.$$

*Set $\mathcal{L}^{\varepsilon}_{c,\Phi}(\alpha,\beta) \coloneqq \inf_{\pi\in\mathcal{U}(\alpha,\beta)}\int c\,\mathrm{d}\pi + \varepsilon B_\Phi(\pi|\xi)$. Assume that Fenchel duality is exact for this constrained problem, as happens in finite-dimensional discretizations and, more generally, under standard compactness and lower-semicontinuity hypotheses. Then*

$$\mathcal{L}^{\varepsilon}_{c,\Phi}(\alpha,\beta) = \sup_{f,g}\int f\mathrm{d}\alpha + \int g\mathrm{d}\beta - \varepsilon\left[\Phi^*\left(\delta\Phi(\xi) + \frac{f\oplus g - c}{\varepsilon}\right) - \Phi^*(\delta\Phi(\xi))\right]. \tag{7.42}$$

*If $(f^\star, g^\star)$ and $\pi^\star$ are optimal and the solution is interior, then $\delta\Phi(\pi^\star) = \delta\Phi(\xi) + \frac{f^\star\oplus g^\star - c}{\varepsilon}$. By contrast, the density-ratio formulation (7.38) has the scalar-integral dual (7.39) and the pointwise density law*

$$\frac{f^\star\oplus g^\star - c}{\varepsilon} \in \partial\varphi\left(\frac{\mathrm{d}\pi^\star}{\mathrm{d}(\alpha\otimes\beta)}\right).$$

*Proof.* Using (7.41), the Bregman-regularized objective can be written, up to a constant independent of $\pi$, as $\varepsilon\Phi(\pi) + \int \big(c - \varepsilon\delta\Phi(\xi)\big)\mathrm{d}\pi - \varepsilon\Phi(\xi) + \varepsilon \int \delta\Phi(\xi)\mathrm{d}\xi$. Introduce dual potentials $(f, g)$ for the two marginal constraints. The inner minimization over nonnegative measures $\pi$ gives

$$\inf_{\pi \geqslant 0} \left\{ \varepsilon\Phi(\pi) + \int \big(c - f \oplus g - \varepsilon\delta\Phi(\xi)\big)\mathrm{d}\pi \right\} = -\varepsilon\Phi^*\left(\delta\Phi(\xi) + \frac{f \oplus g - c}{\varepsilon}\right).$$

Fenchel equality at $\xi$ gives $-\Phi(\xi) + \int \delta\Phi(\xi)\mathrm{d}\xi = \Phi^*(\delta\Phi(\xi))$, which yields (7.42). Equality in Fenchel's inequality gives the optimality condition for $\pi^\star$. The density-ratio dual and density law are exactly those of Proposition 7.22. Placing the two formulas side by side shows the structural difference: Bregman regularization translates the reference measure in the dual coordinate $\delta\Phi$, whereas $\varphi$-regularization applies a scalar nonlinearity to the density with respect to the moving product reference $\alpha \otimes \beta$. $\square$

When $\Phi$ is separable with respect to a fixed dominating measure $\xi_0$, say $\Phi(\pi) = \int h(\mathrm{d}\pi/\mathrm{d}\xi_0)\mathrm{d}\xi_0$ with $\xi \ll \xi_0$, the Bregman optimality condition becomes

$$h'\left(\frac{\mathrm{d}\pi^\star}{\mathrm{d}\xi_0}\right) = h'\left(\frac{\mathrm{d}\xi}{\mathrm{d}\xi_0}\right) + \frac{f^\star \oplus g^\star - c}{\varepsilon}.$$

This is an additive update in entropy coordinates. The density-ratio formulation instead uses the scalar law associated with $\varphi$ relative to $\alpha \otimes \beta$.

**Proposition 7.25** (KL is the common Bregman and $\varphi$ case)**.** *Let $\omega$ be a finite reference measure with a nontrivial measurable subset. Work on probability measures $\alpha = p\omega$ and $\beta = q\omega$ whose densities are bounded above and below away from 0. Assume that $\Phi$ is twice Gateaux differentiable along bounded zero-mass density perturbations, and that $\varphi \in C^2(0, +\infty)$ is convex with $\varphi(1) = 0$. If $B_\Phi(\alpha|\beta) = \mathcal{D}_\varphi(\alpha|\beta)$ for all such $\alpha, \beta$, then there exist $c \geqslant 0$ and $a \in \mathbb{R}$ such that $\varphi(t) = c\, t \log t + a(t-1)$. Hence the common divergence is $c\,\mathrm{KL}(\alpha|\beta)$, and $\Phi(p\omega)$ differs from $c\int p \log p \,\mathrm{d}\omega$ by an affine functional on the positive probability simplex.*

*Proof.* Fix $\beta = q\omega$ and perturb it by $\alpha_t = (q + th)\omega$, where $h$ is bounded, $\int h\mathrm{d}\omega = 0$, and $t$ is small enough that $q + th > 0$. Differentiating twice at $t = 0$ gives $D^2\Phi(q)[h, h] = \varphi''(1) \int \frac{h^2}{q}\mathrm{d}\omega$. Indeed the left-hand side is the second variation of the Bregman error, while

$$\mathcal{D}_\varphi(\alpha_t|\beta) = \int q\,\varphi\left(1 + th/q\right)\mathrm{d}\omega$$

has second derivative $\varphi''(1)\int h^2/q\,\mathrm{d}\omega$. Setting $c = \varphi''(1) \geqslant 0$, the functional $\Psi(p\omega) = \Phi(p\omega) - c\int p \log p\mathrm{d}\omega$ has zero second variation along every zero-mass line segment in the positive simplex. It is therefore affine there, and $B_\Psi = 0$. Thus $B_\Phi = c\,\mathrm{KL}$. Since $\mathcal{D}_\varphi = c\,\mathrm{KL}$, the function $g(t) = \varphi(t) - c\,t \log t$ generates the zero $\varphi$-divergence. Testing on densities for which the ratio $p/q$ takes two values $x < 1 < y$, with weights chosen so that the mean ratio is 1, gives $(y-1)g(x) + (1-x)g(y) = 0$. Hence $g(t)/(t-1)$ is constant on $(0, +\infty) \setminus \{1\}$, so $g(t) = a(t-1)$. This proves the claim. $\square$

Thus the two generalizations lead to different duals and different algorithms. Bregman regularization by $B_\Phi(\pi|\xi)$ keeps the projection geometry of Section 8.1: linear costs tilt the reference in dual coordinates and alternating marginal updates are Bregman projections.

### 7.10 Sinkhorn Divergences

This compact subsection records the debiased entropic discrepancy.

**Entropic bias.** $\alpha_\varepsilon = \underset{\beta}{\operatorname{argmin}}\ \mathcal{L}_c^\varepsilon(\alpha, \beta)$

**Proposition 7.26** (Large-temperature entropic bias)**.** *Assume that $c$ is bounded and continuous. Then $\mathcal{L}_c^\varepsilon(\alpha, \beta) \to \iint c(x, y)\mathrm{d}\alpha(x)\mathrm{d}\beta(y)$ as $\varepsilon \to +\infty$.*

*Proof.* Let $(f_\varepsilon, g_\varepsilon)$ be optimal dual potentials, normalized by $\int g_\varepsilon \mathrm{d}\beta = 0$. The soft $c$-transform equation gives

$$f_\varepsilon(x) = -\varepsilon \log \int \exp\left(\frac{g_\varepsilon(y) - c(x, y)}{\varepsilon}\right)\mathrm{d}\beta(y).$$

For bounded $c$, the oscillations of normalized entropic potentials are bounded uniformly in $\varepsilon$ by the oscillation of $c$. Hence the log-sum-exp expansion is uniform: $f_\varepsilon(x) = -\int (g_\varepsilon(y) - c(x, y))\mathrm{d}\beta(y) + O(\varepsilon^{-1}) = \int c(x, y)\mathrm{d}\beta(y) + O(\varepsilon^{-1})$. At optimality the exponential penalty in the dual integrates to zero, so $\mathcal{L}_c^\varepsilon(\alpha, \beta) = \int f_\varepsilon \mathrm{d}\alpha + \int g_\varepsilon \mathrm{d}\beta = \iint c(x, y)\mathrm{d}\alpha(x)\mathrm{d}\beta(y) + O(\varepsilon^{-1})$. This proves the limit. $\square$

**Sinkhorn divergences.**

**Definition 7.27** (Sinkhorn divergence)**.** For $\varepsilon > 0$, the debiased Sinkhorn divergence associated with the entropic OT value $\mathcal{L}_c^\varepsilon$ is

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \coloneqq \mathcal{L}_c^\varepsilon(\alpha, \beta) - \frac{1}{2}\mathcal{L}_c^\varepsilon(\alpha, \alpha) - \frac{1}{2}\mathcal{L}_c^\varepsilon(\beta, \beta). \tag{7.43}$$

**Lemma 7.28** (Entropic dual cost at optimum)**.** *Let $(f_{\alpha,\beta}, g_{\alpha,\beta})$ be optimal dual potentials, normalized arbitrarily. Then*

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) = \langle f_{\alpha,\beta},\, \alpha\rangle + \langle g_{\alpha,\beta},\, \beta\rangle. \tag{7.44}$$

*Proof.* We first notice that at optimality, the relation $f_{\alpha,\beta} = -\varepsilon \log \int_{\mathcal{Y}} e^{\frac{g_{\alpha,\beta}(y) - c(x,y)}{\varepsilon}}\mathrm{d}\beta(y)$ after taking the exponential, equivalently reads

$$1 = \int_{\mathcal{Y}} e^{\frac{f_{\alpha,\beta}(x) + g_{\alpha,\beta}(y) - c(x,y)}{\varepsilon}}\mathrm{d}\beta(y) \quad\Longrightarrow\quad \int_{\mathcal{X}\times\mathcal{Y}}\left(e^{\frac{f_{\alpha,\beta} \oplus g_{\alpha,\beta} - c}{\varepsilon}} - 1\right)\mathrm{d}(\alpha \otimes \beta) = 0.$$

Substituting this identity in (7.17) gives the result. $\square$

**Proposition 7.29** (Asymptotics of Sinkhorn divergences)**.** *Assume that the two measures are supported on the same space and that $c$ is bounded, continuous, nonnegative and satisfies $c(x,x)=0$. Then $\bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta)\to\mathcal{L}_c(\alpha,\beta)$ when $\varepsilon\to 0$ and*

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta)\to\frac{1}{2}\int -c\mathrm{d}(\alpha-\beta)\otimes\mathrm{d}(\alpha-\beta)\quad \textit{when}\quad \varepsilon\to+\infty.$$

*Proof.* The discrete convergence result above already gives the correct intuition; we now use the standard continuous argument.
**Case $\varepsilon\to 0$.** The first limit follows from the standard $\Gamma$-convergence argument for entropic optimal transport: the entropy term is lower semicontinuous along weakly converging couplings, while any finite-cost coupling can be approximated by couplings with finite entropy. Since $c(x,x)=0$, the two self-costs in the debiased expression converge to zero, and the cross term converges to $\mathcal{L}_c(\alpha,\beta)$.
**Case $\varepsilon\to+\infty$.** We denote by $(f_\varepsilon,g_\varepsilon)$ optimal dual potentials. After normalizing them and using boundedness of $c$, their oscillations stay uniformly bounded, so the following expansion is uniform. The optimality condition on $f_\varepsilon$ (equivalently the Sinkhorn fixed point on $f_\varepsilon$) reads

$$\begin{aligned}f_\varepsilon&=-\varepsilon\log\int\exp\left(\frac{g_\varepsilon(y)-c(\cdot,y)}{\varepsilon}\right)\mathrm{d}\beta(y)=-\varepsilon\log\int\left(1+\frac{g_\varepsilon(y)-c(\cdot,y)}{\varepsilon}+o(1/\varepsilon)\right)\mathrm{d}\beta(y)\\&=-\varepsilon\log\left(1+\frac{1}{\varepsilon}\int(g_\varepsilon(y)-c(\cdot,y))\mathrm{d}\beta(y)+o(1/\varepsilon)\right)=-\int g_\varepsilon\mathrm{d}\beta+\int c(\cdot,y)\mathrm{d}\beta(y)+o(1).\end{aligned}$$

Plugging this relation in the dual expression (7.44) $\mathcal{L}_c^\varepsilon(\alpha,\beta)=\int f_\varepsilon\mathrm{d}\alpha+\int g_\varepsilon\mathrm{d}\beta=\iint c(x,y)\mathrm{d}\alpha(x)\mathrm{d}\beta(y)+o(1)$. Applying this expansion to $(\alpha,\beta)$, $(\alpha,\alpha)$ and $(\beta,\beta)$ gives

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta)\to\int c\,\mathrm{d}\alpha\otimes\mathrm{d}\beta-\frac{1}{2}\int c\,\mathrm{d}\alpha\otimes\mathrm{d}\alpha-\frac{1}{2}\int c\,\mathrm{d}\beta\otimes\mathrm{d}\beta=-\frac{1}{2}\int c\,\mathrm{d}(\alpha-\beta)\otimes\mathrm{d}(\alpha-\beta).$$

□

**Proposition 7.30** (Non-negativity of Sinkhorn divergences)**.** *If $k(x,y)=e^{-c(x,y)/\varepsilon}$ is positive definite, then $\bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta)\geqslant 0$.*

*Proof.* In the following, we denote by $(f_{\alpha,\beta},g_{\alpha,\beta})$ optimal dual potentials for the dual Schrödinger problem between $\alpha$ and $\beta$. We denote by $f_{\alpha,\alpha}=g_{\alpha,\alpha}$ (one can assume they are equal by symmetry) the solution for the problem between $\alpha$ and itself.
Using the suboptimal function $(f_{\alpha,\alpha},g_{\beta,\beta})$ in the dual maximization problem, and using relation (7.44) for the simplified expression of the dual cost, one obtains

$$\mathcal{L}_c^\varepsilon(\alpha,\beta)\geqslant\langle f_{\alpha,\alpha},\,\alpha\rangle+\langle g_{\beta,\beta},\,\beta\rangle-\varepsilon\langle e^{\frac{f_{\alpha,\alpha}\oplus g_{\beta,\beta}-c}{\varepsilon}}-1,\,\alpha\otimes\beta\rangle$$

Moreover $\langle f_{\alpha,\alpha},\,\alpha\rangle=\frac{1}{2}\mathcal{L}_c^\varepsilon(\alpha,\alpha)$, and similarly for $\beta$, so the previous inequality equivalently reads

$$\frac{1}{\varepsilon}\bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta)\geqslant 1-\langle e^{\frac{f_{\alpha,\alpha}\oplus g_{\beta,\beta}-c}{\varepsilon}},\,\alpha\otimes\beta\rangle=1-\langle\tilde\alpha,\,\tilde\beta\rangle_k$$

where $\tilde\alpha=e^{f_{\alpha,\alpha}/\varepsilon}\alpha$, $\tilde\beta=e^{f_{\beta,\beta}/\varepsilon}\beta$ and we introduced the inner product, valid because $k$ is positive definite, $\langle\tilde\alpha,\,\tilde\beta\rangle_k\coloneqq\int k(x,y)\mathrm{d}\tilde\alpha(x)\mathrm{d}\tilde\beta(y)$. The self Sinkhorn fixed point equation, once exponentiated, reads pointwise $e^{f_{\alpha,\alpha}(x)/\varepsilon}\int k(x,y)\mathrm{d}\tilde\alpha(y)=1$ for $\alpha$-a.e. $x$, and hence $\|\tilde\alpha\|_k^2=\langle k(\tilde\alpha),\,\tilde\alpha\rangle=\int e^{f_{\alpha,\alpha}(x)/\varepsilon}k(\tilde\alpha)(x)\mathrm{d}\alpha(x)=1$ and similarly $\|\tilde\beta\|_k^2=1$. Therefore, by Cauchy–Schwarz, one has $1-\langle\tilde\alpha,\,\tilde\beta\rangle_k\geqslant 0$. □

### 7.11 Complex Epsilon

This compact subsection records the complex-temperature fixed point viewpoint.
**Measure fixed point.** Let $\alpha\in\mathcal{P}(\mathcal{X})$ and $\beta\in\mathcal{P}(\mathcal{Y})$. Whenever the following integral operators are well defined, set, for $\varepsilon\in\mathbb{C}\setminus\{0\}$,
$K_\varepsilon(x,y)\coloneqq\exp(-c(x,y)/\varepsilon)$.

$$\mathrm{d}\pi_\varepsilon(x,y)=u(x)K_\varepsilon(x,y)v(y)\,\mathrm{d}\alpha(x)\mathrm{d}\beta(y).\tag{7.45}$$

$$u(x)\int_{\mathcal{Y}}K_\varepsilon(x,y)v(y)\,\mathrm{d}\beta(y)=1\quad\text{for }\alpha\text{-a.e. }x,\qquad v(y)\int_{\mathcal{X}}K_\varepsilon(x,y)u(x)\,\mathrm{d}\alpha(x)=1\quad\text{for }\beta\text{-a.e. }y.\tag{7.46}$$

$$u^{(k+1)}(x)=\left(\int_{\mathcal{Y}}K_\varepsilon(x,y)v^{(k)}(y)\,\mathrm{d}\beta(y)\right)^{-1},\qquad v^{(k+1)}(y)=\left(\int_{\mathcal{X}}K_\varepsilon(x,y)u^{(k+1)}(x)\,\mathrm{d}\alpha(x)\right)^{-1},\tag{7.47}$$

whenever the denominators do not vanish. If a branch of the logarithm is fixed along the local continuation and $u=e^{f/\varepsilon}$, $v=e^{g/\varepsilon}$, this is equivalently the pair of soft-transform equations

$$f(x)=-\varepsilon\log\int_{\mathcal{Y}}\exp\left(\frac{g(y)-c(x,y)}{\varepsilon}\right)\mathrm{d}\beta(y),\qquad g(y)=-\varepsilon\log\int_{\mathcal{X}}\exp\left(\frac{f(x)-c(x,y)}{\varepsilon}\right)\mathrm{d}\alpha(x).$$

For real positive $\varepsilon$, this is the usual Sinkhorn fixed point. For complex $\varepsilon$, the same formulas should be read as local analytic identities, not as a positivity statement.

**Discrete histograms.** If $\alpha = \sum_i \mathrm{a}_i \delta_{x_i}$ and $\beta = \sum_j \mathrm{b}_j \delta_{y_j}$, set $\mathrm{C}_{ij} = c(x_i, y_j)$ and define the complex Gibbs matrix $\mathrm{K}_\varepsilon(\mathrm{C})_{ij} := \exp(-\mathrm{C}_{ij}/\varepsilon)$. $\mathrm{P}_{\varepsilon,ij} = \mathrm{a}_i \mathrm{b}_j u_i \mathrm{K}_\varepsilon(\mathrm{C})_{ij} v_j$, $u_i \sum_j \mathrm{b}_j \mathrm{K}_\varepsilon(\mathrm{C})_{ij} v_j = 1$, $v_j \sum_i \mathrm{a}_i \mathrm{K}_\varepsilon(\mathrm{C})_{ij} u_i = 1$. $\mathrm{P}_\varepsilon = \mathrm{diag}(\mathrm{u})\mathrm{K}_\varepsilon(\mathrm{C})\,\mathrm{diag}(\mathrm{v})$, $\mathrm{P}_\varepsilon \mathbb{1}_m = \mathrm{a}$, $\mathrm{P}_\varepsilon^\top \mathbb{1}_n = \mathrm{b}$.

$$\mathrm{u}^{(k+1)} = \mathrm{a} \oslash \left(\mathrm{K}_\varepsilon(\mathrm{C})\mathrm{v}^{(k)}\right), \qquad \mathrm{v}^{(k+1)} = \mathrm{b} \oslash \left(\mathrm{K}_\varepsilon(\mathrm{C})^\top \mathrm{u}^{(k+1)}\right), \tag{7.48}$$

where $\oslash$ denotes entrywise division. For complex $\varepsilon$, this iteration should be read as a local analytic parametrization of the fixed point, not as a globally convergent algorithm.

**Theorem 7.31** (Carlier's holomorphic continuation of Sinkhorn)**.** *Fix positive histograms $\mathrm{a}^0 \in \Sigma_n$, $\mathrm{b}^0 \in \Sigma_m$, a finite real cost matrix $\mathrm{C}^0$, and a real temperature $\varepsilon_0 > 0$. Let $(\mathrm{f}^0, \mathrm{g}^0)$ be the Sinkhorn potentials at this point, normalized by $\sum_i \mathrm{a}_i^0 f_i^0 = 0$. Then there are complex neighborhoods of $(\varepsilon_0, \mathrm{a}^0, \mathrm{b}^0, \mathrm{C}^0)$, inside the affine constraint $\sum_i \mathrm{a}_i = \sum_j \mathrm{b}_j$, and a unique holomorphic map $(\varepsilon, \mathrm{a}, \mathrm{b}, \mathrm{C}) \mapsto (\mathrm{f}_\varepsilon, \mathrm{g}_\varepsilon) \in \mathbb{C}^n \times \mathbb{C}^m$ satisfying the fixed gauge $\sum_i \mathrm{a}_i^0 f_{\varepsilon,i} = 0$ and solving*

$$\mathrm{P}_{\varepsilon,ij} = \exp\left(\frac{f_{\varepsilon,i} + g_{\varepsilon,j} - C_{ij}}{\varepsilon}\right), \qquad \mathrm{P}_\varepsilon \mathbb{1}_m = \mathrm{a}, \qquad \mathrm{P}_\varepsilon^\top \mathbb{1}_n = \mathrm{b}.$$

*Proof.* Consider the holomorphic map which sends $(\mathrm{f}, \mathrm{g}, \varepsilon, \mathrm{a}, \mathrm{b}, \mathrm{C})$ to the $n$ row residuals, the first $m-1$ column residuals, and the gauge condition,

$$R_i = \sum_j \exp\left(\frac{f_i + g_j - C_{ij}}{\varepsilon}\right) - \mathrm{a}_i, \qquad S_j = \sum_i \exp\left(\frac{f_i + g_j - C_{ij}}{\varepsilon}\right) - \mathrm{b}_j, \qquad G = \sum_i \mathrm{a}_i^0 f_i,$$

where $1 \leqslant i \leqslant n$ and $1 \leqslant j \leqslant m-1$. At the real base point the associated coupling $\mathrm{P}^0$ is strictly positive. Let $(\delta\mathrm{f}, \delta\mathrm{g})$ belong to its kernel and set $\delta P_{ij} = \frac{1}{\varepsilon_0} P_{ij}^0 (\delta f_i + \delta g_j)$. The linearized row sums vanish. The first $m-1$ linearized column sums vanish by definition of the residuals, and the last one vanishes because the total row sum equals the total column sum. Therefore

$$0 = \sum_i \overline{\delta f_i} \sum_j \delta P_{ij} + \sum_j \overline{\delta g_j} \sum_i \delta P_{ij} = \frac{1}{\varepsilon_0} \sum_{i,j} P_{ij}^0 \left|\delta f_i + \delta g_j\right|^2.$$

Since $P_{ij}^0 > 0$, one has $\delta f_i + \delta g_j = 0$ for all $i, j$. Hence $\delta\mathrm{f}$ is constant and $\delta\mathrm{g}$ is the opposite constant. The gauge derivative $\sum_i \mathrm{a}_i^0 \delta f_i = 0$ forces this constant to vanish. The derivative is thus invertible, and the holomorphic implicit-function theorem gives the local holomorphic potentials. The formulas for $\mathrm{u}_\varepsilon$, $\mathrm{v}_\varepsilon$ and $\mathrm{P}_\varepsilon$ are holomorphic compositions of these potentials. ☐

# 8 Entropic Regularization: Convergence

The chapter revisits Sinkhorn convergence through several complementary lenses. Bregman projections explain the alternating-projection geometry, Fortet's order argument gives qualitative fixed-point convergence, and an order-theoretic M-function viewpoint explains why Sinkhorn-like scaling can still converge even when the equations are no longer the gradient of a convex potential.

## 8.1 Sinkhorn Convergence: Bregman View

This compact subsection records the Bregman-projection interpretation of Sinkhorn.

**Alternating KL projections.**

**Definition 8.1** (Bregman divergence)**.** Let $\Phi$ be a differentiable strictly convex function on a convex domain $\Omega$. The Bregman divergence generated by $\Phi$ is $B_\Phi(P|Q) := \Phi(P) - \Phi(Q) - \langle \nabla\Phi(Q), P - Q\rangle$. For the negative entropy $\Phi(P) = \sum_{i,j} P_{i,j} \log P_{i,j}$ on the positive orthant, one obtains $B_\Phi(P|Q) = \mathrm{KL}(P|Q)$ up to the harmless convention at the boundary.

**Linear tilts and Gibbs references.**

**Proposition 8.2** (Linear tilts of Bregman penalties)**.** *Let $\Phi$ be differentiable and strictly convex, and let $B_\Phi$ be its Bregman divergence. Fix a reference point $\mathrm{Q}$ in the interior of the domain. Assume that there exists $\mathrm{Q}^\mathrm{C}$ such that $\nabla\Phi(\mathrm{Q}^\mathrm{C}) = \nabla\Phi(\mathrm{Q}) - \mathrm{C}/\varepsilon$. Then, for all $\mathrm{P}$ in the domain, $\langle \mathrm{P}, \mathrm{C}\rangle + \varepsilon B_\Phi(\mathrm{P}|\mathrm{Q}) = \varepsilon B_\Phi(\mathrm{P}|\mathrm{Q}^\mathrm{C}) + \mathrm{cst}$, where the constant does not depend on $\mathrm{P}$.*

*Proof.* Subtract the two Bregman divergences: $B_\Phi(\mathrm{P}|\mathrm{Q}^\mathrm{C}) - B_\Phi(\mathrm{P}|\mathrm{Q}) = \langle \nabla\Phi(\mathrm{Q}) - \nabla\Phi(\mathrm{Q}^\mathrm{C}), \mathrm{P}\rangle + \mathrm{cst}$. Using $\nabla\Phi(\mathrm{Q}) - \nabla\Phi(\mathrm{Q}^\mathrm{C}) = \mathrm{C}/\varepsilon$ and multiplying by $\varepsilon$ gives the claim. ☐

For the negative entropy $\Phi(\mathrm{P}) = \sum_{i,j} \mathrm{P}_{i,j} \log \mathrm{P}_{i,j}$, one has $B_\Phi = \mathrm{KL}$. Taking $\mathrm{Q} = \mathrm{a} \otimes \mathrm{b}$ gives the tilted reference $\mathrm{K}_{\mathrm{a,b}}^\varepsilon := (\mathrm{a} \otimes \mathrm{b}) \odot e^{-\mathrm{C}/\varepsilon}$. $\langle \mathrm{P}, \mathrm{C}\rangle + \varepsilon \mathrm{KL}(\mathrm{P}|\mathrm{a} \otimes \mathrm{b}) = \varepsilon \mathrm{KL}(\mathrm{P}|\mathrm{K}_{\mathrm{a,b}}^\varepsilon) + \mathrm{cst}$. Thus the unique solution $\mathrm{P}_\varepsilon$ of (7.1) is the KL projection of the tilted Gibbs reference onto $\mathrm{U}(\mathrm{a}, \mathrm{b})$:

$$\mathrm{P}_\varepsilon = \mathrm{Proj}_{\mathrm{U(a,b)}}^{\mathrm{KL}}(\mathrm{K}_{\mathrm{a,b}}^\varepsilon) := \operatorname*{argmin}_{\mathrm{P} \in \mathrm{U(a,b)}} \mathrm{KL}(\mathrm{P}|\mathrm{K}_{\mathrm{a,b}}^\varepsilon). \tag{8.1}$$

**Cyclic projection convergence.**

**Proposition 8.3** (Cyclic Bregman projections on affine constraints)**.** *Let $\Phi$ be a Legendre strictly convex generator on a finite-dimensional convex domain, and let $\mathcal{C}_1, \mathcal{C}_2$ be affine constraint sets whose intersection meets the domain. Define $P_{k+1} = \mathrm{Proj}_{\mathcal{C}_2}^{B_\Phi} \mathrm{Proj}_{\mathcal{C}_1}^{B_\Phi}(P_k)$, starting from an interior point $P_0$. Assume the projections are well-defined and that the iterates remain in a compact subset of the domain. Then $P_k$ converges to the Bregman projection of $P_0$ onto $\mathcal{C}_1 \cap \mathcal{C}_2$. In particular, the KL case converges for positive affine marginal constraints on a bounded transportation polytope.*

*Proof.* We first prove the Pythagorean identity used by the projection argument. For three interior points one has the Bregman three-point formula $B_\Phi(Q|P) = B_\Phi(Q|P^+) + B_\Phi(P^+|P) + \langle \nabla\Phi(P^+) - \nabla\Phi(P), Q - P^+\rangle$. If $P^+ = \mathrm{Proj}_\mathcal{C}^{B_\Phi}(P)$ and $\mathcal{C}$ is affine, the first-order optimality condition for minimizing $R \mapsto B_\Phi(R|P)$ over $R \in \mathcal{C}$ is $\langle \nabla\Phi(P^+) - \nabla\Phi(P), R - P^+\rangle = 0 \quad \forall R \in \mathcal{C}$, because $R - P^+$ ranges over the tangent linear space of $\mathcal{C}$. Taking $R = Q \in \mathcal{C}$ cancels the last term and gives $B_\Phi(Q|P) = B_\Phi(Q|P^+) + B_\Phi(P^+|P) \quad \forall Q \in \mathcal{C}$.

Let $(Z_\ell)_\ell$ be the half-step sequence obtained by alternating projections onto $\mathcal{C}_1$ and $\mathcal{C}_2$, so that $Z_{2k} = P_k$ and $Z_{2k+2} = P_{k+1}$. Fix $Q \in \mathcal{C}_1 \cap \mathcal{C}_2$. Applying the identity at each half-step gives $B_\Phi(Q|Z_\ell) - B_\Phi(Q|Z_{\ell+1}) = B_\Phi(Z_{\ell+1}|Z_\ell) \geqslant 0$. Thus $B_\Phi(Q|Z_\ell)$ decreases and the series $\sum_\ell B_\Phi(Z_{\ell+1}|Z_\ell)$ is finite. The compactness assumption gives cluster points. Since the projection drops tend to zero and $\Phi$ is strictly convex on compact subsets of the domain, $\|Z_{\ell+1} - Z_\ell\| \to 0$. Every cluster point of the even subsequence is therefore also a cluster point of the adjacent odd subsequence. Because these two subsequences lie alternately in the closed affine sets $\mathcal{C}_1$ and $\mathcal{C}_2$, every cluster point belongs to $\mathcal{C}_1 \cap \mathcal{C}_2$.
Let $\bar{P}$ be such a cluster point. For each half-step, the dual displacement $\nabla\Phi(Z_{\ell+1}) - \nabla\Phi(Z_\ell)$ belongs to the normal space of the affine set onto which one projects. Telescoping and using the convergence of $Z_\ell$ gives $\nabla\Phi(\bar{P}) - \nabla\Phi(P_0) \in N_{\mathcal{C}_1} + N_{\mathcal{C}_2} = N_{\mathcal{C}_1 \cap \mathcal{C}_2}$, where the last equality uses that the sets are affine. This is precisely the first-order optimality condition for minimizing $R \mapsto B_\Phi(R|P_0)$ over $R \in \mathcal{C}_1 \cap \mathcal{C}_2$. Thus $\bar{P}$ is the Bregman projection of $P_0$ onto the intersection. Strict convexity gives uniqueness of this minimizer, so all cluster points coincide and the whole sequence converges. The KL statement follows by choosing the negative entropy generator. □

$\mathcal{C}^1_{\mathrm{a}} \coloneqq \{\mathrm{P} \; ; \; \mathrm{P}\mathbb{1}_m = \mathrm{a}\}$ and $\mathcal{C}^2_{\mathrm{b}} \coloneqq \{\mathrm{P} \; ; \; \mathrm{P}^\top\mathbb{1}_n = \mathrm{b}\}$

$$\mathrm{P}^{(\ell+1)} \coloneqq \mathrm{Proj}^{\mathrm{KL}}_{\mathcal{C}^1_{\mathrm{a}}}(\mathrm{P}^{(\ell)}) \quad \text{and} \quad \mathrm{P}^{(\ell+2)} \coloneqq \mathrm{Proj}^{\mathrm{KL}}_{\mathcal{C}^2_{\mathrm{b}}}(\mathrm{P}^{(\ell+1)}). \tag{8.2}$$

**Row and column scalings.**

**Proposition 8.4** (KL projections are scalings)**.** *One has* $\mathrm{Proj}^{\mathrm{KL}}_{\mathcal{C}^1_{\mathrm{a}}}(\mathrm{P}) = \mathrm{diag}\left(\frac{\mathrm{a}}{\mathrm{P}\mathbb{1}_m}\right)\mathrm{P}$ *and* $\mathrm{Proj}^{\mathrm{KL}}_{\mathcal{C}^2_{\mathrm{b}}}(\mathrm{P}) = \mathrm{P}\,\mathrm{diag}\left(\frac{\mathrm{b}}{\mathrm{P}^\top\mathbb{1}_n}\right)$.

*Proof.* Consider the problem along each row or column vector to impose a fixed sum $s \in \mathbb{R}_+$ $\min_p \{\mathrm{KL}(p|q) \; ; \; \langle p, \mathbb{1}\rangle = s\}$. The Lagrange multiplier equation for this problem reads $\log(p/q) + \lambda\mathbb{1} = 0 \implies p = uq$ where $u = e^{-\lambda} > 0$. The constraint $\langle p, \mathbb{1}\rangle = s$ is equivalent to $\langle uq, \mathbb{1}\rangle = s$, i.e. $u = s/\sum_i q_i$, which gives the desired scaling formula $p = sq/\sum_i q_i$. □

$$\mathrm{P}^{(2\ell)} \coloneqq \mathrm{diag}(\mathrm{u}^{(\ell)})\mathrm{K}\,\mathrm{diag}(\mathrm{v}^{(\ell)}),$$

$$\begin{aligned} &\mathrm{P}^{(2\ell+1)} \coloneqq \mathrm{diag}(\mathrm{u}^{(\ell+1)})\mathrm{K}\,\mathrm{diag}(\mathrm{v}^{(\ell)}) \\ \text{and} \quad &\mathrm{P}^{(2\ell+2)} \coloneqq \mathrm{diag}(\mathrm{u}^{(\ell+1)})\mathrm{K}\,\mathrm{diag}(\mathrm{v}^{(\ell+1)}) \end{aligned}$$

### 8.2 Sinkhorn Convergence: Monotone Point of View

This compact subsection records the order-theoretic route to Sinkhorn convergence.

**Fortet monotonicity.** The double soft transform is order preserving after quotienting constants.

**Proposition 8.5** (Monotone fixed-point route to Sinkhorn convergence)**.** *Let $c$ be bounded and continuous on compact spaces, and define the double Sinkhorn map, normalized by subtracting its $\alpha$-mean, $\mathcal{A}(f) \coloneqq (f^{c,\varepsilon})^{\bar{c},\varepsilon} - \int (f^{c,\varepsilon})^{\bar{c},\varepsilon}\mathrm{d}\alpha$. The map $\mathcal{A}$ is order preserving on the quotient by additive constants. If a representative of the initial class is chosen so that $f_0 \leqslant \mathcal{A}(f_0)$, then representatives of the iterates $f_{k+1} = \mathcal{A}(f_k)$ can be chosen to increase pointwise, remain uniformly bounded in oscillation, and converge to a fixed point. Since constants are free, any bounded initialization can be shifted downward to satisfy the subsolution inequality. The fixed point is the entropic potential, hence the associated Sinkhorn scalings converge.*

*Proof.* The soft $c$-transform is order reversing: if $g \leqslant g'$, then $-\varepsilon \log \int e^{(g-c)/\varepsilon}\mathrm{d}\beta \geqslant -\varepsilon \log \int e^{(g'-c)/\varepsilon}\mathrm{d}\beta$. The composition of two order-reversing transforms is therefore order preserving. The transform also commutes with additive constants in the projective sense, which is why the argument is naturally stated for equivalence classes modulo constants. Starting from a subsolution representative gives $f_0 \leqslant f_1$, and order preservation gives representatives satisfying $f_k \leqslant f_{k+1}$ for all $k$. Soft-transform oscillation bounds, controlled by $\sup c - \inf c$, prevent escape to infinity after normalization. Monotone pointwise convergence, compactness of equicontinuous soft transforms, and continuity of $\mathcal{A}$ then give a fixed point. Uniqueness of entropic potentials up to constants, Proposition 7.2 and the dual uniqueness statement above identify this fixed point with the Sinkhorn solution. □

**Extension beyond KL.** $\Phi \coloneqq \varphi^{*,\geqslant 0}$.

$$g^{\bar{c},\varepsilon,\varphi}(x) \in \operatorname*{argmin}_{u \in \mathbb{R}} \left\{\varepsilon \int_{\mathcal{Y}} \Phi\left(\frac{u + g(y) - c(x,y)}{\varepsilon}\right)\mathrm{d}\beta(y) - u\right\},$$

$$1 \in \int_{\mathcal{Y}} \partial\Phi\left(\frac{g^{\bar{c},\varepsilon,\varphi}(x) + g(y) - c(x,y)}{\varepsilon}\right)\mathrm{d}\beta(y), \qquad 1 \in \int_{\mathcal{X}} \partial\Phi\left(\frac{f(x) + f^{c,\varepsilon,\varphi}(y) - c(x,y)}{\varepsilon}\right)\mathrm{d}\alpha(x),$$

where the integral of intervals is understood in the usual Aumann sense. When $\Phi$ is differentiable and $\rho_\varphi \coloneqq \Phi'$, this becomes the normalization relations

$$\int_{\mathcal{Y}} \rho_\varphi\left(\frac{g^{\bar{c},\varepsilon,\varphi}(x) + g(y) - c(x,y)}{\varepsilon}\right)\mathrm{d}\beta(y) = 1, \qquad \int_{\mathcal{X}} \rho_\varphi\left(\frac{f(x) + f^{c,\varepsilon,\varphi}(y) - c(x,y)}{\varepsilon}\right)\mathrm{d}\alpha(x) = 1.$$

For KL, $\rho_\varphi(s) = e^s$ and these are exactly Sinkhorn row and column normalizations. For Burg or quadratic penalties, the same equations hold with $\rho_\varphi(s) = (1-s)^{-1}$ or $\rho_\varphi(s) = (1+s)_+$, respectively, on the appropriate domains.

**Proposition 8.6** (Monotonicity of generalized double soft transforms)**.** *Assume that the scalar minimizer sets defining $g^{\bar{c},\varepsilon,\varphi}$ and $f^{c,\varepsilon,\varphi}$ are nonempty compact intervals, and choose either the minimal or the maximal minimizer in every scalar problem. Then each selected generalized soft $c$-transform is order reversing and additively anti-homogeneous:*

$$g \leqslant g' \Rightarrow g^{\bar{c},\varepsilon,\varphi} \geqslant g'^{\bar{c},\varepsilon,\varphi}, \qquad (g+\lambda)^{\bar{c},\varepsilon,\varphi} = g^{\bar{c},\varepsilon,\varphi} - \lambda.$$

*The same statements hold for $f \mapsto f^{c,\varepsilon,\varphi}$. $\mathcal{A}_\varphi(f) \coloneqq (f^{c,\varepsilon,\varphi})^{\bar{c},\varepsilon,\varphi}$ is order preserving and additively homogeneous: $f \leqslant f' \Rightarrow \mathcal{A}_\varphi(f) \leqslant \mathcal{A}_\varphi(f')$, $\mathcal{A}_\varphi(f+\lambda) = \mathcal{A}_\varphi(f) + \lambda$. In particular, this applies without mentioning selections whenever the scalar minimizers are unique.*

*Proof.* Since $\Phi(s) = \sup_{r\geqslant 0}\{sr - \varphi(r)\}$, the function $\Phi$ is convex and nondecreasing. For fixed $x$, set

$$H_g^x(u) := \varepsilon \int_{\mathcal{Y}} \Phi\Big(\frac{u + g(y) - c(x,y)}{\varepsilon}\Big)\, \mathrm{d}\beta(y) - u.$$

If $g \leqslant g'$, write $\Delta(y) = g'(y) - g(y) \geqslant 0$. The difference

$$H_{g'}^x(u) - H_g^x(u) = \varepsilon \int_{\mathcal{Y}} \Big[\Phi\Big(\frac{u + g(y) + \Delta(y) - c(x,y)}{\varepsilon}\Big) - \Phi\Big(\frac{u + g(y) - c(x,y)}{\varepsilon}\Big)\Big]\, \mathrm{d}\beta(y)$$

is nondecreasing in $u$: for every $\delta \geqslant 0$, the function $s \mapsto \Phi(s+\delta) - \Phi(s)$ is nondecreasing by convexity of $\Phi$. Let $I = \operatorname{argmin} H_g^x$ and $I' = \operatorname{argmin} H_{g'}^x$. If $a = \min I$ and $a' = \min I'$ with $a' > a$, then the two optimality inequalities $H_g^x(a) \leqslant H_g^x(a')$, $\quad H_{g'}^x(a') \leqslant H_{g'}^x(a)$ imply $H_{g'}^x(a') - H_g^x(a') \leqslant H_{g'}^x(a) - H_g^x(a)$, contradicting monotonicity unless equality holds throughout. In the equality case $a$ is also a minimizer of $H_{g'}^x$, contradicting the definition of $a' = \min I'$. Hence $\min I' \leqslant \min I$. The same argument with $b = \max I$ and $b' = \max I'$ shows $\max I' \leqslant \max I$. Thus both extremal selections are order reversing. Finally, $H_{g+\lambda}^x(u) = H_g^x(u+\lambda) + \lambda$, so the minimizer interval is shifted by $-\lambda$, which proves additive anti-homogeneity. The proof for $f \mapsto f^{c,\varepsilon,\varphi}$ is identical. Composing two order-reversing anti-homogeneous selections gives an order-preserving additively homogeneous double transform. $\square$

**Variation seminorm and topical maps.**

**Definition 8.7** (Variation seminorm)**.** For a bounded real-valued function $h$, the variation seminorm is $\|h\|_V := \sup h - \inf h$. It vanishes exactly on constant functions, hence becomes a norm after quotienting by additive constants.

**Proposition 8.8** (Topical maps are variation-nonexpansive)**.** *Let $E$ be a vector space of real-valued bounded functions, ordered pointwise, and write $\|\cdot\|_V$ for the variation seminorm of Definition 8.7. Let $\mathcal{T} : E \to E$ be monotone and additively homogeneous, $f \leqslant g \Rightarrow \mathcal{T}(f) \leqslant \mathcal{T}(g)$, $\qquad \mathcal{T}(f+\lambda) = \mathcal{T}(f) + \lambda \quad \forall \lambda \in \mathbb{R}$. Then $\|\mathcal{T}(f) - \mathcal{T}(g)\|_V \leqslant \|f - g\|_V$. The same conclusion holds for order-reversing maps satisfying $\mathcal{T}(f+\lambda) = \mathcal{T}(f) - \lambda$.*

*Proof.* Set $a = \inf(f-g)$ and $b = \sup(f-g)$. Then $g+a \leqslant f \leqslant g+b$. If $\mathcal{T}$ is order preserving and additively homogeneous, applying $\mathcal{T}$ gives $\mathcal{T}(g)+a \leqslant \mathcal{T}(f) \leqslant \mathcal{T}(g)+b$. Hence every value of $\mathcal{T}(f) - \mathcal{T}(g)$ lies in $[a,b]$, so its oscillation is at most $b - a = \|f-g\|_V$. For an order-reversing, additively anti-homogeneous map, the same inequalities give $\mathcal{T}(g) - b \leqslant \mathcal{T}(f) \leqslant \mathcal{T}(g) - a$, and the oscillation bound is identical. $\square$

**Corollary 8.9** (Soft transforms are nonexpansive)**.** *For every $\varepsilon > 0$, the soft $c$-transforms (7.19)–(7.20) are 1-Lipschitz for the variation seminorm.*

*Proof.* The soft transform is order reversing and satisfies $(g+\lambda)^{\bar{c},\varepsilon} = g^{\bar{c},\varepsilon} - \lambda$, and similarly for the other block. Proposition 8.8 applies to each block, and the composition of two 1-Lipschitz maps is 1-Lipschitz. $\square$

**General $\varphi$-divergence regularization.** Proposition 8.6 shows that the order argument is not tied to the logarithmic KL formula. For the $\varphi$-divergence regularizers of Section 7.9, each selected one-sided generalized soft $c$-transform is order reversing and additively anti-homogeneous: increasing one potential decreases the updated one, and adding a constant subtracts the same constant.
$\mathcal{A}_\varphi(f) := (f^{c,\varepsilon,\varphi})^{\bar{c},\varepsilon,\varphi}$ $\|\mathcal{A}_\varphi(f) - \mathcal{A}_\varphi(h)\|_V \leqslant \|f-h\|_V$. Thus the Fortet monotonicity mechanism and the basic nonexpansiveness estimate extend to general convex density-ratio penalties. What remains special to KL is the stronger Hilbert-metric contraction of Section 8.5, which uses the strictly positive Gibbs kernel to obtain a linear rate.

### 8.3 Monotone Clearing Beyond Variational Sinkhorn

This compact subsection records monotone fixed-point mechanisms beyond variational Sinkhorn.

**Two-block clearing maps.** $Q(z) = \big(Q^\alpha(u,v), Q^\beta(u,v)\big) \in \mathbb{R}^n \times \mathbb{R}^m$ $Q_\ell\big(T_\ell(z), z_{-\ell}\big) = 0, \qquad 1 \leqslant \ell \leqslant n+m.$

**Definition 8.10** (Z-functions and M-functions)**.** Let $D \subset \mathbb{R}^N$ be an order interval. A map $Q : D \to \mathbb{R}^N$ is diagonally isotone if $Q_\ell(z_\ell, z_{-\ell})$ is nondecreasing in $z_\ell$ for every $\ell$. It is a Z-function if increasing the other coordinates cannot increase the $\ell$-th component: $z_{-\ell} \leqslant z'_{-\ell} \quad\Longrightarrow\quad Q_\ell(z_\ell, z_{-\ell}) \geqslant Q_\ell(z_\ell, z'_{-\ell})$. For a $C^1$ map this means $\partial Q_\ell/\partial z_k \leqslant 0$ for $k \neq \ell$. An M-function is a Z-function which is inverse isotone: $Q(z) \leqslant Q(z') \quad\Longrightarrow\quad z \leqslant z'$. A vector $z$ is a subsolution if $Q(z) \leqslant 0$, and a supersolution if $Q(z) \geqslant 0$.

**Theorem 8.11** (Monotone convergence of coordinate clearing)**.** *Let $D = [\underline{z}, \overline{z}] \subset \mathbb{R}^N$ be a closed order interval, with $\underline{z}$ a subsolution and $\overline{z}$ a supersolution. Assume that $Q : D \to \mathbb{R}^N$ is continuous, diagonally isotone and an M-function. Assume also that, for every $z \in D$ and every $\ell$, the scalar clearing equation $Q_\ell(\xi, z_{-\ell}) = 0$ has a unique solution $\xi \in [\underline{z}_\ell, \overline{z}_\ell]$. Then $Q(z) = 0$ has a unique solution $z^\star \in D$. The Jacobi coordinate-clearing iterates starting from any subsolution in $D$ increase monotonically to $z^\star$, while those starting from any supersolution in $D$ decrease monotonically to $z^\star$.*

*Proof.* Let $T$ be the coordinate-clearing map. If $z \in D$ is a subsolution, then $Q_\ell(z_\ell, z_{-\ell}) \leqslant 0 = Q_\ell(T_\ell(z), z_{-\ell})$. Diagonal isotonicity and uniqueness of the scalar zero give $z_\ell \leqslant T_\ell(z)$ for every $\ell$, hence $z \leqslant T(z)$. Since $Q$ is a Z-function, $Q_\ell(T(z)) \leqslant Q_\ell(T_\ell(z), z_{-\ell}) = 0$, so $T(z)$ is again a subsolution. Since every subsolution lies below every supersolution by inverse isotonicity, $T(z) \leqslant \overline{z}$, and therefore $T(z) \in D$. The supersolution argument is the same with all inequalities reversed. The lower and upper iterates are thus monotone and trapped in the compact interval $D$. Their limits exist, and passing to the limit in the identities $Q_\ell(z_\ell^{k+1}, z_{-\ell}^k) = 0$ gives $Q(z^\star) = 0$. If $z$ and $z'$ were two zeros in $D$, then $Q(z) \leqslant Q(z')$ and $Q(z') \leqslant Q(z)$; inverse isotonicity gives $z = z'$. $\square$

**Proposition 8.12** (Smooth M-matrix certificate)**.** *Let $D \subset \mathbb{R}^N$ be a convex order interval and let $Q : D \to \mathbb{R}^N$ be $C^1$. Assume that $DQ(z)$ has nonpositive off-diagonal entries for every $z \in D$. If there exists $\lambda \gg 0$ such that $\lambda^\top DQ(z) \gg 0 \qquad$ for all $z \in D$, or equivalently*

$$\lambda_\ell \frac{\partial Q_\ell}{\partial z_\ell}(z) > \sum_{k \neq \ell} \lambda_k \left|\frac{\partial Q_k}{\partial z_\ell}(z)\right|, \qquad 1 \leqslant \ell \leqslant N,$$

*then $Q$ is an M-function on $D$.*

*Proof.* The off-diagonal sign gives the Z-property by integrating the partial derivatives along coordinatewise increasing segments. The displayed inequalities imply that each $DQ(z)$ is a strictly weighted column diagonally dominant Z-matrix, hence a nonsingular M-matrix; in particular $DQ(z)^{-1} \geqslant 0$ for each fixed $z$. For two points $z, z' \in D$, set $A = \int_0^1 DQ\big(z' + t(z - z')\big)\,\mathrm{d}t$. The matrix $A$ has the same Z-sign pattern and the same strict weighted diagonal dominance, so it is again a nonsingular M-matrix and $A^{-1} \geqslant 0$. Since $Q(z) - Q(z') = A(z - z')$, the implication $Q(z) \leqslant Q(z')$ gives $z - z' = A^{-1}(Q(z) - Q(z')) \leqslant 0$. This is inverse isotonicity. The positive diagonal entries also give diagonal isotonicity. ☐

**Sinkhorn as the canonical $M_0$-system.** $P_{ij} = r_i \mathrm{K}_{ij} s_j$, $r_i = e^{u_i}$, $s_j = e^{-v_j}$. $Q_i^\alpha(u, v) = \sum_j \mathrm{K}_{ij} e^{u_i - v_j} - a_i$, $Q_j^\beta(u, v) = b_j - \sum_i \mathrm{K}_{ij} e^{u_i - v_j}$. $r_i^+ = \frac{a_i}{\sum_j \mathrm{K}_{ij} s_j}$, $s_j^+ = \frac{b_j}{\sum_i \mathrm{K}_{ij} r_i}$.

$$\frac{\partial Q_i^\alpha}{\partial u_i} = \sum_j P_{ij}, \qquad \frac{\partial Q_i^\alpha}{\partial v_j} = -P_{ij}, \qquad \frac{\partial Q_j^\beta}{\partial v_j} = \sum_i P_{ij}, \qquad \frac{\partial Q_j^\beta}{\partial u_i} = -P_{ij}.$$

### 8.4 Sinkhorn Convergence: Sublinear Robust Rate

Sinkhorn is cyclic coordinate ascent on the smooth dual objective $\mathcal{D}_\varepsilon$ defined in (7.18); equivalently, it alternates KL projections on the two marginal constraint sets. We state it for balanced entropic OT, which is the specialization needed here.

**Proposition 8.13** (Pinsker inequality)**.** *If $p, q \in \Sigma_n$, then $\|p - q\|_1^2 \leqslant 2\,\mathrm{KL}(p|q)$.*

*Proof.* Let $A = \{i : p_i \geqslant q_i\}$ and set $a = \sum_{i \in A} p_i$, $b = \sum_{i \in A} q_i$. Then $a - b = \frac{1}{2}\|p - q\|_1$. Applying data processing for relative entropy to the partition $(A, A^c)$ gives $\mathrm{KL}(p|q) \geqslant a \log \frac{a}{b} + (1 - a) \log \frac{1-a}{1-b}$. For fixed $b \in (0, 1)$, the function $h(a) = a \log \frac{a}{b} + (1 - a) \log \frac{1-a}{1-b} - 2(a - b)^2$ satisfies $h(b) = h'(b) = 0$ and $h''(a) = \frac{1}{a} + \frac{1}{1-a} - 4 \geqslant 0$, because $a(1 - a) \leqslant 1/4$. Hence the binary relative entropy is at least $2(a - b)^2 = \frac{1}{2}\|p - q\|_1^2$. The boundary cases follow by approximation. ☐

**Proposition 8.14** (A compact $O(1/k)$ dual rate)**.** *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact, that $c$ is bounded, and write $R = \sup c - \inf c$. Let $(f_k, g_k)$ be Sinkhorn dual iterates normalized by $\int f_k \mathrm{d}\alpha = 0$, and let $\Delta_k := \mathcal{D}_\varepsilon(f^\star, g^\star) - \mathcal{D}_\varepsilon(f_k, g_k)$ be the dual suboptimality gap for the entropic dual objective (7.18). Then there exists a numerical constant $C$ such that $\Delta_k \leqslant \frac{CR^2}{\varepsilon(k+1)}$.*

*Proof.* First, the soft $c$-transform bounds the oscillation of every normalized iterate and every normalized optimum: $\|f_k\|_V + \|g_k\|_V + \|f^\star\|_V + \|g^\star\|_V \leqslant CR$. Here $\|h\|_V := \sup h - \inf h$ is the variation seminorm, the same projective norm used in Hilbert's metric in Section 8.5. It is natural because dual potentials can be shifted by constants without changing the coupling.
Second, each Sinkhorn half-step is an exact KL projection. The Pythagorean identity for KL projections gives the ascent identity $\mathcal{D}_\varepsilon(f_{k+1}, g_{k+1}) - \mathcal{D}_\varepsilon(f_k, g_k) = \varepsilon\big[\mathrm{KL}(\pi^\star|\pi_k) - \mathrm{KL}(\pi^\star|\pi_{k+1})\big]$, where $\pi_k = e^{(f_k \oplus g_k - c)/\varepsilon}\alpha \otimes \beta$ and $\pi^\star$ is the optimal entropic coupling. The KL drop controls the marginal residuals through Pinsker's inequality, Proposition 8.13. Third, convexity of the exponential dual objective gives a one-step estimate of the form $\Delta_k^2 \leqslant \frac{CR^2}{\varepsilon}\big(\mathcal{D}_\varepsilon(f_{k+1}, g_{k+1}) - \mathcal{D}_\varepsilon(f_k, g_k)\big)$. This is the usual Bregman-projection estimate: the dual gap is controlled by the product of a bounded dual radius and the marginal residual corrected by the next projection, while the residual squared is controlled by the KL drop. Summing the reciprocal inequality obtained from the last display yields $\frac{1}{\Delta_{k+1}} - \frac{1}{\Delta_k} \geqslant \frac{\varepsilon}{CR^2}$, and therefore $\Delta_k \leqslant CR^2/(\varepsilon(k + 1))$. ☐

**Corollary 8.15** (Approximating unregularized OT by regularized dual costs)**.** *Consider discrete histograms $\mathrm{a} \in \Sigma_n$, $\mathrm{b} \in \Sigma_m$ and a finite cost matrix $\mathrm{C}$. Let $\mathcal{D}_{\varepsilon,k}$ be the KL-normalized entropic dual value after $k$ Sinkhorn cycles, and let $\Delta_k$ be its dual gap. Define the entropy-corrected lower bound $L_{\varepsilon,k} := \mathcal{D}_{\varepsilon,k} - \varepsilon\mathrm{H}(\mathrm{a}) - \varepsilon\mathrm{H}(\mathrm{b})$, $\mathrm{H}(\mathrm{a}) = -\sum_i a_i \log a_i$. Then $0 \leqslant \mathrm{L}_\mathrm{C}(\mathrm{a}, \mathrm{b}) - L_{\varepsilon,k} \leqslant \varepsilon \log(nm) + \Delta_k$. Under Proposition 8.14, the intermediate condition is $k + 1 \geqslant 2CR^2/(\varepsilon\delta)$. With the above choice of $\varepsilon$, it is sufficient to take $k + 1 \geqslant \frac{4CR^2 \log(nm)}{\delta^2}$, hence $k = O(R^2 \log(nm)/\delta^2)$, up to constants and logarithmic stabilization factors.*

*Proof.* The KL-normalized objective differs from the entropy convention (7.1) by the constant $\varepsilon\mathrm{H}(\mathrm{a}) + \varepsilon\mathrm{H}(\mathrm{b})$ on the transport polytope, because $\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) = -\mathrm{H}(\mathrm{P}) + \mathrm{H}(\mathrm{a}) + \mathrm{H}(\mathrm{b})$. Let $E_\varepsilon$ be the optimum of the entropy-regularized objective $\langle \mathrm{P}, \mathrm{C}\rangle - \varepsilon\mathrm{H}(\mathrm{P})$. Since $0 \leqslant \mathrm{H}(\mathrm{P}) \leqslant \log(nm)$ for any coupling matrix, $\mathrm{L}_\mathrm{C}(\mathrm{a}, \mathrm{b}) - \varepsilon \log(nm) \leqslant E_\varepsilon \leqslant \mathrm{L}_\mathrm{C}(\mathrm{a}, \mathrm{b})$. The corrected iterate satisfies $L_{\varepsilon,k} = E_\varepsilon - \Delta_k$, which gives the displayed value bound. The final iteration estimate follows by combining $\Delta_k \leqslant CR^2/(\varepsilon(k + 1))$ with the target $\Delta_k \leqslant \delta/2$. ☐

### 8.5 Sinkhorn Convergence: Linear Hilbert Metric Rate

This compact subsection separates the projective contraction argument from its dual-potential form.

**Projective contraction.** The first block gives the Hilbert-metric contraction mechanism.

**Definition 8.16** (Hilbert metric)**.** On $\mathbb{R}^n_{+,*}$, Hilbert's projective metric is

$$\forall\,(\mathrm{u}, \mathrm{u}') \in (\mathbb{R}^n_{+,*})^2, \quad d_\mathcal{H}(\mathrm{u}, \mathrm{u}') := \|\log(\mathrm{u}) - \log(\mathrm{u}')\|_V. \tag{8.3}$$

where, for vectors, $\|z\|_V = \max_i z_i - \min_i z_i$.

**Proposition 8.17** (Hilbert metric on the projective cone)**.** *The function $d_\mathcal{H}$ defines a complete distance on the projective cone $\mathbb{R}^n_{+,*}/\sim$, where $\mathrm{u} \sim \mathrm{u}'$ means that $\mathrm{u} = s\mathrm{u}'$ for some $s > 0$.*

*Proof.* The map $\mathrm{u} \mapsto \log \mathrm{u}$ identifies $\mathbb{R}^n_{+,*}/\sim$ with the quotient vector space $\mathbb{R}^n/\operatorname{Span}(\mathbb{1}_n)$, because multiplying $\mathrm{u}$ by $s > 0$ adds the constant vector $\log(s)\mathbb{1}_n$. The variation seminorm $\|z\|_V = \max_i z_i - \min_i z_i$ vanishes exactly on constant vectors, so it induces a norm on this quotient. Symmetry, the triangle inequality and separation therefore follow from the corresponding norm properties. Completeness follows because $\mathbb{R}^n/\operatorname{Span}(\mathbb{1}_n)$ is finite-dimensional and all finite-dimensional normed spaces are complete. ☐

**Theorem 8.18** (Birkhoff contraction theorem)**.** *Let $\mathrm{K} \in \mathbb{R}^{n\times m}_{+,*}$, then for $(\mathrm{v}, \mathrm{v}') \in (\mathbb{R}^m_{+,*})^2$*

$$d_\mathcal{H}(\mathrm{K}\mathrm{v}, \mathrm{K}\mathrm{v}') \leqslant \lambda(\mathrm{K}) d_\mathcal{H}(\mathrm{v}, \mathrm{v}') \text{ where } \begin{cases} \lambda(\mathrm{K}) := \frac{\sqrt{\eta(\mathrm{K})} - 1}{\sqrt{\eta(\mathrm{K})} + 1} < 1 \\ \eta(\mathrm{K}) := \max\limits_{i,j,k,\ell} \frac{\mathrm{K}_{i,k}\mathrm{K}_{j,\ell}}{\mathrm{K}_{j,k}\mathrm{K}_{i,\ell}}. \end{cases}$$

*Proof.* For a positive linear map $A$ on a cone, define its projective diameter $\Delta(A) := \sup_{u,v>0} d_{\mathcal{H}}(Au, Av)$. Then $d_{\mathcal{H}}(Au, Av) \leqslant \tanh(\Delta(A)/4) d_{\mathcal{H}}(u, v)$. Indeed, after quotienting by positive scalings, write $r_k = u_k/v_k$ and normalize so that $e^{-h/2} \leqslant r_k \leqslant e^{h/2}$, where $h = d_{\mathcal{H}}(u, v)$. The ratio between two coordinates of $Au/Av$ is a quotient of two weighted averages of the numbers $r_k$. A two-point extremal argument shows that the largest possible contraction is obtained when the mass of the two weights is placed on the two endpoints $e^{-h/2}$ and $e^{h/2}$; the cross-ratio bound defining $\Delta(A)$ then gives

$$d_{\mathcal{H}}(Au, Av) \leqslant 2 \log \frac{e^{\Delta(A)/4} e^{h/2} + e^{-\Delta(A)/4} e^{-h/2}}{e^{\Delta(A)/4} e^{-h/2} + e^{-\Delta(A)/4} e^{h/2}} \leqslant \tanh(\Delta(A)/4) h.$$

For the matrix K, its projective diameter is $\Delta(\mathrm{K}) = \log \eta(\mathrm{K})$, $\quad \eta(\mathrm{K}) = \max_{i,j,k,\ell} \frac{\mathrm{K}_{i,k} \mathrm{K}_{j,\ell}}{\mathrm{K}_{j,k} \mathrm{K}_{i,\ell}}$. Therefore $\tanh(\Delta(\mathrm{K})/4) = (\sqrt{\eta(\mathrm{K})} - 1)/(\sqrt{\eta(\mathrm{K})} + 1)$, which is the claimed contraction factor. □

**Theorem 8.19** (Linear convergence of Sinkhorn)**.** *One has* $(\mathrm{u}^{(\ell)}, \mathrm{v}^{(\ell)}) \to (\mathrm{u}^\star, \mathrm{v}^\star)$ *and*

$$d_{\mathcal{H}}(\mathrm{u}^{(\ell)}, \mathrm{u}^\star) = O(\lambda(\mathrm{K})^{2\ell}), \quad d_{\mathcal{H}}(\mathrm{v}^{(\ell)}, \mathrm{v}^\star) = O(\lambda(\mathrm{K})^{2\ell}). \tag{8.4}$$

*One also has*

$$d_{\mathcal{H}}(\mathrm{u}^{(\ell)}, \mathrm{u}^\star) \leqslant \frac{d_{\mathcal{H}}(\mathrm{P}^{(\ell)} \mathbb{1}_m, \mathrm{a})}{1 - \lambda(\mathrm{K})} \quad \textit{and} \quad d_{\mathcal{H}}(\mathrm{v}^{(\ell)}, \mathrm{v}^\star) \leqslant \frac{d_{\mathcal{H}}(\mathrm{P}^{(\ell),\top} \mathbb{1}_n, \mathrm{b})}{1 - \lambda(\mathrm{K})}, \tag{8.5}$$

*where we denoted* $\mathrm{P}^{(\ell)} := \mathrm{diag}(\mathrm{u}^{(\ell)}) \mathrm{K} \, \mathrm{diag}(\mathrm{v}^{(\ell)})$*. Lastly, one has*

$$\| \log(\mathrm{P}^{(\ell)}) - \log(\mathrm{P}^\star) \|_\infty \leqslant d_{\mathcal{H}}(\mathrm{u}^{(\ell)}, \mathrm{u}^\star) + d_{\mathcal{H}}(\mathrm{v}^{(\ell)}, \mathrm{v}^\star) \tag{8.6}$$

*where* $\mathrm{P}^\star$ *is the unique solution of* (7.1)*.*

*Proof.* Notice that for any $(\mathrm{v}, \mathrm{v}') \in (\mathbb{R}^m_{+,*})^2$, one has $d_{\mathcal{H}}(\mathrm{v}, \mathrm{v}') = d_{\mathcal{H}}(\mathrm{v}/\mathrm{v}', \mathbb{1}_m) = d_{\mathcal{H}}(\mathbb{1}_m/\mathrm{v}, \mathbb{1}_m/\mathrm{v}')$, since indeed $d_{\mathcal{H}}(\mathrm{a}/\mathrm{v}, \mathrm{a}/\mathrm{v}') = d_{\mathcal{H}}(\mathrm{v}, \mathrm{v}')$.
This shows that

$$d_{\mathcal{H}}(\mathrm{u}^{(\ell+1)}, \mathrm{u}^\star) = d_{\mathcal{H}} \left( \frac{\mathrm{a}}{\mathrm{K}\mathrm{v}^{(\ell)}}, \frac{\mathrm{a}}{\mathrm{K}\mathrm{v}^\star} \right) = d_{\mathcal{H}}(\mathrm{K}\mathrm{v}^{(\ell)}, \mathrm{K}\mathrm{v}^\star) \leqslant \lambda(\mathrm{K}) d_{\mathcal{H}}(\mathrm{v}^{(\ell)}, \mathrm{v}^\star).$$

where we used Theorem 8.18. This shows (8.4). One also has, using the triangular inequality,

$$\begin{aligned} d_{\mathcal{H}}(\mathrm{u}^{(\ell)}, \mathrm{u}^\star) &\leqslant d_{\mathcal{H}}(\mathrm{u}^{(\ell+1)}, \mathrm{u}^{(\ell)}) + d_{\mathcal{H}}(\mathrm{u}^{(\ell+1)}, \mathrm{u}^\star) \leqslant d_{\mathcal{H}} \left( \frac{\mathrm{a}}{\mathrm{K}\mathrm{v}^{(\ell)}}, \mathrm{u}^{(\ell)} \right) + \lambda(\mathrm{K}) d_{\mathcal{H}}(\mathrm{u}^{(\ell)}, \mathrm{u}^\star) \\ &= d_{\mathcal{H}} \left( \mathrm{a}, \mathrm{u}^{(\ell)} \odot (\mathrm{K}\mathrm{v}^{(\ell)}) \right) + \lambda(\mathrm{K}) d_{\mathcal{H}}(\mathrm{u}^{(\ell)}, \mathrm{u}^\star), \end{aligned}$$

which gives the first part of (8.5) since $\mathrm{u}^{(\ell)} \odot (\mathrm{K}\mathrm{v}^{(\ell)}) = \mathrm{P}^{(\ell)} \mathbb{1}_m$ (the second one being similar).
The primal contraction follows from the same projective estimate. □

**Dual-potential form of the contraction.** $\|f_k - f^\star\|_V = O(\lambda^k)$ and $\|g_k - g^\star\|_V = O(\lambda^k)$

$$\| \log \frac{\mathrm{d}\pi_k}{\mathrm{d}\pi^\star} \|_\infty = \|(f_k - f^\star) \oplus (g_k - g^\star)\|_\infty \leqslant \|f_k - f^\star\|_V + \|g_k - g^\star\|_V$$

where the contraction ratio is the Birkhoff factor of the Gibbs kernel $\mathrm{K}_\varepsilon = e^{-c/\varepsilon}$. Namely, with $\eta = \eta(\mathrm{K}_\varepsilon)$ as in Theorem 8.18 and $R := \sup c - \inf c$,
$\lambda = \frac{\sqrt{\eta} - 1}{\sqrt{\eta} + 1} \leqslant \tanh(R/(2\varepsilon)) < 1$. $\|f_k - f^\star\|_V \leqslant \frac{\| \log \frac{\mathrm{d}\pi_{k,1}}{\mathrm{d}\alpha} \|_\infty}{1 - \lambda}$

### 8.6 Entropic Optimal Transport between Gaussians

**Proposition 8.20** (Quadratic closure of Sinkhorn iterates)**.** *Let* $\beta = \mathcal{N}(\mathbf{m}_\beta, \mathbf{\Sigma}_\beta)$ *on* $\mathbb{R}^d$ *and take* $c(x, y) = \|x - y\|^2$*. If* $g(y)$ *is a quadratic polynomial such that the Gaussian integral below is finite, then the soft transform*

$$f(x) = -\varepsilon \log \int \exp\left( \frac{g(y) - \|x - y\|^2}{\varepsilon} \right) \mathrm{d}\beta(y)$$

*is a quadratic polynomial in* $x$*. In particular, starting Sinkhorn from* $g_0 = 0$ *gives*

$$f_1(x) = \frac{\varepsilon}{2} \log \det \left( \mathrm{Id} + \frac{2\mathbf{\Sigma}_\beta}{\varepsilon} \right) + \varepsilon \langle x - \mathbf{m}_\beta, \, (\varepsilon \mathrm{Id} + 2\mathbf{\Sigma}_\beta)^{-1} (x - \mathbf{m}_\beta) \rangle.$$

*Proof.* The exponent is the sum of a quadratic polynomial in $y$ and the logarithm of the Gaussian density of $\beta$. Completing the square in $y$ evaluates the integral as a positive constant times the exponential of a quadratic polynomial in $x$. Taking $-\varepsilon \log$ therefore gives a quadratic polynomial.
For $g_0 = 0$, let $Y \sim \beta$. The Gaussian identity

$$\mathbb{E} \exp\left( -\frac{\|x - Y\|^2}{\varepsilon} \right) = \det\left( \mathrm{Id} + \frac{2\mathbf{\Sigma}_\beta}{\varepsilon} \right)^{-1/2} \exp\left( -\langle x - \mathbf{m}_\beta, \, (\varepsilon \mathrm{Id} + 2\mathbf{\Sigma}_\beta)^{-1} (x - \mathbf{m}_\beta) \rangle \right)$$

gives the displayed expression. □

**Proposition 8.21** (Balanced entropic OT between Gaussians). *Let $\alpha = \mathcal{N}(\mathbf{m}_\alpha, \mathbf{\Sigma}_\alpha)$ and $\beta = \mathcal{N}(\mathbf{m}_\beta, \mathbf{\Sigma}_\beta)$ with positive-definite covariances, and let $\mathbf{\Sigma}_\alpha^{1/2}\mathbf{\Sigma}_\beta^{1/2} = U \operatorname{diag}(\sigma_i) V^\top$ be a singular-value decomposition. For the balanced objective $\min_{\pi\in\mathcal{U}(\alpha,\beta)} \int \|x - y\|^2 \mathrm{d}\pi(x, y) + \varepsilon \operatorname{KL}(\pi|\alpha\otimes\beta)$, the optimizer is Gaussian with cross-covariance $K_\varepsilon = \mathbf{\Sigma}_\alpha^{1/2} U \operatorname{diag}(s_i) V^\top \mathbf{\Sigma}_\beta^{1/2}$, $\quad s_i = \frac{\sqrt{\varepsilon^2+16\sigma_i^2}-\varepsilon}{4\sigma_i}$. The optimal value is*

$$\|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \operatorname{tr}(\mathbf{\Sigma}_\alpha) + \operatorname{tr}(\mathbf{\Sigma}_\beta) + \sum_i \left(-2\sigma_i s_i - \frac{\varepsilon}{2}\log(1 - s_i^2)\right).$$

*As $\varepsilon \downarrow 0$, $s_i \to 1$ and the full covariance contribution, including the two trace terms, converges to $\mathcal{B}(\mathbf{\Sigma}_\alpha, \mathbf{\Sigma}_\beta)^2$.*

*Proof.* Let $(X, Y)$ be any coupling with finite second moments and cross-covariance $K = \mathbb{E}\big[(X - \mathbf{m}_\alpha)(Y - \mathbf{m}_\beta)^\top\big]$. Replacing $(X, Y)$ by the Gaussian vector with the same mean and covariance leaves the quadratic cost unchanged. Since the marginals are fixed, $\operatorname{KL}(\pi|\alpha\otimes\beta) = -h(X, Y) + h(\alpha) + h(\beta)$, where $h$ denotes differential entropy when it is finite. Among laws with a fixed covariance, the Gaussian maximizes entropy; if the entropy is not finite, the relative entropy is already $+\infty$. Thus the Gaussian replacement cannot increase the objective, and it is enough to optimize over Gaussian couplings.
Any such coupling has covariance

$$\begin{pmatrix} \mathbf{\Sigma}_\alpha & K \\ K^\top & \mathbf{\Sigma}_\beta \end{pmatrix}.$$

Write $K = \mathbf{\Sigma}_\alpha^{1/2} S \mathbf{\Sigma}_\beta^{1/2}$. The block covariance constraint is equivalent to the singular values of $S$ being at most one, and finite entropy forces them to be strictly smaller than one. The cost depends on $K$ through $\|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \operatorname{tr}(\mathbf{\Sigma}_\alpha) + \operatorname{tr}(\mathbf{\Sigma}_\beta) - 2\operatorname{tr}(K)$, while $\operatorname{KL}(\pi|\alpha\otimes\beta) = -\frac{1}{2}\log\det(\mathrm{Id} - SS^\top)$. By von Neumann's trace inequality, the minimizer aligns $S$ with the singular vectors of $\mathbf{\Sigma}_\alpha^{1/2}\mathbf{\Sigma}_\beta^{1/2}$, so $S = U\operatorname{diag}(s_i)V^\top$. The problem separates into scalar minimizations $\min_{0\leqslant s<1} -2\sigma_i s - \frac{\varepsilon}{2}\log(1 - s^2)$. The first-order condition is $2\sigma_i = \varepsilon s/(1 - s^2)$, whose positive solution is the displayed $s_i$. Substitution gives the value formula. Since $s_i \to 1$ and $\varepsilon \log(1 - s_i^2) \to 0$ as $\varepsilon \downarrow 0$, the spectral sum converges to $-2\sum_i \sigma_i$. The identity

$$\sum_i \sigma_i = \operatorname{tr}\Big((\mathbf{\Sigma}_\alpha^{1/2}\mathbf{\Sigma}_\beta\mathbf{\Sigma}_\alpha^{1/2})^{1/2}\Big)$$

then gives the Bures–Wasserstein covariance contribution. $\square$

**Corollary 8.22** (Gaussian Sinkhorn divergence and smoothed Bures term). *Let $\alpha = \mathcal{N}(\mathbf{m}_\alpha, \mathbf{\Sigma}_\alpha)$ and $\beta = \mathcal{N}(\mathbf{m}_\beta, \mathbf{\Sigma}_\beta)$ have positive-definite covariances. For $r > 0$, define*

$$\tau_\varepsilon(r) := \frac{\sqrt{\varepsilon^2 + 16r^2} - \varepsilon}{4r}, \qquad \psi_\varepsilon(r) := -2r\,\tau_\varepsilon(r) - \frac{\varepsilon}{2}\log\big(1 - \tau_\varepsilon(r)^2\big).$$

*If $\sigma_i(\Sigma, \Lambda)$ denotes the singular values of $\Sigma^{1/2}\Lambda^{1/2}$ and $\lambda_i(\Sigma)$ the eigenvalues of $\Sigma$, then the debiased Sinkhorn divergence (7.43) is $\bar{\mathcal{L}}^{\varepsilon}_{\|\cdot-\cdot\|^2}(\alpha, \beta) = \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2 + \mathcal{B}_\varepsilon(\mathbf{\Sigma}_\alpha, \mathbf{\Sigma}_\beta)^2$, where the Gaussian covariance contribution is the debiased smoothed Bures term*

$$\mathcal{B}_\varepsilon(\Sigma, \Lambda)^2 := \sum_i \psi_\varepsilon\big(\sigma_i(\Sigma, \Lambda)\big) - \frac{1}{2}\sum_i \psi_\varepsilon\big(\lambda_i(\Sigma)\big) - \frac{1}{2}\sum_i \psi_\varepsilon\big(\lambda_i(\Lambda)\big).$$

*Moreover $\mathcal{B}_\varepsilon(\Sigma, \Lambda)^2 \to \mathcal{B}(\Sigma, \Lambda)^2$ as $\varepsilon \downarrow 0$, where $\mathcal{B}$ is the Bures–Wasserstein metric of Proposition 2.35.*

*Proof.* Proposition 8.21 writes the raw entropic value as the squared mean displacement plus trace terms and a spectral sum. With the notation above, the spectral part is exactly $\sum_i \psi_\varepsilon(\sigma_i(\mathbf{\Sigma}_\alpha, \mathbf{\Sigma}_\beta))$. Applying the same formula to the self-costs $(\alpha, \alpha)$ and $(\beta, \beta)$ replaces these singular values by the eigenvalues of $\mathbf{\Sigma}_\alpha$ and $\mathbf{\Sigma}_\beta$. In the polarization formula (7.43), the trace terms cancel: $\operatorname{tr}\mathbf{\Sigma}_\alpha + \operatorname{tr}\mathbf{\Sigma}_\beta - \frac{1}{2}(2\operatorname{tr}\mathbf{\Sigma}_\alpha) - \frac{1}{2}(2\operatorname{tr}\mathbf{\Sigma}_\beta) = 0$, while the polarization of the squared mean terms leaves exactly $\|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2$. Since $\tau_\varepsilon(r) \to 1$ and $\varepsilon\log(1 - \tau_\varepsilon(r)^2) \to 0$, one has $\psi_\varepsilon(r) \to -2r$. The limit is therefore $\operatorname{tr}\Sigma + \operatorname{tr}\Lambda - 2\sum_i \sigma_i(\Sigma, \Lambda) = \mathcal{B}(\Sigma, \Lambda)^2$, which is the Bures formula (2.18). $\square$

**Proposition 8.23** (One-dimensional Gaussian Sinkhorn rate). *Consider $\alpha = \beta = \mathcal{N}(0, 1)$ on $\mathbb{R}$ with $c(x, y) = (x - y)^2$. If a dual potential has the form $g_q(y) = qy^2 + \mathrm{cst}$, then one soft transform has quadratic coefficient $T_\varepsilon(q) = 1 - \frac{1}{1-q+\varepsilon/2}$, $\quad q < 1 + \varepsilon/2$, and one full Sinkhorn cycle acts as $q \mapsto T_\varepsilon(T_\varepsilon(q))$. The fixed point $q_\star = T_\varepsilon(q_\star)$ is determined by $A_\star^2 - \frac{\varepsilon}{2}A_\star - 1 = 0$, $\quad A_\star := 1 - q_\star + \frac{\varepsilon}{2} = \frac{\varepsilon+\sqrt{\varepsilon^2+16}}{4}$.*

$$\rho_\varepsilon = A_\star^{-4} = \left(\frac{4}{\varepsilon + \sqrt{\varepsilon^2 + 16}}\right)^4.$$

*Proof.* Completing the square in

$$\int \exp\left(\frac{qy^2 - (x - y)^2}{\varepsilon}\right) \mathrm{d}\mathcal{N}(0, 1)(y)$$

gives the coefficient $T_\varepsilon(q)$. The fixed-point equation $q_\star = 1 - 1/A_\star$, together with $q_\star = 1 + \varepsilon/2 - A_\star$, gives $A_\star^2 - \frac{\varepsilon}{2}A_\star - 1 = 0$. The positive solution is the displayed $A_\star$. Since $T_\varepsilon'(q) = -\frac{1}{(1-q+\varepsilon/2)^2}$, the derivative of the full-cycle map at the fixed point is $T_\varepsilon'(q_\star)^2 = A_\star^{-4}$. $\square$

### 8.7 Continuous $\varepsilon$-Sinkhorn Flow

This compact subsection records the parabolic limit and its Gaussian closure.

**Parabolic Monge–Ampere limit.** The first block records the continuous limiting flow.

**Definition 8.24** (Continuous $\varepsilon$-Sinkhorn flow)**.** Let $\alpha = e^{-F}\mathrm{d}x$ and $\beta = e^{-G}\mathrm{d}x$ be smooth positive probability densities on the unit-volume flat torus $\mathbb{T}^d$, normalized with respect to Lebesgue measure. A gauge-fixed potential $u_t : \mathbb{T}^d \to \mathbb{R}$, with $\int_{\mathbb{T}^d} u_t \mathrm{d}x = 0$, follows the continuous $\varepsilon$-Sinkhorn flow if

$$\partial_t u_t(x) = \log\det\big(\mathrm{Id} + \nabla^2 u_t(x)\big) - G\big(x + \nabla u_t(x)\big) + F(x) - \bar{r}_t, \tag{CES}$$

where $\mathrm{Id} + \nabla^2 u_t \succ 0$ and $\bar{r}_t := \int_{\mathbb{T}^d} \Big[\log\det\big(\mathrm{Id} + \nabla^2 u_t(x)\big) - G\big(x + \nabla u_t(x)\big) + F(x)\Big]\mathrm{d}x$.

The subtraction of $\bar{r}_t$ fixes the additive constant of the potential, just as the usual Sinkhorn potentials are defined only up to a common additive gauge. With the quadratic cost convention, $u_t$ represents one member of the Sinkhorn potential pair $(u_t, v_t)$, while the companion target potential $v_t$ is recovered, at the discrete level, by the corresponding soft $c$-transform.

**Proposition 8.25** (Scaled log-Sinkhorn limit)**.** *Let $\varepsilon_k = 1/k$, and let $u_m^{(k)}$ be the mean-zero source log-potential after $m$ Sinkhorn updates at temperature $\varepsilon_k$. Define the time interpolation $u^{(k)}(t) := u^{(k)}_{\lfloor kt \rfloor}$. Assume that, on every compact time interval, $u^{(k)}(t)$ converges smoothly to $u_t$, that $\mathrm{Id} + \nabla^2 u_t \succ 0$, and that the Laplace expansion below holds uniformly along the sequence. Then $u_t$ solves the continuous $\varepsilon$-Sinkhorn flow* (CES).

*Proof.* Berman's normalized log-Sinkhorn update can be written in the form

$$u^{(k)}_{m+1} - u^{(k)}_m = k^{-1}\left[\log \rho_{ku^{(k)}_m} - \int_{\mathbb{T}^d} \log \rho_{ku^{(k)}_m} \mathrm{d}x\right],$$

where the subtraction enforces the gauge $\int u^{(k)}_{m+1}\mathrm{d}x = 0$. The relevant Laplace expansion is $\rho_{ku}(x) = \det(\mathrm{Id} + \nabla^2 u(x)) e^{F(x) - G(x+\nabla u(x))}\big(1 + O(k^{-1})\big)$, uniformly for the smooth potentials under consideration. Taking logarithms gives $\log \rho_{ku}(x) = \log\det(\mathrm{Id} + \nabla^2 u(x)) - G(x + \nabla u(x)) + F(x) + O(k^{-1})$. Dividing the increment by the time step $1/k$, passing to the smooth limit $u^{(k)}(t) \to u_t$, and keeping the mean-zero normalization gives exactly (CES). □

**Proposition 8.26** (Stationary continuous Sinkhorn potentials)**.** *Assume that a smooth stationary solution of* (CES) *satisfies $\mathrm{Id} + \nabla^2 u \succ 0$, and define $T(x) = x + \nabla u(x)$. Then $T_\# \alpha = \beta$. Conversely, any smooth map of the form $T = \mathrm{Id} + \nabla u$ with $\mathrm{Id} + \nabla^2 u \succ 0$ and $T_\#\alpha = \beta$ solves the stationary equation, up to the additive gauge of $u$.*

*Proof.* At stationarity, the bracket in (CES) is a constant $c$: $\log\det\big(\mathrm{Id} + \nabla^2 u(x)\big) - G(T(x)) + F(x) = c$. Exponentiating, and using $\nabla T = \mathrm{Id} + \nabla^2 u$, gives $e^{-G(T(x))}\det(\nabla T(x)) = e^{-F(x)}e^c$. Because $T$ is homotopic to the identity and $\mathrm{Id} + \nabla^2 u \succ 0$, it is an orientation-preserving local diffeomorphism of degree one; hence it is a diffeomorphism of the torus. Since $\alpha$ and $\beta$ have total mass one, integration and the change-of-variables formula give $e^c = 1$. The same identity then becomes precisely the Jacobian equation for $T_\#(e^{-F}\mathrm{d}x) = e^{-G}\mathrm{d}x$. The converse follows by reversing this argument. □

**Gaussian closure.** $T_t(x) := x + \nabla u_t(x) = q_t + B_t(x - \mathbf{m}_\alpha), \qquad B_t \in \mathbb{S}^d_{++}$.

$$F(x) = \frac{1}{2}\langle x - \mathbf{m}_\alpha, \mathbf{\Sigma}_\alpha^{-1}(x - \mathbf{m}_\alpha)\rangle + \mathrm{cst}, \qquad G(y) = \frac{1}{2}\langle y - \mathbf{m}_\beta, \mathbf{\Sigma}_\beta^{-1}(y - \mathbf{m}_\beta)\rangle + \mathrm{cst},$$

$\dot{B}_t = \mathbf{\Sigma}_\alpha^{-1} - B_t\mathbf{\Sigma}_\beta^{-1}B_t, \qquad \dot{q}_t = -B_t\mathbf{\Sigma}_\beta^{-1}(q_t - \mathbf{m}_\beta)$. $\mathbf{\Sigma}_t = B_t\mathbf{\Sigma}_\alpha B_t, \qquad \dot{\mathbf{\Sigma}}_t = \dot{B}_t\mathbf{\Sigma}_\alpha B_t + B_t\mathbf{\Sigma}_\alpha\dot{B}_t$. $B\mathbf{\Sigma}_\beta^{-1}B = \mathbf{\Sigma}_\alpha^{-1}, \qquad q = \mathbf{m}_\beta$,
$\partial_t u_t(x) = \log\big(1 + u_t''(x)\big) - G\big(x + u_t'(x)\big) + F(x) - \bar{r}_t$,

# 9 Statistical Optimal Transport

## 9.1 Law of Large Numbers and Central Limit Theorem

This compact subsection separates empirical consistency from fluctuation estimates.

**Empirical laws.** The first block states the empirical law-of-large-numbers principle.

$$\hat{\alpha}_n := \frac{1}{n}\sum_{i=1}^n \delta_{X_i}. \tag{9.1}$$

The ordinary law of large numbers says that empirical averages converge to expectations. In measure language this means that $\hat{\alpha}_n$ converges weakly toward $\alpha$, because testing $\hat{\alpha}_n$ against a bounded continuous function $\varphi$ gives the sample average $n^{-1}\sum_i \varphi(X_i)$.

**Proposition 9.1** (Empirical law of large numbers in $\mathcal{W}_p$)**.** *Let $(\mathcal{X}, d)$ be a Polish metric space, let $p \geqslant 1$, and let $\alpha \in \mathcal{P}_p(\mathcal{X})$. Let $(X_i)_{i \geqslant 1}$ be i.i.d. random variables with law $\alpha$, and define $\hat{\alpha}_n$ by* (9.1)*. Then*

$$\hat{\alpha}_n \rightharpoonup \alpha \quad \text{and} \quad \mathcal{W}_p(\hat{\alpha}_n, \alpha) \longrightarrow 0$$

*almost surely. Moreover, $\mathbb{E}\, \mathcal{W}_p(\hat{\alpha}_n, \alpha)^p \longrightarrow 0$.*

*Proof.* Fix a reference point $x_0 \in \mathcal{X}$, and write $r(x) = d(x, x_0)$. Since $\mathcal{X}$ is Polish, the weak topology on $\mathcal{P}(\mathcal{X})$ admits a countable convergence-determining class $(\varphi_k)_{k \geqslant 1} \subset C_b(\mathcal{X})$. For each fixed $k$, the strong law of large numbers gives $\int \varphi_k \mathrm{d}\hat{\alpha}_n = \frac{1}{n}\sum_{i=1}^n \varphi_k(X_i) \longrightarrow \mathbb{E}\varphi_k(X_1) = \int \varphi_k \mathrm{d}\alpha$ almost surely. Intersecting the corresponding probability-one events over the countable set of indices gives, on a single event of probability one, convergence against every $\varphi_k$. By the convergence-determining property, this is exactly $\hat{\alpha}_n \rightharpoonup \alpha$.

The moment condition $\alpha \in \mathcal{P}_p(\mathcal{X})$ means $\int r^p \mathrm{d}\alpha < +\infty$. Applying the strong law again to the integrable random variable $r(X_1)^p$ gives $\int r^p \mathrm{d}\hat{\alpha}_n = \frac{1}{n}\sum_{i=1}^n r(X_i)^p \longrightarrow \int r^p \mathrm{d}\alpha$ almost surely. The characterization of the $\mathcal{W}_p$ topology recalled after Proposition 3.37 states that weak convergence plus convergence of the $p$-th moments is equivalent to $\mathcal{W}_p$-convergence on $\mathcal{P}_p(\mathcal{X})$. Hence $\mathcal{W}_p(\hat{\alpha}_n, \alpha) \to 0$ almost surely.

Set $A_n = \int r^p \mathrm{d}\hat{\alpha}_n$ and $M = \int r^p \mathrm{d}\alpha$. The triangle inequality through the Dirac mass $\delta_{x_0}$, followed by $(a+b)^p \leqslant 2^{p-1}(a^p+b^p)$, gives the deterministic bound $\mathcal{W}_p(\hat{\alpha}_n, \alpha)^p \leqslant 2^{p-1}\big(A_n + M\big)$. The family $(A_n)_n$ is uniformly integrable. Indeed, by the de la Vallée–Poussin criterion, choose a convex superlinear function $\Psi$ such that $\mathbb{E}\Psi(r(X_1)^p) < +\infty$; Jensen's inequality gives $\mathbb{E}\Psi(A_n) \leqslant \frac{1}{n}\sum_{i=1}^n \mathbb{E}\Psi(r(X_i)^p) = \mathbb{E}\Psi(r(X_1)^p)$. Thus $(A_n + M)_n$, and hence $(\mathcal{W}_p(\hat{\alpha}_n, \alpha)^p)_n$, is uniformly integrable. Together with almost-sure convergence to zero, this implies $\mathbb{E}\,\mathcal{W}_p(\hat{\alpha}_n, \alpha)^p \to 0$. ☐

**Central-limit fluctuations.**

**Proposition 9.2** (Berry–Esseen bound in $\mathcal{W}_1$)**.** *Let $(X_i)_{i=1}^n$ be i.i.d. real random variables such that $\mathbb{E}X_i = 0$, $\mathbb{E}X_i^2 = 1$ and $\mathbb{E}|X_i|^3 < +\infty$. If $\alpha_n$ is the law of $n^{-1/2}\sum_i X_i$ and $\gamma$ is the standard Gaussian law, then $\mathcal{W}_1(\alpha_n, \gamma) \leqslant \frac{C\,\mathbb{E}|X_1|^3}{\sqrt{n}}$, where $C$ is a universal constant.*

*Proof.* By Kantorovich–Rubinstein duality,

$$\mathcal{W}_1(\alpha_n, \gamma) = \sup_{\mathrm{Lip}(h)\leqslant 1} |\mathbb{E}h(S_n) - \mathbb{E}h(G)|\,, \qquad S_n = n^{-1/2}\sum_i X_i,\ G \sim \gamma.$$

For each such $h$, solve Stein's equation $f_h'(x) - x f_h(x) = h(x) - \mathbb{E}h(G)$. The solution satisfies uniform first- and second-derivative bounds depending only on the Lipschitz constant of $h$. Hence $\mathbb{E}h(S_n) - \mathbb{E}h(G) = \mathbb{E}[f_h'(S_n) - S_n f_h(S_n)]$. Write $S_n^{(i)} = S_n - X_i/\sqrt{n}$. Since $S_n^{(i)}$ is independent of $X_i$, a Taylor expansion of $f_h'$ and $f_h$ around $S_n^{(i)}$ cancels the first- and second-order terms by $\mathbb{E}X_i = 0$ and $\mathbb{E}X_i^2 = 1$. The remainder is bounded by $C\mathbb{E}|X_i/\sqrt{n}|^3$. Summing over $i$ gives the displayed $n^{-1/2}$ rate. Sharper constants and higher-order transport-distance refinements are available, but are omitted from the compact handout. ☐

### 9.2 Sample Complexity

This compact subsection records the main statistical rates for empirical OT-type discrepancies.

**Unregularized OT.**

**Proposition 9.3** (Dyadic partition bound for $\mathcal{W}_1$)**.** *Let $\mathcal{Q}_j$ be nested dyadic partitions of $[0,1]^d$ into half-open cubes of side length $2^{-j}$, with the obvious convention on the outer boundary. For all $\alpha, \beta \in \mathcal{P}([0,1]^d)$ and every integer $J \geqslant 0$, $\mathcal{W}_1(\alpha, \beta) \leqslant \sqrt{d}\sum_{j=0}^{J-1} 2^{-j}\sum_{Q\in\mathcal{Q}_{j+1}} |\alpha(Q) - \beta(Q)| + \sqrt{d}\,2^{-J}$.*

*Proof.* The proof is constructive: one decomposes the two measures into pieces which can be coupled at different spatial scales. Write $\Omega = [0,1]^d$. We build positive measures $(\alpha_j, \beta_j)_{j=0}^J$ such that $\sum_{j=0}^J \alpha_j = \alpha$, $\sum_{j=0}^J \beta_j = \beta$, $\alpha_j(Q) = \beta_j(Q)$ for every $Q \in \mathcal{Q}_j$. Assuming such a decomposition for a moment, each pair $(\alpha_j, \beta_j)$ can be coupled inside the cells of $\mathcal{Q}_j$. Indeed, on every cell $Q \in \mathcal{Q}_j$ with positive mass, take the product coupling between the normalized restrictions of $\alpha_j$ and $\beta_j$ to $Q$, multiplied by the common mass $\alpha_j(Q) = \beta_j(Q)$. Summing over cells and over $j$ gives a coupling of $\alpha$ and $\beta$. Since the diameter of a cell in $\mathcal{Q}_j$ is at most $\sqrt{d}\,2^{-j}$, this coupling has cost bounded by $\mathcal{W}_1(\alpha, \beta) \leqslant \sqrt{d}\sum_{j=0}^J 2^{-j}\alpha_j(\Omega)$.
Start at the finest scale. For $Q \in \mathcal{Q}_J$, define $\alpha_J$ and $\beta_J$ on $Q$ by keeping the common part of the two masses:

$$\alpha_J(A \cap Q) = \frac{\alpha(Q) \wedge \beta(Q)}{\alpha(Q)}\,\alpha(A \cap Q), \qquad \beta_J(A \cap Q) = \frac{\alpha(Q) \wedge \beta(Q)}{\beta(Q)}\,\beta(A \cap Q),$$

with the convention that the corresponding term is zero when the denominator vanishes. For $j = J-1, \ldots, 1$, assume that the pieces at all finer levels $k > j$ have been constructed and define the residuals $\alpha_j' = \alpha - \sum_{k=j+1}^J \alpha_k$, $\beta_j' = \beta - \sum_{k=j+1}^J \beta_k$. On each $Q \in \mathcal{Q}_j$, set $\alpha_j$ and $\beta_j$ equal to the common part of $\alpha_j'$ and $\beta_j'$ inside $Q$, exactly as above. Finally set $\alpha_0 = \alpha - \sum_{k=1}^J \alpha_k$ and $\beta_0 = \beta - \sum_{k=1}^J \beta_k$. Each step removes only common residual mass, so all measures are positive, and by construction $\alpha_j(Q) = \beta_j(Q)$ on all cells $Q \in \mathcal{Q}_j$.
For notational consistency, set $\alpha_0' = \alpha_0$, $\beta_0' = \beta_0$, $\alpha_J' = \alpha$, and $\beta_J' = \beta$. Let $0 \leqslant j < J$ and $R \in \mathcal{Q}_j$. Since all pieces constructed at levels finer than $j+1$ have equal mass on every child $Q \in \mathcal{Q}_{j+1}$, the residual imbalance on $Q$ just before the matching at level $j+1$ is the original imbalance: $\alpha_{j+1}'(Q) - \beta_{j+1}'(Q) = \alpha(Q) - \beta(Q)$. After removing the common part at level $j+1$, the remaining source residual inside $Q$ has mass $\alpha_j'(Q) = \alpha_{j+1}'(Q) - \alpha_{j+1}(Q) = \big(\alpha_{j+1}'(Q) - \beta_{j+1}'(Q)\big)_+$. Because $\alpha_j$ is itself a common submeasure of the residual $\alpha_j'$, the mass matched at level $j$ inside $R$ satisfies

$$\alpha_j(R) = \beta_j(R) \leqslant \alpha_j'(R) = \sum_{Q\subset R}\big(\alpha_{j+1}'(Q) - \beta_{j+1}'(Q)\big)_+ \leqslant \sum_{Q\subset R}|\alpha(Q) - \beta(Q)|\,,$$

where the sums are over $Q \in \mathcal{Q}_{j+1}$. Summing over $R \in \mathcal{Q}_j$ gives $\alpha_j(\Omega) \leqslant \sum_{Q\in\mathcal{Q}_{j+1}}|\alpha(Q) - \beta(Q)|$. The last level satisfies $\alpha_J(\Omega) \leqslant 1$. Substituting these bounds into the cost estimate proves the claim. ☐

**Proposition 9.4** (Empirical OT has intrinsic-dimension value rates)**.** *Let $\alpha$ and $\beta$ be probability measures supported on $[0,1]^d$, and let $\hat{\alpha}_n$ and $\hat{\beta}_m$ be independent empirical measures. Then $\mathbb{E}\left|\mathcal{W}_1(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{W}_1(\alpha, \beta)\right| \lesssim_d r_d(n) + r_d(m)$, where, for $N \geqslant 2$,*

$$r_d(N) = \begin{cases} N^{-1/2}, & d = 1, \\ (\log N)N^{-1/2}, & d = 2, \\ N^{-1/d}, & d \geqslant 3. \end{cases}$$

*For $N = 1$, the value of $r_d(1)$ is absorbed in the implicit constant. In particular, for $d \geqslant 3$, the worst-case one-sample empirical error for exact $\mathcal{W}_1$ has the characteristic high-dimensional scale $N^{-1/d}$. More generally, for $\mathcal{W}_p$ on regular $d'$-dimensional supports, the corresponding high-dimensional rate is $N^{-1/d'}$ when $d' > 2p$, under the usual volume-growth and moment assumptions. Thus the relevant dimension is the intrinsic dimension of the support rather than necessarily the ambient dimension.*

*Proof.* By the triangle inequality, $\left|\mathcal{W}_1(\hat\alpha_n,\hat\beta_m)-\mathcal{W}_1(\alpha,\beta)\right| \leqslant \mathcal{W}_1(\hat\alpha_n,\alpha)+\mathcal{W}_1(\hat\beta_m,\beta)$. It is therefore enough to bound $\mathbb{E}\,\mathcal{W}_1(\hat\alpha_N,\alpha)$. Apply Proposition 9.3 to $(\hat\alpha_N,\alpha)$. If $Q\in\mathcal{Q}_{j+1}$, then $N\hat\alpha_N(Q)$ is a binomial random variable with parameter $\alpha(Q)$, and hence $\mathbb{E}\,|\hat\alpha_N(Q)-\alpha(Q)|\leqslant\sqrt{\mathbb{E}\,|\hat\alpha_N(Q)-\alpha(Q)|^2}\leqslant\sqrt{\frac{\alpha(Q)}{N}}$. Summing over the $2^{(j+1)d}$ cells and using the Cauchy–Schwarz inequality gives

$$\sum_{Q\in\mathcal{Q}_{j+1}}\mathbb{E}\,|\hat\alpha_N(Q)-\alpha(Q)|\leqslant\frac{1}{\sqrt N}\sum_{Q\in\mathcal{Q}_{j+1}}\sqrt{\alpha(Q)}\leqslant\frac{2^{(j+1)d/2}}{\sqrt N}.$$

Therefore, for every $J\geqslant 0$, $\mathbb{E}\,\mathcal{W}_1(\hat\alpha_N,\alpha)\lesssim_d 2^{-J}+\frac{1}{\sqrt N}\sum_{j=0}^{J-1}2^{j(d/2-1)}$. If $d=1$, the sum is uniformly bounded and we choose $J$ such that $2^{-J}\lesssim N^{-1/2}$. If $d=2$, the sum is $J$, and choosing $2^J\simeq N^{1/2}$ gives $(\log N)N^{-1/2}$. If $d\geqslant 3$, the sum is dominated by its last term; choosing $2^J\simeq N^{1/d}$ gives $2^{-J}+N^{-1/2}2^{J(d/2-1)}\simeq N^{-1/d}$. This proves the displayed $\mathcal{W}_1$ rates. The extension to a regular $d'$-dimensional support uses the same proof with nested partitions whose cells have diameter $2^{-j}$ and cardinality $O(2^{jd'})$; the general $\mathcal{W}_p$ statement follows from the standard multiscale estimates of. The elementary dyadic proof above is not sharp in the critical case $d=2$: the optimal rate is $(\log N/N)^{1/2}$, obtained by finer matching/Fourier arguments. □

### Lower bounds and minimax optimality.

**Proposition 9.5** (Minimax lower bound for bounded densities)**.** *Let $d\geqslant 3$. Let $\mathcal{A}$ be the class of probability measures on $[0,1]^d$ admitting a density $f$ with respect to Lebesgue measure such that $\frac12\leqslant f(x)\leqslant\frac32$ for a.e. $x\in[0,1]^d$. There exists $c_d>0$ such that, for every $N\geqslant 1$, $\inf_{\tilde\alpha_N}\sup_{\alpha\in\mathcal{A}}\mathbb{E}_\alpha\,\mathcal{W}_1(\tilde\alpha_N,\alpha)\geqslant c_d\,N^{-1/d}$. Here the infimum is over all estimators based on $N$ i.i.d. samples from $\alpha$. In particular, the high-dimensional rate of Proposition 9.4 is minimax sharp for $\mathcal{W}_1$, even when one allows arbitrary estimators rather than the empirical measure.*

*Proof.* We use a standard hypercube construction. Choose a smooth function $\psi$ compactly supported in $(0,1)^d$, with $\int\psi=0$, $\|\psi\|_\infty\leqslant 1$, and whose positive and negative parts are separated in the first coordinate. Then there exists a 1-Lipschitz function $\varphi$, also compactly supported in $(0,1)^d$, such that $\int\varphi\psi\mathrm{d}x>0$. We extend both functions by zero outside $(0,1)^d$.
Let $m$ be an integer, $h=1/m$, and partition $[0,1]^d$ into $M=m^d$ cubes $Q_k=z_k+h[0,1)^d$. Define the rescaled bumps

$$\psi_k(x)=\psi\left(\frac{x-z_k}{h}\right)\mathbf{1}_{Q_k}(x).$$

For $\theta=(\theta_1,\dots,\theta_M)\in\{-1,1\}^M$ and a fixed small $\eta>0$, set $f_\theta(x)=1+\eta\sum_{k=1}^M\theta_k\psi_k(x),\qquad \alpha_\theta=f_\theta\mathrm{d}x$. For $\eta$ small enough, all these densities belong to $\mathcal{A}$, and they integrate to one because every $\psi_k$ has zero mean.
If $\theta$ and $\theta'$ differ on a set $S$ of coordinates, the dual formula for $\mathcal{W}_1$ gives an additive lower bound. Indeed, rescale $\varphi$ as

$$\varphi_k(x)=h\,\varphi\left(\frac{x-z_k}{h}\right).$$

Each $\varphi_k$ is supported inside $Q_k$, and the supports are disjoint up to boundaries; hence any signed sum $\sum_k s_k\varphi_k$, $s_k\in\{-1,0,1\}$, is 1-Lipschitz. Testing $\alpha_\theta-\alpha_{\theta'}$ against the signed sum $s_k=\operatorname{sign}(\theta_k-\theta_k')$ on $S$ yields $\mathcal{W}_1(\alpha_\theta,\alpha_{\theta'})\geqslant c\,\eta\,h^{d+1}\operatorname{Ham}(\theta,\theta')$, where Ham denotes Hamming distance and $c>0$ depends only on the chosen bump.
Neighboring vertices of the hypercube are statistically close. If $\theta^{(k)}$ is obtained from $\theta$ by flipping only the $k$-th sign, then the one-sample Kullback–Leibler divergence is bounded by $\mathrm{KL}(\alpha_\theta|\alpha_{\theta^{(k)}})\leqslant C\eta^2h^d$, because the two densities differ only on $Q_k$, the perturbation has size $O(\eta)$, and the densities are uniformly bounded away from zero. For $N$ independent samples the divergence is multiplied by $N$. Choose $m$ so that $m^d\simeq N$. Then $Nh^d$ is bounded, and, by taking $\eta$ sufficiently small, Pinsker's inequality gives a uniform bound $\mathrm{TV}\left(\alpha_\theta^{\otimes N},\alpha_{\theta^{(k)}}^{\otimes N}\right)\leqslant q<1$. Assouad's lemma therefore implies

$$\inf_{\tilde\alpha_N}\sup_{\theta\in\{-1,1\}^M}\mathbb{E}_\theta\,\mathcal{W}_1(\tilde\alpha_N,\alpha_\theta)\geqslant c'\eta\,Mh^{d+1}(1-q)=c'\eta(1-q)h\simeq N^{-1/d}.$$

Decreasing the constant handles the finitely many small values of $N$, and the claim follows. □

### Leveraging smoothness.

**Proposition 9.6** (Smoothed plug-in rates)**.** *Let $d\geqslant 3$, $s>0$, and let $\alpha=f\mathrm{d}x$, $\beta=g\mathrm{d}x$ be probability measures on $[0,1]^d$. Assume that $f$ and $g$ are bounded above and below by positive constants and belong to a Hölder–Zygmund, equivalently Besov $B^s_{\infty,\infty}$ ball of radius $L$. Let $\hat\alpha_n$ and $\hat\beta_m$ be independent empirical measures. There exist normalized nonnegative density estimators $\tilde\alpha_n=\tilde f_n\mathrm{d}x,\qquad \tilde\beta_m=\tilde g_m\mathrm{d}x$, obtained for instance by a projected wavelet estimator at resolution $h_N\simeq N^{-1/(d+2s)}$, such that*

$$\mathbb{E}\left|\mathcal{W}_1(\tilde\alpha_n,\tilde\beta_m)-\mathcal{W}_1(\alpha,\beta)\right|\leqslant C\left(n^{-\frac{s+1}{d+2s}}+m^{-\frac{s+1}{d+2s}}\right).$$

*The constant depends on $d,s,L$, on the chosen wavelet system and on the lower and upper density bounds. Thus any positive smoothness improves on the empirical $N^{-1/d}$ exponent when $d>2$; if $s$ is of order $d$, the exponent is of order $1/3$, and as $s\to\infty$ it approaches the parametric exponent $1/2$.*

*Proof sketch.* It is enough, by the triangle inequality, to bound $\mathbb{E}\,\mathcal{W}_1(\tilde\alpha_N,\alpha)$. For densities bounded below and above, $W_1$ is controlled by a negative smoothness norm of the density difference; in a wavelet notation one may write, schematically, $\mathcal{W}_1(\tilde\alpha_N,\alpha)\lesssim\|\tilde f_N-f\|_{B^{-1}_{1,1}}$. Truncating the empirical wavelet expansion at scale $J$, or equivalently smoothing at bandwidth $h=2^{-J}$, gives two terms. The deterministic approximation error of an $s$-smooth density is $O(2^{-J(s+1)})$ in the negative $B^{-1}$ norm, while the stochastic contribution of the retained coefficients is $N^{-1/2}\sum_{j\leqslant J}2^{j(d/2-1)}\simeq N^{-1/2}2^{J(d/2-1)}\qquad(d\geqslant 3)$. Balancing these two quantities gives $2^J\simeq N^{1/(d+2s)}$ and hence the rate $N^{-(s+1)/(d+2s)}$. Projecting the linear wavelet estimate onto the cone of probability densities in the same negative Besov norm preserves the order of this bound. Applying the same estimate to both marginals proves the displayed two-sample bound. The full minimax theorem also covers $W_p$, Besov classes, lower bounds and endpoint logarithms in dimensions 1 and 2. □

**MMD.**

**Proposition 9.7** (MMD has a parametric value rate)**.** *Let $k$ be a bounded positive definite kernel with RKHS $\mathcal{H}_k$, and define $\mathrm{MMD}_k(\alpha,\beta)=\|\int k(x,\cdot)\mathrm{d}(\alpha-\beta)(x)\|_{\mathcal{H}_k}$. If $\hat\alpha_n$ and $\hat\beta_m$ are independent empirical measures, then*

$$\mathbb{E}\left|\mathrm{MMD}_k(\hat\alpha_n,\hat\beta_m)-\mathrm{MMD}_k(\alpha,\beta)\right|\leqslant\kappa\left(\frac{1}{\sqrt{n}}+\frac{1}{\sqrt{m}}\right)$$

*when $k(x,x)\leqslant\kappa^2$.*

*Proof.* Let $\Phi(x)=k(x,\cdot)$ be the feature map and $m_\alpha=\mathbb{E}\Phi(X)$. The reverse triangle inequality for the RKHS norm gives $\left|\mathrm{MMD}_k(\hat\alpha_n,\hat\beta_m)-\mathrm{MMD}_k(\alpha,\beta)\right|\leqslant\mathrm{MMD}_k(\hat\alpha_n,\alpha)+\mathrm{MMD}_k(\hat\beta_m,\beta)$. Independence cancels the cross terms after taking the squared norm and expectation, giving $\mathbb{E}\,\mathrm{MMD}_k(\hat\alpha_n,\alpha)^2=\frac{1}{n}\mathbb{E}\|\Phi(X)-m_\alpha\|^2_{\mathcal{H}_k}=\frac{1}{n}\Big(\mathbb{E}k(X,X)-\mathbb{E}k(X,X')\Big)$. The same estimate applies to $\hat\beta_m$, and Jensen's inequality together with $k(x,x)\leqslant\kappa^2$ gives the displayed bound. □

**Entropic OT.**

**Proposition 9.8** (Sinkhorn divergences interpolate the rates)**.** *Let $c(x,y)=\|x-y\|^2/2$, and assume that $\alpha$ and $\beta$ are $\sigma^2$-subgaussian probability measures on $\mathbb{R}^d$, in the sense that $\int e^{\|x\|^2/(2d\sigma^2)}\mathrm{d}\alpha(x)\leqslant 2,\qquad \int e^{\|y\|^2/(2d\sigma^2)}\mathrm{d}\beta(y)\leqslant 2$. Set*

$$q_d:=\left\lceil\frac{5d}{2}\right\rceil+6,\qquad b_d:=\left\lceil\frac{5d}{4}\right\rceil+3,\qquad \Lambda_{d,\sigma}(\varepsilon):=\varepsilon\left(1+\frac{\sigma^{q_d}}{\varepsilon^{b_d}}\right).$$

*Let $\hat\alpha_n$ and $\hat\beta_m$ be independent empirical measures. Then, for every $\varepsilon>0$,*

$$\mathbb{E}\left|\bar{\mathcal{L}}^\varepsilon_c(\hat\alpha_n,\hat\beta_m)-\bar{\mathcal{L}}^\varepsilon_c(\alpha,\beta)\right|\leqslant C_d\,\Lambda_{d,\sigma}(\varepsilon)\left(\frac{1}{\sqrt{n}}+\frac{1}{\sqrt{m}}\right).$$

*In particular, for fixed $\varepsilon>0$ the empirical fluctuation is parametric, while for small $\varepsilon$ the prefactor deteriorates only polynomially in $1/\varepsilon$, not as $e^{C/\varepsilon}$. Compactly supported measures are covered by this statement after a common translation of the two supports, with $\sigma$ controlled by the diameter of their union.*

*Proof.* The original proof of Genevay–Chizat–Bach–Cuturi–Peyré also reduces the problem to an empirical process indexed by entropic dual potentials, but it bounds these potentials on a compact domain in a way that produces an exponential factor $e^{D^2/\varepsilon}$. The refinement of Mena–Weed keeps the same reduction and replaces this step by polynomial Hölder estimates for the quadratic-cost Sinkhorn potentials.

For probability measures $\rho,\eta$, write the dual objective, following (7.18), as

$$\mathcal{A}^{\rho,\eta}_\varepsilon(u,v):=\int u\mathrm{d}\rho+\int v\mathrm{d}\eta-\varepsilon\iint\left(\exp\left(\frac{u(x)+v(y)-c(x,y)}{\varepsilon}\right)-1\right)\mathrm{d}\rho(x)\mathrm{d}\eta(y).$$

Following the extension argument of Mena–Weed, choose optimal potentials in the normalized class for which the Sinkhorn equations hold pointwise on $\mathbb{R}^d$. Thus, at an optimal pair $(u,v)$,

$$\int\exp\left(\frac{u(x)+v(y)-c(x,y)}{\varepsilon}\right)\mathrm{d}\eta(y)=1\qquad\text{for all } x.$$

Equivalently, the potentials are fixed points of the soft $c$-transform, but only the dual normalization is needed below. Let $(u,v)$ and $(u_n,v_n)$ be optimal dual potentials for $(\rho,\eta)$ and $(\rho_n,\eta)$. By optimality,

$$\mathcal{A}^{\rho,\eta}_\varepsilon(u_n,v_n)\leqslant\mathcal{A}^{\rho,\eta}_\varepsilon(u,v),\qquad \mathcal{A}^{\rho_n,\eta}_\varepsilon(u,v)\leqslant\mathcal{A}^{\rho_n,\eta}_\varepsilon(u_n,v_n).$$

Hence the difference of the two optimal values is bounded by evaluating the two objectives at either pair of potentials. For an optimal pair $(u,v)$ associated with $(\rho,\eta)$, the preceding Sinkhorn equation cancels the exponential term in the change of the first marginal: $\mathcal{A}^{\rho,\eta}_\varepsilon(u,v)-\mathcal{A}^{\rho_n,\eta}_\varepsilon(u,v)=\int u\mathrm{d}(\rho-\rho_n)$. The same identity holds for $(u_n,v_n)$, now using the fact that this pair satisfies the same normalization with respect to $\eta$. $|\mathcal{L}^\varepsilon_c(\rho_n,\eta)-\mathcal{L}^\varepsilon_c(\rho,\eta)|\leqslant 2\sup_{w\in\mathcal{F}_{\varepsilon,\sigma}}\left|\int w\mathrm{d}(\rho_n-\rho)\right|$, where $\mathcal{F}_{\varepsilon,\sigma}$ denotes the class of first-marginal entropic potentials arising from pairs of $\sigma^2$-subgaussian marginals.

For $\varepsilon=1$, Mena–Weed prove that the normalized quadratic-cost potentials belong to a polynomial-growth Hölder class. More precisely, after subtracting the quadratic part $x\mapsto\|x\|^2/2$, these potentials have derivatives up to order $s_0=\lceil d/2\rceil+1$ whose local Hölder norms are bounded by polynomial functions of the subgaussian proxy. Denote by $\mathcal{F}^s_{\sigma,L}$ such a class, where $L$ is a random envelope controlled by the empirical subgaussian moment. Its empirical $L_2$-covering numbers satisfy a bound of the form $\log N(\delta,\mathcal{F}^s_{\sigma,L},L_2(\rho_n))\leqslant C_d\,L^{d/(2s)}\delta^{-d/s}(1+\sigma^{2d})$. This holds for $s>s_0-1$, with $\mathbb{E}L\leqslant C$ after the normalization of the subgaussian moment. Since $d/(2s_0)<1$, Dudley's entropy integral is finite, and the standard Rademacher symmetrization and chaining bound yields $\mathbb{E}\sup_{w\in\mathcal{F}_{1,\sigma}}\left|\int w\mathrm{d}(\hat\rho_n-\rho)\right|\leqslant C_d(1+\sigma^{q_d})\,n^{-1/2}$. This is exactly the step which avoids the exponential $e^{D^2/\varepsilon}$ factor: one controls the regularity class of the dual potentials polynomially instead of bounding their sup-norm through the worst value of the Gibbs kernel.

The general $\varepsilon$ case follows by scaling. If $T_\varepsilon(x)=x/\sqrt{\varepsilon}$ and $\rho^\varepsilon=(T_\varepsilon)_\sharp\rho$, then $\mathcal{L}^\varepsilon_c(\rho,\eta)=\varepsilon\,\mathcal{L}^1_c(\rho^\varepsilon,\eta^\varepsilon)$. The push-forward of a $\sigma^2$-subgaussian law by $T_\varepsilon$ is $(\sigma^2/\varepsilon)$-subgaussian, so the standard-deviation proxy $\sigma$ is replaced by $\sigma/\sqrt{\varepsilon}$.

$$\varepsilon\,C_d\left(1+\left(\frac{\sigma}{\sqrt{\varepsilon}}\right)^{q_d}\right)n^{-1/2}\leqslant C_d\,\varepsilon\left(1+\frac{\sigma^{q_d}}{\varepsilon^{b_d}}\right)n^{-1/2},$$

where $b_d = \lceil q_d/2 \rceil = \lceil 5d/4 \rceil + 3$ is the rounded exponent used in. Thus $\mathbb{E}\,|\mathcal{L}_c^\varepsilon(\hat\rho_n, \eta) - \mathcal{L}_c^\varepsilon(\rho, \eta)| \leqslant C_d\,\Lambda_{d,\sigma}(\varepsilon)\, n^{-1/2}$. Changing both marginals is handled by the triangle inequality,

$$\left|\mathcal{L}_c^\varepsilon(\hat\alpha_n, \hat\beta_m) - \mathcal{L}_c^\varepsilon(\alpha, \beta)\right| \leqslant \left|\mathcal{L}_c^\varepsilon(\hat\alpha_n, \hat\beta_m) - \mathcal{L}_c^\varepsilon(\alpha, \hat\beta_m)\right| + \left|\mathcal{L}_c^\varepsilon(\alpha, \hat\beta_m) - \mathcal{L}_c^\varepsilon(\alpha, \beta)\right|,$$

Empirical measures drawn from a subgaussian law have such a random proxy, with moments controlled by the population proxy. Integrating over this proxy gives the Mena–Weed two-sample extension. The argument does not require the two empirical marginals to have the same sample size, so the two contributions scale as $n^{-1/2}$ and $m^{-1/2}$.
Finally, use the definition $\bar{\mathcal{L}}_c^\varepsilon(\alpha, \nu) = \mathcal{L}_c^\varepsilon(\alpha, \nu) - \frac{1}{2}\mathcal{L}_c^\varepsilon(\alpha, \alpha) - \frac{1}{2}\mathcal{L}_c^\varepsilon(\nu, \nu)$. The cross term is controlled by the two-marginal estimate. For the self term, use the telescoping inequality

$$|\mathcal{L}_c^\varepsilon(\hat\alpha_n, \hat\alpha_n) - \mathcal{L}_c^\varepsilon(\alpha, \alpha)| \leqslant |\mathcal{L}_c^\varepsilon(\hat\alpha_n, \hat\alpha_n) - \mathcal{L}_c^\varepsilon(\alpha, \hat\alpha_n)| + |\mathcal{L}_c^\varepsilon(\alpha, \hat\alpha_n) - \mathcal{L}_c^\varepsilon(\alpha, \alpha)|\,.$$

Each term is a one-marginal perturbation in the same pathwise sense: the other marginal may itself be empirical, but it is part of the common random subgaussian class. Thus the self term $\mathcal{L}_c^\varepsilon(\hat\alpha_n, \hat\alpha_n) - \mathcal{L}_c^\varepsilon(\alpha, \alpha)$ is bounded at the same order as the cross term involving $\alpha$, and similarly for $\beta$. Summing these three contributions gives the displayed estimate, up to changing $C_d$. ☐

**Sample complexity of estimating OT maps.**

$$u_{n,m}^\varepsilon(x) := (\mathrm{g}_{n,m}^\varepsilon)^{\bar c, \varepsilon}(x) = -\varepsilon \log \sum_j b_j \exp\left(\frac{\mathrm{g}_{n,m,j}^\varepsilon - c_2(x, Y_j)}{\varepsilon}\right), \qquad T_{n,m}^\varepsilon(x) := x - \nabla u_{n,m}^\varepsilon(x).$$

$$\varphi_{n,m}^\varepsilon(x) := \frac{1}{2}\|x\|^2 - u_{n,m}^\varepsilon(x) = \varepsilon \log \sum_j b_j \exp\left(\frac{\langle x, Y_j\rangle + \mathrm{g}_{n,m,j}^\varepsilon - \frac{1}{2}\|Y_j\|^2}{\varepsilon}\right), \qquad T_{n,m}^\varepsilon = \nabla \varphi_{n,m}^\varepsilon.$$

$$T_{n,m}^\varepsilon(x) = \sum_j \omega_j^\varepsilon(x) Y_j, \qquad \omega_j^\varepsilon(x) = \frac{b_j \exp\big((\mathrm{g}_{n,m,j}^\varepsilon - c_2(x, Y_j))/\varepsilon\big)}{\sum_k b_k \exp\Big((\mathrm{g}_{n,m,k}^\varepsilon - c_2(x, Y_k))/\varepsilon\Big)}. \tag{9.2}$$

$\omega_j^\varepsilon(X_i) = \frac{P_{ij}^\varepsilon}{a_i}, \qquad T_{n,m}^\varepsilon(X_i) = \frac{1}{a_i}\sum_j P_{ij}^\varepsilon Y_j$. Thus the extrapolated soft-transform map coincides on the empirical support with the barycentric projection (11.16) of the optimal entropic plan.

**Proposition 9.9** (Consistency of empirical barycentric maps)**.** *Let $\Omega \subset \mathbb{R}^d$ be compact, let $\alpha, \beta \in \mathcal{P}_2(\Omega)$, and assume that $\alpha$ is absolutely continuous. Let $T$ be the unique Brenier map from $\alpha$ to $\beta$ for the cost $c_2(x, y) = \|x - y\|^2/2$, represented by a Borel subgradient selection $T(x) \in \partial\varphi(x)$ of a Brenier potential $\varphi$. Let $\hat\alpha_n = \frac{1}{n}\sum_{i=1}^n \delta_{X_i}, \qquad \hat\beta_m = \frac{1}{m}\sum_{j=1}^m \delta_{Y_j}$ satisfy $\hat\alpha_n \to \alpha$, $\hat\beta_m \to \beta$ in $\mathcal{W}_2$. Let $\varepsilon_{n,m} \geqslant 0$ satisfy $\varepsilon_{n,m} \to 0$ and $\varepsilon_{n,m}\log(nm) \to 0$. For each $n, m$, let $P_{n,m}^\varepsilon$ be an optimal coupling between $\hat\alpha_n$ and $\hat\beta_m$: unregularized if $\varepsilon = 0$, entropic if $\varepsilon > 0$. Its barycentric projection, defined on the support of $\hat\alpha_n$, is $\bar T_{n,m}^\varepsilon(X_i) := n\sum_j P_{ij}^\varepsilon Y_j$ satisfies $\int \|\bar T_{n,m}^\varepsilon(x) - T(x)\|^2 \mathrm{d}\hat\alpha_n(x) = \frac{1}{n}\sum_i \|\bar T_{n,m}^\varepsilon(X_i) - T(X_i)\|^2 \longrightarrow 0$. If $\beta$ is also absolutely continuous, the analogous backward barycentric projections converge in $L^2(\hat\beta_m)$ to the Brenier map $S : \beta \to \alpha$.*

*Proof.* The assumptions imply $\mathcal{L}_{c_2}(\hat\alpha_n, \hat\beta_m) \to \mathcal{L}_{c_2}(\alpha, \beta)$. If $\varepsilon = 0$, $P_{n,m}^\varepsilon$ is exactly optimal. If $\varepsilon > 0$, compare its entropic objective with an unregularized optimal empirical plan $Q_{n,m}$. Since $Q_{n,m}$ is supported on an $n \times m$ grid and $\hat\alpha_n \otimes \hat\beta_m$ assigns mass $1/(nm)$ to each pair, $\mathrm{KL}(Q_{n,m}|\hat\alpha_n \otimes \hat\beta_m) \leqslant \log(nm)$, so $\int c_2\, \mathrm{d}P_{n,m}^\varepsilon \leqslant \mathcal{L}_{c_2}(\hat\alpha_n, \hat\beta_m) + \varepsilon\log(nm)$. Hence the transport costs of $P_{n,m}^\varepsilon$ converge to the population optimum. By compactness, every weak limit point of the plans $P_{n,m}^\varepsilon$ has marginals $\alpha, \beta$ and optimal quadratic cost. Brenier uniqueness therefore forces the whole sequence to converge weakly to $(\mathrm{Id}, T)_\sharp\alpha$.
Disintegrate $P_{n,m}^\varepsilon$ with respect to its first marginal and denote its barycentric projection by $\bar T_{n,m}^\varepsilon$. Jensen's inequality gives $\int \|\bar T_{n,m}^\varepsilon(x) - T(x)\|^2 \mathrm{d}\hat\alpha_n(x) \leqslant \iint \|y - T(x)\|^2 \mathrm{d}P_{n,m}^\varepsilon(x, y)$. At every differentiability point of $\varphi$, the subdifferential is the singleton $\{\nabla\varphi(x)\}$. Since the target is compact, all selected subgradients remain bounded, and the closed-graph property of the subdifferential implies that any nearby selected subgradient converges to $\nabla\varphi(x)$. Convex functions are differentiable Lebesgue-almost everywhere. Since $\alpha$ is absolutely continuous, the bounded function $(x, y) \mapsto \|y - T(x)\|^2$ is continuous $(\mathrm{Id}, T)_\sharp\alpha$-almost everywhere. The portmanteau theorem for bounded functions whose discontinuity set has zero limiting mass gives $\iint \|y - T(x)\|^2 \mathrm{d}P_{n,m}^\varepsilon(x, y) \longrightarrow \int \|T(x) - T(x)\|^2 \mathrm{d}\alpha(x) = 0$. The proof for the backward map is identical when the optimal plan is also induced by the Brenier map from $\beta$ to $\alpha$. ☐

$\mathbb{E}\|\widehat T_n - T_0\|_{L^2(\alpha)}^2 \lesssim n^{-\frac{2s}{2(s-1)+d}}$ up to logarithmic factors,

$$\mathbb{E}\|T_{\varepsilon,(n,n)} - T_0\|_{L^2(\alpha)}^2 \lesssim \varepsilon^{-d/2}\log(n) n^{-1/2} + \varepsilon^{(\bar s+1)/2} + \varepsilon^2 \mathcal{I}_0(\alpha, \beta),$$

where $\mathcal{I}_0(\alpha, \beta)$ is the Fisher-information quantity appearing in Proposition 7.13. The first term is the statistical fluctuation of the entropic map, whereas the last two terms are regularization-bias terms that vanish only as $\varepsilon \to 0$.
$\mathbb{E}\|T_{\varepsilon,(n,n)} - T_0\|_{L^2(\alpha)}^2 \lesssim (1 + \mathcal{I}_0(\alpha, \beta)) n^{-\frac{\bar s+1}{2(d+\bar s+1)}} \log n$.

**Sliced Wasserstein.** For a direction $\theta \in \mathbb{S}^{d-1}$, let $P_\theta(x) = \langle\theta, x\rangle$ be the one-dimensional projection and let $\sigma$ denote the normalized surface measure on the sphere. For $\alpha, \beta \in \mathcal{P}_p(\mathbb{R}^d)$, the sliced Wasserstein distance is
$\mathrm{SW}_p(\alpha, \beta)^p := \int_{\mathbb{S}^{d-1}} \mathcal{W}_p\left((P_\theta)_\sharp\alpha, (P_\theta)_\sharp\beta\right)^p \mathrm{d}\sigma(\theta)$.

**Theorem 9.10** (Dimension-free empirical rate for sliced Wasserstein)**.** *Let $\alpha, \beta \in \mathcal{P}_p(\mathbb{R}^d)$, and let $\hat\alpha_n, \hat\beta_m$ be independent empirical measures. Assume that all one-dimensional projections satisfy the uniform empirical bounds*

$$\sup_{\theta \in \mathbb{S}^{d-1}} \mathbb{E}\,\mathcal{W}_p\left((P_\theta)_\sharp\hat\alpha_n, (P_\theta)_\sharp\alpha\right)^p \leqslant \frac{A_\alpha}{n^{p/2}}, \qquad \sup_{\theta \in \mathbb{S}^{d-1}} \mathbb{E}\,\mathcal{W}_p\left((P_\theta)_\sharp\hat\beta_m, (P_\theta)_\sharp\beta\right)^p \leqslant \frac{A_\beta}{m^{p/2}}.$$

*Then* $\mathbb{E}\left|\mathrm{SW}_p(\hat\alpha_n,\hat\beta_m)-\mathrm{SW}_p(\alpha,\beta)\right| \leqslant \frac{A_\alpha^{1/p}}{\sqrt{n}}+\frac{A_\beta^{1/p}}{\sqrt{m}}$. *In particular, for* $p=1$ *and supports contained in a Euclidean ball of radius* $R$, *the assumption holds with a constant depending only on* $R$. *For* $p=2$, *the same* $n^{-1/2}$ *rate for* $\mathrm{SW}_2$ *holds whenever the projected laws satisfy a uniform one-dimensional quantile-process bound, for instance when their quantile functions* $Q_{\alpha,\theta}$, $Q_{\beta,\theta}$ *and densities* $h_{\alpha,\theta}$, $h_{\beta,\theta}$ *obey*

$$\sup_{\theta\in\mathbb{S}^{d-1}}\int_0^1 \frac{u(1-u)}{h_{\alpha,\theta}(Q_{\alpha,\theta}(u))^2}+\frac{u(1-u)}{h_{\beta,\theta}(Q_{\beta,\theta}(u))^2}\,\mathrm{d}u<+\infty.$$

*Proof.* The triangle inequality for $\mathrm{SW}_p$ gives $\left|\mathrm{SW}_p(\hat\alpha_n,\hat\beta_m)-\mathrm{SW}_p(\alpha,\beta)\right| \leqslant \mathrm{SW}_p(\hat\alpha_n,\alpha)+\mathrm{SW}_p(\hat\beta_m,\beta)$. By the definition above, Jensen's inequality and the assumed one-dimensional bounds,

$$\mathbb{E}\,\mathrm{SW}_p(\hat\alpha_n,\alpha)\leqslant\left(\int_{\mathbb{S}^{d-1}}\mathbb{E}\,\mathcal{W}_p\left((P_\theta)_\sharp\hat\alpha_n,(P_\theta)_\sharp\alpha\right)^p\mathrm{d}\sigma(\theta)\right)^{1/p}\leqslant\frac{A_\alpha^{1/p}}{\sqrt{n}},$$

and the same argument applies to $\beta$. For $p=1$, compact support gives the displayed one-dimensional empirical bound by the Dvoretzky–Kiefer–Wolfowitz inequality and the identity $\mathcal{W}_1(\gamma,\eta)=\int_{\mathbb{R}}|F_\gamma-F_\eta|\mathrm{d}s$. For $p=2$, the displayed quantile condition is the standard sufficient condition behind the one-dimensional bound $\mathbb{E}\,\mathcal{W}_2^2(\hat\gamma_n,\gamma)=O(n^{-1})$, uniformly over the projected laws; it follows from the classical quantile-process expansion. □

### 9.3 Bias and Variance of OT

$B_{n,m}(\mathcal{V}) \coloneqq \mathbb{E}\mathcal{V}(\hat\alpha_n,\hat\beta_m)-\mathcal{V}(\alpha,\beta)$ $Z_{n,m}(\mathcal{V})\coloneqq\mathcal{V}(\hat\alpha_n,\hat\beta_m)-\mathbb{E}\mathcal{V}(\hat\alpha_n,\hat\beta_m)$,

**Literature map.** The shape of the limit depends sharply on the analytic regularity of the transport value. For exact OT on a finite or countable space, the map from empirical weights to optimal cost is convex and piecewise affine.

**Bias versus centered fluctuation.**

$$\mathcal{V}(\hat\alpha_n,\hat\beta_m)-\mathcal{V}(\alpha,\beta)=\underbrace{\mathbb{E}\mathcal{V}(\hat\alpha_n,\hat\beta_m)-\mathcal{V}(\alpha,\beta)}_{\text{bias}}+\underbrace{\mathcal{V}(\hat\alpha_n,\hat\beta_m)-\mathbb{E}\mathcal{V}(\hat\alpha_n,\hat\beta_m)}_{\text{centered fluctuation}}$$

**Proposition 9.11** (Finite-space bias and CLT for exact OT)**.** *Let* $a\in\Sigma_n$ *and* $b\in\Sigma_m$ *have positive entries, and let* $\mathrm{C}\in\mathbb{R}^{n\times m}$. *Recall the discrete OT value* $\mathrm{L}_{\mathrm{C}}(a,b)\coloneqq\min_{\mathrm{P}\in\mathrm{U}(a,b)}\langle\mathrm{C},\,\mathrm{P}\rangle$. *Let* $\hat a_N$ *be the empirical histogram of* $N$ *independent samples from* $a$, *while* $b$ *is fixed. Denote by* $\mathcal{D}^\star(a,b)\coloneqq\operatorname{argmax}_{f_i+g_j\leqslant\mathrm{C}_{ij}}\langle f,\,a\rangle+\langle g,\,b\rangle$ *the set of optimal dual vectors, modulo the gauge* $(f,g)\mapsto(f+\lambda\mathbb{1},g-\lambda\mathbb{1})$. *If* $G_a$ *is a centered Gaussian vector with covariance* $\mathbb{E}\,G_aG_a^\top=\mathrm{diag}(a)-aa^\top$, *then* $\sqrt{N}\big(\mathrm{L}_{\mathrm{C}}(\hat a_N,b)-\mathrm{L}_{\mathrm{C}}(a,b)\big)\implies\sup_{(f,g)\in\mathcal{D}^\star(a,b)}\langle f,\,G_a\rangle$. *Moreover,* $\sqrt{N}\big(\mathbb{E}\mathrm{L}_{\mathrm{C}}(\hat a_N,b)-\mathrm{L}_{\mathrm{C}}(a,b)\big)\longrightarrow\mathbb{E}\sup_{(f,g)\in\mathcal{D}^\star(a,b)}\langle f,\,G_a\rangle$, *and the rescaled variance converges to the variance of the same limit. If the source dual potential* $f^\star$ *is unique up to constants, the limit is Gaussian with variance*

$$\sigma^2_{\mathrm{OT}}=\sum_i a_i\left(f_i^\star-\sum_k a_kf_k^\star\right)^2.$$

*In this differentiable case, the first-order bias vanishes:* $\mathbb{E}\mathrm{L}_{\mathrm{C}}(\hat a_N,b)-\mathrm{L}_{\mathrm{C}}(a,b)=o(N^{-1/2})$.

*Proof.* The multinomial central limit theorem gives $\sqrt{N}(\hat a_N-a)\Longrightarrow G_a,\qquad G_a\in\{\xi:\langle\mathbb{1},\,\xi\rangle=0\}$. The dual formulation writes the map $a'\mapsto\mathrm{L}_{\mathrm{C}}(a',b)$ as the supremum, over the feasible polyhedron $f_i+g_j\leqslant\mathrm{C}_{ij}$, of the affine functions $a'\mapsto\langle f,\,a'\rangle+\langle g,\,b\rangle$. Its directional derivative at $a$, in any tangent direction $h$ with $\langle\mathbb{1},\,h\rangle=0$, is therefore $D_a\mathrm{L}_{\mathrm{C}}(a,b)[h]=\sup_{(f,g)\in\mathcal{D}^\star(a,b)}\langle f,\,h\rangle$, which is the finite-dimensional form of Danskin's theorem already used in Proposition 11.12. The directional delta method gives the displayed distributional limit. After fixing a gauge, the relevant dual face is bounded, so the support functions above grow at most linearly in the multinomial fluctuation. Uniform integrability then yields convergence of the first two moments, hence the bias and variance statements. If $f^\star$ is unique modulo constants, the support function reduces to the linear form $\langle f^\star,\,G_a\rangle$. Its expectation is zero, and the covariance $\mathrm{diag}(a)-aa^\top$ gives exactly the displayed centered second moment of $f^\star$. □

$$r_{N,M}\big(\mathrm{L}_{\mathrm{C}}(\hat a_N,\hat b_M)-\mathrm{L}_{\mathrm{C}}(a,b)\big)\Longrightarrow\sup_{(f,g)\in\mathcal{D}^\star(a,b)}\left(\sqrt{1-\lambda}\,\langle f,\,G_a\rangle+\sqrt{\lambda}\,\langle g,\,G_b\rangle\right),$$

$$\frac{1}{N}\sum_i a_i(f_i^\star-\bar f^\star)^2+\frac{1}{M}\sum_j b_j(g_j^\star-\bar g^\star)^2,\qquad\bar f^\star=\sum_i a_if_i^\star,\quad\bar g^\star=\sum_j b_jg_j^\star.$$

**Proposition 9.12** (Finite-space bias and CLT for entropic OT)**.** *Fix* $\varepsilon>0$, *positive histograms* $a,b$, *and a finite cost matrix* $\mathrm{C}$. *Consider the finite-dimensional KL-normalized entropic value* $\mathcal{V}_{\mathrm{C},\varepsilon}(a,b)\coloneqq\min_{\mathrm{P}\in\mathrm{U}(a,b)}\langle\mathrm{C},\,\mathrm{P}\rangle+\varepsilon\mathrm{KL}(\mathrm{P}|a\otimes b)$. *This is the discrete version of the value* $\mathcal{V}_{c,\varepsilon}$ *used in Proposition 11.13. Let* $(f_\varepsilon,g_\varepsilon)$ *be normalized entropic dual potentials for* $\mathcal{V}_{\mathrm{C},\varepsilon}(a,b)$. *If* $\hat a_N$ *is the empirical histogram of* $N$ *samples from* $a$, *then*

$$\sqrt{N}\big(\mathcal{V}_{\mathrm{C},\varepsilon}(\hat a_N,b)-\mathcal{V}_{\mathrm{C},\varepsilon}(a,b)\big)\quad\Longrightarrow\quad\mathcal{N}(0,\sigma^2_{\varepsilon,a}),$$

*and*

$$\mathbb{E}\mathcal{V}_{\mathrm{C},\varepsilon}(\hat a_N,b)-\mathcal{V}_{\mathrm{C},\varepsilon}(a,b)=O(N^{-1}),\qquad N\ \mathrm{Var}\left(\mathcal{V}_{\mathrm{C},\varepsilon}(\hat a_N,b)\right)\to\sigma^2_{\varepsilon,a},$$

*where*

$$\sigma^2_{\varepsilon,a}=\sum_i a_i\left((f_\varepsilon)_i-\sum_k a_k(f_\varepsilon)_k\right)^2.$$

*For two independent empirical histograms, the first-order asymptotic variance is*

$$\frac{\sigma^2_{\varepsilon,a}}{N}+\frac{\sigma^2_{\varepsilon,b}}{M},\qquad\sigma^2_{\varepsilon,b}=\sum_j b_j\left((g_\varepsilon)_j-\sum_\ell b_\ell(g_\varepsilon)_\ell\right)^2.$$

*Proof.* For positive histograms and fixed $\varepsilon > 0$, the entropic problem has a unique optimizer, and the KL-normalized value is smooth on a neighborhood of $(a, b)$ inside the relative interiors of the simplices. Proposition 11.13 gives $D_a \mathcal{V}_{C,\varepsilon}(a, b)[h] = \langle f_\varepsilon, h\rangle$. Applying the ordinary delta method to the multinomial CLT $\sqrt{N}(\hat{a}_N - a) \Rightarrow G_a$ gives the one-sample limit $\langle f_\varepsilon, G_a\rangle$, whose variance is the displayed covariance formula. The only boundary nuisance is that an empirical histogram may have a zero entry. Since $a_i > 0$ for all $i$, this event has probability at most $\sum_i (1 - a_i)^N$, hence is exponentially small and does not affect the first-order limit or the $N^{-1}$ bias. On the complementary event, Taylor's formula applies in the relative interior: the first-order term is centered and the Hessian is locally bounded, so the plug-in bias is $O(N^{-1})$. The variance convergence follows from the same expansion and from bounded moments of the multinomial fluctuation. The two-sample formula follows by applying the argument independently to the two marginals. □

For the entropy-only convention $\mathrm{L}_{\mathrm{C}}^{\varepsilon}$ used in Section 7.1, one has, for positive histograms,
$\mathrm{L}_{\mathrm{C}}^{\varepsilon}(a, b) = \mathcal{V}_{C,\varepsilon}(a, b) - \varepsilon \mathrm{H}(a) - \varepsilon \mathrm{H}(b)$. Thus the marginal entropy derivatives described after Proposition 11.13 must be added to the potentials: the derivative with respect to the source marginal is represented on the simplex tangent space by $f_\varepsilon + \varepsilon \log a$, up to an irrelevant additive constant.

**Three error terms when entropy estimates exact OT.** If the target is the exact cost $\mathcal{L}_c(\alpha, \beta)$ but the statistic uses the entropic value $\mathcal{L}_c^\varepsilon$, three errors should be kept separate:

$$\mathcal{L}_c^\varepsilon(\hat\alpha_n, \hat\beta_m) - \mathcal{L}_c(\alpha, \beta) = \underbrace{\mathcal{L}_c^\varepsilon(\alpha, \beta) - \mathcal{L}_c(\alpha, \beta)}_{\text{regularization bias}} + \underbrace{\mathbb{E}\mathcal{L}_c^\varepsilon(\hat\alpha_n, \hat\beta_m) - \mathcal{L}_c^\varepsilon(\alpha, \beta)}_{\text{statistical bias}} + \underbrace{\mathcal{L}_c^\varepsilon(\hat\alpha_n, \hat\beta_m) - \mathbb{E}\mathcal{L}_c^\varepsilon(\hat\alpha_n, \hat\beta_m)}_{\text{centered fluctuation}}.$$

The same decomposition applies to the debiased value $\bar{\mathcal{L}}_c^\varepsilon$ by replacing $\mathcal{L}_c^\varepsilon$ everywhere by $\bar{\mathcal{L}}_c^\varepsilon$. This identity is trivial algebraically but important statistically.

**What changes in continuous spaces.** The finite-dimensional formulas above are not merely toy models; they are the cleanest template for the general mechanism. Whenever an OT value is Hadamard differentiable and admits a unique sufficiently regular dual potential $f^\star$, the empirical-process CLT and the delta method give a first-order Gaussian limit with one-sample variance $\mathrm{Var}_\alpha(f^\star(X))$.

### 9.4 Sketching Sinkhorn in Linear Time

This compact subsection records feature-sketch approximations of the Sinkhorn kernel.

**PSD kernels and ordinary feature sketches.** The RKHS/MMD section, Section 6.2, already used positive semidefinite kernels to define Hilbertian discrepancies between measures. Recall that a symmetric kernel $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is positive semidefinite if, for every finite family $(x_i)_{i=1}^n$, the Gram matrix $(k(x_i, x_j))_{ij}$ is positive semidefinite.
$k(x, y) = \int_Z \langle \varphi(x, z), \varphi(y, z)\rangle_{\mathbb{R}^p} \, d\rho(z)$, $\quad \varphi : \mathcal{X} \times Z \to \mathbb{R}^p$, $\widetilde{k}_r(x, y) = \frac{1}{r}\sum_{\ell=1}^r \langle \varphi(x, z_\ell), \varphi(y, z_\ell)\rangle_{\mathbb{R}^p}$.

**Application to Sinkhorn kernels.** $\alpha = \sum_{i=1}^n a_i \delta_{x_i}$, $\quad \beta = \sum_{j=1}^m b_j \delta_{y_j}$, $\quad K_{ij} = k_\varepsilon(x_i, y_j) := e^{-c(x_i,y_j)/\varepsilon}$. $u = \frac{a}{Kv}$, $\quad v = \frac{b}{K^\top u}$, $(\Phi_X)_{i,(\ell,q)} = r^{-1/2}\varphi_q(x_i, z_\ell)$, $\quad (\Phi_Y)_{j,(\ell,q)} = r^{-1/2}\varphi_q(y_j, z_\ell)$. $\widetilde{K}_{ij} = (\Phi_X \Phi_Y^\top)_{ij} = \frac{1}{r}\sum_{\ell=1}^r \langle \varphi(x_i, z_\ell), \varphi(y_j, z_\ell)\rangle_{\mathbb{R}^p} \simeq K_{ij}$. $\widetilde{K}v = \Phi_X(\Phi_Y^\top v)$, $\quad \widetilde{K}^\top u = \Phi_Y(\Phi_X^\top u)$,

**Proposition 9.13** (Finite positivity and logarithmic accuracy)**.** *Let $K_{ij} = \mathbb{E}h_{ij}(Z) > 0$ for $1 \leqslant i \leqslant n$, $1 \leqslant j \leqslant m$, where $|h_{ij}(Z)| \leqslant M$ almost surely. Let $\widetilde{K}_{ij} = \frac{1}{r}\sum_{\ell=1}^r h_{ij}(Z_\ell)$, $\quad Z_1, \ldots, Z_r \sim \rho$. Then, for every $\eta > 0$, $\mathbb{P}\left(\max_{i,j}\left|\widetilde{K}_{ij} - K_{ij}\right| > \eta\right) \leqslant 2nm \exp\left(-\frac{r\eta^2}{2M^2}\right)$. In particular, if $\kappa := \min_{i,j} K_{ij} > 0$, $\eta < \kappa$, and the event above does not occur, then $\widetilde{K}$ is entrywise positive. If moreover $\eta \leqslant \kappa/2$, then the sketched cost $\widetilde{c}_{ij} = -\varepsilon \log \widetilde{K}_{ij}$ satisfies $\max_{i,j}|\widetilde{c}_{ij} - c_{ij}| \leqslant \frac{2\varepsilon\eta}{\kappa}$. If, more sharply, the relative variables $\zeta_{ij}(Z) := h_{ij}(Z)/K_{ij}$ satisfy $|\zeta_{ij}(Z)| \leqslant \psi$ almost surely, then for every $0 < \delta \leqslant 1/2$, $\mathbb{P}\left(\max_{i,j}\left|\frac{\widetilde{K}_{ij}}{K_{ij}} - 1\right| > \delta\right) \leqslant 2nm \exp\left(-\frac{r\delta^2}{2\psi^2}\right)$. On the complementary event, $\widetilde{K}_{ij} > 0$ and $\max_{i,j}|\widetilde{c}_{ij} - c_{ij}| \leqslant 2\varepsilon\delta$.*

*Proof.* For a fixed pair $(i, j)$, Hoeffding's inequality applied to bounded variables in $[-M, M]$ gives $\mathbb{P}\left(\left|\widetilde{K}_{ij} - K_{ij}\right| > \eta\right) \leqslant 2\exp\left(-\frac{r\eta^2}{2M^2}\right)$. A union bound over the $nm$ entries gives the first claim. If $\max_{ij}\left|\widetilde{K}_{ij} - K_{ij}\right| \leqslant \eta < \kappa$, then $\widetilde{K}_{ij} \geqslant \kappa - \eta > 0$. Writing $\widetilde{K}_{ij} = K_{ij}(1 + s_{ij})$, one has $|s_{ij}| \leqslant \eta/\kappa \leqslant 1/2$, hence $|\log(1 + s_{ij})| \leqslant 2|s_{ij}|$, which proves the absolute logarithmic bound. Applying the same Hoeffding argument to $\zeta_{ij}(Z)$, whose expectation is 1, gives the relative estimate. On the corresponding event, $\widetilde{K}_{ij} = K_{ij}(1 + s_{ij})$ with $|s_{ij}| \leqslant \delta \leqslant 1/2$, so positivity and the bound $\varepsilon|\log(1 + s_{ij})| \leqslant 2\varepsilon\delta$ follow. □

$r_*(X, \eta, \varepsilon_0) \lesssim \left(\eta R^2 + \log(n/\varepsilon_0)\right)^d$

**Positive features and total positivity.**

$$\mathrm{DNN}_n = \left\{A \in \mathbb{R}^{n\times n} \; ; \; A \succeq 0, \; A_{ij} \geqslant 0\right\}, \qquad \mathrm{CP}_n = \left\{BB^\top \; ; \; B \in \mathbb{R}_+^{n\times q} \text{ for some } q \geqslant 1\right\}.$$

$k(x, y) = \int_Z \langle \varphi(x, z), \varphi(y, z)\rangle_{\mathbb{R}^p} \, d\rho(z)$, $\quad \varphi(x, z) \in \mathbb{R}_+^p$,

**Proposition 9.14** (Basic closure of totally positive kernels)**.** *Nonnegative sums and pointwise products of totally positive kernels are totally positive. In particular, positive autocorrelations $k(x, y) = \int g(u-x)g(u-y)\, du$, $\quad g \geqslant 0$, and all nonnegative mixtures of such kernels are totally positive.*

*Proof.* For sums, concatenate the nonnegative features and rescale them by the square roots of the nonnegative coefficients. For products, tensor the nonnegative features:

$$\langle \varphi_1(x, z_1), \varphi_1(y, z_1)\rangle\langle \varphi_2(x, z_2), \varphi_2(y, z_2)\rangle = \langle \varphi_1(x, z_1) \otimes \varphi_2(x, z_2), \varphi_1(y, z_1) \otimes \varphi_2(y, z_2)\rangle.$$

The tensor feature remains entrywise nonnegative. The autocorrelation formula is already a positive feature representation with latent variable $u$ and scalar feature $g(u - x)$. □

**Proposition 9.15** (Generalized Gaussian positive features). *Let* $0 < p \leqslant 2$, $\varepsilon > 0$, *and* $k_{p,\varepsilon}(x,y) = \exp\left(-\frac{\|x-y\|^p}{\varepsilon}\right)$, $\quad x, y \in \mathbb{R}^d$. *Set* $a = p/2 \in (0,1]$. *Let* $S_a$ *be the positive* $a$*-stable random variable normalized by* $\mathbb{E}\left(e^{-tS_a}\right) = e^{-t^a}$, $\quad t \geqslant 0$, *and set* $\Lambda = \varepsilon^{-1/a} S_a$. *Denote by* $\nu_{a,\varepsilon}$ *the law of* $\Lambda$. *Then* $\exp\left(-\frac{t^a}{\varepsilon}\right) = \int_0^{+\infty} e^{-\lambda t}\, d\nu_{a,\varepsilon}(\lambda)$, $\quad t \geqslant 0$. *Equivalently, for a nonnegative random variable* $\Lambda \sim \nu_{a,\varepsilon}$, $k_{p,\varepsilon}(x,y) = \mathbb{E}_\Lambda \exp\left(-\Lambda\|x-y\|^2\right)$. *Fix a centering point* $x_0 \in \mathbb{R}^d$, *let* $Z \sim \mathcal{N}(0, \mathrm{Id}_d)$, *independent of* $\Lambda$, *and set* $q_x(\Lambda) = \sqrt{2\Lambda}\,(x - x_0)$, $\quad \varphi_p(x, \Lambda, Z) = \exp\left(\langle Z, q_x(\Lambda)\rangle - \|q_x(\Lambda)\|^2\right)$. *Then* $k_{p,\varepsilon}(x,y) = \mathbb{E}_{\Lambda,Z}\left(\varphi_p(x,\Lambda,Z)\varphi_p(y,\Lambda,Z)\right)$. *Thus* $k_{p,\varepsilon}$ *is totally positive in the complete-positive sense. In particular, if* $(\Lambda_\ell, Z_\ell)_{\ell=1}^r$ *are i.i.d. copies of* $(\Lambda, Z)$, *then* $\widetilde{k}_{p,\varepsilon}(x,y) = \sum_{\ell=1}^r \varphi_\ell(x)\varphi_\ell(y)$, $\quad \varphi_\ell(x) = r^{-1/2}\varphi_p(x, \Lambda_\ell, Z_\ell)$, *is an unbiased nonnegative feature sketch of* $k_{p,\varepsilon}$. *For* $p = 2$, *the mixing law reduces to the Dirac mass* $\nu_{1,\varepsilon} = \delta_{1/\varepsilon}$, *which recovers the usual Gaussian feature formula.*

*Proof.* The function $s \mapsto e^{-s^a}$ is completely monotone for $0 < a \leqslant 1$. Bernstein's theorem therefore gives the Laplace-transform representation of a positive $a$-stable random variable $S_a$. The scaling $\Lambda = \varepsilon^{-1/a} S_a$ gives $\mathbb{E}e^{-t\Lambda} = \exp\left(-\frac{t^a}{\varepsilon}\right)$, which is the displayed representation by $\nu_{a,\varepsilon}$. Applying this identity with $t = \|x-y\|^2$ gives the Gaussian-mixture formula.
Conditionally on $\Lambda = \lambda$, the Gaussian moment-generating function gives

$$
\begin{aligned}
\mathbb{E}_Z\left(\varphi_p(x,\lambda,Z)\varphi_p(y,\lambda,Z)\right) &= \exp\left(-\|q_x(\lambda)\|^2 - \|q_y(\lambda)\|^2\right)\mathbb{E}_Z\left(\exp\langle Z, q_x(\lambda) + q_y(\lambda)\rangle\right) \\
&= \exp\left(-\frac{1}{2}\|q_x(\lambda) - q_y(\lambda)\|^2\right) = \exp\left(-\lambda\|x-y\|^2\right).
\end{aligned}
$$

Taking the expectation over $\Lambda$ gives $k_{p,\varepsilon}$. Since $\varphi_p$ is positive, this is a positive-feature representation; hence every finite Gram matrix is completely positive. The Monte-Carlo sketch is just the empirical average of this representation. $\square$

$k_\lambda(x,y) = \lambda + \cos^2\left(\frac{x-y}{2}\right)$, $\quad \lambda > 0$. $\langle H, K\rangle = \frac{21}{4} - \frac{5\sqrt{5}}{2} < 0$.

**Positive sketches for Sinkhorn.** $k_{p,\varepsilon}(x,y) = \exp\left(-\frac{\|x-y\|^p}{\varepsilon}\right)$. For $p \geqslant 1$, this is the usual $p$-Wasserstein power cost; for $0 < p < 1$, it should be read simply as a concave power transport cost. The factor $1/p$, often used in the definition of $\mathcal{W}_p^p$, only rescales $\varepsilon$.
$\varphi_\ell(x) = r^{-1/2}\varphi_p(x, \Lambda_\ell, Z_\ell)$, $\quad 1 \leqslant \ell \leqslant r$.

$$
\widetilde{k}_{p,\varepsilon}(x,y) = \sum_{\ell=1}^r \varphi_\ell(x)\varphi_\ell(y) \geqslant 0, \qquad \mathbb{E}[\widetilde{k}_{p,\varepsilon}(x,y)] = k_{p,\varepsilon}(x,y).
$$

$\widetilde{P} = \mathrm{diag}(u)\Phi_X\Phi_Y^\top \mathrm{diag}(v) = LR^\top$, $\quad L = \mathrm{diag}(u)\Phi_X$, $\quad R = \mathrm{diag}(v)\Phi_Y$.
**Connection with linear attention.**

# 10 Generalized Wasserstein Distances

## 10.1 Unbalanced OT

This compact subsection summarizes transport models with relaxed marginal constraints.
**Relaxed formulation.** For nonnegative measures $(\alpha, \beta) \in \mathcal{M}_+(\mathcal{X}) \times \mathcal{M}_+(\mathcal{Y})$, a generic relaxed formulation is

$$
\mathrm{UW}_c(\alpha,\beta) = \inf_{\pi \in \mathcal{M}_+(\mathcal{X}\times\mathcal{Y})} \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\mathrm{d}\pi(x,y) + \mathcal{D}_{\psi_1}(\pi_1|\alpha) + \mathcal{D}_{\psi_2}(\pi_2|\beta), \tag{10.1}
$$

where $\psi_1, \psi_2$ are convex entropy functions. Exact conservation $(\pi_1, \pi_2) = (\alpha, \beta)$ is replaced by a cost for changing the marginals. $\mathrm{UW}_{c,\tau}(\alpha,\beta) = \inf_{\pi \geqslant 0} \int c\mathrm{d}\pi + \tau\mathcal{D}_{\bar{\psi}_1}(\pi_1|\alpha) + \tau\mathcal{D}_{\bar{\psi}_2}(\pi_2|\beta)$. These figures should be read as pictures of a single relaxed plan $\pi$: the same nonnegative measure determines both the transported coupling and its two relaxed marginals. The small-$\tau$ result below formalizes the opposite regime, where transport becomes negligible compared with local mass variation.
**Proposition 10.1** (Small-transport-scale limit for marginal penalties). *Assume that* $\alpha, \beta$ *are finite measures on a compact metric space* $\mathcal{X}$, *that* $c$ *is continuous,* $c \geqslant 0$, *and* $c(x,y) = 0$ *if and only if* $x = y$. *Assume also that the marginal divergences are nonnegative, weak-* lower semicontinuous, and have weak-* compact sublevel sets on* $\mathcal{M}_+(\mathcal{X})$. *Then*

$$
\lim_{\tau \downarrow 0} \frac{1}{\tau}\mathrm{UW}_{c,\tau}(\alpha,\beta) = \inf_{\rho \in \mathcal{M}_+(\mathcal{X})} \mathcal{D}_{\bar{\psi}_1}(\rho|\alpha) + \mathcal{D}_{\bar{\psi}_2}(\rho|\beta).
$$

*The right-hand side is the infimal gluing divergence obtained by matching the two measures through a common zero-transport marginal* $\rho$. *In the dominated case, if* $\alpha = a\lambda$, $\beta = b\lambda$, *and the minimizing common marginal is absolutely continuous,* $\rho = r\lambda$, *this divergence decouples pointwise as*

$$
\int \mathfrak{m}_{\bar{\psi}_1,\bar{\psi}_2}(a(x), b(x))\mathrm{d}\lambda(x), \qquad \mathfrak{m}_{\bar{\psi}_1,\bar{\psi}_2}(a,b) := \inf_{r \geqslant 0} a\,\bar{\psi}_1(r/a) + b\,\bar{\psi}_2(r/b),
$$

*with the usual recession conventions when* $a = 0$ *or* $b = 0$. *For superlinear entropies, and in particular for KL, finite energy forces this dominated form. Thus, when* $\bar{\psi}_1 = \bar{\psi}_2$ *is the KL entropy,* $\inf_{\rho \in \mathcal{M}_+(\mathcal{X})} \mathrm{KL}(\rho|\alpha) + \mathrm{KL}(\rho|\beta) = \int(\sqrt{a} - \sqrt{b})^2\mathrm{d}\lambda$. *Thus the KL marginal relaxation contains the squared Hellinger distance as its local mass-variation limit.*

*Proof.* For the upper bound, restrict to diagonal plans $\pi = (\mathrm{Id}, \mathrm{Id})_\sharp\rho$, whose transport cost is zero and whose two marginals are both $\rho$.
For the lower bound, let $\tau_n \downarrow 0$ and let $\pi_n$ be almost minimizing plans with bounded scaled values $\tau_n^{-1}\mathrm{UW}_{c,\tau_n}(\alpha,\beta)$. Since the divergences are nonnegative, $\int c\mathrm{d}\pi_n = O(\tau_n)$, hence $\int c\mathrm{d}\pi_n \to 0$. The bounded scaled values also put the two marginals in compact divergence sublevel sets. Since a coupling has the same total mass as each marginal, the couplings are tight on $\mathcal{X}\times\mathcal{X}$. Up to subsequences, $\pi_n \rightharpoonup \pi_0$. Lower semicontinuity of the transport cost yields $\int c\mathrm{d}\pi_0 = 0$, so $\pi_0$ is concentrated on the diagonal. Its two marginals are therefore equal to a common measure $\rho$. Lower semicontinuity of the marginal divergences gives $\liminf_n \frac{1}{\tau_n}\mathrm{UW}_{c,\tau_n}(\alpha,\beta) \geqslant \mathcal{D}_{\bar{\psi}_1}(\rho|\alpha) + \mathcal{D}_{\bar{\psi}_2}(\rho|\beta)$, and optimizing over $\rho$ gives the lower bound.

In the dominated case, writing $\rho = r\lambda$ gives $\mathcal{D}_{\bar\psi_1}(\rho|\alpha)+\mathcal{D}_{\bar\psi_2}(\rho|\beta) = \int a\,\bar\psi_1(r/a)+b\,\bar\psi_2(r/b)\mathrm{d}\lambda$, so the minimization over $\rho$ decouples into the scalar envelope $\mathfrak{m}_{\bar\psi_1,\bar\psi_2}$. For KL, no singular part is admissible when $\alpha$ and $\beta$ are dominated by $\lambda$. The pointwise objective is $r\log(r/a) - r + a + r\log(r/b) - r + b$. Its optimality condition is $\log(r/a) + \log(r/b) = 0$, hence $r = \sqrt{ab}$, and the minimum is $a + b - 2\sqrt{ab} = (\sqrt{a} - \sqrt{b})^2$. □

**Proposition 10.2** (Dual of unbalanced optimal transport)**.** *Under the usual Fenchel–Rockafellar qualification assumptions, one has equality between* (10.1) *and* $\mathrm{UW}_c(\alpha,\beta) = \sup_{f\oplus g\leqslant c} -\mathcal{D}^*_{\psi_1}(-f|\alpha) - \mathcal{D}^*_{\psi_2}(-g|\beta)$.

*Proof.* Use the variational formula (6.9) for the dual of a divergence and introduce the marginal variables through continuous potentials: $\inf_{\pi\geqslant 0}\sup_{f,g}\int c\mathrm{d}\pi + \int -f\mathrm{d}\pi_1 + \int -g\mathrm{d}\pi_2 - \mathcal{D}^*_{\psi_1}(-f|\alpha) - \mathcal{D}^*_{\psi_2}(-g|\beta)$. Exchanging the infimum and the supremum gives $\sup_{f,g} -\mathcal{D}^*_{\psi_1}(-f|\alpha) - \mathcal{D}^*_{\psi_2}(-g|\beta) + \inf_{\pi\geqslant 0}\int \big(c - (f\oplus g)\big)\mathrm{d}\pi$. The last infimum is 0 when $f\oplus g \leqslant c$ and $-\infty$ otherwise, which gives the displayed dual. □

**Reverse and homogeneous formulations.**

$$\begin{aligned}\int c(x,y)\mathrm{d}\pi(x,y) &+ \mathcal{D}_{\psi_1}(\pi_1|\alpha) + \mathcal{D}_{\psi_2}(\pi_2|\beta)\\ &= \int \left(c(x,y) + \psi_1\!\left(\frac{\mathrm{d}\pi_1}{\mathrm{d}\alpha}(x)\right)\frac{\mathrm{d}\alpha}{\mathrm{d}\pi_1}(x) + \psi_2\!\left(\frac{\mathrm{d}\pi_2}{\mathrm{d}\beta}(y)\right)\frac{\mathrm{d}\beta}{\mathrm{d}\pi_2}(y)\right)\mathrm{d}\pi(x,y).\end{aligned}$$

$$L_c(r,s) := c + r\psi_1(1/r) + s\psi_2(1/s), \tag{10.2}$$

with the usual recession convention at $r = 0$ or $s = 0$. If $\alpha = F\pi_1 + \alpha^\perp$ and $\beta = G\pi_2 + \beta^\perp$ are the Lebesgue decompositions of the reference marginals with respect to the transported marginals, then the reverse formulation reads $\mathrm{UW}_c(\alpha,\beta) = \inf_{\pi\geqslant 0}\int L_{c(x,y)}(F(x),G(y))\mathrm{d}\pi(x,y) + \psi_1(0)\alpha^\perp(\mathcal{X}) + \psi_2(0)\beta^\perp(\mathcal{Y})$.

$$H_c(r,s) := \inf_{\theta>0} \theta L_c(r/\theta, s/\theta), \tag{10.3}$$

$$\mathrm{HW}_c(\alpha,\beta) = \inf_{\pi\geqslant 0}\int H_{c(x,y)}(F(x),G(y))\mathrm{d}\pi(x,y) + \psi_1(0)\alpha^\perp(\mathcal{X}) + \psi_2(0)\beta^\perp(\mathcal{Y}). \tag{10.4}$$

**Proposition 10.3** (Homogenization does not change the unbalanced cost)**.** *One has* $\mathrm{HW}_c(\alpha,\beta) = \mathrm{UW}_c(\alpha,\beta)$.

*Proof.* The inequality $\mathrm{HW} \leqslant \mathrm{UW}$ follows from $H_c \leqslant L_c$ by taking $\theta = 1$. Conversely, take a feasible measure $\pi$ in the homogeneous formulation. By definition of the perspective transform, for every $(x,y)$ and every $\eta > 0$ there exists a scale $\theta(x,y) > 0$ such that $H_{c(x,y)}(F(x),G(y)) + \eta \geqslant \theta(x,y)L_{c(x,y)}\big(F(x)/\theta(x,y), G(y)/\theta(x,y)\big)$. Replacing $\pi$ by the rescaled measure $\tilde\pi = \theta\pi$ and the densities by $F/\theta$ and $G/\theta$ gives an admissible competitor for the reverse formulation with cost no larger than the homogeneous cost plus $\eta\pi(\mathcal{X}\times\mathcal{Y})$. Letting $\eta \to 0$ yields $\mathrm{UW} \leqslant \mathrm{HW}$. The singular terms are unchanged because the same rescaling is performed before taking the Lebesgue decomposition of the marginals. □

$$\begin{aligned}\mathrm{HW}_c(\alpha,\beta) = \inf_{\substack{\lambda\in\mathcal{M}_+(\mathcal{X}\times\mathcal{Y})\\ u,v\geqslant 0}} \Big\{ &\int_{\mathcal{X}\times\mathcal{Y}} H_{c(x,y)}\big(u(x,y),v(x,y)\big)\mathrm{d}\lambda(x,y)\ ;\\ &(\mathrm{p}_1)_\#(u\lambda) = \alpha, \qquad (\mathrm{p}_2)_\#(v\lambda) = \beta\Big\}.\end{aligned} \tag{10.5}$$

**Conic lifting.** Assume now that $\mathcal{X} = \mathcal{Y}$ and $\psi_1 = \psi_2 = \psi$. The last formulation lifts the problem to the cone space $\mathfrak{C}[\mathcal{X}] := (\mathcal{X}\times\mathbb{R}_+)/\sim$, where all points $(x,0)$ are identified at the apex. For an exponent $p \geqslant 1$, define $\Delta((x,r),(y,s)) := H_{c(x,y)}(r^p,s^p)^{1/p}$.

- $\mathcal{D}_\psi = \mathrm{KL}$, $p = 2$, and $c(x,y) = -\log\cos^2(d(x,y)\wedge\pi/2)$ give the Hellinger–Kantorovich or Wasserstein–Fisher–Rao cone metric $\Delta((x,r),(y,s))^2 = r^2 + s^2 - 2rs\cos(d(x,y)\wedge\pi/2)$.
- $\mathcal{D}_\psi = \mathrm{KL}$, $p = 2$, and $c(x,y) = d(x,y)^2$ give the Gaussian Hellinger cone metric $\Delta((x,r),(y,s))^2 = r^2 + s^2 - 2rse^{-d(x,y)^2/2}$.
- $\mathcal{D}_\psi = \mathrm{TV}$, $p = 1$, and $c(x,y) = d(x,y)$ give the partial-transport cone cost $\Delta((x,r),(y,s)) = r + s - (r\wedge s)(2 - d(x,y))_+$.

For a finite measure $\eta$ on the cone, define its weighted base projection $\mathsf{P}_p\eta \in \mathcal{M}_+(\mathcal{X})$ by $\int_{\mathcal{X}}\varphi(x)\mathrm{d}(\mathsf{P}_p\eta)(x) = \int_{\mathfrak{C}[\mathcal{X}]}\varphi(x)r^p\mathrm{d}\eta(x,r)$ for every bounded Borel $\varphi$. This is well defined at the apex because the weight $r^p$ vanishes there.

$$\mathrm{CW}(\alpha,\beta) = \inf_{\gamma\in\mathcal{M}_+(\mathfrak{C}[\mathcal{X}]^2)}\left\{\int \Delta((x,r),(y,s))^p\mathrm{d}\gamma\ ;\ \mathsf{P}_p\gamma_1 = \alpha,\quad \mathsf{P}_p\gamma_2 = \beta\right\}.$$

**Theorem 10.4** (Cone formulation of unbalanced OT)**.** *One has* $\mathrm{UW} = \mathrm{HW} = \mathrm{CW}$. *If* $\Delta$ *is a distance, then* $\mathrm{CW}^{1/p}$ *is a distance between nonnegative measures.*

*Proof.* The equality $\mathrm{UW} = \mathrm{HW}$ is Proposition 10.3.
Let $(\lambda,u,v)$ be feasible in (10.5). Define the cone coupling $\gamma = \big((x,y)\mapsto((x,u(x,y)^{1/p}),(y,v(x,y)^{1/p}))\big)_\#\lambda$. Then $\mathsf{P}_p\gamma_1 = \alpha$ and $\mathsf{P}_p\gamma_2 = \beta$ by the weighted marginal constraints. Moreover $\int\Delta((x,r),(y,s))^p\mathrm{d}\gamma = \int H_{c(x,y)}(u(x,y),v(x,y))\mathrm{d}\lambda(x,y)$. Taking the infimum gives $\mathrm{CW} \leqslant \mathrm{HW}$.
Conversely, let $\gamma$ be feasible for CW. Choose arbitrary representatives of cone points at the apex. This does not affect the weighted constraints, and it does not change the cost because, by the recession convention, $H_{c(x,y)}(0,s)$ and $H_{c(x,y)}(r,0)$ do not depend on the arbitrary spatial representative attached to the zero radius. Let $\lambda$ be the projection of $\gamma$ onto the spatial variables $(x,y)$, and disintegrate $\gamma = \gamma_{x,y}\mathrm{d}\lambda(x,y)$ with respect to this projection. Set $u(x,y) = \int r^p\mathrm{d}\gamma_{x,y}(r,s)$, $\quad v(x,y) = \int s^p\mathrm{d}\gamma_{x,y}(r,s)$. The cone marginal constraints imply $(\mathrm{p}_1)_\#(u\lambda) = \alpha$ and $(\mathrm{p}_2)_\#(v\lambda) = \beta$. Since $H_c$ is convex in its two arguments, as a perspective of

the convex reverse cost $L_c$, Jensen's inequality gives, for $\lambda$-a.e. $(x,y)$, $H_{c(x,y)}(u(x,y),v(x,y)) \leqslant \int H_{c(x,y)}(r^p,s^p)\mathrm{d}\gamma_{x,y}(r,s)$. After integration and using $\Delta((x,r),(y,s))^p = H_{c(x,y)}(r^p,s^p)$, this yields $\mathrm{HW} \leqslant \mathrm{CW}$. Hence $\mathrm{UW} = \mathrm{HW} = \mathrm{CW}$.
Assume now that $\Delta$ is a distance on $\mathfrak{C}[\mathcal{X}]$. Nonnegativity and symmetry are immediate. The triangle inequality is the standard gluing property of the homogeneous cone construction: one applies the gluing lemma to almost optimal cone couplings and then Minkowski's inequality for the metric $\Delta$; the linear weighted projection constraints are preserved under this gluing. Finally, if $\mathrm{CW}(\alpha,\beta)=0$, any minimizing sequence is supported in the limit on the diagonal of the cone, because $\Delta$ is a distance. On this diagonal one has $x=y$ and $r=s$, so the two weighted base projections coincide. Therefore $\alpha=\beta$. $\square$

**Entropic KL relaxation.** $\inf_{\pi\in\mathcal{M}_+(\mathcal{X}\times\mathcal{Y})}\int c\mathrm{d}\pi + \mathcal{D}_{\psi_1}(\pi_1|\alpha) + \mathcal{D}_{\psi_2}(\pi_2|\beta) + \varepsilon\mathcal{D}_\varphi(\pi|\alpha\otimes\beta)$.

$$\sup_{f,g} -\mathcal{D}^*_{\psi_1}(-f|\alpha) - \mathcal{D}^*_{\psi_2}(-g|\beta) - \varepsilon\mathcal{D}^*_\varphi\left(\frac{f\oplus g - c}{\varepsilon}\middle|\alpha\otimes\beta\right).$$

For $\mathcal{D}_\varphi = \mathrm{KL}$, the primal-dual relation is $\mathrm{d}\pi = e^{(f\oplus g - c)/\varepsilon}\mathrm{d}\alpha\mathrm{d}\beta$. If, in addition, $\mathcal{D}_{\psi_1} = \mathcal{D}_{\psi_2} = \tau\,\mathrm{KL}$, the dual specializes to

$$\sup_{f,g} -\tau\int(e^{-f/\tau}-1)\mathrm{d}\alpha - \tau\int(e^{-g/\tau}-1)\mathrm{d}\beta - \varepsilon\iint\left(e^{(f\oplus g - c)/\varepsilon}-1\right)\mathrm{d}\alpha\mathrm{d}\beta,$$

$$f \leftarrow \rho\, g^{\bar{c},\varepsilon}, \qquad g \leftarrow \rho\, f^{c,\varepsilon}, \qquad \rho := \frac{\tau}{\tau+\varepsilon},$$

where the soft $c$-transforms are those of Definition 7.16. Equivalently,

$$\begin{aligned} f(x) &= -\frac{\tau\varepsilon}{\tau+\varepsilon}\log\int_{\mathcal{Y}}\exp\left(\frac{g(y)-c(x,y)}{\varepsilon}\right)\mathrm{d}\beta(y),\\ g(y) &= -\frac{\tau\varepsilon}{\tau+\varepsilon}\log\int_{\mathcal{X}}\exp\left(\frac{f(x)-c(x,y)}{\varepsilon}\right)\mathrm{d}\alpha(x). \end{aligned}$$

$$u_i \leftarrow \left(\frac{\mathrm{a}_i}{(Kv)_i}\right)^\rho, \qquad v_j \leftarrow \left(\frac{\mathrm{b}_j}{(K^\top u)_j}\right)^\rho, \qquad \mathrm{P} = \mathrm{diag}(u)K\,\mathrm{diag}(v).$$

**Metric contraction of the damped updates.** For balanced entropic OT, potentials are only defined up to the gauge $(f,g)\sim(f+\lambda,g-\lambda)$, so Hilbert's projective metric is natural. The KL marginal penalties break this invariance: under the usual finite-dual assumptions, the dual potentials are unique up to $\alpha$- and $\beta$-null sets, and the natural distance is the ordinary $L^\infty$ distance on potentials.
$d_T\big((u,v),(u',v')\big) := \max\{\|\log u - \log u'\|_\infty, \|\log v - \log v'\|_\infty\}$,
**Proposition 10.5** (Linear contraction of unbalanced Sinkhorn)**.** *Assume that $c$ is bounded, or more generally that the soft transforms below are finite on the bounded potentials under consideration. Let $\rho = \tau/(\tau+\varepsilon) < 1$, and define $d_\infty((f,g),(f',g')) := \max\{\|f-f'\|_\infty, \|g-g'\|_\infty\}$. The damped soft-transform map*

$$\mathcal{T}_{\mathrm{GS}}(f,g) := (\tilde{f},\tilde{g}), \qquad \tilde{f} = \rho\, g^{\bar{c},\varepsilon}, \qquad \tilde{g} = \rho\, \tilde{f}^{c,\varepsilon},$$

*is a contraction: $d_\infty\big(\mathcal{T}_{\mathrm{GS}}(f,g),\mathcal{T}_{\mathrm{GS}}(f',g')\big) \leqslant \rho\, d_\infty((f,g),(f',g'))$. $(f^\star,g^\star)$ and the iterates satisfy $d_\infty((f^{(k)},g^{(k)}),(f^\star,g^\star)) \leqslant \rho^k d_\infty((f^{(0)},g^{(0)}),(f^\star,g^\star))$. Equivalently, for $u^{(k)} = e^{f^{(k)}/\varepsilon}$ and $v^{(k)} = e^{g^{(k)}/\varepsilon}$, $d_T((u^{(k)},v^{(k)}),(u^\star,v^\star)) \leqslant \rho^k d_T((u^{(0)},v^{(0)}),(u^\star,v^\star))$.*

*Proof.* The soft $c$-transform is 1-Lipschitz in $L^\infty$. Indeed, if $\|g-g'\|_\infty \leqslant \delta$, then $g-\delta \leqslant g' \leqslant g+\delta$. Since $g\mapsto g^{\bar{c},\varepsilon}$ is order reversing and satisfies $(g+\lambda)^{\bar{c},\varepsilon} = g^{\bar{c},\varepsilon}-\lambda$, this implies $g^{\bar{c},\varepsilon}-\delta \leqslant (g')^{\bar{c},\varepsilon} \leqslant g^{\bar{c},\varepsilon}+\delta$. Hence $\|g^{\bar{c},\varepsilon}-(g')^{\bar{c},\varepsilon}\|_\infty \leqslant \|g-g'\|_\infty$, and the same argument applies to $f\mapsto f^{c,\varepsilon}$.
Let $(\tilde{f},\tilde{g}) = \mathcal{T}_{\mathrm{GS}}(f,g)$ and $(\tilde{f}',\tilde{g}') = \mathcal{T}_{\mathrm{GS}}(f',g')$. With $D = d_\infty((f,g),(f',g'))$, the previous Lipschitz estimate gives $\|\tilde{f}-\tilde{f}'\|_\infty \leqslant \rho\|g-g'\|_\infty \leqslant \rho D$, and then $\|\tilde{g}-\tilde{g}'\|_\infty \leqslant \rho\|\tilde{f}-\tilde{f}'\|_\infty \leqslant \rho^2 D \leqslant \rho D$. This proves the contraction. The Banach fixed-point theorem gives existence, uniqueness and the displayed linear rate. $\square$

**Partial optimal transport.** $0\leqslant m \leqslant \min\{\alpha(\mathcal{X}),\beta(\mathcal{Y})\}$,

$$\mathrm{POT}_m(\alpha,\beta) := \inf_{\substack{\pi\in\mathcal{M}_+(\mathcal{X}\times\mathcal{Y})\\ \pi_1\leqslant\alpha,\ \pi_2\leqslant\beta\\ \pi(\mathcal{X}\times\mathcal{Y})=m}} \int c\,\mathrm{d}\pi.$$

Thus only a submeasure of $\alpha$ is transported onto a submeasure of $\beta$; the remaining mass is left unmatched. The corresponding Lagrangian form is obtained by adding total-variation penalties.
$\inf_{\pi\in\mathcal{M}_+(\mathcal{X}\times\mathcal{Y})}\int c\,\mathrm{d}\pi + \lambda\|\alpha-\pi_1\|_{\mathrm{TV}} + \lambda\|\beta-\pi_2\|_{\mathrm{TV}}$. Assume $c\geqslant 0$. Allowing transported marginals larger than the available marginals does not improve the value: excess transported mass can be trimmed, which decreases the transport cost and cannot increase the sum of the two total-variation penalties. Hence an optimal plan may be chosen with $\pi_1\leqslant\alpha$ and $\pi_2\leqslant\beta$. If $m = \pi(\mathcal{X}\times\mathcal{Y})$, the penalty then reduces to
$\lambda\big(\alpha(\mathcal{X})-m\big) + \lambda\big(\beta(\mathcal{Y})-m\big)$,
**Proposition 10.6** (TV penalization selects fixed-mass partial OT)**.** *Assume that $c\geqslant 0$ and set $A=\alpha(\mathcal{X})$, $B=\beta(\mathcal{Y})$, $M=\min(A,B)$. For $\lambda\geqslant 0$, define the TV-penalized partial value*

$$\mathrm{POT}^{\mathrm{TV}}_\lambda(\alpha,\beta) := \inf_{\substack{\pi\in\mathcal{M}_+(\mathcal{X}\times\mathcal{Y})\\ \pi_1\leqslant\alpha,\ \pi_2\leqslant\beta}}\left\{\int c\,\mathrm{d}\pi + \lambda\big(A+B-2\pi(\mathcal{X}\times\mathcal{Y})\big)\right\}.$$

*Then*
$$\mathrm{POT}^{\mathrm{TV}}_\lambda(\alpha,\beta)=\lambda(A+B)+\inf_{0\leqslant m\leqslant M}\left\{\mathrm{POT}_m(\alpha,\beta)-2\lambda m\right\}.$$
*If $\pi_\lambda$ minimizes $\mathrm{POT}^{\mathrm{TV}}_\lambda$ and $m_\lambda=\pi_\lambda(\mathcal{X}\times\mathcal{Y})$, then $m_\lambda$ minimizes $m\mapsto \mathrm{POT}_m(\alpha,\beta)-2\lambda m$ and $\pi_\lambda$ minimizes $\mathrm{POT}_{m_\lambda}(\alpha,\beta)$. Conversely, if $2\lambda$ is a subgradient at $m$ of the convex function $m\mapsto \mathrm{POT}_m(\alpha,\beta)$ extended by $+\infty$ outside $[0,M]$, then every minimizer of $\mathrm{POT}_m(\alpha,\beta)$ is a minimizer of $\mathrm{POT}^{\mathrm{TV}}_\lambda(\alpha,\beta)$. In particular, every prescribed mass $m\in[0,M]$ is selected by at least one multiplier $\lambda\geqslant 0$, possibly together with a whole interval of masses when the value function has a flat exposed face.*

*Proof.* Write $V(m)=\mathrm{POT}_m(\alpha,\beta)$. The function $V$ is convex on $[0,M]$: if $\pi_0,\pi_1$ are admissible competitors of masses $m_0,m_1$, then $(1-t)\pi_0+t\pi_1$ is admissible for mass $(1-t)m_0+tm_1$ and has the corresponding convex-combination cost. It is also nondecreasing because any competitor of mass $m_1>m_0$ can be rescaled by $m_0/m_1$ to give a competitor of mass $m_0$ with no larger cost.
For any admissible $\pi$ in the definition of $\mathrm{POT}^{\mathrm{TV}}_\lambda$, let $m=\pi(\mathcal{X}\times\mathcal{Y})$. The plan is admissible for $V(m)$, hence its objective is at least $V(m)+\lambda(A+B-2m)$. Taking the infimum over $\pi$ gives the lower bound by the displayed right-hand side. Conversely, for each $m$ and each minimizing sequence for $V(m)$, the same plans are admissible in $\mathrm{POT}^{\mathrm{TV}}_\lambda$ and have objective tending to $V(m)+\lambda(A+B-2m)$. Taking the infimum over $m$ gives the reverse inequality.
The optimizer statement follows from the same inequalities. If $\pi_\lambda$ minimizes the penalized problem with mass $m_\lambda$, then equality in the lower bound forces both $V(m_\lambda)-2\lambda m_\lambda$ to be minimal over $[0,M]$ and $\pi_\lambda$ to be optimal at fixed mass $m_\lambda$. Conversely, $2\lambda\in\partial V(m)$ means $V(m')-2\lambda m'\geqslant V(m)-2\lambda m\quad$ for all $m'\in[0,M]$. Thus any minimizer of $V(m)$ attains the penalized infimum. Since a finite convex function on an interval has a nonempty subdifferential at every point after adding the normal cone of the constraint interval, such a $\lambda$ exists for every $m\in[0,M]$; monotonicity of $V$ ensures that one may choose $\lambda\geqslant 0$. ☐

**Proposition 10.7** (Metric case of fixed-mass partial OT)**.** *Let $(\mathcal{X},d)$ be a metric space, let $p\geqslant 1$, set $c=d^p$, and fix $m>0$. On the class of nonnegative measures with total mass exactly $m$ and finite pth moment, $\mathcal{M}^p_m(\mathcal{X}):=\left\{\alpha\in\mathcal{M}_+(\mathcal{X})\,;\,\alpha(\mathcal{X})=m,\ \int d(x_0,x)^p\mathrm{d}\alpha(x)<+\infty\right\}$ for one, hence any, base point $x_0$, the quantity $d_m(\alpha,\beta):=\mathrm{POT}_m(\alpha,\beta)^{1/p}$ is a distance. More precisely, if $\bar\alpha=\alpha/m$ and $\bar\beta=\beta/m$, then $d_m(\alpha,\beta)=m^{1/p}\,\mathcal{W}_p(\bar\alpha,\bar\beta)$. If instead $\mathrm{POT}_m$ is considered on a larger class of measures with mass at least $m$, then it is not a distance in general for $m>0$: different measures can share a common submeasure of mass $m$, giving zero partial cost.*

*Proof.* If $\alpha(\mathcal{X})=\beta(\mathcal{X})=m$ and $\pi$ is admissible in $\mathrm{POT}_m(\alpha,\beta)$, then $\pi_1\leqslant\alpha$ and both measures have mass $m$, hence $\pi_1=\alpha$. Similarly $\pi_2=\beta$. Thus the admissible plans are exactly the ordinary couplings between $\alpha$ and $\beta$. Dividing such a coupling by $m$ gives a probability coupling between $\bar\alpha$ and $\bar\beta$, and conversely multiplying any coupling of $\bar\alpha,\bar\beta$ by $m$ gives a coupling of $\alpha,\beta$. Therefore $\mathrm{POT}_m(\alpha,\beta)=m\,\mathcal{W}_p(\bar\alpha,\bar\beta)^p$. The metric properties of $d_m$ follow from those of $\mathcal{W}_p$ in Proposition 3.29.
For the last claim, take three distinct points $x,y,z$ in a metric space and a number $r>0$. The measures $\alpha=m\delta_x+r\delta_y$ and $\beta=m\delta_x+r\delta_z$ are different but the partial plan $m\delta_{(x,x)}$ is admissible and has zero cost. Hence $\mathrm{POT}_m(\alpha,\beta)=0$ although $\alpha\neq\beta$. ☐

Thus the TV-penalized formulation is the Lagrangian envelope of constrained partial OT: increasing $\lambda$ rewards transporting more mass, while small $\lambda$ makes deletion and creation cheaper. As the prescribed transported mass decreases, the plan keeps only the least costly facing portions of the two marginals.

### 10.2 Sliced Wasserstein Distances

Sliced distances replace high-dimensional OT by many one-dimensional projected problems.

**Projection principle.** For $\theta\in\mathbb{S}^{d-1}$, write $P_\theta(x)=\langle\theta,\,x\rangle$. The push-forwards $(P_\theta)_\sharp\alpha$ and $(P_\theta)_\sharp\beta$ are one-dimensional measures.
**Spherical averaging.**

**Definition 10.8** (Sliced Wasserstein distance)**.** Let $\sigma$ be the uniform probability measure on the sphere $\mathbb{S}^{d-1}$. The sliced $p$-Wasserstein distance is
$$\mathrm{SW}_p(\alpha,\beta)^p:=\int_{\mathbb{S}^{d-1}}\mathcal{W}_p\left((P_\theta)_\sharp\alpha,(P_\theta)_\sharp\beta\right)^p\mathrm{d}\sigma(\theta). \tag{10.6}$$

**Proposition 10.9** (Metric properties of sliced Wasserstein)**.** *For $p\geqslant 1$, $\mathrm{SW}_p$ is a distance on $\mathcal{P}_p(\mathbb{R}^d)$. Moreover, $\mathrm{SW}_p$ metrizes weak convergence together with convergence of the pth moment. Finally, $\mathrm{SW}_p(\alpha,\beta)\leqslant\mathcal{W}_p(\alpha,\beta)$, and, for $p=2$ with the uniform probability measure on the sphere, $\mathrm{SW}_2(\alpha,\beta)^2\leqslant\frac{1}{d}\mathcal{W}_2(\alpha,\beta)^2$. Conversely, if $d\geqslant 2$ and $\alpha,\beta\in\mathcal{P}_1(\mathbb{R}^d)$ are supported in $B_R:=\left\{x\in\mathbb{R}^d\,;\,\|x\|\leqslant R\right\}$, then there is a constant $C_d$ depending only on $d$ such that $\mathcal{W}_1(\alpha,\beta)\leqslant C_dR^{\frac{d}{d+1}}\,\mathrm{SW}_1(\alpha,\beta)^{\frac{1}{d+1}}$. Moreover, the sharp dimension-dependent comparison for $p=1$ has the form $\mathcal{W}_1(\alpha,\beta)\leqslant c_dR^{\frac{d-1}{d}}\,\mathrm{SW}_1(\alpha,\beta)^{\frac{1}{d}}$.*

*Proof.* Non-negativity and symmetry follow from the one-dimensional Wasserstein distance. For the triangle inequality, apply the triangle inequality of $\mathcal{W}_p$ for each direction $\theta$ and then Minkowski's inequality in $L^p(\mathbb{S}^{d-1})$.
If $\mathrm{SW}_p(\alpha,\beta)=0$, then $(P_\theta)_\sharp\alpha=(P_\theta)_\sharp\beta$ for almost every direction. By continuity of characteristic functions this holds for all directions, and the Cramér–Wold theorem implies $\alpha=\beta$. This proves separation.
The bound $\mathrm{SW}_p\leqslant\mathcal{W}_p$ follows because $P_\theta$ is 1-Lipschitz, so $\mathcal{W}_p((P_\theta)_\sharp\alpha,(P_\theta)_\sharp\beta)\leqslant\mathcal{W}_p(\alpha,\beta)$ for every $\theta$. For $p=2$, using any coupling $\pi$ between $\alpha$ and $\beta$, $\int_{\mathbb{S}^{d-1}}\int|\langle x-y,\,\theta\rangle|^2\mathrm{d}\pi(x,y)\mathrm{d}\sigma(\theta)=\frac{1}{d}\int\|x-y\|^2\mathrm{d}\pi(x,y)$. Optimizing over $\pi$ gives the sharper inequality.
The weak-convergence statement follows from the same Cramér–Wold mechanism plus the moment condition: convergence in $\mathrm{SW}_p$ gives convergence of almost all one-dimensional projections and tightness of the $p$th moments; conversely, weak convergence with $p$th-moment convergence implies convergence of projected $\mathcal{W}_p$ distances and dominated convergence on the sphere.
The reverse estimates on compact sets are deeper and use the Radon structure of slices. Bonnotte's Lemma 5.1.4 gives the first bound by mollifying a Kantorovich–Rubinstein test function, representing the regularized test through one-dimensional projections, bounding the resulting action by $\mathrm{SW}_1$, and optimizing the smoothing scale. The scaling in $R$ follows by applying the unit-ball estimate to $(x\mapsto x/R)_\sharp\alpha$ and $(x\mapsto x/R)_\sharp\beta$. The sharper exponent $1/d$ is the main comparison theorem of Carlier, Figalli, Mérigot and Wang; their examples also show that this exponent cannot be improved in general. ☐

**Proposition 10.10** (First-order comparison along Brenier perturbations)**.** *Let $\alpha\in\mathcal{P}_2(\mathbb{R}^d)$ be absolutely continuous, and let $\varphi\in C^2(\mathbb{R}^d)$ be such that $\nabla\varphi\in L^2(\alpha;\mathbb{R}^d)$ and $\nabla^2\varphi$ is bounded. For $|t|$ small enough, set $T_t(x):=x+t\nabla\varphi(x),\qquad \alpha_t:=(T_t)_\sharp\alpha$. Then $T_t$ is the quadratic Brenier map from $\alpha$ to $\alpha_t$ for $|t|$ small enough, and $\mathcal{W}_2(\alpha,\alpha_t)^2=t^2\int_{\mathbb{R}^d}\|\nabla\varphi(x)\|^2\mathrm{d}\alpha(x)$. Moreover,*

$\mathrm{SW}_2(\alpha,\alpha_t)^2 \leqslant \frac{t^2}{d}\int_{\mathbb{R}^d}\|\nabla\varphi(x)\|^2\mathrm{d}\alpha(x)$. *Thus the sliced infinitesimal speed is at most $d^{-1/2}$ times the Wasserstein infinitesimal speed along smooth Brenier perturbations.*

*Proof.* The map $T_t$ is the gradient of $\psi_t(x) := \frac{1}{2}\|x\|^2 + t\varphi(x)$. Since $\nabla^2\varphi$ is bounded, $\nabla^2\psi_t = \mathrm{Id} + t\nabla^2\varphi$ is positive semidefinite for $|t|$ small enough. Hence $\psi_t$ is convex and $T_t = \nabla\psi_t$ is a Brenier map. Its graph is cyclically monotone, so the Brenier optimality criterion gives the optimality of $(\mathrm{Id}, T_t)_\sharp\alpha$, and $\mathcal{W}_2(\alpha,\alpha_t)^2 = \int\|x - T_t(x)\|^2\mathrm{d}\alpha(x) = t^2\int\|\nabla\varphi(x)\|^2\mathrm{d}\alpha(x)$. For a fixed direction $\theta$, the coupling $\left(P_\theta, P_\theta\circ T_t\right)_\sharp\alpha$ is admissible between $(P_\theta)_\sharp\alpha$ and $(P_\theta)_\sharp\alpha_t$. Therefore $\mathcal{W}_2\left((P_\theta)_\sharp\alpha,(P_\theta)_\sharp\alpha_t\right)^2 \leqslant t^2\int|\langle\theta,\,\nabla\varphi(x)\rangle|^2\,\mathrm{d}\alpha(x)$. Integrating over the sphere and using $\int_{\mathbb{S}^{d-1}}|\langle\theta,\,w\rangle|^2\,\mathrm{d}\sigma(\theta) = \frac{1}{d}\|w\|^2$ gives the sliced estimate. □

**Proposition 10.11** (Translations separate sliced length from $\mathcal{W}_2$)**.** *Let $\tau_h(x) = x + h$ for $h \in \mathbb{R}^d$. For any $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$, $\mathrm{SW}_2^{\mathrm{len}}\left(\alpha,(\tau_h)_\sharp\alpha\right) = \mathrm{SW}_2\left(\alpha,(\tau_h)_\sharp\alpha\right) = \frac{\|h\|}{\sqrt{d}}$, whereas $\mathcal{W}_2\left(\alpha,(\tau_h)_\sharp\alpha\right) = \|h\|$.*

*Proof.* For each direction $\theta$, the projected measure $(P_\theta)_\sharp(\tau_h)_\sharp\alpha$ is the translate of $(P_\theta)_\sharp\alpha$ by the scalar $\langle\theta,\,h\rangle$. Hence $\mathcal{W}_2\left((P_\theta)_\sharp\alpha,(P_\theta)_\sharp(\tau_h)_\sharp\alpha\right)^2 = |\langle\theta,\,h\rangle|^2$. After integration on the sphere, $\mathrm{SW}_2\left(\alpha,(\tau_h)_\sharp\alpha\right)^2 = \int_{\mathbb{S}^{d-1}}|\langle\theta,\,h\rangle|^2\,\mathrm{d}\sigma(\theta) = \frac{\|h\|^2}{d}$. For the translation curve $\alpha_s = (\tau_{sh})_\sharp\alpha$, the same computation gives $\mathrm{SW}_2(\alpha_s,\alpha_t) = |s-t|\frac{\|h\|}{\sqrt{d}}$. Every partition therefore has $\mathrm{SW}_2$-length exactly $\|h\|/\sqrt{d}$, which gives the upper bound on $\mathrm{SW}_2^{\mathrm{len}}$. The opposite inequality follows from the general fact that any metric distance is bounded by the length of every connecting curve.
Finally, the transport map $\tau_h = \nabla(\frac{1}{2}\|x\|^2 + \langle h,\,x\rangle)$ is the gradient of a convex function. By the Brenier cyclic-monotonicity criterion, the translation coupling is optimal for the quadratic cost, and hence the $\mathcal{W}_2$ distance is $\|h\|$. □

**Definition 10.12** (Max-sliced Wasserstein)**.** The max-sliced distance replaces the average over directions by the most discriminating one: $\mathrm{MaxSW}_p(\alpha,\beta) := \sup_{\theta\in\mathbb{S}^{d-1}}\mathcal{W}_p((P_\theta)_\sharp\alpha,(P_\theta)_\sharp\beta)$.

**Subspace-sliced variants.**

**Definition 10.13** (Subspace-sliced Wasserstein)**.** Fix $1 \leqslant k \leqslant d$. Subspace-sliced variants replace one-dimensional lines by $k$-dimensional orthogonal projections. If $U \in \mathbb{R}^{d\times k}$ satisfies $U^\top U = \mathrm{Id}_k$, then $\mathrm{SW}_{p,k}(\alpha,\beta)^p := \int\mathcal{W}_p((U^\top)_\sharp\alpha,(U^\top)_\sharp\beta)^p\mathrm{d}U$, where $\mathrm{d}U$ denotes the normalized invariant measure on the Stiefel manifold $\mathrm{St}(d,k) = \{U \in \mathbb{R}^{d\times k} : U^\top U = \mathrm{Id}_k\}$, and $\mathrm{MaxSW}_{p,k}(\alpha,\beta) := \sup_{U^\top U=\mathrm{Id}_k}\mathcal{W}_p((U^\top)_\sharp\alpha,(U^\top)_\sharp\beta)$. The case $k = 1$ recovers ordinary sliced and max-sliced Wasserstein, while $k = d$ recovers the original Wasserstein distance.

**Proposition 10.14** (Basic bounds for sliced variants)**.** *Let $p \geqslant 1$ and let $\alpha,\beta \in \mathcal{P}_p(\mathbb{R}^d)$. With normalized spherical and Stiefel measures, $\mathrm{SW}_p(\alpha,\beta) \leqslant \mathrm{MaxSW}_p(\alpha,\beta) \leqslant \mathcal{W}_p(\alpha,\beta)$. For $k$-dimensional subspace projections, $\mathrm{SW}_{p,k}(\alpha,\beta) \leqslant \mathrm{MaxSW}_{p,k}(\alpha,\beta) \leqslant \mathcal{W}_p(\alpha,\beta)$.*

*Proof.* The first inequality in each line follows because an $L^p$ average over a probability space is bounded by the corresponding supremum. The second inequality follows because orthogonal projections are 1-Lipschitz: pushing any admissible coupling between $\alpha$ and $\beta$ through a projection gives an admissible coupling for the projected measures with no larger transport cost. Optimizing over couplings and then averaging or maximizing over the projection gives the result. □

**Min-sliced lifted transport plans.** $\pi_\theta = \frac{1}{n}\sum_{i=1}^n\delta_{(x_i,y_{\sigma_\theta(i)})}$ $\mathrm{MSWGG}_2(\alpha,\beta)^2 := \min_{\theta\in\mathbb{S}^{d-1}}\int\|x-y\|^2\mathrm{d}\pi_\theta(x,y)$. $\mathcal{W}_2^2(\alpha,\beta) \leqslant \int\|x-y\|^2\mathrm{d}\pi_\theta(x,y)$, $\qquad\mathcal{W}_2^2(\alpha,\beta) \leqslant \mathrm{MSWGG}_2(\alpha,\beta)^2$.

### 10.3 Quotient Wasserstein and Wasserstein–Procrustes

This compact subsection separates the abstract quotient construction from the rigid-motion example.

**Quotient distances.** The first block defines the quotient metric before specializing it.

**Definition 10.15** (Quotient Wasserstein distance)**.** Let a group $\mathcal{G}$ act on a metric space $(\mathcal{X},d)$ by isometries, and assume that the action preserves finite $p$th moments. Write $g_\sharp\alpha$ for the push-forward of a measure $\alpha$ by $g \in \mathcal{G}$, and $[\alpha] := \{g_\sharp\alpha : g \in \mathcal{G}\}$ for its orbit. The quotient $p$-Wasserstein distance between the orbits of $\alpha,\beta \in \mathcal{P}_p(\mathcal{X})$ is

$$\mathcal{W}_{p,\mathcal{G}}([\alpha],[\beta]) := \inf_{g,h\in\mathcal{G}}\mathcal{W}_p(g_\sharp\alpha,h_\sharp\beta) = \inf_{g\in\mathcal{G}}\mathcal{W}_p(\alpha,g_\sharp\beta).$$

**Proposition 10.16** (Metric property on the quotient)**.** *If $\mathcal{G}$ acts by isometries and preserves $\mathcal{P}_p(\mathcal{X})$, then $\mathcal{W}_{p,\mathcal{G}}$ is a pseudo-distance on $\mathcal{P}_p(\mathcal{X})$. If, in addition, the infimum is attained between any two orbits and all orbits are closed, then it is a distance on the orbit space $\mathcal{P}_p(\mathcal{X})/\mathcal{G}$. This applies in particular to continuous actions of compact groups.*

*Proof.* Isometry of the action gives $\mathcal{W}_p(g_\sharp\alpha,g_\sharp\nu) = \mathcal{W}_p(\alpha,\nu)$, which proves the second formula in Definition 10.15. Non-negativity and symmetry are inherited from $\mathcal{W}_p$. For the triangle inequality, choose $g_1,g_2 \in \mathcal{G}$. Then

$$\mathcal{W}_p(\alpha,(g_1)_\sharp\beta) + \mathcal{W}_p(\beta,(g_2)_\sharp\gamma) = \mathcal{W}_p(\alpha,(g_1)_\sharp\beta) + \mathcal{W}_p((g_1)_\sharp\beta,(g_1g_2)_\sharp\gamma).$$

Applying the triangle inequality for $\mathcal{W}_p$ and then taking infima over $g_1,g_2$ gives the result. If the infimum is attained and the quotient distance is zero, there exist $g,h \in \mathcal{G}$ such that $\mathcal{W}_p(g_\sharp\alpha,h_\sharp\beta) = 0$, hence $g_\sharp\alpha = h_\sharp\beta$. Closed orbits then imply $[\alpha] = [\beta]$. For compact groups, continuity of the action and compactness give the required attainment. □

**Rigid motions and Wasserstein–Procrustes.** $\inf_{R\in\mathrm{O}(d),\,t\in\mathbb{R}^d,\,\pi\in\mathcal{U}(\alpha,\beta)}\int\|Rx+t-y\|^2\mathrm{d}\pi(x,y)$. This is an extrinsic counterpart of the Gromov–Wasserstein viewpoint developed in Section 12.3. Procrustes alignment searches only over ambient rigid motions, whereas GW is invariant under all measure-preserving isometries of the metric-measure spaces.
$\mathcal{GW}((\mathbb{R}^d,\|\cdot\|,\alpha),(\mathbb{R}^d,\|\cdot\|,\beta)) = \mathcal{GW}((\mathbb{R}^d,\|\cdot\|,g_\sharp\alpha),(\mathbb{R}^d,\|\cdot\|,h_\sharp\beta))$. $\mathcal{GW}((\mathbb{R}^d,\|\cdot\|,g_\sharp\alpha),(\mathbb{R}^d,\|\cdot\|,h_\sharp\beta)) \leqslant 2\,\mathcal{W}_p(g_\sharp\alpha,h_\sharp\beta)$.

$$\mathcal{GW}((\mathbb{R}^d,\|\cdot\|,\alpha),(\mathbb{R}^d,\|\cdot\|,\beta)) \leqslant 2\,\mathcal{W}_{p,\mathrm{E}(d)}([\alpha],[\beta]), \tag{10.7}$$

for the same exponent $p$ and $\Delta(u,v) = |u-v|$. Thus a good Wasserstein–Procrustes registration forces small GW distortion.

$$\mathrm{P}^{(k)} \in \operatorname{argmin}_{\mathrm{P}\in \mathrm{U(a,b)}} \sum_{i,j} \mathrm{P}_{ij}\|R^{(k)}x_i + t^{(k)} - y_j\|^2. \tag{10.8}$$

$$(R^{(k+1)}, t^{(k+1)}) \in \operatorname{argmin}_{R\in \mathrm{O}(d),\, t\in\mathbb{R}^d} \sum_{i,j} \mathrm{P}^{(k)}_{ij}\|Rx_i + t - y_j\|^2, \tag{10.9}$$

**Proposition 10.17** (Rigid update for a fixed coupling)**.** *Let $\alpha = \sum_i a_i\delta_{x_i}$ and $\beta = \sum_j b_j\delta_{y_j}$, and fix $\mathrm{P} \in \mathrm{U(a,b)}$. Define $\bar{x} = \sum_i a_i x_i, \qquad \bar{y} = \sum_j b_j y_j, \qquad M_{\mathrm{P}} = \sum_{i,j} \mathrm{P}_{ij}(y_j - \bar{y})(x_i - \bar{x})^\top$. If $M_{\mathrm{P}} = U\Sigma V^\top$ is a singular value decomposition, then the minimizers over the full orthogonal group of $\min_{R\in\mathrm{O}(d),\, t\in\mathbb{R}^d} \sum_{i,j} \mathrm{P}_{ij}\|Rx_i + t - y_j\|^2$ are given by $R^\star = UV^\top$ and $t^\star = \bar{y} - R^\star\bar{x}$, up to the usual non-uniqueness when $M_{\mathrm{P}}$ is rank deficient. If one imposes $R \in \mathrm{SO}(d)$, set $D = \mathrm{diag}(1,\dots,1,\det(UV^\top)), \qquad R^\star = UDV^\top, \qquad t^\star = \bar{y} - R^\star\bar{x}$.*

*Proof.* For fixed $R$, differentiating with respect to $t$ gives $t = \bar{y} - R\bar{x}$. Substituting this value centers the variables and leaves $\sum_{i,j} \mathrm{P}_{ij}\|R(x_i - \bar{x}) - (y_j - \bar{y})\|^2$. The two quadratic terms independent of $R$ are fixed, so the problem is equivalent to maximizing $\sum_{i,j} \mathrm{P}_{ij}\langle R(x_i - \bar{x}),\, y_j - \bar{y}\rangle = \mathrm{tr}(R^\top M_{\mathrm{P}})$. Von Neumann's trace inequality gives $\mathrm{tr}(R^\top U\Sigma V^\top) \leqslant \mathrm{tr}(\Sigma)$ for $R \in \mathrm{O}(d)$, with equality for $R = UV^\top$. Under the constraint $R \in \mathrm{SO}(d)$, the same argument applies with the additional determinant constraint on $U^\top RV$, which gives the displayed diagonal correction. □

## 10.4 Vector Quantiles and Linear Optimal Transport

This compact subsection records the linearized coordinates induced by a reference measure.

**Vector quantiles.** Assume that $\rho$ is absolutely continuous. For a target law $\alpha$ with finite second moment, its vector quantile relative to $\rho$ is the Brenier map
$T_\alpha = \nabla\varphi_\alpha, \qquad (T_\alpha)_\sharp\rho = \alpha,\ \min_{T_\sharp\rho=\alpha} \int \|x - T(x)\|^2 \mathrm{d}\rho(x)$.

**Linearized Wasserstein coordinates.**

$$\alpha \mapsto T_\alpha - \mathrm{Id} \in L^2(\rho;\mathbb{R}^d), \qquad \mathrm{LOT}_\rho(\alpha,\beta) = \|T_\alpha - T_\beta\|_{L^2(\rho)}. \tag{10.10}$$

If one of the two targets equals the reference, the linearized distance is exact: for instance, $\mathrm{LOT}_\rho(\rho,\alpha) = \|T_\alpha - \mathrm{Id}\|_{L^2(\rho)} = \mathcal{W}_2(\rho,\alpha)$. For a family $(\alpha_s)_s$ with weights $(\lambda_s)_s$, the linearized barycenter is obtained by averaging maps,
$\bar{T} = \sum_s \lambda_s T_{\alpha_s}, \qquad \bar{\alpha}_{\mathrm{LOT}} = \bar{T}_\sharp\rho$. This is exact in one dimension, where quantile functions linearize $\mathcal{W}_2$, and it is especially useful when many barycenters with changing weights must be evaluated quickly.

**Principal components in linear OT coordinates.** $z_i \coloneqq T_{\alpha_i} - \mathrm{Id}, \qquad \bar{z} \coloneqq \frac{1}{N}\sum_{i=1}^N z_i$. $\mathcal{C}_{\mathrm{LOT}}h \coloneqq \frac{1}{N}\sum_{i=1}^N \langle z_i - \bar{z},\, h\rangle_{L^2(\rho)}(z_i - \bar{z})$, $e_k = \frac{1}{\sqrt{N\lambda_k}} \sum_{i=1}^N v_i^{(k)}(z_i - \bar{z})$ $a_{i,k} \coloneqq \langle z_i - \bar{z},\, e_k\rangle_{L^2(\rho)}$. $T_a(x) \coloneqq x + \bar{z}(x) + \sum_{k=1}^m a_k e_k(x), \qquad \alpha_a \coloneqq (T_a)_\sharp\rho$.

**Proposition 10.18** (Local stability of linear OT)**.** *Assume that the measures are supported on a fixed convex compact set, with densities bounded above and below, and that the Brenier maps from $\rho$ are regular. Then, for $\alpha, \beta$ in a sufficiently small regular neighborhood of $\rho$, $\mathcal{W}_2(\alpha,\beta) \leqslant \mathrm{LOT}_\rho(\alpha,\beta)$ and $\mathrm{LOT}_\rho(\alpha,\beta) \leqslant C\,\mathcal{W}_2(\alpha,\beta)^\eta$ for constants $C > 0$ and $\eta \in (0,1]$ depending on regularity.*

*Proof.* The first inequality is immediate: $(T_\alpha, T_\beta)_\sharp\rho$ is a feasible coupling between $\alpha$ and $\beta$. The reverse local estimate is a standard stability statement for the Monge–Ampère equation under the stated regularity assumptions: changes in the target measure control changes in the Brenier potential in Hölder norms, hence control $T_\alpha - T_\beta$ in $L^2(\rho)$. In simple one-dimensional settings, quantile functions make this exact with $\eta = 1$. In several dimensions one should not read the statement as a global Lipschitz estimate in $\mathcal{W}_2$. Quantitative stability results for semi-discrete and Monge–Ampère maps give Hölder exponents depending on the dimension, density bounds, support geometry and regularity; see for instance the estimates of Mérigot, Delalande and Chazal. Under stronger smooth perturbations of uniformly convex smooth densities, elliptic regularity can give Lipschitz dependence in stronger function norms, but converting those controls to Wasserstein perturbations generally loses powers. □

## 10.5 Spectral and Robust Wasserstein Distances

**Definition 10.19** (Monotone spectral gauge)**.** A monotone spectral gauge on positive semidefinite matrices is a convex, positively 1-homogeneous map $\gamma : \mathbb{S}^d_+ \to \mathbb{R}_+$ such that $\gamma(M) = 0$ only for $M = 0$, $\gamma(QMQ^\top) = \gamma(M)$ for every orthogonal matrix $Q$, and $0 \preceq M \preceq N \implies \gamma(M) \leqslant \gamma(N)$.

**Definition 10.20** (Spectral Wasserstein distance)**.** Let $\gamma$ be a monotone spectral gauge. For a coupling $\pi \in \mathcal{U}(\alpha,\beta)$, define its displacement covariance $M_\pi \coloneqq \int (x-y)(x-y)^\top \mathrm{d}\pi(x,y)$. The spectral Wasserstein distance associated with $\gamma$ is

$$\mathcal{W}_\gamma(\alpha,\beta)^2 \coloneqq \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \gamma(M_\pi). \tag{10.11}$$

The special case $\gamma(M) = \mathrm{tr}(M)$ gives the usual quadratic Wasserstein distance $\mathcal{W}_2$. The spectral gauge $\gamma(M) = \lambda_{\max}(M)$ instead measures the worst transported variance direction.

$$\mathcal{W}_{2,A}(\alpha,\beta)^2 \coloneqq \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \int (x-y)^\top A(x-y)\mathrm{d}\pi(x,y) = \mathcal{W}_2((A^{1/2})_\sharp\alpha, (A^{1/2})_\sharp\beta)^2. \tag{10.12}$$

$$\mathcal{B}_\gamma \coloneqq \{A \succeq 0\ ;\ \mathrm{tr}(AM) \leqslant \gamma(M) \quad \text{for all } M \succeq 0\}, \tag{10.13}$$

**Proposition 10.21** (Robust representation and metric equivalence)**.** *Assume, for simplicity, that the measures are compactly supported and that $\gamma$ is closed and finite on the positive semidefinite cone. Then $\mathcal{W}_\gamma(\alpha,\beta)^2 = \sup_{A\in\mathcal{B}_\gamma} \mathcal{W}_{2,A}(\alpha,\beta)^2$. If there exist constants $0 < a \leqslant b < +\infty$ such that $a\mathrm{Id} \in \mathcal{B}_\gamma$ and $\mathcal{B}_\gamma \subset \{A\ ;\ 0 \preceq A \preceq b\mathrm{Id}\}$, equivalently $a\,\mathrm{tr}(M) \leqslant \gamma(M) \leqslant b\,\mathrm{tr}(M) \qquad (M \succeq 0)$, then $\sqrt{a}\,\mathcal{W}_2(\alpha,\beta) \leqslant \mathcal{W}_\gamma(\alpha,\beta) \leqslant \sqrt{b}\,\mathcal{W}_2(\alpha,\beta)$. In particular, $\mathcal{W}_\gamma$ is a distance. When $\gamma$ is the restriction of a norm to the positive semidefinite cone, these bounds hold automatically in finite dimension, so $\mathcal{W}_\gamma$ is equivalent to $\mathcal{W}_2$ on measures with finite second moments.*

*Proof.* Using the polar representation of $\gamma$, $\mathcal{W}_\gamma(\alpha,\beta)^2 = \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \sup_{A\in\mathcal{B}_\gamma} \operatorname{tr}(AM_\pi)$. The coupling set is convex and compact for weak convergence under compact support. Since $\gamma$ is a finite gauge on a finite-dimensional cone and vanishes only at the origin, it is equivalent to the trace norm on the slice $\operatorname{tr}(M)=1$, so $\mathcal{B}_\gamma$ is convex and compact. The map $(\pi, A)\mapsto \operatorname{tr}(AM_\pi)$ is affine in each variable and continuous. Sion's minimax theorem gives

$$\inf_{\pi}\sup_{A\in\mathcal{B}_\gamma} \operatorname{tr}(AM_\pi) = \sup_{A\in\mathcal{B}_\gamma}\inf_{\pi} \operatorname{tr}(AM_\pi) = \sup_{A\in\mathcal{B}_\gamma} \mathcal{W}_{2,A}(\alpha,\beta)^2.$$

For fixed $A \succeq 0$, $\mathcal{W}_{2,A}$ is the Wasserstein pseudodistance associated with the seminorm $x\mapsto \|A^{1/2}x\|$. Since all terms are nonnegative, the robust identity also gives $\mathcal{W}_\gamma(\alpha,\beta) = \sup_{A\in\mathcal{B}_\gamma} \mathcal{W}_{2,A}(\alpha,\beta)$. A supremum of pseudodistances is symmetric and satisfies the triangle inequality.
If $a\mathrm{Id}\in\mathcal{B}_\gamma$ and $A\preceq b\mathrm{Id}$ for all $A\in\mathcal{B}_\gamma$, then $a\,\mathcal{W}_2(\alpha,\beta)^2 = \mathcal{W}_{2,a\mathrm{Id}}(\alpha,\beta)^2 \leqslant \mathcal{W}_\gamma(\alpha,\beta)^2 \leqslant b\,\mathcal{W}_2(\alpha,\beta)^2$, which proves definiteness and equivalence with $\mathcal{W}_2$. The equivalence between these operator bounds and $a\operatorname{tr}(M)\leqslant\gamma(M)\leqslant b\operatorname{tr}(M)$ follows directly from the polar formula. In finite dimension, any norm restricted to the positive semidefinite cone is equivalent to the trace norm on that cone. The finite-second-moment case follows by truncation when these norm-equivalence bounds hold. □

**Definition 10.22** (Subspace robust Wasserstein)**.** For $1\leqslant k\leqslant d$, the Paty–Cuturi subspace robust Wasserstein distance is

$$\mathrm{SRW}_{2,k}(\alpha,\beta) := \sup_{U^\top U=\mathrm{Id}_k} \mathcal{W}_2((U^\top)_\sharp\alpha,(U^\top)_\sharp\beta) = \sup_{P^2=P=P^\top,\ \operatorname{tr}(P)=k} \mathcal{W}_{2,P}(\alpha,\beta).$$

$\gamma_k(M) = \sum_{\ell=1}^k \lambda_\ell(M)$, $\mathcal{B}_{\gamma_k} = \{A\ ;\ 0\preceq A\preceq \mathrm{Id},\ \operatorname{tr}(A)\leqslant k\}$. Thus $k=d$ gives $\gamma_d(M)=\operatorname{tr}(M)$ and recovers $\mathcal{W}_2$. The convex hull of rank-$k$ projectors is
$\{A\ ;\ 0\preceq A\preceq\mathrm{Id},\ \operatorname{tr}(A)=k\}$, and, since $M\succeq 0$, the associated support function is the same Ky Fan gauge. Thus $\mathcal{W}_{\gamma_k}$ is the convexified spectral counterpart of $\mathrm{SRW}_{2,k}$, while $\mathrm{SRW}_{2,k}$ keeps the original non-convex rank constraint.

### 10.6 Conditional Wasserstein Distances

**Definition 10.23** (Conditional couplings and conditional OT)**.** Let $(S,\lambda)$ be a standard Borel probability space of conditions and let $(\Omega,d)$ be a Polish metric space. For $p\geqslant 1$, denote by $\mathcal{P}_{p,\lambda}(S\times\Omega)$ the set of probability measures on $S\times\Omega$ whose first marginal is $\lambda$ and whose disintegration $\alpha(\mathrm{d}s,\mathrm{d}x) = \alpha_s(\mathrm{d}x)\lambda(\mathrm{d}s)$ satisfies $\int_S\int_\Omega d(x,x_0)^p\,\mathrm{d}\alpha_s(x)\mathrm{d}\lambda(s) < +\infty$ for some, hence every, $x_0\in\Omega$. For $\alpha,\beta\in\mathcal{P}_{p,\lambda}(S\times\Omega)$, a conditional coupling is a measure $\Pi(\mathrm{d}s,\mathrm{d}x,\mathrm{d}y) = \pi_s(\mathrm{d}x,\mathrm{d}y)\lambda(\mathrm{d}s)$ such that $\pi_s\in\mathcal{U}(\alpha_s,\beta_s)$ for $\lambda$-a.e. $s$. The set of such couplings is denoted $\mathcal{U}_\lambda(\alpha,\beta)$. For a Borel family of lower-semicontinuous costs $c_s:\Omega\times\Omega\to\mathbb{R}\cup\{+\infty\}$ bounded from below, the conditional OT value is

$$\begin{aligned}\mathcal{L}_c^\lambda(\alpha,\beta) &:= \inf_{\Pi\in\mathcal{U}_\lambda(\alpha,\beta)} \int_S\int_{\Omega\times\Omega} c_s(x,y)\mathrm{d}\pi_s(x,y)\mathrm{d}\lambda(s)\\ &= \int_S \inf_{\pi_s\in\mathcal{U}(\alpha_s,\beta_s)}\int_{\Omega\times\Omega} c_s(x,y)\mathrm{d}\pi_s(x,y)\mathrm{d}\lambda(s).\end{aligned} \tag{10.14}$$

**Definition 10.24** (Conditional Wasserstein distance)**.** For $c_s(x,y) = d(x,y)^p$, define

$$\mathcal{W}_{p,\lambda}(\alpha,\beta) := \left(\int_S \mathcal{W}_p^p(\alpha_s,\beta_s)\mathrm{d}\lambda(s)\right)^{1/p}. \tag{10.15}$$

**Proposition 10.25** (Metric property)**.** *For $p\geqslant 1$, $\mathcal{W}_{p,\lambda}$ is a distance on $\mathcal{P}_{p,\lambda}(S\times\Omega)$.*

*Proof.* Non-negativity and symmetry follow from the corresponding properties of $\mathcal{W}_p$ on each fiber. If $\mathcal{W}_{p,\lambda}(\alpha,\beta)=0$, then $\mathcal{W}_p(\alpha_s,\beta_s)=0$ for $\lambda$-a.e. $s$, hence $\alpha_s=\beta_s$ for $\lambda$-a.e. $s$ and the disintegrated measures $\alpha$ and $\beta$ coincide. For the triangle inequality, let $\gamma\in\mathcal{P}_{p,\lambda}(S\times\Omega)$. Since $\mathcal{W}_p(\alpha_s,\gamma_s)\leqslant\mathcal{W}_p(\alpha_s,\beta_s)+\mathcal{W}_p(\beta_s,\gamma_s)$ for a.e. $s$, Minkowski's inequality in $L^p(S,\lambda)$ gives $\mathcal{W}_{p,\lambda}(\alpha,\gamma)\leqslant\mathcal{W}_{p,\lambda}(\alpha,\beta)+\mathcal{W}_{p,\lambda}(\beta,\gamma)$. □

**Proposition 10.26** (Conditional geodesics)**.** *Assume that $(\Omega,d)$ is a geodesic Polish space and let $\alpha^0,\alpha^1\in\mathcal{P}_{p,\lambda}(S\times\Omega)$. For $\lambda$-a.e. $s$, choose a constant-speed $\mathcal{W}_p$ geodesic $t\mapsto\alpha_s^t$ between $\alpha_s^0$ and $\alpha_s^1$, measurably in $s$, and set $\alpha^t(\mathrm{d}s,\mathrm{d}x)=\alpha_s^t(\mathrm{d}x)\lambda(\mathrm{d}s)$. Then $t\mapsto\alpha^t$ is a constant-speed geodesic for $\mathcal{W}_{p,\lambda}$.*

*Proof.* For $0\leqslant r\leqslant t\leqslant 1$, the fiberwise geodesic property gives $\mathcal{W}_p(\alpha_s^r,\alpha_s^t) = (t-r)\,\mathcal{W}_p(\alpha_s^0,\alpha_s^1)$ for $\lambda$-a.e. $s$. Integrating the $p$th power over $S$ gives $\mathcal{W}_{p,\lambda}(\alpha^r,\alpha^t) = (t-r)\,\mathcal{W}_{p,\lambda}(\alpha^0,\alpha^1)$. Hence the conditional curve has constant speed and realizes the distance between its endpoints. In Euclidean space one may take, for each $s$, an optimal plan $\pi_s\in\mathcal{U}(\alpha_s^0,\alpha_s^1)$ and set $\alpha_s^t = ((1-t)\mathrm{p}_1+t\mathrm{p}_2)_\sharp\pi_s$, where $\mathrm{p}_1,\mathrm{p}_2$ are the two coordinate projections. On a general geodesic space, the same construction uses a measurable family of optimal dynamical plans; when geodesics are non-unique, different measurable selections can produce different conditional geodesics. □

## 11 Generalized OT Problems

### 11.1 OT Barycenters

This compact subsection summarizes Wasserstein barycenters and their discrete approximations.

**Fréchet means.** $\min_{\alpha\in\mathcal{M}_+^1(\mathcal{X})} \sum_{s=1}^S \lambda_s\mathcal{L}_c(\alpha,\beta_s)$.

**Fixed-support restriction.** $\beta_s = \sum_{j=1}^{n_s} b_{s,j}\delta_{x_{s,j}}$, $\quad \mathrm{b}_s = (b_{s,j})_{j=1}^{n_s}\in\Sigma_{n_s}$. $(\mathrm{C}_s)_{ij} = c(y_i,x_{s,j})\in\mathbb{R}^{n\times n_s}$.

$$\min_{\mathrm{a}\in\Sigma_n}\sum_{s=1}^S \lambda_s \mathrm{L}_{\mathrm{C}_s}(\mathrm{a},\mathrm{b}_s), \tag{11.1}$$

where $\mathrm{a}=(a_i)_i$ are the weights on the prescribed barycenter support.

**Proposition 11.1** (Convexity of the OT cost)**.** *The map $(\alpha,\beta)\mapsto\mathcal{L}_c(\alpha,\beta)$ is convex.*

*Proof.* Let $(\alpha_0,\beta_0)$ and $(\alpha_1,\beta_1)$ be two pairs of probability measures and let $t\in[0,1]$. For $\eta>0$, choose couplings $\pi_i\in\mathcal{U}(\alpha_i,\beta_i)$ such that $\int c\mathrm{d}\pi_i\leqslant\mathcal{L}_c(\alpha_i,\beta_i)+\eta \qquad (i=0,1)$. Then $\pi_t=(1-t)\pi_0+t\pi_1$ is a coupling between $(1-t)\alpha_0+t\alpha_1$ and $(1-t)\beta_0+t\beta_1$. Hence $\mathcal{L}_c((1-t)\alpha_0+t\alpha_1,(1-t)\beta_0+t\beta_1)\leqslant(1-t)\mathcal{L}_c(\alpha_0,\beta_0)+t\mathcal{L}_c(\alpha_1,\beta_1)+\eta$. Letting $\eta\to0$ gives the claim. □

**One-dimensional case.**

**Proposition 11.2** (Quantile barycenters on the line)**.** *For $\mathcal{X}=\mathbb{R}$ and $c(x,y)=|x-y|^2$, the quantile function of a Wasserstein barycenter is the weighted average of the input quantile functions: $\mathcal{C}_{\alpha^\star}^{-1}(r)=\sum_{s=1}^S\lambda_s\mathcal{C}_{\beta_s}^{-1}(r), \qquad r\in[0,1]$.*

*Proof.* The one-dimensional formula (2.11) gives $\sum_s\lambda_s\mathcal{W}_2^2(\alpha,\beta_s)=\int_0^1\sum_s\lambda_s\left|\mathcal{C}_\alpha^{-1}(r)-\mathcal{C}_{\beta_s}^{-1}(r)\right|^2\mathrm{d}r$. The minimization decouples pointwise in $r$. For each fixed $r$, the minimizer of $z\mapsto\sum_s\lambda_s|z-\mathcal{C}_{\beta_s}^{-1}(r)|^2$ is the weighted average $\sum_s\lambda_s\mathcal{C}_{\beta_s}^{-1}(r)$. This function is nondecreasing because it is a positive weighted sum of nondecreasing quantile functions, hence it is a valid quantile function. □

**Sliced and Radon barycenters.** Slicing gives a scalable surrogate for high-dimensional barycenters by applying the preceding one-dimensional quantile formula in every projection direction. For measures on $\mathbb{R}^d$, define the sliced barycenter by replacing $\mathcal{W}_2$ with $\mathrm{SW}_2$, introduced in Definition 10.8 and interpreted through the Radon transform in Section 10.2:

$$\min_{\alpha\in\mathcal{M}_+^1(\mathbb{R}^d)}\sum_{s=1}^S\lambda_s\,\mathrm{SW}_2(\alpha,\beta_s)^2=\min_{\alpha\in\mathcal{M}_+^1(\mathbb{R}^d)}\int_{\mathbb{S}^{d-1}}\sum_{s=1}^S\lambda_s\,\mathcal{W}_2\left((P_\theta)_\sharp\alpha,(P_\theta)_\sharp\beta_s\right)^2\mathrm{d}\sigma(\theta).$$

$$\min_{(\gamma_\theta)_\theta}\int_{\mathbb{S}^{d-1}}\sum_{s=1}^S\lambda_s\,\mathcal{W}_2(\gamma_\theta,\beta_{s,\theta})^2\mathrm{d}\sigma(\theta),\qquad\beta_{s,\theta}\coloneqq(P_\theta)_\sharp\beta_s.$$

$\mathcal{C}_{\gamma_\theta}^{-1}(r)=\sum_{s=1}^S\lambda_s\mathcal{C}_{\beta_{s,\theta}}^{-1}(r), \qquad r\in[0,1]$.

$$R^\dagger h(x)=c_d\int_{\mathbb{S}^{d-1}}\int_{\mathbb{R}}e^{\mathrm{i}\omega\langle\theta,\,x\rangle}|\omega|^{d-1}\widehat{h}(\theta,\omega)\mathrm{d}\omega\,\mathrm{d}\sigma(\theta),$$

**Gaussian case.**

**Sinkhorn for barycenters.**

$$\min_{\mathrm{a}\in\Sigma_n}\sum_{s=1}^S\lambda_s\mathrm{L}_{\mathrm{C}_s}^\varepsilon(\mathrm{a},\mathrm{b}_s)\tag{11.2}$$

$$\min_{(\mathrm{P}_s)_s}\left\{\varepsilon\sum_s\lambda_s\mathrm{KL}(\mathrm{P}_s|\mathrm{K}_s)\ ;\ \forall s,\ {\mathrm{P}_s}^\top\mathbb{1}_n=\mathrm{b}_s,\quad\mathrm{P}_1\mathbb{1}_{n_1}=\ldots=\mathrm{P}_S\mathbb{1}_{n_S}\right\},\tag{11.3}$$

where $\mathrm{K}_s\coloneqq e^{-\mathrm{C}_s/\varepsilon}$. The barycenter a is implicitly encoded in the common row marginal $\mathrm{a}=\mathrm{P}_1\mathbb{1}_{n_1}=\ldots=\mathrm{P}_S\mathbb{1}_{n_S}$.

$$\mathrm{P}_s=\mathrm{diag}(\mathrm{u}_s)\mathrm{K}_s\,\mathrm{diag}(\mathrm{v}_s),\tag{11.4}$$

$$\forall\,s\in[\![1,S]\!],\qquad\mathrm{v}_s^{(\ell+1)}\coloneqq\frac{\mathrm{b}_s}{{\mathrm{K}_s}^\top\mathrm{u}_s^{(\ell)}},\tag{11.5}$$

$$\forall\,s\in[\![1,S]\!],\qquad\mathrm{u}_s^{(\ell+1)}\coloneqq\frac{\mathrm{a}^{(\ell+1)}}{\mathrm{K}_s\mathrm{v}_s^{(\ell+1)}},\tag{11.6}$$

$$\text{where}\quad\mathrm{a}^{(\ell+1)}\coloneqq\prod_s(\mathrm{K}_s\mathrm{v}_s^{(\ell+1)})^{\lambda_s}.\tag{11.7}$$

**Proposition 11.3** (Dual of entropic barycenters)**.** *The optimal scalings in (11.4) can be written as $(\mathrm{u}_s,\mathrm{v}_s)=(e^{\mathrm{f}_s/\varepsilon},e^{\mathrm{g}_s/\varepsilon})$, where $(\mathrm{f}_s,\mathrm{g}_s)_s$ solve the dual problem*

$$\max_{(\mathrm{f}_s,\mathrm{g}_s)_s}\left\{\sum_s\lambda_s\left(\langle\mathrm{g}_s,\,\mathrm{b}_s\rangle-\varepsilon\langle\mathrm{K}_se^{\mathrm{g}_s/\varepsilon},\,e^{\mathrm{f}_s/\varepsilon}\rangle\right)\ ;\ \sum_s\lambda_s\mathrm{f}_s=0\right\}.\tag{11.8}$$

*Proof.* Introduce Lagrange multipliers in (11.3):

$$\min_{(\mathrm{P}_s)_s,\mathrm{a}}\max_{(\mathrm{f}_s,\mathrm{g}_s)_s}\sum_s\lambda_s\Big(\varepsilon\mathrm{KL}(\mathrm{P}_s|\mathrm{K}_s)+\langle\mathrm{a}-\mathrm{P}_s\mathbb{1}_{n_s},\,\mathrm{f}_s\rangle+\langle\mathrm{b}_s-{\mathrm{P}_s}^\top\mathbb{1}_n,\,\mathrm{g}_s\rangle\Big).$$

Strong duality holds, so one can exchange the minimum and maximum. The minimization with respect to a gives the constraint $\sum_s\lambda_s\mathrm{f}_s=0$, and the minimization with respect to $\mathrm{P}_s$ gives the Legendre transform of $\mathrm{KL}(\cdot|\mathrm{K}_s)$:

$$\max_{(\mathrm{f}_s,\mathrm{g}_s)_s}\sum_s\lambda_s\left[\langle\mathrm{g}_s,\,\mathrm{b}_s\rangle-\varepsilon\mathrm{KL}^*\left(\frac{\mathrm{f}_s\oplus\mathrm{g}_s}{\varepsilon}\middle|\mathrm{K}_s\right)\right],\qquad\sum_s\lambda_s\mathrm{f}_s=0.$$

The separable conjugate is

$$\mathrm{KL}^*(\mathrm{U}|\mathrm{K})=\sum_{i,j}\mathrm{K}_{i,j}\left(e^{\mathrm{U}_{i,j}}-1\right),\tag{11.9}$$

because for $k>0$, $\sup_{r\geqslant0}ur-\big(r\log(r/k)-r+k\big)=k(e^u-1)$, and the case $k=0$ follows by lower semicontinuity. Dropping constants independent of $(\mathrm{f}_s,\mathrm{g}_s)_s$ gives (11.8). The coordinate maximization in $\mathrm{g}_s$ gives (11.5); the block maximization in all $(\mathrm{f}_s)_s$ gives the common marginal (11.7) and then (11.6). □

**Wasserstein-over-Wasserstein and barycenters.** $\int_{\mathcal{P}(\Omega)} \mathcal{L}_c(\beta_0,\alpha)\mathrm{d}\mathfrak{A}(\alpha) < +\infty$, $\mathcal{B}_c(\mathfrak{A}) := \mathrm{Argmin}_{\beta\in\mathcal{P}(\Omega)} \int_{\mathcal{P}(\Omega)} \mathcal{L}_c(\beta,\alpha)\mathrm{d}\mathfrak{A}(\alpha)$.
$\widetilde{\alpha}_{\mathfrak{A}} := \underset{\beta\in\mathcal{P}(\Omega)}{\mathrm{argmin}} \int_{\mathcal{P}(\Omega)} \mathcal{L}_c(\beta,\alpha)\mathrm{d}\mathfrak{A}(\alpha)$,

**Proposition 11.4** (Law of large numbers for barycenters over measures)**.** *Let $\Omega$ be a compact metric space, let $c : \Omega\times\Omega \to \mathbb{R}$ be continuous, and let $\mathfrak{A} \in \mathcal{P}(\mathcal{P}(\Omega))$. Let $\alpha_1, \alpha_2, \ldots$ be independent random probability measures with common law $\mathfrak{A}$, and set $\widehat{\mathfrak{A}}_p := \frac{1}{p}\sum_{i=1}^p \delta_{\alpha_i} \in \mathcal{P}(\mathcal{P}(\Omega))$. Then, almost surely,*

$$\widehat{\mathfrak{A}}_p \rightharpoonup \mathfrak{A} \quad \text{in } \mathcal{P}(\mathcal{P}(\Omega)), \qquad \bar{\alpha}_{\widehat{\mathfrak{A}}_p} \rightharpoonup \bar{\alpha}_{\mathfrak{A}} \quad \text{in } \mathcal{P}(\Omega).$$

*Moreover, if $\mathcal{B}_c(\mathfrak{A}) = \{\widetilde{\alpha}_{\mathfrak{A}}\}$ is a singleton and if $\widetilde{\alpha}_{\widehat{\mathfrak{A}}_p} \in \mathcal{B}_c(\widehat{\mathfrak{A}}_p)$ is any empirical barycenter, then, almost surely,*

$$\widetilde{\alpha}_{\widehat{\mathfrak{A}}_p} \rightharpoonup \widetilde{\alpha}_{\mathfrak{A}} \quad \text{in } \mathcal{P}(\Omega).$$

*Proof.* Set $K = \mathcal{P}(\Omega)$. Since $\Omega$ is compact metric, $K$ is compact metric for weak convergence, and so is $\mathcal{P}(K)$. The space $\mathcal{C}(K)$ is separable for the uniform norm. Applying the scalar strong law to a countable dense family of test functions and then using uniform approximation gives, almost surely, for every $\Phi \in \mathcal{C}(K)$, $\int \Phi(\alpha)\mathrm{d}\widehat{\mathfrak{A}}_p(\alpha) = \frac{1}{p}\sum_{i=1}^p \Phi(\alpha_i) \longrightarrow \int \Phi(\alpha)\mathrm{d}\mathfrak{A}(\alpha)$ This is exactly $\widehat{\mathfrak{A}}_p \rightharpoonup \mathfrak{A}$ in $\mathcal{P}(K)$.
For the collapsed mixtures, take $f \in \mathcal{C}(\Omega)$ and define $\Phi_f(\alpha) = \int_\Omega f\mathrm{d}\alpha$. This function is continuous on $\mathcal{P}(\Omega)$. Hence

$$\int_\Omega f\mathrm{d}\bar{\alpha}_{\widehat{\mathfrak{A}}_p} = \int_{\mathcal{P}(\Omega)} \Phi_f(\alpha)\mathrm{d}\widehat{\mathfrak{A}}_p(\alpha) \longrightarrow \int_{\mathcal{P}(\Omega)} \Phi_f(\alpha)\mathrm{d}\mathfrak{A}(\alpha) = \int_\Omega f\mathrm{d}\bar{\alpha}_{\mathfrak{A}},$$

which proves $\bar{\alpha}_{\widehat{\mathfrak{A}}_p} \rightharpoonup \bar{\alpha}_{\mathfrak{A}}$.
Since $c$ is continuous on the compact product $\Omega\times\Omega$, the value $(\beta,\alpha) \mapsto \mathcal{L}_c(\beta,\alpha)$ is continuous on $K^2$. Therefore the map $\beta \mapsto h_\beta$, where $h_\beta(\alpha) = \mathcal{L}_c(\beta,\alpha)$, is continuous from the compact space $K$ to $\mathcal{C}(K)$. Its image $\mathcal{H} := \{h_\beta : \beta \in K\}$ is compact in $\mathcal{C}(K)$. The convergence of $\widehat{\mathfrak{A}}_p$ to $\mathfrak{A}$ is uniform over $\mathcal{H}$: given $\eta > 0$, cover $\mathcal{H}$ by finitely many $\eta$-balls in $\|\cdot\|_\infty$, use weak convergence for the centers, and bound the error on the balls by the total variation of $\widehat{\mathfrak{A}}_p - \mathfrak{A}$. Hence the empirical objectives $F_p(\beta) := \int \mathcal{L}_c(\beta,\alpha)\mathrm{d}\widehat{\mathfrak{A}}_p(\alpha)$ converge uniformly on $K$ to $F(\beta) := \int \mathcal{L}_c(\beta,\alpha)\mathrm{d}\mathfrak{A}(\alpha)$. Let $\beta_p \in \mathcal{B}_c(\widehat{\mathfrak{A}}_p)$. Compactness gives a subsequence $\beta_{p_k} \rightharpoonup \beta$. Uniform convergence, continuity of $F$, and optimality of $\beta_{p_k}$ give, for any $\gamma \in K$, $F(\beta) = \lim_k F_{p_k}(\beta_{p_k}) \leqslant \lim_k F_{p_k}(\gamma) = F(\gamma)$, so $\beta \in \mathcal{B}_c(\mathfrak{A})$. If this set is the singleton $\{\widetilde{\alpha}_{\mathfrak{A}}\}$, every converging subsequence has the same limit, and therefore the whole sequence $\widetilde{\alpha}_{\widehat{\mathfrak{A}}_p}$ converges to $\widetilde{\alpha}_{\mathfrak{A}}$. □

**Toward central limit theorems on Wasserstein space.** There are nevertheless important settings where such results can be proved. In one dimension, the quantile representation linearizes $\mathcal{W}_2$, so barycenter fluctuations can be studied through empirical averages of quantile functions.

### 11.2 Multimarginal OT

This compact subsection collects the basic multimarginal formulation and its structured cases.

**Definition and basic structure.** The multi-marginal formulation replaces a coupling between two measures by a joint distribution with several prescribed marginals. Given measures $(\alpha_s)_{s=1}^S$ on spaces $(\mathcal{X}_s)_{s=1}^S$ and a cost $c : \mathcal{X}_1\times\cdots\times\mathcal{X}_S \to \mathbb{R}$, the problem reads
$\inf_{\pi\in\mathcal{U}(\alpha_1,\ldots,\alpha_S)} \int_{\mathcal{X}_1\times\cdots\times\mathcal{X}_S} c(x_1,\ldots,x_S)\mathrm{d}\pi(x_1,\ldots,x_S)$, where $\mathcal{U}(\alpha_1,\ldots,\alpha_S)$ is the set of probability measures whose $s$-th marginal is $\alpha_s$. This is still a linear program in the discrete setting, but its ambient tensor has size $\prod_s n_s$.

**Monge structure and splitting-set twist.**

**Definition 11.5** (Twist on splitting sets)**.** Fix $x_1 \in \mathcal{X}_1$. A set $M \subset \mathcal{X}_2\times\cdots\times\mathcal{X}_S$ is a $c$-splitting set at $x_1$ if there exist functions $u_s : \mathcal{X}_s \to \mathbb{R}\cup\{-\infty\}$, for $s = 2,\ldots,S$, such that $\sum_{s=2}^S u_s(x_s) \leqslant c(x_1,x_2,\ldots,x_S)$ for all $(x_2,\ldots,x_S)$, with equality on $M$. Assume $c$ is differentiable in $x_1$. The cost is twisted on splitting sets if, for every $x_1$ and every $c$-splitting set $M$ at $x_1$, the map $(x_2,\ldots,x_S) \longmapsto \nabla_{x_1}c(x_1,x_2,\ldots,x_S)$ is injective on $M$.

**Proposition 11.6** (Multi-marginal Monge structure)**.** *Assume that $\mathcal{X}_s \subset \mathbb{R}^d$, that $c$ is continuous and differentiable with respect to $x_1$, and that $c$ is twisted on splitting sets. Suppose that Kantorovich dual maximizers $(\varphi_s)_{s=1}^S$ exist and that $\varphi_1$ is differentiable $\alpha_1$-a.e. If $\alpha_1$ is absolutely continuous, then every optimal plan $\pi^\star \in \mathcal{U}(\alpha_1,\ldots,\alpha_S)$ is concentrated on the graph of maps $\pi^\star = (\mathrm{Id}, T_2,\ldots,T_S)_\sharp\alpha_1, \qquad (T_s)_\sharp\alpha_1 = \alpha_s$. In particular, under these hypotheses the optimizer is unique.*

*Proof.* Let $(\varphi_s)_{s=1}^S$ be optimal dual potentials, so that $\sum_s \varphi_s(x_s) \leqslant c(x_1,\ldots,x_S)$ with equality on $\mathrm{supp}(\pi^\star)$. Fix a point $x_1$ where $\varphi_1$ is differentiable and consider the fiber $M(x_1) = \{(x_2,\ldots,x_S) : (x_1,x_2,\ldots,x_S) \in \mathrm{supp}(\pi^\star)\}$. Indeed, set $u_2 = \varphi_2 + \varphi_1(x_1)$ and $u_s = \varphi_s$ for $s \geqslant 3$. Dual feasibility gives $\sum_{s=2}^S u_s(x_s) \leqslant c(x_1,x_2,\ldots,x_S)$, and equality holds on $M(x_1)$. If $(x_2,\ldots,x_S) \in M(x_1)$, the function $z \longmapsto c(z,x_2,\ldots,x_S) - \sum_{s=2}^S \varphi_s(x_s)$ touches $\varphi_1$ from above at $z = x_1$. Differentiating at this contact point gives $\nabla\varphi_1(x_1) = \nabla_{x_1}c(x_1,x_2,\ldots,x_S)$. All points in the fiber therefore have the same value of $\nabla_{x_1}c$. Twist on splitting sets makes the fiber a singleton for $\alpha_1$-a.e. $x_1$. Disintegrating $\pi^\star$ with respect to its first marginal gives Dirac conditional measures, hence measurable maps $(T_2,\ldots,T_S)$. If $\pi^1$ and $\pi^2$ are two optimal plans, their average is also optimal. The conditional measure of this average over $x_1$ is the average of the two Dirac conditionals, and it must again be a Dirac mass by the preceding argument. Hence the two Dirac masses coincide for $\alpha_1$-a.e. $x_1$, proving uniqueness. □

**Coulomb cost and density-functional theory.** $c_{\mathrm{Coul}}(x_1,\ldots,x_N) := \sum_{1\leqslant i<j\leqslant N} \frac{1}{\|x_i-x_j\|}$,

$$V_{\mathrm{ee}}^{\mathrm{SCE}}[\rho] := \inf_{\pi\in\mathcal{U}(\alpha,\ldots,\alpha)} \int_{(\mathbb{R}^3)^N} c_{\mathrm{Coul}}(x_1,\ldots,x_N)\mathrm{d}\pi(x_1,\ldots,x_N).$$

$\pi = (\mathrm{Id}, T_2,\ldots,T_N)_\sharp\alpha, \qquad (T_i)_\sharp\alpha = \alpha,$

**Proposition 11.7** (Cyclic co-motion plans)**.** *Let $\alpha \in \mathcal{P}(\mathbb{R}^3)$ and let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be measurable with $T_\sharp\alpha = \alpha$ and $T^N = \mathrm{Id}$ $\alpha$-a.e. Set $T^0 = \mathrm{Id}$ and $\pi_T = (T^0, T^1, \dots, T^{N-1})_\sharp\alpha$. Then $\pi_T \in \mathcal{U}(\alpha, \dots, \alpha)$. If $R_\sigma(x_1, \dots, x_N) = (x_{\sigma(1)}, \dots, x_{\sigma(N)})$ for $\sigma \in \mathrm{Perm}(N)$, then $\bar{\pi}_T := \frac{1}{N!}\sum_{\sigma\in\mathrm{Perm}(N)}(R_\sigma)_\sharp\pi_T$ is a symmetric admissible plan and*

$$\int c_{\mathrm{Coul}}\mathrm{d}\bar{\pi}_T = \int c_{\mathrm{Coul}}\mathrm{d}\pi_T = \int_{\mathbb{R}^3} \sum_{0\leqslant i<j\leqslant N-1} \frac{1}{\|T^i(x) - T^j(x)\|}\mathrm{d}\alpha(x).$$

*Proof.* Since $T_\sharp\alpha = \alpha$, all iterates $T^i$ preserve $\alpha$, so every marginal of $\pi_T$ is $\alpha$. The plan $\bar{\pi}_T$ is an average of coordinate permutations of $\pi_T$, hence has the same marginals and is invariant under coordinate permutations. The Coulomb cost is symmetric in its arguments, so its integral is unchanged by each $R_\sigma$. The last identity follows by evaluating $c_{\mathrm{Coul}}$ on the graph $(x, T(x), \dots, T^{N-1}(x))$. ☐

Proposition 11.6 explains the general mechanism that can force graph solutions, while Proposition 11.7 records the additional equal-marginal symmetry used by co-motion maps. Thus the DFT problem is both a central application and a warning that multi-marginal OT is richer than a naive deterministic matching problem.

**Multi-marginal formulation of barycenters.** $c_{\mathrm{bar}}(x_1, \dots, x_S) = \min_{x\in\mathbb{R}^d}\sum_{s=1}^S \lambda_s\|x - x_s\|^2$.

**Proposition 11.8** (Multi-marginal formula for quadratic barycenters)**.** *Let $\beta_1, \dots, \beta_S \in \mathcal{P}_2(\mathbb{R}^d)$ and $\lambda \in \Sigma_S$. Define $B(x_1, \dots, x_S) = \sum_{s=1}^S \lambda_s x_s$, $c_{\mathrm{bar}}(x_1, \dots, x_S) = \min_x \sum_s \lambda_s\|x - x_s\|^2$. If $\pi^\star$ solves the multi-marginal OT problem with marginals $(\beta_s)_s$ and cost $c_{\mathrm{bar}}$, then $\alpha^\star = B_\sharp\pi^\star$ is a Wasserstein barycenter. Conversely, every barycenter is obtained this way from an optimal multi-marginal plan.*

*Proof.* For any candidate barycenter $\alpha$ and couplings $\pi_s \in \mathcal{U}(\alpha, \beta_s)$, glue the couplings along their common $\alpha$ marginal to obtain a joint law of $(X, Y_1, \dots, Y_S)$. Conditioning on $(Y_s)_s$ and minimizing over $X$ gives $\sum_s \lambda_s\mathbb{E}\|X - Y_s\|^2 \geqslant \mathbb{E}\min_x \sum_s \lambda_s\|x - Y_s\|^2 = \mathbb{E}c_{\mathrm{bar}}(Y_1, \dots, Y_S)$. Taking the infimum over the couplings gives that the barycenter value is at least the multi-marginal value. Conversely, from an optimal multi-marginal plan $\pi^\star$, set $X = B(Y_1, \dots, Y_S)$. The couplings between $X$ and each $Y_s$ are feasible for the barycenter problem and attain exactly the multi-marginal cost, proving equality and the formula. If $\alpha^\star$ is any barycenter, choose optimal couplings between $\alpha^\star$ and each $\beta_s$ and glue them along the common $\alpha^\star$ marginal. Since the barycenter and multi-marginal values are equal, the conditional minimization inequality above must be an equality. Thus $X = B(Y_1, \dots, Y_S)$ almost surely for the induced optimal multi-marginal plan, and $\alpha^\star = B_\sharp\pi^\star$. ☐

**Corollary 11.9** (Gaussian and discrete barycenters)**.** *Quadratic Wasserstein barycenters of Gaussian measures are Gaussian. If the input measures are discrete, then there exists a barycenter supported on the set of weighted averages $\sum_s \lambda_s x_{s,i_s}$ of one support point from each input; in particular, if the $s$-th input has $n_s$ atoms, a barycenter exists with at most $\prod_s n_s$ atoms.*

*Proof.* Let the input Gaussians have means $\mathbf{m}_s$ and covariances $\mathbf{\Sigma}_s$. For any candidate barycenter $\alpha$ with mean $\mathbf{m}$ and covariance $\mathbf{\Sigma}$, Gelbrich's inequality, proved later in Theorem 15.12, gives $\mathcal{W}_2^2(\alpha, \beta_s) \geqslant \|\mathbf{m} - \mathbf{m}_s\|^2 + \mathcal{B}(\mathbf{\Sigma}, \mathbf{\Sigma}_s)^2$, with equality for the Gaussian law with mean $\mathbf{m}$ and covariance $\mathbf{\Sigma}$. Therefore the barycenter objective is bounded below by a function depending only on $(\mathbf{m}, \mathbf{\Sigma})$, and this lower bound is attained by the Gaussian measure with the minimizing mean and covariance. Hence at least one barycenter is Gaussian, and uniqueness in the usual nondegenerate setting gives the Gaussian barycenter mentioned above. For discrete inputs, any multi-marginal optimizer is supported on the finite product of the input supports, and $B$ maps this product to at most $\prod_s n_s$ points. ☐

**Entropic regularization of multi-marginal OT.** $\inf_{\pi\in\mathcal{U}(\alpha_1,\dots,\alpha_S)} \int c\mathrm{d}\pi + \varepsilon\,\mathrm{KL}(\pi|\alpha_1 \otimes \cdots \otimes \alpha_S)$.

$$\mathrm{d}\pi^\star(x_1, \dots, x_S) = \exp\left(\frac{\sum_s f_s(x_s) - c(x_1, \dots, x_S)}{\varepsilon}\right)\prod_s \mathrm{d}\alpha_s(x_s),$$

$c(x_1, \dots, x_S) = \sum_{(r,s)\in E} c_{r,s}(x_r, x_s)$,

### 11.3 Low-Rank Optimal Transport

**Definition 11.10** (Low-rank factored couplings)**.** Let $\mathrm{a} \in \Sigma_n$, $\mathrm{b} \in \Sigma_m$ and let $r \geqslant 1$. A rank-$r$ factored coupling is a triple $(Q, R, g)$ such that $Q \in \mathbb{R}_+^{n\times r}$, $R \in \mathbb{R}_+^{m\times r}$, $g \in \Sigma_r$, with $Q\mathbb{1}_r = \mathrm{a}$, $R\mathbb{1}_r = \mathrm{b}$, $Q^\top\mathbb{1}_n = R^\top\mathbb{1}_m = g$. It induces the coupling

$$P(Q, R, g) = Q\,\mathrm{diag}(g)^{-1}R^\top, \qquad P_{i,j} = \sum_{k=1}^r \frac{Q_{i,k}R_{j,k}}{g_k}, \tag{11.10}$$

where columns with $g_k = 0$ are discarded before applying the formula.

**Proposition 11.11** (Factored couplings and nonnegative rank)**.** *For every feasible triple $(Q, R, g)$ in Definition 11.10, the matrix $P(Q, R, g)$ belongs to $\mathrm{U}(\mathrm{a}, \mathrm{b})$ and has nonnegative rank at most $r$. Conversely, every coupling $P \in \mathrm{U}(\mathrm{a}, \mathrm{b})$ with nonnegative rank at most $r$ admits a representation of the form (11.10), after possibly deleting zero latent components.*

*Proof.* The marginal constraints follow by direct summation: $P\mathbb{1}_m = Q\,\mathrm{diag}(g)^{-1}R^\top\mathbb{1}_m = Q\,\mathrm{diag}(g)^{-1}g = Q\mathbb{1}_r = \mathrm{a}$, and similarly $P^\top\mathbb{1}_n = \mathrm{b}$. Since $P$ is a sum of $r$ nonnegative rank-one matrices $Q_{:,k}R_{:,k}^\top/g_k$, its nonnegative rank is at most $r$.
Conversely, suppose $P = UV^\top$ with $U \in \mathbb{R}_+^{n\times r}$ and $V \in \mathbb{R}_+^{m\times r}$. Set $q_k = \sum_i U_{i,k}$ and $s_k = \sum_j V_{j,k}$. Components with $q_k s_k = 0$ do not contribute and can be removed. Define $g_k = q_k s_k$, $Q_{i,k} = U_{i,k}s_k$ and $R_{j,k} = V_{j,k}q_k$. Since $P$ has total mass one, $g \in \Sigma_r$. Moreover $Q\mathbb{1}_r = P\mathbb{1}_m = \mathrm{a}$, $R\mathbb{1}_r = P^\top\mathbb{1}_n = \mathrm{b}$, and both column marginals are equal to $g$. Finally, $\sum_k \frac{Q_{i,k}R_{j,k}}{g_k} = \sum_k \frac{U_{i,k}s_k V_{j,k}q_k}{q_k s_k} = P_{i,j}$. ☐

For a cost matrix $C \in \mathbb{R}^{n\times m}$, the low-rank constrained OT value is therefore

$\min_{(Q,R,g)} \sum_{k=1}^{r} \frac{Q_{:,k}{}^{\top} C R_{:,k}}{g_k}$ subject to Definition 11.10.

$$\min_{(Q,R,g)} \sum_{k=1}^{r} \frac{Q_{:,k}{}^{\top} C R_{:,k}}{g_k} + \varepsilon \mathrm{KL}(Q|\mathrm{a}\otimes g) + \varepsilon \mathrm{KL}(R|\mathrm{b}\otimes g), \tag{11.11}$$

where the constraints are those of Definition 11.10.

## 11.4 Capacity-Constrained Optimal Transport

Let $\alpha \in \mathcal{M}_+^1(\mathcal{X})$, $\beta \in \mathcal{M}_+^1(\mathcal{Y})$ and let $\kappa : \mathcal{X}\times\mathcal{Y} \to [0,+\infty]$ be a measurable capacity. The capacity-constrained transport value is

$$\mathcal{L}_c^{\kappa}(\alpha,\beta) := \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \left\{ \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\mathrm{d}\pi(x,y) \; ; \; \pi \ll \alpha\otimes\beta, \quad \frac{\mathrm{d}\pi}{\mathrm{d}(\alpha\otimes\beta)}(x,y) \leqslant \kappa(x,y) \right\}. \tag{11.12}$$

The product coupling $\alpha\otimes\beta$ is feasible whenever $\kappa \geqslant 1$. Thus the constraint is not meant to promote the product plan; rather, it prevents the optimizer from using any pair more than the prescribed density ratio.
For discrete measures $\alpha = \sum_i a_i\delta_{x_i}$ and $\beta = \sum_j b_j \delta_{y_j}$, a capacity is an upper matrix $U \in \mathbb{R}_+^{n\times m}$. The density-ratio discretization of (11.12) is $U_{i,j} = \kappa_{i,j} a_i b_j$, and the finite-dimensional problem is the linear program

$$\min_{P\in\mathrm{U}(\mathrm{a},\mathrm{b})} \langle C,\, P\rangle \quad \text{subject to} \quad 0 \leqslant P_{i,j} \leqslant U_{i,j} \quad \forall (i,j). \tag{11.13}$$

$$\min_{P\in\mathrm{U}(\mathrm{a},\mathrm{b}),\, 0\leqslant P\leqslant U} \langle C,\, P\rangle + \varepsilon \mathrm{KL}(P|\mathrm{a}\otimes\mathrm{b}). \tag{11.14}$$

## 11.5 Metric learning and inverse OT

### *11.5.1 Differentiating OT losses*

This compact subsubsection records the sensitivity formulas used for inverse problems.

**First variations.** The first block gives the marginal and cost derivatives.

**Proposition 11.12** (First variations of unregularized OT)**.** *Let $\alpha \in \mathcal{P}(\mathcal{X})$ and $\beta\in\mathcal{P}(\mathcal{Y})$, where $\mathcal{X},\mathcal{Y}$ are compact metric spaces, and let $c \in \mathcal{C}(\mathcal{X}\times\mathcal{Y})$. Define $\mathcal{V}_c(\alpha,\beta) := \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\mathrm{d}\pi(x,y)$. Let $\mathcal{O}_c(\alpha,\beta)$ be the set of optimal couplings and let $\mathcal{D}_c(\alpha,\beta)$ be the set of optimal dual potentials $(f,g)$ with $f(x)+g(y) \leqslant c(x,y)$. If $\chi$ is a signed measure with $\chi(\mathcal{X}) = 0$ and $\alpha_t = \alpha + t\chi$ is a probability measure for $0\leqslant t\leqslant t_0$, then*

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0^+} \mathcal{V}_c(\alpha_t,\beta) = \sup_{(f,g)\in\mathcal{D}_c(\alpha,\beta)} \int_{\mathcal{X}} f(x)\mathrm{d}\chi(x).$$

*If $h\in\mathcal{C}(\mathcal{X}\times\mathcal{Y})$ and $c_t = c + th$, then*

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0^+} \mathcal{V}_{c_t}(\alpha,\beta) = \inf_{\pi\in\mathcal{O}_c(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{Y}} h(x,y)\mathrm{d}\pi(x,y).$$

*In particular, if the normalized optimal potential $f^\star$ and the optimal plan $\pi^\star$ are unique, then $\frac{\delta\mathcal{V}_c}{\delta\alpha} = f^\star$, $\quad \frac{\delta\mathcal{V}_c}{\delta c} = \pi^\star$. The second identity means that the first variation with respect to the function $c$ is represented by the optimal measure $\pi^\star$ on $\mathcal{X}\times\mathcal{Y}$.*

*Proof.* Kantorovich duality writes $\mathcal{V}_c(\alpha,\beta) = \sup_{f\oplus g\leqslant c} \int f\mathrm{d}\alpha + \int g\mathrm{d}\beta$. This is a supremum of affine functions of $\alpha$, so Danskin's theorem gives the one-sided directional derivative as the supremum over active maximizers, namely the optimal dual potentials. The condition $\chi(\mathcal{X}) = 0$ makes the formula independent of the additive gauge $(f,g)\mapsto(f+\lambda, g-\lambda)$.
For the cost variable, $\mathcal{V}_{c_t}(\alpha,\beta) = \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \int c\mathrm{d}\pi + t\int h\mathrm{d}\pi$ is an infimum of affine functions of $t$. Danskin's theorem for a minimum gives the right directional derivative as the infimum of $\int h\mathrm{d}\pi$ over the active minimizers. If the active dual potential or coupling is unique, the corresponding directional derivative is linear in the perturbation, which is the displayed first variation. □

$f^\star \in \partial_{\mathrm{a}} \mathrm{L}_{\mathrm{C}}(\mathrm{a},\mathrm{b}), \qquad \mathrm{P}^\star \in \partial_{\mathrm{C}}^{\sup} \mathrm{L}_{\mathrm{C}}(\mathrm{a},\mathrm{b}).$

**Proposition 11.13** (First variations of entropic OT)**.** *Let $\varepsilon > 0$ and define the KL-normalized entropic value $\mathcal{V}_{c,\varepsilon}(\alpha,\beta) := \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \int c\mathrm{d}\pi + \varepsilon\,\mathrm{KL}(\pi|\alpha\otimes\beta)$. Assume that the optimal entropic potentials $(f_\varepsilon, g_\varepsilon)$ exist and are normalized so that the following density has marginals $\alpha$ and $\beta$: $\mathrm{d}\pi_\varepsilon(x,y) = \exp\left(\frac{f_\varepsilon(x)+g_\varepsilon(y)-c(x,y)}{\varepsilon}\right)\mathrm{d}\alpha(x)\mathrm{d}\beta(y)$ This defines the unique optimal coupling. For the same perturbations $\alpha_t = \alpha + t\chi$ and $c_t = c + th$ as in Proposition 11.12,*

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0^+} \mathcal{V}_{c,\varepsilon}(\alpha_t,\beta) = \int f_\varepsilon \mathrm{d}\chi, \qquad \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0^+} \mathcal{V}_{c_t,\varepsilon}(\alpha,\beta) = \int h\mathrm{d}\pi_\varepsilon.$$

*Equivalently, $\frac{\delta\mathcal{V}_{c,\varepsilon}}{\delta\alpha} = f_\varepsilon$, $\quad \frac{\delta\mathcal{V}_{c,\varepsilon}}{\delta c} = \pi_\varepsilon$. In finite dimension, for positive histograms, these are ordinary derivatives on the relative interior of the simplices.*

*Proof.* The cost derivative follows directly from the primal envelope theorem, because the entropic optimizer is unique. For the measure derivative, use the dual formula (7.18). At an optimal pair, the soft-transform equations imply the row normalization $\int_{\mathcal{Y}} \exp\left(\frac{f_\varepsilon(x)+g_\varepsilon(y)-c(x,y)}{\varepsilon}\right)\mathrm{d}\beta(y) = 1$ for all $x$. Differentiating the dual objective with respect to $\alpha$ at fixed optimal potentials gives $\int f_\varepsilon\mathrm{d}\chi - \varepsilon\int_{\mathcal{X}\times\mathcal{Y}} \left(e^{(f_\varepsilon(x)+g_\varepsilon(y)-c(x,y))/\varepsilon} - 1\right)\mathrm{d}\chi(x)\mathrm{d}\beta(y)$. The second term vanishes by the previous normalization identity, leaving $\int f_\varepsilon \mathrm{d}\chi$. The gauge ambiguity of $(f_\varepsilon, g_\varepsilon)$ again disappears because $\chi(\mathcal{X}) = 0$. □

For a finite-dimensional parametrization $c_\theta$ or $C_\theta$, the entropic formula gives the backpropagation rule
$\partial_\theta \mathcal{V}_{c_\theta,\varepsilon} = \int \partial_\theta c_\theta(x,y)\mathrm{d}\pi_\varepsilon(x,y)$, where $\pi_\varepsilon$ is the entropic optimizer. For the unregularized value $\mathcal{V}_{c_\theta}$, uniqueness of the optimal plan $\pi^\star$ gives $\partial_\theta\mathcal{V}_{c_\theta} = \int \partial_\theta c_\theta \mathrm{d}\pi^\star$.
$\mathrm{L}_{\mathrm{C}}^{\varepsilon}(\mathrm{a},\mathrm{b}) = \mathcal{V}_{\mathrm{C},\varepsilon}(\mathrm{a},\mathrm{b}) - \varepsilon\mathrm{H}(\mathrm{a}) - \varepsilon\mathrm{H}(\mathrm{b}),$

**Inverse Optimal Transport.** $\inf_{\pi\in\mathcal{U}(\alpha,\beta)}\int c(x,y)\mathrm{d}\pi(x,y)$. A useful statistical methodology is to measure the suboptimality of the observed plan through a Fenchel–Young loss. Write the score as $s=-c$ and define the convex regularized prediction value $G_\varepsilon(s)=\sup_{\pi\in\mathcal{U}(\alpha,\beta)}\int s\mathrm{d}\pi-\varepsilon\,\mathrm{KL}(\pi|\alpha\otimes\beta)$. $\mathcal{L}_\varepsilon(c;\widehat{\pi})=G_\varepsilon(-c)+G^*_\varepsilon(\widehat{\pi})+\int c\mathrm{d}\widehat{\pi}$ is nonnegative by Fenchel's inequality and vanishes exactly when $\widehat{\pi}\in\partial G_\varepsilon(-c)$, i.e. when $\widehat{\pi}$ satisfies the regularized optimality conditions for $c$. Entropic regularization is important here because it makes the forward map smoother and provides gradients with respect to $c$, at the price of a bias that must be controlled statistically.

In the discrete unregularized case, this loss reduces to the optimality gap of the observed coupling. For $\widehat{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})$ and a cost matrix $C$, denote it by
$\mathcal{L}_{\mathrm{iOT}}(C;\widehat{P})=\langle C,\,\widehat{P}\rangle-\min_{P\in\mathrm{U}(\mathrm{a},\mathrm{b})}\langle C,\,P\rangle$. $C_\theta=\sum_{r=1}^R\theta_r C^{(r)},\qquad \theta\in\Theta$, where $\Theta$ is convex and the matrices $C^{(r)}$ encode features, graph distances or a Mahalanobis parameterization.
$c_A(x,y)=\langle Ax,\,y\rangle,\qquad A\in\mathbb{R}^{d\times d}$. For empirical measures $\alpha=\frac{1}{n}\sum_i\delta_{x_i}$ and $\beta=\frac{1}{n}\sum_j\delta_{y_j}$, this gives the cost matrix $C(A)_{i,j}=\langle Ax_i,\,y_j\rangle$, For a fixed matrix $A$, the forward prediction is the optimal face
$\mathcal{P}_A:=\operatorname*{argmin}_{P\in\mathrm{U}(\mathbb{1}_n/n,\mathbb{1}_n/n)}\langle C(A),\,P\rangle$. $\|x-y\|^2=\|x\|^2+\|y\|^2-2\langle x,\,y\rangle$, $A_t=-\operatorname{diag}(1+t,1-t),\qquad -1\leqslant t\leqslant 1,$

$$\mathcal{L}_{\mathrm{iOT}}(C(A_t);\widehat{P})=\langle C(A_t),\,\widehat{P}\rangle-\min_{P\in\mathrm{U}(\mathbb{1}_n/n,\mathbb{1}_n/n)}\langle C(A_t),\,P\rangle,\qquad C(A_t)_{i,j}=\langle A_t x_i,\,y_j\rangle.$$

**Proposition 11.14** (Convex dual-gap formulation of inverse OT)**.** *Let $\widehat{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})$ be an observed coupling and let $C_\theta$ depend affinely on $\theta\in\Theta$, where $\Theta$ is convex. The condition that $\widehat{P}$ is optimal for the cost $C_\theta$ is equivalent to the existence of dual potentials $(f,g)$ such that $f_i+g_j\leqslant(C_\theta)_{i,j}$ and $\sum_{i,j}\widehat{P}_{i,j}\big((C_\theta)_{i,j}-f_i-g_j\big)=0$.*

$$\min_{\theta\in\Theta,f,g}R(\theta)+\lambda\sum_{i,j}\widehat{P}_{i,j}\big((C_\theta)_{i,j}-f_i-g_j\big)\quad\textit{subject to}\quad f_i+g_j\leqslant(C_\theta)_{i,j}\quad\forall i,j.\tag{11.15}$$

*Proof.* For a fixed cost $C_\theta$, Kantorovich duality gives $\min_{P\in\mathrm{U}(\mathrm{a},\mathrm{b})}\langle C_\theta,\,P\rangle=\max_{f_i+g_j\leqslant(C_\theta)_{i,j}}\langle f,\,\mathrm{a}\rangle+\langle g,\,\mathrm{b}\rangle$. Since $\widehat{P}$ has marginals $(\mathrm{a},\mathrm{b})$, every dual feasible pair satisfies $\langle C_\theta,\,\widehat{P}\rangle-\langle f,\,\mathrm{a}\rangle-\langle g,\,\mathrm{b}\rangle=\sum_{i,j}\widehat{P}_{i,j}\big((C_\theta)_{i,j}-f_i-g_j\big)\geqslant 0$. It vanishes if and only if $\widehat{P}$ reaches the dual value and is therefore optimal. If $C_\theta$ is affine and $\Theta$ and $R$ are convex, the constraints and objective in (11.15) are convex, proving the relaxation claim. □

$\widehat{P}_{i,j}\approx a_i b_j\exp\left(\frac{f_i+g_j-(C_\theta)_{i,j}}{\varepsilon}\right)$,

## 11.6 Weak Optimal Transport

This compact subsection records barycentric projection and weak transport costs.

**Barycentric projection of a coupling.**

**Definition 11.15** (Barycentric projection of a coupling)**.** Let $\alpha,\beta\in\mathcal{P}_1(\mathbb{R}^d)$ and let $\pi\in\mathcal{U}(\alpha,\beta)$. Disintegrate $\pi$ with respect to its first marginal as $\pi(\mathrm{d}x,\mathrm{d}y)=\pi_x(\mathrm{d}y)\alpha(\mathrm{d}x)$. Since $\beta$ has finite first moment, the conditional mean is finite for $\alpha$-a.e. $x$, and the barycentric projection of $\pi$ is the map

$$\bar{T}_\pi(x):=\int_{\mathbb{R}^d}y\mathrm{d}\pi_x(y),\qquad\bar{\beta}_\pi:=(\bar{T}_\pi)_\sharp\alpha.\tag{11.16}$$

**Proposition 11.16** (Barycentric projection of a quadratic optimal plan)**.** *Let $\pi\in\mathcal{U}(\alpha,\beta)$ be optimal for the quadratic cost $\|x-y\|^2$ between $\alpha,\beta\in\mathcal{P}_2(\mathbb{R}^d)$, and define $\bar{T}_\pi$ and $\bar{\beta}_\pi$ by (11.16). Then $(\mathrm{Id},\bar{T}_\pi)_\sharp\alpha$ is an optimal coupling between $\alpha$ and $\bar{\beta}_\pi$. Equivalently, $\bar{T}_\pi$ is a quadratic optimal transport map from $\alpha$ to the projected target $\bar{\beta}_\pi$.*

*Proof.* By Theorem 3.23, $\pi$ is concentrated on a $c$-cyclically monotone set $\Gamma$ for $c(x,y)=\|x-y\|^2$. For the quadratic cost, and since it is enough to check cyclic permutations, this means that every finite cycle $(x_i,y_i)_{i=1}^m\subset\Gamma$ satisfies $\sum_{i=1}^m\langle x_i,\,y_i\rangle\geqslant\sum_{i=1}^m\langle x_i,\,y_{i+1}\rangle,\qquad y_{m+1}=y_1$. After changing the disintegration on an $\alpha$-negligible set, $\pi_x$ is supported on the section $\Gamma_x=\{y\,;\,(x,y)\in\Gamma\}$ for $\alpha$-a.e. $x$. Choose $x_1,\dots,x_m$ in this full-measure set and independently sample $Y_i\sim\pi_{x_i}$. Applying the cyclic inequality to $(x_i,Y_i)$ and taking expectations gives $\sum_{i=1}^m\langle x_i,\,\bar{T}_\pi(x_i)\rangle\geqslant\sum_{i=1}^m\langle x_i,\,\bar{T}_\pi(x_{i+1})\rangle$. Thus $(\mathrm{Id},\bar{T}_\pi)_\sharp\alpha$ is concentrated on a cyclically monotone graph. By the cyclic-monotonicity characterization of quadratic optimality, this plan is optimal between its two marginals, namely $\alpha$ and $\bar{\beta}_\pi$. □

**Weak transport costs.**

$$\mathrm{WOT}_C(\alpha,\beta):=\inf_{\pi\in\mathcal{U}(\alpha,\beta)}\int C(x,\pi_x)\mathrm{d}\alpha(x).\tag{11.17}$$

**Proposition 11.17** (Weak Kantorovich duality)**.** *Assume that $\mathcal{X},\mathcal{Y}$ are compact metric spaces and that $C(x,\nu)$ is lower semicontinuous, bounded from below and convex in $\nu$, with the standard qualification assumptions ensuring Fenchel–Rockafellar duality. For $g\in\mathcal{C}(\mathcal{Y})$ define the weak $C$-transform*

$$g^C(x):=\inf_{\nu\in\mathcal{P}(\mathcal{Y})}\left\{C(x,\nu)-\int g(y)\mathrm{d}\nu(y)\right\}.$$

*Then*

$$\mathrm{WOT}_C(\alpha,\beta)=\sup_{g\in\mathcal{C}(\mathcal{Y})}\left\{\int g^C(x)\mathrm{d}\alpha(x)+\int g(y)\mathrm{d}\beta(y)\right\}.$$

*When $C(x,\nu)=\int c(x,y)\mathrm{d}\nu(y)$, this reduces to the usual Kantorovich dual with $g^C(x)=\inf_y(c(x,y)-g(y))$.*

*Proof.* For any coupling $\pi$ and any $g \in \mathcal{C}(\mathcal{Y})$, the definition of $g^C$ gives $C(x, \pi_x) \geqslant g^C(x) + \int g(y) \mathrm{d}\pi_x(y)$. After integration with respect to $\alpha$, the second term becomes $\int g \mathrm{d}\beta$ because the second marginal of $\pi$ is $\beta$. This proves weak duality.
For the reverse inequality, consider the convex minimization over probability kernels $x \mapsto \pi_x$ with the affine constraint $\int \pi_x \mathrm{d}\alpha(x) = \beta$. Fenchel–Rockafellar duality gives a continuous Lagrange multiplier $g$ for this marginal constraint. Minimizing the Lagrangian over each conditional law gives exactly the pointwise term $g^C(x)$, while the multiplier contributes $\int g \mathrm{d}\beta$. The compactness, lower semicontinuity, convexity and qualification assumptions ensure no duality gap. This is the weak-cost analogue of Proposition 4.2. □

**Proposition 11.18** (Barycentric weak transport is weaker than $\mathcal{W}_2$)**.** *Let $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$ and define $C_{\mathrm{bar}}(x, \nu) = \|x - \int y \mathrm{d}\nu(y)\|^2$. Equivalently, for a coupling $\pi$, the integrand is $\|x - \bar{T}_\pi(x)\|^2$. Then $\mathcal{W}_{C_{\mathrm{bar}}}(\alpha, \beta) \leqslant \mathcal{W}_2^2(\alpha, \beta)$.*

*Proof.* Let $\pi$ be any coupling and disintegrate it as $\pi_x \alpha$. By Jensen's inequality, $\|x - \bar{T}_\pi(x)\|^2 \leqslant \int \|x - y\|^2 \mathrm{d}\pi_x(y)$. Integrating in $x$ gives $\int C_{\mathrm{bar}}(x, \pi_x) \mathrm{d}\alpha(x) \leqslant \int \|x - y\|^2 \mathrm{d}\pi(x, y)$. Taking the infimum over $\pi$ proves the claim. □

The barycentric cost is the canonical example to keep in mind: admissibility still constrains the full conditional laws to have second marginal $\beta$, but the objective only charges the displacement from $x$ to $\bar{T}_\pi(x)$ and ignores the conditional variance around this barycenter.

*11.6.1 Martingale Optimal Transport*

**Definition 11.19** (Martingale couplings and martingale OT)**.** Let $\alpha, \beta \in \mathcal{P}_1(\mathbb{R}^d)$. A coupling $\pi \in \mathcal{U}(\alpha, \beta)$ is a martingale coupling if, for the disintegration $\pi(\mathrm{d}x, \mathrm{d}y) = \pi_x(\mathrm{d}y)\alpha(\mathrm{d}x)$,

$$\int_{\mathbb{R}^d} y \mathrm{d}\pi_x(y) = x \qquad \text{for } \alpha\text{-a.e. } x. \tag{11.18}$$

Equivalently, $\bar{T}_\pi = \mathrm{Id}$ in (11.16). The set of such couplings is denoted $\mathcal{U}_{\mathrm{mart}}(\alpha, \beta) := \left\{\pi \in \mathcal{U}(\alpha, \beta) \; ; \; \bar{T}_\pi = \mathrm{Id} \quad \alpha\text{-a.e.}\right\}$.

For a cost $c$, the martingale OT problem minimizes $\int c \mathrm{d}\pi$ over $\mathcal{U}_{\mathrm{mart}}(\alpha, \beta)$, with value $+\infty$ when the admissible set is empty.
The terminology comes from probability: if $(X, Y) \sim \pi$, then (11.18) is exactly $\mathbb{E}[Y|X] = X$. Hence martingale OT is a Kantorovich problem with the usual two marginal constraints plus a barycentric constraint on the conditional laws.
$(\mathrm{Id}, \bar{T}_\pi)_\sharp \alpha$

**Stochastic orders.**

**Proposition 11.20** (Strassen's theorem for stochastic order)**.** *Let $\alpha, \beta \in \mathcal{P}(\mathbb{R})$ and define the classical stochastic order by*

$$\alpha \preceq_{\mathrm{st}} \beta \quad \Longleftrightarrow \quad \int \varphi \, \mathrm{d}\alpha \leqslant \int \varphi \, \mathrm{d}\beta \quad \textit{for every bounded increasing } \varphi.$$

*Then $\alpha \preceq_{\mathrm{st}} \beta$ if and only if there exists $\pi \in \mathcal{U}(\alpha, \beta)$ concentrated on $\{(x, y) \; ; \; x \leqslant y\}$.*

*Proof.* A coupling supported on $x \leqslant y$ immediately gives the integral inequality for every increasing $\varphi$. Conversely, testing approximate indicators of intervals $(t, +\infty)$ gives $F_\alpha(t) \geqslant F_\beta(t)$ for all continuity points of the distribution functions. If $U$ is uniform on $(0, 1)$ and $F_\alpha^{-1}, F_\beta^{-1}$ are the generalized quantiles, then $X = F_\alpha^{-1}(U)$ and $Y = F_\beta^{-1}(U)$ have laws $\alpha$ and $\beta$, and the inequality between distribution functions gives $X \leqslant Y$ almost surely. □

**Convex order and martingale feasibility.** For martingale OT, the pointwise order constraint is replaced by a barycentric constraint on conditional laws. The corresponding order is the convex order. For $\alpha, \beta \in \mathcal{P}_1(\mathbb{R}^d)$,

$$\alpha \preceq_{\mathrm{cx}} \beta \quad \Longleftrightarrow \quad \int \varphi \, \mathrm{d}\alpha \leqslant \int \varphi \, \mathrm{d}\beta \quad \text{for every convex } \varphi \text{ with finite integrals.}$$

**Theorem 11.21** (Strassen's martingale theorem)**.** *Let $\alpha, \beta \in \mathcal{P}_1(\mathbb{R}^d)$. Then $\alpha \preceq_{\mathrm{cx}} \beta \iff \mathcal{U}_{\mathrm{mart}}(\alpha, \beta) \neq \emptyset$. Equivalently, $\beta$ is above $\alpha$ in convex order if and only if there exists a coupling $\pi(\mathrm{d}x, \mathrm{d}y) = \pi_x(\mathrm{d}y)\alpha(\mathrm{d}x)$ such that $\int y \mathrm{d}\pi_x(y) = x$ for $\alpha$-almost every $x$.*

*Proof.* If $\pi$ is a martingale coupling, then Jensen's inequality gives, for every convex $\varphi$,

$$\varphi(x) = \varphi\Big(\int y \mathrm{d}\pi_x(y)\Big) \leqslant \int \varphi(y) \mathrm{d}\pi_x(y),$$

and integration in $x$ gives $\int \varphi \mathrm{d}\alpha \leqslant \int \varphi \mathrm{d}\beta$.
Conversely, assume $\alpha \preceq_{\mathrm{cx}} \beta$. In the compactly supported case, one proves existence by separating $\beta$ from the closed convex set of all second marginals reachable from $\alpha$ by martingale kernels. If $\beta$ were not in this set, a separating test function $\psi$ would satisfy

$$\int \psi \mathrm{d}\beta > \sup \left\{ \int \psi \mathrm{d}\eta \; : \; \eta \in \mathcal{P}_1(\mathbb{R}^d), \; \mathcal{U}_{\mathrm{mart}}(\alpha, \eta) \neq \emptyset \right\}.$$

For a fixed $x$, optimizing over conditional laws with barycenter $x$ gives the concave envelope $\operatorname{conc} \psi(x)$; hence the right-hand side is $\int \operatorname{conc} \psi \mathrm{d}\alpha$. Since $\alpha \preceq_{\mathrm{cx}} \beta$ is equivalently the reverse inequality for concave functions, $\int \operatorname{conc} \psi \mathrm{d}\alpha \geqslant \int \operatorname{conc} \psi \mathrm{d}\beta \geqslant \int \psi \mathrm{d}\beta$, a contradiction. The finite-first-moment case follows by the standard truncation and tightness argument in Strassen's theorem. □

$\mathcal{N}(m, \Sigma_0) \preceq_{\mathrm{cx}} \mathcal{N}(m, \Sigma_1) \iff \Sigma_1 - \Sigma_0 \succeq 0.$

# 12 Beyond Comparing Measures

## 12.1 Vector and Matrix-Valued Measures

This compact subsection records how transport ideas extend to vector and matrix-valued measures.

**Positive vector-valued measures.**

**Definition 12.1** (Positive vector-valued measure)**.** A positive $\mathbb{R}^m_+$-valued measure on $\mathcal{X}$ is a tuple $\alpha = (\alpha^1, \dots, \alpha^m) \in \mathcal{M}_+(\mathcal{X}; \mathbb{R}^m_+)$, where each component $\alpha^k$ is a nonnegative finite measure.

To keep the notation readable, first assume that the endpoints and the curve have densities. The direct analogue of Benamou–Brenier fixes a vector density $u_t(x) \in \mathbb{R}^m_+$ and a spatial flux $V_t(x) = (V_{t,1}, \dots, V_{t,d}) \in (\mathbb{R}^m)^d$, where $V^k_{t,\ell}$ is the momentum of channel $k$ in spatial direction $\ell$.

$$\mathcal{W}^2_\Phi(\alpha_0, \alpha_1) := \inf_{u,V} \int_0^1 \int_{\mathcal{X}} \Phi(u_t(x), V_t(x))\, \mathrm{d}x\, \mathrm{d}t \tag{12.1}$$

$$\partial_t u_t + \nabla_x \cdot V_t = 0, \qquad (\nabla_x \cdot V_t)^k = \sum_{\ell=1}^d \partial_{x_\ell} V^k_{t,\ell}. \tag{12.2}$$

**Proposition 12.2** (Diagonal positive vector Benamou–Brenier)**.** *Assume that $\alpha^k_0, \alpha^k_1 \in \mathcal{M}_+(\mathcal{X})$ have the same mass $m_k$ for every $k$. For the diagonal mobility $\mathsf{M}_{\mathrm{diag}}$, the value of (12.1) is*

$$\mathcal{W}^2_{\mathrm{diag}}(\alpha_0, \alpha_1) = \sum_{k: m_k > 0} m_k\, \mathcal{W}^2_2\left(\frac{\alpha^k_0}{m_k}, \frac{\alpha^k_1}{m_k}\right),$$

*with the convention that zero-mass channels contribute zero.*

*Proof.* For $\mathsf{M}_{\mathrm{diag}}(u) = \mathrm{diag}(u_1, \dots, u_m)$, the action separates as $\sum_{\ell=1}^d V_\ell^\top \mathsf{M}_{\mathrm{diag}}(u)^\dagger V_\ell = \sum_{k=1}^m \frac{|V^k|^2}{u^k}$, where $V^k = (V^k_1, \dots, V^k_d)$ is the spatial momentum of channel $k$, and the scalar perspective convention is used. The constraint (12.2) also separates into $\partial_t u^k + \nabla \cdot V^k = 0$. The minimization therefore splits into $m$ independent scalar Benamou–Brenier problems. If $m_k = 0$, nonnegativity and conservation force the whole channel to vanish. If $m_k > 0$, normalizing $\rho^k_t = u^k_t / m_k$ and $p^k_t = V^k_t / m_k$ factors the channel action as $m_k \int |p^k_t|^2 / \rho^k_t$, hence the scalar value is $m_k\, \mathcal{W}^2_2(\alpha^k_0 / m_k, \alpha^k_1 / m_k)$. Summing over the channels proves the claim. □

**Positive matrix-valued measures.**

**Definition 12.3** (Positive matrix-valued measure)**.** Write $\mathbb{S}^m$ for real symmetric matrices and $\mathbb{S}^m_+$ for the positive semidefinite cone. A positive $\mathbb{S}^m_+$-valued measure is an element $\mathcal{A} \in \mathcal{M}_+(\mathcal{X}; \mathbb{S}^m_+)$.

If $\mathcal{A}$ has density $A(x) \succeq 0$, then $\operatorname{tr} A(x)$ is the scalar amount of mass at $x$, while, wherever $\operatorname{tr} A(x) > 0$, the normalized matrix $A(x)/\operatorname{tr} A(x)$ records an internal covariance or orientation. This is the matrix analogue of the positive vector case: diagonal matrices encode nonnegative vector components, and non-diagonal matrices add a local eigenbasis.

$$\mathcal{W}^2_{\mathrm{mat}}(\mathcal{A}_0, \mathcal{A}_1) := \inf_{A,P} \int_0^1 \int_{\mathcal{X}} \sum_{\ell=1}^d \operatorname{tr}\left(P^\top_{t,\ell} A^\dagger_t P_{t,\ell}\right) \mathrm{d}x \mathrm{d}t \tag{12.3}$$

$$\partial_t A_t + \nabla_x \cdot P_t = 0, \qquad \nabla_x \cdot P_t = \sum_{\ell=1}^d \partial_{x_\ell} P_{t,\ell}. \tag{12.4}$$

**Proposition 12.4** (Diagonal matrix subproblem)**.** *Assume that the endpoints are diagonal in a fixed orthonormal basis, $\mathcal{A}_i = \mathrm{diag}(\alpha^1_i, \dots, \alpha^m_i)$, $\quad i = 0, 1$, and that $\alpha^k_0(\mathcal{X}) = \alpha^k_1(\mathcal{X}) = m_k$ for every $k$. If one restricts the admissible curves in (12.3) to remain diagonal in that basis, $A_t = \mathrm{diag}(u^1_t, \dots, u^m_t)$, $\quad P_{t,\ell} = \mathrm{diag}(V^1_{t,\ell}, \dots, V^m_{t,\ell})$, then the value of this restricted matrix problem is*

$$\sum_{k: m_k > 0} m_k\, \mathcal{W}^2_2\left(\frac{\alpha^k_0}{m_k}, \frac{\alpha^k_1}{m_k}\right),$$

*with zero contribution from zero-mass channels. Thus the commuting matrix submodel is exactly the diagonal positive vector-valued Benamou–Brenier model of Proposition 12.2.*

*Proof.* The continuity equation (12.4) is diagonal entry by diagonal entry and gives $\partial_t u^k + \nabla \cdot V^k = 0$. Moreover, $\sum_{\ell=1}^d \operatorname{tr}\left(P^\top_{t,\ell} A^\dagger_t P_{t,\ell}\right) = \sum_{k=1}^m \frac{|V^k_t|^2}{u^k_t}$ with the same scalar perspective convention as before. The admissible set and the action are therefore exactly those of the diagonal vector model. □

## 12.2 Wasserstein over Wasserstein

**Proposition 12.5** (Wasserstein spaces as ground spaces)**.** *If $(\mathcal{X}, d)$ is a Polish metric space, then $\mathcal{P}_p(\mathcal{X})$ endowed with $\mathcal{W}_p$ is Polish. If $\mathcal{X}$ is compact, then $\mathcal{P}(\mathcal{X})$ is compact for the Wasserstein topology, and the construction can be iterated to form $\mathcal{P}(\mathcal{P}(\mathcal{X}))$, $\mathcal{P}(\mathcal{P}(\mathcal{P}(\mathcal{X})))$, and so on.*

*Proof.* This is a standard structural theorem for Wasserstein spaces. Completeness follows by representing a $\mathcal{W}_p$-Cauchy sequence through almost optimally glued couplings, which gives a Cauchy random sequence whose law is the desired limit; separability follows by approximating measures with finitely supported measures on a countable dense subset and rational weights. If $\mathcal{X}$ is compact, Prokhorov compactness gives compactness of $\mathcal{P}(\mathcal{X})$ for weak convergence, and Proposition 3.37 identifies this topology with any Wasserstein topology. □

We denote elements of $\mathcal{P}_2(\mathcal{X})$ by $\alpha, \beta, \ldots$. Elements of $\mathcal{P}_2(\mathcal{P}_2(\mathcal{X}))$ are denoted by fraktur letters, for instance $\mathfrak{A}, \mathfrak{B}$; they are probability laws over probability measures, or random probability measures.

$$\mathfrak{A} = (\zeta \mapsto \alpha_\zeta)_\sharp \gamma. \tag{12.5}$$

If $\gamma = \sum_{i=1}^K a_i \delta_{\zeta_i}$, then
$\mathfrak{A} = \sum_{i=1}^K a_i \delta_{\alpha_{\zeta_i}}$.

**Definition 12.6** (Collapsed, or barycentric, mixture)**.** For $\mathfrak{A} \in \mathcal{P}(\mathcal{P}_2(\mathcal{X}))$, the collapsed, or barycentric, mixture associated with $\mathfrak{A}$ is the measure $\bar{\alpha}_{\mathfrak{A}}$ defined by

$$\int_{\mathcal{X}} f(x) \mathrm{d}\bar{\alpha}_{\mathfrak{A}}(x) = \int_{\mathcal{P}_2(\mathcal{X})} \left( \int_{\mathcal{X}} f(x) \mathrm{d}\alpha(x) \right) \mathrm{d}\mathfrak{A}(\alpha), \tag{12.6}$$

for bounded continuous $f$.

$$\mathbb{W}_2^2(\mathfrak{A}, \mathfrak{B}) \coloneqq \inf_{\Pi \in \mathcal{U}(\mathfrak{A}, \mathfrak{B})} \int_{\mathcal{P}_2(\mathcal{X}) \times \mathcal{P}_2(\mathcal{X})} \mathcal{W}_2^2(\alpha, \beta) \mathrm{d}\Pi(\alpha, \beta). \tag{12.7}$$

For Gaussian mixtures, this separates two levels of geometry. A mixture $\sum_i a_i \mathcal{N}(m_i, \Sigma_i)$ can either be viewed as the collapsed measure on $\mathcal{X}$, or as the component law
$\mathfrak{A} = \sum_i a_i \delta_{\mathcal{N}(m_i, \Sigma_i)}$ $\mathfrak{A} = \sum_i a_i \delta_{\mathcal{N}(m_i, \Sigma_i)}$, $\quad \mathfrak{B} = \sum_j b_j \delta_{\mathcal{N}(n_j, \Lambda_j)}$, $C_{ij} = \|m_i - n_j\|^2 + \mathcal{B}(\Sigma_i, \Lambda_j)^2$. Let $\Pi^\star$ be an optimal coupling between the weights $a$ and $b$. If $A_{ij}$ denotes the Brenier linear part from $\Sigma_i$ to $\Lambda_j$, then each active component pair is interpolated by
$m_{ij,t} = (1-t) m_i + t n_j, \quad \Sigma_{ij,t} = \big((1-t)\mathrm{Id} + t A_{ij}\big) \Sigma_i \big((1-t)\mathrm{Id} + t A_{ij}\big)$, $\bar{\alpha}_t = \sum_{i,j} \Pi^\star_{ij} \mathcal{N}(m_{ij,t}, \Sigma_{ij,t})$.

**Proposition 12.7** (Collapsing is non-expansive)**.** *Let $\mathfrak{A}, \mathfrak{B} \in \mathcal{P}_2(\mathcal{P}_2(\mathcal{X}))$, and let $\bar{\alpha}_{\mathfrak{A}}$ and $\bar{\beta}_{\mathfrak{B}}$ be the collapsed mixtures defined by (12.6). Then $\mathcal{W}_2(\bar{\alpha}_{\mathfrak{A}}, \bar{\beta}_{\mathfrak{B}}) \leqslant \mathbb{W}_2(\mathfrak{A}, \mathfrak{B})$.*

*Proof.* Fix $\Pi \in \mathcal{U}(\mathfrak{A}, \mathfrak{B})$. For every $(\alpha, \beta)$ choose, by a standard measurable selection argument and up to an arbitrarily small error, a coupling $\pi_{\alpha,\beta} \in \mathcal{U}(\alpha, \beta)$ whose quadratic cost is $\mathcal{W}_2^2(\alpha, \beta)$. Integrating this Markov kernel against $\Pi$ gives a coupling $\bar{\pi}$ between $\bar{\alpha}_{\mathfrak{A}}$ and $\bar{\beta}_{\mathfrak{B}}$. Its cost satisfies $\int_{\mathcal{X} \times \mathcal{X}} d(x,y)^2 \mathrm{d}\bar{\pi}(x,y) \leqslant \int_{\mathcal{P}_2(\mathcal{X})^2} \mathcal{W}_2^2(\alpha, \beta) \mathrm{d}\Pi(\alpha, \beta)$ up to the arbitrary selection error. Taking first the infimum over $\bar{\pi}$ and then over $\Pi$ proves the claim. $\square$

### 12.3 Gromov–Wasserstein

This compact subsection summarizes comparison of metric-measure spaces up to relabeling.

**Discrete formulation.** Optimal transport needs a ground cost C to compare histograms (a, b), and thus cannot be used directly if the histograms are not defined on the same underlying space, or if one cannot pre-register these spaces to define a ground cost. To address this issue, one can instead use a weaker requirement: two matrices $\mathrm{D} \in \mathbb{R}^{n \times n}$ and $\mathrm{D}' \in \mathbb{R}^{m \times m}$ are available and represent relationships between the points on which the histograms are defined. A typical scenario is when these matrices are powers of distance matrices.

$$\mathrm{GW}((\mathrm{a}, \mathrm{D}), (\mathrm{b}, \mathrm{D}'))^p \coloneqq \min_{\mathrm{P} \in \mathrm{U}(\mathrm{a}, \mathrm{b})} \mathcal{E}_{\mathrm{D}, \mathrm{D}'}(\mathrm{P}) \coloneqq \sum_{i,j,i',j'} \Delta(\mathrm{D}_{i,i'}, \mathrm{D}'_{j,j'})^p \mathrm{P}_{i,j} \mathrm{P}_{i',j'}, \tag{12.8}$$

where $p \geqslant 1$ and $\Delta$ is a distance on $\mathbb{R}$, typically $\Delta(u,v) = |u - v|$.
When the matrices $\mathrm{D}, \mathrm{D}'$ are genuine distance matrices, the general construction below shows that GW satisfies the triangle inequality and defines a distance between metric spaces equipped with a probability distribution, up to measure-preserving isometries.

**General setting.**

**Definition 12.8** (Metric-measure space)**.** A metric-measure space is a triple $\mathbb{X} = (\mathcal{X}, d_{\mathcal{X}}, \alpha)$, where $(\mathcal{X}, d_{\mathcal{X}})$ is a metric space and $\alpha$ is a probability measure on $\mathcal{X}$.

The general setting corresponds to computing couplings between metric-measure spaces $\mathbb{X} = (\mathcal{X}, d_{\mathcal{X}}, \alpha)$ and $\mathbb{Y} = (\mathcal{Y}, d_{\mathcal{Y}}, \beta)$, where the distance and the measure are both part of the data.

$$\mathcal{GW}(\mathbb{X}, \mathbb{Y})^p \coloneqq \min_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{\mathcal{X}^2 \times \mathcal{Y}^2} \Delta(d_{\mathcal{X}}(x, x'), d_{\mathcal{Y}}(y, y'))^p \mathrm{d}\pi(x,y) \mathrm{d}\pi(x', y'). \tag{12.9}$$

**Proposition 12.9** (Fixed-space GW is controlled by Wasserstein)**.** *Let $\alpha, \beta \in \mathcal{P}_p(\mathcal{X})$ be probability measures on the same metric space $(\mathcal{X}, d_{\mathcal{X}})$, and take $\Delta(u,v) = |u - v|$ in (12.9). Then $\mathcal{GW}((\mathcal{X}, d_{\mathcal{X}}, \alpha), (\mathcal{X}, d_{\mathcal{X}}, \beta)) \leqslant 2\, \mathcal{W}_p(\alpha, \beta)$. The Wasserstein distance on the right is computed with the same ground metric $d_{\mathcal{X}}$. The factor 2 comes from the normalization in (12.9); with the alternative convention $\Delta(u,v) = |u - v|/2$, the displayed constant becomes 1.*

*Proof.* Let $\pi$ be any coupling between $\alpha$ and $\beta$. For two independent pairs $(X, Y), (X', Y') \sim \pi$, the reverse triangle inequality gives $\big| d_{\mathcal{X}}(X, X') - d_{\mathcal{X}}(Y, Y') \big| \leqslant d_{\mathcal{X}}(X, Y) + d_{\mathcal{X}}(X', Y')$. Taking the $L^p$ norm and using Minkowski gives a bound by $2 (\int d_{\mathcal{X}}(x,y)^p \mathrm{d}\pi)^{1/p}$. Optimizing over $\pi$ proves the claim. $\square$

**Definition 12.10** (Isometric metric-measure spaces)**.** Two metric-measure spaces $\mathbb{X} = (\mathcal{X}, d_{\mathcal{X}}, \alpha)$ and $\mathbb{Y} = (\mathcal{Y}, d_{\mathcal{Y}}, \beta)$ are isometric if there exists a measurable map $\varphi : \mathrm{supp}(\alpha) \to \mathrm{supp}(\beta)$ such that $\varphi_\sharp \alpha = \beta$, $\varphi(\mathrm{supp}(\alpha)) = \mathrm{supp}(\beta)$, and $d_{\mathcal{Y}}(\varphi(x), \varphi(x')) = d_{\mathcal{X}}(x, x')$ for all $x, x' \in \mathrm{supp}(\alpha)$.

**Theorem 12.11** (Gromov–Wasserstein metric modulo isometries)**.** *For compact metric-measure spaces, $p \geqslant 1$ and $\Delta(u,v) = |u - v|$, $\mathcal{GW}$ defines a distance up to measure-preserving isometries.*

*Proof.* If $\mathcal{GW}(\mathbb{X},\mathbb{Y})=0$ and $\pi$ is an optimal plan, then $d_{\mathcal{X}}(x,x')=d_{\mathcal{Y}}(y,y')$ holds $\pi\otimes\pi$-almost everywhere. By continuity, this equality holds on $\mathrm{supp}(\pi)^2$. $d_\pi((x,y),(x',y')) := \frac{1}{2}d_{\mathcal{X}}(x,x')+\frac{1}{2}d_{\mathcal{Y}}(y,y')$. The first projection $\psi:(x,y)\mapsto x$ is measure-preserving. For $((x,y),(x',y'))\in\mathrm{supp}(\pi)^2$, $d_{\mathcal{X}}(\psi(x,y),\psi(x',y'))=d_{\mathcal{X}}(x,x')=d_{\mathcal{Y}}(y,y')=d_\pi((x,y),(x',y'))$, so $\psi$ is an isometry and therefore injective. To see surjectivity onto $\mathrm{supp}(\alpha)$, take $x\in\mathrm{supp}(\alpha)$. Since $\psi_\sharp\pi=\alpha$, there is a sequence $(x_k,y_k)\in\mathrm{supp}(\pi)$ with $x_k\to x$. The equality of distances on $\mathrm{supp}(\pi)$ makes $(y_k)_k$ Cauchy, and compactness gives a convergent subsequence with limit $(x,y)\in\mathrm{supp}(\pi)$. The same argument for the second projection shows that the support space is also isometric to $\mathbb{Y}$.
For the triangle inequality, let $\pi$ be an optimal coupling between $\mathbb{X}$ and $\mathbb{Y}$, and $\xi$ an optimal coupling between $\mathbb{Y}$ and $\mathbb{Z}=(Z,d_Z,\gamma)$. By the gluing lemma, take $\sigma$ on $\mathcal{X}\times\mathcal{Y}\times Z$ whose $(\mathcal{X},\mathcal{Y})$ and $(\mathcal{Y},Z)$ marginals are $\pi$ and $\xi$. Let $\rho=(P_{\mathcal{X},Z})_\sharp\sigma$, and write $\bar\sigma=\sigma\otimes\sigma$ for the product law of two independent triples $(x,y,z)$ and $(x',y',z')$. Then $\rho$ is feasible between $\mathbb{X}$ and $\mathbb{Z}$, and

$$\begin{aligned}\mathcal{GW}(\mathbb{X},\mathbb{Z}) &\leqslant \left(\int |d_{\mathcal{X}}(x,x')-d_Z(z,z')|^p\,\mathrm{d}\bar\sigma\right)^{1/p}\\ &\leqslant \left(\int |d_{\mathcal{X}}(x,x')-d_{\mathcal{Y}}(y,y')|^p\,\mathrm{d}\bar\sigma\right)^{1/p}+\left(\int |d_{\mathcal{Y}}(y,y')-d_Z(z,z')|^p\,\mathrm{d}\bar\sigma\right)^{1/p}\\ &= \mathcal{GW}(\mathbb{X},\mathbb{Y})+\mathcal{GW}(\mathbb{Y},\mathbb{Z}),\end{aligned}$$

where the second inequality uses the pointwise triangle inequality followed by Minkowski's inequality. Symmetry and non-negativity are immediate. ☐

**Proposition 12.12** (Marginal stability and empirical GW rates)**.** *Let $\mathbb{X}=(\mathcal{X},d_{\mathcal{X}},\alpha)$ and $\mathbb{Y}=(\mathcal{Y},d_{\mathcal{Y}},\beta)$ be compact metric-measure spaces, with $\Delta(u,v)=|u-v|$ and exponent $p\geqslant 1$. For any probability measures $\tilde\alpha$ on $\mathcal{X}$ and $\tilde\beta$ on $\mathcal{Y}$, set $\widetilde{\mathbb{X}}=(\mathcal{X},d_{\mathcal{X}},\tilde\alpha)$, $\widetilde{\mathbb{Y}}=(\mathcal{Y},d_{\mathcal{Y}},\tilde\beta)$. Then, deterministically, $\left|\mathcal{GW}(\widetilde{\mathbb{X}},\widetilde{\mathbb{Y}})-\mathcal{GW}(\mathbb{X},\mathbb{Y})\right|\leqslant 2\,\mathcal{W}_p^{\mathcal{X}}(\tilde\alpha,\alpha)+2\,\mathcal{W}_p^{\mathcal{Y}}(\tilde\beta,\beta)$, where $\mathcal{W}_p^{\mathcal{X}}$ and $\mathcal{W}_p^{\mathcal{Y}}$ denote Wasserstein distances computed with $d_{\mathcal{X}}$ and $d_{\mathcal{Y}}$. If $\tilde\alpha=\hat\alpha_n$ and $\tilde\beta=\hat\beta_m$ are independent empirical measures built from iid samples with laws $\alpha$ and $\beta$, and if $\widehat{\mathbb{X}}_n=(\mathcal{X},d_{\mathcal{X}},\hat\alpha_n)$, $\widehat{\mathbb{Y}}_m=(\mathcal{Y},d_{\mathcal{Y}},\hat\beta_m)$, then $\mathbb{E}\left|\mathcal{GW}(\widehat{\mathbb{X}}_n,\widehat{\mathbb{Y}}_m)-\mathcal{GW}(\mathbb{X},\mathbb{Y})\right|\leqslant 2\mathbb{E}\,\mathcal{W}_p^{\mathcal{X}}(\hat\alpha_n,\alpha)+2\mathbb{E}\,\mathcal{W}_p^{\mathcal{Y}}(\hat\beta_m,\beta)$. In particular, if $\mathcal{X}$ and $\mathcal{Y}$ are regular compact subsets of Euclidean spaces of intrinsic dimensions $d_{\mathcal{X}}$ and $d_{\mathcal{Y}}$, and the high-dimensional Wasserstein regime $d_{\mathcal{X}},d_{\mathcal{Y}}>2p$ applies, then $\mathbb{E}\left|\mathcal{GW}(\widehat{\mathbb{X}}_n,\widehat{\mathbb{Y}}_m)-\mathcal{GW}(\mathbb{X},\mathbb{Y})\right|=O(n^{-1/d_{\mathcal{X}}}+m^{-1/d_{\mathcal{Y}}})$. For $p=1$ on subsets of $[0,1]^d$, the rates, including the low-dimensional logarithmic cases, are those of Proposition 9.4.*

*Proof.* The triangle inequality of Theorem 12.11 gives $\mathcal{GW}(\widetilde{\mathbb{X}},\widetilde{\mathbb{Y}})-\mathcal{GW}(\mathbb{X},\mathbb{Y})\leqslant\mathcal{GW}(\widetilde{\mathbb{X}},\mathbb{X})+\mathcal{GW}(\mathbb{Y},\widetilde{\mathbb{Y}})$. Exchanging the two pairs gives the same bound for the opposite difference, hence the absolute value. The two terms on the right compare metric-measure spaces with the same underlying metric and only different measures, so Proposition 12.9 bounds them by $2\,\mathcal{W}_p^{\mathcal{X}}(\tilde\alpha,\alpha)$ and $2\,\mathcal{W}_p^{\mathcal{Y}}(\tilde\beta,\beta)$. Substituting $\tilde\alpha=\hat\alpha_n$ and $\tilde\beta=\hat\beta_m$ and taking expectations gives the second display. The Euclidean rate then follows from the empirical Wasserstein rates recalled in Section 9.2. ☐

**Proposition 12.13** (Gromov–Wasserstein geodesics)**.** *Let $\mathbb{X}_0=(\mathcal{X}_0,d_{\mathcal{X}_0},\alpha_0)$ and $\mathbb{X}_1=(\mathcal{X}_1,d_{\mathcal{X}_1},\alpha_1)$ be compact metric-measure spaces, take $\Delta(u,v)=|u-v|$ in (12.9), and let $\pi^\star$ be an optimal coupling. Define, on $Z=\mathcal{X}_0\times\mathcal{X}_1$, $d_t((x_0,x_1),(x_0',x_1')) := (1-t)d_{\mathcal{X}_0}(x_0,x_0')+td_{\mathcal{X}_1}(x_1,x_1')$, $\mathbb{X}_t=(Z,d_t,\pi^\star)$. At $t=0$ and $t=1$, and possibly in degenerate cases, one quotients $Z$ by the zero-distance relation associated with $d_t$. Then $t\mapsto\mathbb{X}_t$ is a constant-speed geodesic: $\mathcal{GW}(\mathbb{X}_s,\mathbb{X}_t)=|t-s|\,\mathcal{GW}(\mathbb{X}_0,\mathbb{X}_1)\quad\forall s,t\in[0,1]$.*

*Proof.* Write $D=\mathcal{GW}(\mathbb{X}_0,\mathbb{X}_1)$. For $s<t$, couple $\mathbb{X}_s$ and $\mathbb{X}_t$ by the diagonal coupling induced by the identity on $Z$ and the measure $\pi^\star$. For two independent points $z=(x_0,x_1)$ and $z'=(x_0',x_1')$ sampled from $\pi^\star$, $d_t(z,z')-d_s(z,z')=(t-s)\big(d_{\mathcal{X}_1}(x_1,x_1')-d_{\mathcal{X}_0}(x_0,x_0')\big)$. Using this feasible coupling gives $\mathcal{GW}(\mathbb{X}_s,\mathbb{X}_t)\leqslant(t-s)D$. The same construction with the projections from $Z$ to $\mathcal{X}_0$ and $\mathcal{X}_1$ gives $\mathcal{GW}(\mathbb{X}_0,\mathbb{X}_t)\leqslant tD$ and $\mathcal{GW}(\mathbb{X}_t,\mathbb{X}_1)\leqslant(1-t)D$. The triangle inequality for $\mathcal{GW}$ then yields, for $0\leqslant s\leqslant t\leqslant 1$, $D\leqslant\mathcal{GW}(\mathbb{X}_0,\mathbb{X}_s)+\mathcal{GW}(\mathbb{X}_s,\mathbb{X}_t)+\mathcal{GW}(\mathbb{X}_t,\mathbb{X}_1)\leqslant sD+\mathcal{GW}(\mathbb{X}_s,\mathbb{X}_t)+(1-t)D$, so $\mathcal{GW}(\mathbb{X}_s,\mathbb{X}_t)\geqslant(t-s)D$. This proves equality. ☐

**Proposition 12.14** (Mémoli profile lower bound)**.** *Let $\mathbb{X}=(\mathcal{X},d_{\mathcal{X}},\alpha)$ and $\mathbb{Y}=(\mathcal{Y},d_{\mathcal{Y}},\beta)$ be compact metric-measure spaces and take $\Delta(u,v)=|u-v|$ in (12.9), with the same exponent $p\geqslant 1$. For each $x\in\mathcal{X}$ and $y\in\mathcal{Y}$, define the distance-profile measures on $\mathbb{R}_+$ by $\alpha_x := (d_{\mathcal{X}}(x,\cdot))_\sharp\alpha$, $\beta_y := (d_{\mathcal{Y}}(y,\cdot))_\sharp\beta$. Let $\mathsf{E}_{\mathbb{X}}=(x\mapsto\alpha_x)_\sharp\alpha$ and $\mathsf{E}_{\mathbb{Y}}=(y\mapsto\beta_y)_\sharp\beta$, which are probability measures on $\mathcal{P}(\mathbb{R}_+)$. Then $\mathcal{W}_p\big(\mathsf{E}_{\mathbb{X}},\mathsf{E}_{\mathbb{Y}}\big)\leqslant\mathcal{GW}(\mathbb{X},\mathbb{Y})$. Here the left-hand distance is taken on the space $\mathcal{P}(\mathbb{R}_+)$ of profile measures. Its ground cost is the one-dimensional Wasserstein distance $\mathcal{W}_p$.*

*Proof.* Fix any $\pi\in\mathcal{U}(\alpha,\beta)$. It induces a coupling $(x,y)\mapsto(\alpha_x,\beta_y)$ between $\mathsf{E}_{\mathbb{X}}$ and $\mathsf{E}_{\mathbb{Y}}$, hence $\mathcal{W}_p\big(\mathsf{E}_{\mathbb{X}},\mathsf{E}_{\mathbb{Y}}\big)^p\leqslant\int_{\mathcal{X}\times\mathcal{Y}}\mathcal{W}_p(\alpha_x,\beta_y)^p\,\mathrm{d}\pi(x,y)$. For fixed $(x,y)$, the map $(x',y')\mapsto(d_{\mathcal{X}}(x,x'),d_{\mathcal{Y}}(y,y'))$ pushes the same coupling $\pi$ to a coupling between $\alpha_x$ and $\beta_y$. Therefore $\mathcal{W}_p(\alpha_x,\beta_y)^p\leqslant\int_{\mathcal{X}\times\mathcal{Y}}|d_{\mathcal{X}}(x,x')-d_{\mathcal{Y}}(y,y')|^p\,\mathrm{d}\pi(x',y')$. Integrating in $(x,y)$ gives $\mathcal{W}_p\big(\mathsf{E}_{\mathbb{X}},\mathsf{E}_{\mathbb{Y}}\big)^p\leqslant\int_{\mathcal{X}^2\times\mathcal{Y}^2}|d_{\mathcal{X}}(x,x')-d_{\mathcal{Y}}(y,y')|^p\,\mathrm{d}\pi(x,y)\mathrm{d}\pi(x',y')$. Taking the infimum over $\pi$ and then the $p$-th root proves the claim. ☐

**Relation with Wasserstein–Procrustes.**

**Proposition 12.15** (Wasserstein–Procrustes upper certificate)**.** *Let $\alpha,\beta$ be probability measures on $\mathbb{R}^d$, equipped with the Euclidean distance, and take $\Delta(u,v)=|u-v|$ in (12.9). If $\mathcal{W}_{p,\mathrm{E}(d)}$ denotes the quotient Wasserstein distance of Definition 10.15 for the Euclidean group, then*

$$\mathcal{GW}((\mathbb{R}^d,\|\cdot\|,\alpha),(\mathbb{R}^d,\|\cdot\|,\beta))\leqslant 2\,\mathcal{W}_{p,\mathrm{E}(d)}([\alpha],[\beta]). \tag{12.10}$$

*Proof.* Let $g,h\in\mathrm{E}(d)$ be rigid motions. They preserve all pairwise Euclidean distances, so $(\mathbb{R}^d,\|\cdot\|,\alpha)$ is isometric to $(\mathbb{R}^d,\|\cdot\|,g_\sharp\alpha)$, and similarly for $\beta$ and $h_\sharp\beta$. Hence $\mathcal{GW}$ is unchanged by pushing the two measures forward by $g$ and $h$. Applying Proposition 12.9 to $g_\sharp\alpha$ and $h_\sharp\beta$ gives $\mathcal{GW}((\mathbb{R}^d,\|\cdot\|,\alpha),(\mathbb{R}^d,\|\cdot\|,\beta))\leqslant 2\,\mathcal{W}_p(g_\sharp\alpha,h_\sharp\beta)$. Taking the infimum over $g,h\in\mathrm{E}(d)$ gives (12.10). ☐

$$\mathcal{W}_p(\mathsf{E}_{\mathbb{X}},\mathsf{E}_{\mathbb{Y}})\leqslant\mathcal{GW}(\mathbb{X},\mathbb{Y})\leqslant 2\,\mathcal{W}_{p,\mathrm{E}(d)}([\alpha],[\beta]), \tag{12.11}$$

where $\mathbb{X}=(\mathbb{R}^d,\|\cdot\|,\alpha)$ and $\mathbb{Y}=(\mathbb{R}^d,\|\cdot\|,\beta)$. The left term is intrinsic and inexpensive, while the right term is an extrinsic rigid-registration certificate.

**Entropic regularization and iterative solver.** For the common squared distortion $\Delta(u,v)^2=(u-v)^2$, one often computes a stationary point of the entropic relaxation

$$\min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \mathcal{E}_{\mathrm{D},\mathrm{D}'}(\mathrm{P})-\varepsilon\mathrm{H}(\mathrm{P}). \tag{12.12}$$

$$\mathrm{P}^{(\ell+1)} \coloneqq \min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \langle \mathrm{P},\, \mathrm{C}^{(\ell)}\rangle-\varepsilon\mathrm{H}(\mathrm{P}), \qquad \mathrm{C}^{(\ell)} \coloneqq \mathrm{D}^{\odot 2}\mathrm{a}\,\mathbb{1}_m^\top+\mathbb{1}_n(\mathrm{D}'^{\odot 2}\mathrm{b})^\top-2\mathrm{D}\,\mathrm{P}^{(\ell)}\,\mathrm{D}'^\top. \tag{12.13}$$

**Adding features.**

$$\mathrm{FGW}_{\lambda,p}((\mathrm{a},\mathrm{D}),(\mathrm{b},\mathrm{D}'))^p \coloneqq \min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} (1-\lambda)\sum_{i,j} M_{ij}\mathrm{P}_{ij}+\lambda\sum_{i,j,i',j'}\Delta(\mathrm{D}_{ii'},\mathrm{D}'_{jj'})^p\mathrm{P}_{ij}\mathrm{P}_{i'j'}.$$

**Hausdorff and Gromov–Hausdorff viewpoints.** If $A,B$ are compact subsets of a common metric space $(Z,d_Z)$, their Hausdorff distance is

$$d_{\mathrm{H}}^Z(A,B)=\max\left\{\sup_{a\in A}\inf_{b\in B} d_Z(a,b), \sup_{b\in B}\inf_{a\in A} d_Z(a,b)\right\}.$$

$d_{\mathrm{GH}}(\mathcal{X},\mathcal{Y})=\inf_{Z,\varphi,\psi} d_{\mathrm{H}}^Z(\varphi(\mathcal{X}),\psi(\mathcal{Y}))$.

## 12.4 Quantum Optimal Transport

This compact subsection collects the finite-dimensional quantum OT objects.

**Finite-dimensional states and couplings.**

**Definition 12.16** (Hermitian and density matrices)**.** Let $\mathbb{H}_n$ be the real vector space of $n\times n$ Hermitian matrices, $\mathbb{H}_n^+=\{A\in\mathbb{H}_n\;;\;A\succeq 0\}$, $\quad \mathbb{H}_n^{+,1}=\{A\in\mathbb{H}_n^+\;;\;\mathrm{tr}(A)=1\}$. Elements of $\mathbb{H}_n^{+,1}$ are density matrices.

A joint quantum state between $\mathbb{C}^n$ and $\mathbb{C}^m$ is a matrix $T\in\mathbb{H}_{nm}^+$ acting on $\mathbb{C}^n\otimes\mathbb{C}^m$. Its marginals are the partial traces, defined by duality through

$$\mathrm{tr}(F\ \mathrm{Tr}_B\,T)=\mathrm{tr}((F\otimes\mathrm{Id}_m)T), \qquad \mathrm{tr}(G\ \mathrm{Tr}_A\,T)=\mathrm{tr}((\mathrm{Id}_n\otimes G)T), \tag{12.14}$$

for all $F\in\mathbb{H}_n$ and $G\in\mathbb{H}_m$. Thus $\mathrm{Tr}_B(T)\in\mathbb{H}_n^+$ and $\mathrm{Tr}_A(T)\in\mathbb{H}_m^+$ play exactly the role of the two marginals of a classical coupling.

**Definition 12.17** (Finite-dimensional quantum OT)**.** Let $A\in\mathbb{H}_n^{+,1}$, $B\in\mathbb{H}_m^{+,1}$ and let $C\in\mathbb{H}_{nm}$ be a Hermitian cost observable. The quantum OT value is the semidefinite program

$$\mathrm{QOT}_C(A,B)\coloneqq\min_{T\in\mathbb{H}_{nm}^+}\{\mathrm{tr}(CT)\;;\;\mathrm{Tr}_B(T)=A,\ \mathrm{Tr}_A(T)=B\}. \tag{12.15}$$

**Proposition 12.18** (Quantum Kantorovich duality)**.** *For $A\in\mathbb{H}_n^{+,1}$ and $B\in\mathbb{H}_m^{+,1}$, the dual of (12.15) is*

$$\mathrm{QOT}_C(A,B)=\max_{F\in\mathbb{H}_n,\ G\in\mathbb{H}_m}\{\mathrm{tr}(FA)+\mathrm{tr}(GB):\ F\otimes\mathrm{Id}_m+\mathrm{Id}_n\otimes G\preceq C\}. \tag{12.16}$$

*If $A$ and $B$ are positive definite, strong duality follows directly from Slater's condition; the semidefinite case follows by restriction to the supports of $A$ and $B$ or by approximation.*

*Proof.* Introduce Hermitian Lagrange multipliers $F$ and $G$ for the two marginal constraints. The Lagrangian is

$$\mathrm{tr}(CT)+\mathrm{tr}(F(A-\mathrm{Tr}_B\,T))+\mathrm{tr}(G(B-\mathrm{Tr}_A\,T))=\mathrm{tr}(FA)+\mathrm{tr}(GB)+\mathrm{tr}((C-F\otimes\mathrm{Id}_m-\mathrm{Id}_n\otimes G)T),$$

where (12.14) was used in the last equality. Minimizing over $T\succeq 0$ gives a finite lower bound if and only if $C-F\otimes\mathrm{Id}_m-\mathrm{Id}_n\otimes G\succeq 0$, in which case the infimum in $T$ is 0. When $A,B\succ 0$, the coupling $A\otimes B$ is strictly feasible, so Slater's theorem gives equality of primal and dual values and dual attainment. The general finite-dimensional semidefinite case is obtained by approximation $A_\delta=(1-\delta)A+\delta\mathrm{Id}_n/n$, $B_\delta=(1-\delta)B+\delta\mathrm{Id}_m/m$ and by compactness, or equivalently by reducing to the supports of $A$ and $B$. □

**Entropic regularization and Bregman iterations.**

**Definition 12.19** (von Neumann quantum entropy)**.** For a density matrix or positive semidefinite matrix $T$, the shifted von Neumann entropy functional used here is

$$H(T)=\mathrm{tr}(T(\log T-\mathrm{Id})), \qquad \nabla H(T)=\log T,$$

with the convention $0\log 0=0$ on eigenvalues. This is the convex negative quantum entropy; on trace-one states it differs from the physical entropy $-\mathrm{tr}(T\log T)$ by a sign and an additive constant.

For $\varepsilon>0$ define

$$\mathrm{QOT}_C^\varepsilon(A,B)=\min_{T\succeq 0}\{\mathrm{tr}(CT)+\varepsilon H(T):\ \mathrm{Tr}_B(T)=A,\mathrm{Tr}_A(T)=B\}. \tag{12.17}$$

**Proposition 12.20** (Entropic quantum OT duality)**.** *Assume $A\succ 0$, $B\succ 0$ and $\varepsilon>0$. Then (12.17) has a unique positive minimizer. Its dual is*

$$\mathrm{QOT}_C^\varepsilon(A,B)=\max_{F\in\mathbb{H}_n,\ G\in\mathbb{H}_m}\left\{\mathrm{tr}(FA)+\mathrm{tr}(GB)-\varepsilon\ \mathrm{tr}\exp\left(\frac{F\otimes\mathrm{Id}_m+\mathrm{Id}_n\otimes G-C}{\varepsilon}\right)\right\}. \tag{12.18}$$

*At optimality, primal and dual variables are linked by the Gibbs formula*

$$T_e(F,G) = \exp\left(\frac{F\otimes \mathrm{Id}_m + \mathrm{Id}_n \otimes G - C}{\varepsilon}\right), \tag{12.19}$$

*with* $\mathrm{Tr}_B(T_e) = A$ *and* $\mathrm{Tr}_A(T_e) = B$.

*Proof.* The feasible set is compact and nonempty, and it contains the positive definite point $A\otimes B$. The trace entropy is strictly convex on positive semidefinite matrices, hence the regularized primal has a unique minimizer. Slater's condition justifies the Lagrange dual computation. The Fenchel identity $\sup_{T\succeq 0}\ \mathrm{tr}(YT) - \varepsilon H(T) = \varepsilon\ \mathrm{tr}\exp(Y/\varepsilon)$ is the matrix analogue of the scalar exponential conjugacy. Applying it to the Lagrangian of (12.17), with $Y = F\otimes \mathrm{Id}_m + \mathrm{Id}_n\otimes G - C$, gives (12.18). The stationarity condition of this Fenchel identity gives (12.19); differentiating the dual objective with respect to $F$ and $G$ yields the two marginal equations. □

$$D_H(T|K) = \mathrm{tr}(T(\log T - \log K) - T + K).$$

$\mathcal{M}_A = \{T\succeq 0 : \mathrm{Tr}_B(T) = A\}, \qquad \mathcal{M}_B = \{T\succeq 0 : \mathrm{Tr}_A(T) = B\}.$

**Proposition 12.21** (Exact Bregman projections)**.** *Assume* $A, B \succ 0$ *and let* $K = \exp(-C/\varepsilon)$*. The minimizer of* (12.17) *is equivalently the minimizer of* $D_H(T|K)$ *over* $\mathcal{M}_A\cap\mathcal{M}_B$*. Moreover, if a current positive definite matrix has Gibbs form* $T_e(F,G)$*, then its Bregman projection onto* $\mathcal{M}_A$ *has the form* $T_e(F^+,G)$*, where* $F^+$ *is chosen so that* $\mathrm{Tr}_B\, T_e(F^+,G) = A$*. The projection onto* $\mathcal{M}_B$ *is analogous. Thus, when each one-block marginal equation is solved exactly, alternating Bregman projections are equivalent to alternating block maximization of the dual* (12.18)*.*

*Proof.* Since $\log K = -C/\varepsilon$, the identity $\mathrm{tr}(CT) + \varepsilon H(T) = \varepsilon D_H(T|K) - \varepsilon\,\mathrm{tr}(K)$ holds, so the primal minimizer is the constrained Bregman projection of $K$ up to an additive constant. For the projection of a positive definite matrix $S$ onto $\mathcal{M}_A$, the affine set contains the positive definite point $A\otimes \mathrm{Id}_m/m$; the entropy derivative $\log T - \log S$ is singular at the boundary, so the projection lies in the interior of the positive cone. Its Lagrangian first variation is $\log T - \log S - \Lambda\otimes \mathrm{Id}_m = 0$ for a Hermitian multiplier $\Lambda$. Hence $T = \exp(\log S + \Lambda\otimes\mathrm{Id}_m)$. If $S = T_e(F,G)$, this is again of the form $T_e(F+\varepsilon\Lambda, G)$. The multiplier is fixed by the marginal equation $\mathrm{Tr}_B(T) = A$. The same argument applies to $\mathcal{M}_B$. Finally, the first-order optimality condition for maximizing (12.18) over one block is exactly the corresponding marginal equation, so the Bregman and block-dual views coincide. □

$\mathrm{Tr}_B\, T_e(F,G) = A, \qquad \mathrm{Tr}_A\, T_e(F,G) = B$

**Gurvits scaling and quantum Sinkhorn.**

$$T_s(F,G) = \exp\left(\frac{Z}{2\varepsilon}\right)\exp(-C/\varepsilon)\exp\left(\frac{Z}{2\varepsilon}\right) = (U\otimes V)K(U\otimes V), \qquad Z = F\otimes\mathrm{Id}_m + \mathrm{Id}_n\otimes G, \tag{12.20}$$

where $U = \exp(F/(2\varepsilon))$, $V = \exp(G/(2\varepsilon))$ and $K = \exp(-C/\varepsilon)$. If $[Z,C] = 0$, then $T_s(F,G) = T_e(F,G)$; otherwise this is a Strang-type symmetric surrogate.

Fix a Choi convention and let $\mathcal{K} : \mathbb{H}_m \to \mathbb{H}_n$ be the completely positive map represented by the positive Choi matrix $K$; let $\mathcal{K}^\star$ be its Hilbert–Schmidt adjoint. Up to the transpose dictated by the chosen Choi convention, the marginal equations for the symmetric coupling take the operator-scaling form

$U\,\mathcal{K}(V^2)\,U = A, \qquad V\,\mathcal{K}^\star(U^2)\,V = B.$

$$\begin{aligned} R_V &= \mathcal{K}(V^2), \quad & U &\leftarrow R_V^{-1/2}\big(R_V^{1/2} A R_V^{1/2}\big)^{1/2} R_V^{-1/2},\\ S_U &= \mathcal{K}^\star(U^2), \quad & V &\leftarrow S_U^{-1/2}\big(S_U^{1/2} B S_U^{1/2}\big)^{1/2} S_U^{-1/2}. \end{aligned} \tag{12.21}$$

These inverse square roots are well-defined when $K\succ 0$ and $U, V, A, B \succ 0$. This is Gurvits/operator scaling with prescribed targets; when all matrices are diagonal it reduces to classical Sinkhorn scaling, and when the targets are proportional to identities it matches the usual bistochastic operator-scaling normalization, up to the conventional trace normalization.

# 13 Dynamic Optimal Transport

## 13.1 Evolutions over the Space of Measures

This compact subsection fixes the Eulerian and Lagrangian language for evolving measures.

**Lagrangian and Eulerian descriptions.**

$$\frac{\mathrm{d}x(t)}{\mathrm{d}t} = v_t(x(t)), \tag{13.1}$$

$$\frac{\partial \alpha_t}{\partial t} + \mathrm{div}(v_t\alpha_t) = 0. \tag{13.2}$$

$$\int_0^1\int_{\mathbb{R}^d} \left(\partial_t\varphi(t,x) + \langle v_t(x),\, \nabla_x\varphi(t,x)\rangle\right)\mathrm{d}\alpha_t(x)\mathrm{d}t = 0. \tag{13.3}$$

This equation is obtained from (13.2) by integration by parts. Hence, for smooth positive densities, the classical and weak formulations are equivalent; the weak formulation is useful because it still makes sense for discrete measures whose particles evolve according to (13.1).

**Proposition 13.1** (Lagrangian flows solve the continuity equation)**.** *Consider a smooth flow* $T_t : \mathbb{R}^d\to\mathbb{R}^d$ *and define* $\alpha_t = (T_t)_\sharp\alpha_0$*. Define the Eulerian velocity field by* $v_t(T_t(y)) = \partial_t T_t(y)$*. Then* $(\alpha_t, v_t)$ *solves the continuity equation in the weak sense* (13.3)*. In particular, if* $\alpha_0 = \frac{1}{n}\sum_i \delta_{x_i(0)}$ *is empirical, then* $\alpha_t = \frac{1}{n}\sum_i\delta_{x_i(t)}$ *is empirical as well, with particle velocities* $\dot{x}_i(t) = v_t(x_i(t))$*.*

*Proof.* Let $\varphi(t,x)$ be a smooth test function vanishing at $t=0$ and $t=1$. Since $\alpha_t=(T_t)_\sharp\alpha_0$, $\frac{\mathrm{d}}{\mathrm{d}t}\int\varphi(t,x)\mathrm{d}\alpha_t(x) = \frac{\mathrm{d}}{\mathrm{d}t}\int\varphi(t,T_t(y))\mathrm{d}\alpha_0(y)$. The chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\int\varphi(t,T_t(y))\mathrm{d}\alpha_0(y)=\int\left(\partial_t\varphi(t,T_t(y))+\langle\nabla_x\varphi(t,T_t(y)),\,\partial_tT_t(y)\rangle\right)\mathrm{d}\alpha_0(y).$$

Using the definition of $v_t$ and the push-forward relation, this equals

$$\int\left(\partial_t\varphi(t,x)+\langle\nabla_x\varphi(t,x),\,v_t(x)\rangle\right)\mathrm{d}\alpha_t(x).$$

Integrating in time and using the boundary values of $\varphi$ gives (13.3). □

**From measure evolutions to vector fields.** For a given evolution $(\alpha_t)_t$, there are typically infinitely many velocity fields $v_t$ satisfying

$$\partial_t\alpha_t+\mathrm{div}(\alpha_tv_t)=0. \tag{13.4}$$

$\mathcal{H}_\alpha=\{v\;;\;\mathrm{div}(\alpha v)=0\}$.

**Dacorogna–Moser inversion.**

$$v_t=-\frac{1}{\alpha_t}\nabla\Delta^{-1}(\partial_t\alpha_t), \tag{13.5}$$

$\alpha_t=(1-t)\alpha_0+t\alpha_1=\rho_t\,\mathrm{d}x$, $\quad\rho_t=(1-t)\rho_0+t\rho_1$. $\mathrm{div}\,w=\rho_0-\rho_1$, $\quad w\cdot n=0$ on $\partial\Omega$, for instance $w=-\nabla\varphi$ with $\Delta\varphi=\rho_1-\rho_0$ and Neumann boundary condition. Then
$v_t=\frac{w}{\rho_t}$ satisfies $\partial_t\rho_t+\mathrm{div}(\rho_tv_t)=0$. The flow $\partial_tT_t=v_t\circ T_t$, $T_0=\mathrm{Id}$, therefore transports $\rho_0\mathrm{d}x$ onto $\rho_t\mathrm{d}x$, and $T_1$ solves the prescribed-Jacobian problem $\rho_1(T_1(x))\det(\nabla T_1(x))=\rho_0(x)$.

**Least-square inversion and gradient structure.**

$$\min_v\frac{1}{2}\int_0^1\int_{\mathbb{R}^d}\|v_t(x)\|^2\,\mathrm{d}\alpha_t(x)\mathrm{d}t\quad\text{subject to}\quad\partial_t\alpha_t+\mathrm{div}(\alpha_tv_t)=0. \tag{13.6}$$

**Proposition 13.2** (Least-square velocities are gradients)**.** *Assume that $\alpha_t=\rho_t\,\mathrm{d}x$ is a smooth positive density curve and that boundary terms vanish. The minimizer of (13.6), if it exists, is a gradient field $v_t=\nabla\varphi_t$, where $\varphi_t$, unique up to an additive constant, solves the weighted Poisson equation*

$$-\mathrm{div}(\rho_t\nabla\varphi_t)=\partial_t\rho_t,\qquad v_t=-\nabla\Delta_{\alpha_t}^{-1}(\partial_t\alpha_t),\quad\Delta_{\alpha_t}\varphi=\mathrm{div}(\alpha_t\nabla\varphi). \tag{13.7}$$

*Proof.* Introduce a Lagrange multiplier $\varphi_t$ for the continuity equation. The constrained problem has the formal saddle formulation

$$\min_v\max_\varphi\int_0^1\left[\frac{1}{2}\int_{\mathbb{R}^d}\|v_t(x)\|^2\,\mathrm{d}\alpha_t(x)+\int_{\mathbb{R}^d}\varphi_t(x)\big(\mathrm{div}(\alpha_tv_t)(x)+\partial_t\alpha_t(x)\big)\mathrm{d}x\right]\mathrm{d}t.$$

Integrating by parts in the divergence term gives, for each $t$,

$$\int\left(\frac{1}{2}\|v_t\|^2-\langle\nabla\varphi_t,\,v_t\rangle\right)\mathrm{d}\alpha_t+\int\varphi_t\,\partial_t\alpha_t.$$

The pointwise minimizer in $v_t$ is therefore $v_t=\nabla\varphi_t$. Substituting this into the constraint $\partial_t\rho_t+\mathrm{div}(\rho_tv_t)=0$ gives the weighted Poisson equation in (13.7). The inverse notation is just a shorthand for solving this equation on zero-mean right-hand sides, modulo additive constants. □

## 13.2 Benamou–Brenier dynamic formulation of OT

This compact subsection records the dynamic formulation of quadratic OT.

**Benamou–Brenier formulation.**

**Theorem 13.3** (Benamou–Brenier)**.** *For probability measures $\alpha_0,\alpha_1\in\mathcal{P}_2(\mathbb{R}^d)$,*

$$\mathcal{W}_2^2(\alpha_0,\alpha_1)=\inf_{(\alpha_t,v_t)}\int_0^1\int_{\mathbb{R}^d}\|v_t(x)\|^2\,\mathrm{d}\alpha_t(x)\mathrm{d}t, \tag{13.8}$$

*where the infimum is over $(\alpha_t,v_t)$ solving $\partial_t\alpha_t+\nabla\cdot(\alpha_tv_t)=0$ with $\alpha_{t=0}=\alpha_0$ and $\alpha_{t=1}=\alpha_1$. If $\alpha_0$ has a density and $T$ is the optimal Monge map $T_\sharp\alpha_0=\alpha_1$, the minimizer is*

$$\alpha_t=((1-t)\mathrm{Id}+tT)_\sharp\alpha_0,\qquad v_t\big((1-t)x+tT(x)\big)=T(x)-x. \tag{13.9}$$

*Proof.* For the inequality "dynamic $\leqslant$ static", assume first that a Monge map $T$ exists and define $(\alpha_t,v_t)$ by (13.9). Since the Lagrangian velocity $T(x)-x$ is independent of $t$, $\int_0^1\int\|v_t\|^2\,\mathrm{d}\alpha_t\mathrm{d}t=\int\|T(x)-x\|^2\,\mathrm{d}\alpha_0(x)$, so the dynamic cost is no larger than the static Monge cost. Without a Monge map, the same construction is made with an optimal coupling $\pi$: sample $(X,Y)\sim\pi$ and move along the straight path $\gamma_{X,Y}(t)=(1-t)X+tY$. This path measure has action $\int\|x-y\|^2\mathrm{d}\pi(x,y)$; projecting path velocities onto their conditional mean at time $t$ gives an admissible Eulerian velocity with no larger action, so the dynamic value is no larger than the Kantorovich value.

Conversely, for a smooth deterministic path, take the flow $T_t$ defined by $\dot{T}_t=v_t\circ T_t$ and $T_0=\mathrm{Id}$. Then $\alpha_t=(T_t)_\sharp\alpha_0$ and $(T_1)_\sharp\alpha_0=\alpha_1$. Jensen's inequality gives $\|T_1(x)-x\|^2\leqslant\int_0^1\|v_t(T_t(x))\|^2\mathrm{d}t$. After integration with respect to $\alpha_0$, the Monge cost is bounded above by the dynamic action. For general finite-energy solutions of the continuity equation, the superposition principle lifts the curve to a probability measure on absolutely continuous paths; applying Jensen's inequality pathwise gives a coupling of the endpoints whose quadratic cost is no larger than the action. Thus the Kantorovich value is bounded above by the dynamic value. □

**Convex moment-based reformulation.** Although (13.8) is not jointly convex in $(\alpha_t, v_t)$, it becomes convex after replacing velocities by the momentum measure $m_t = v_t\alpha_t$ and using the perspective action. In the absolutely continuous case $\alpha_t = \rho_t\,\mathrm{d}x$ and $m_t(x) = \rho_t(x)v_t(x)$, this reads

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{\substack{\partial_t\rho_t + \operatorname{div} m_t = 0 \\ \rho_{t=0}\mathrm{d}x = \alpha_0,\ \rho_{t=1}\mathrm{d}x = \alpha_1}} \int_0^1 \int_{\mathbb{R}^d} \frac{\|m_t(x)\|^2}{\rho_t(x)}\,\mathrm{d}x\,\mathrm{d}t, \tag{13.10}$$

**Dual Hamilton–Jacobi formulation.** The momentum formulation also has a useful dual. It turns the least-action problem into a Hamilton–Jacobi subsolution inequality for a scalar potential, with equality on the part of space-time actually visited by the optimal curve.

**Proposition 13.4** (Dual Benamou–Brenier problem)**.** *Assume, for simplicity, that the densities are smooth, compactly supported, and that boundary terms vanish. Then the convex dynamic value (13.10) has the dual formulation*

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \sup_{\varphi} \left\{ \int_{\mathbb{R}^d} \varphi_1\,\mathrm{d}\alpha_1 - \int_{\mathbb{R}^d} \varphi_0\,\mathrm{d}\alpha_0\ ;\ \partial_t\varphi_t + \frac{1}{4}\|\nabla\varphi_t\|^2 \leqslant 0 \right\}, \tag{13.11}$$

*where the supremum is over smooth test potentials $\varphi : [0,1]\times\mathbb{R}^d \to \mathbb{R}$. In the general metric-measure statement, the same formula is understood after relaxation: $\varphi$ is a Hamilton–Jacobi subsolution, for instance in the viscosity sense, and finite-dimensional discretizations follow directly from Fenchel–Rockafellar duality. If $(\rho, m)$ and $\varphi$ are smooth primal and dual optimizers, then*

$$m_t = \frac{\rho_t}{2}\nabla\varphi_t, \qquad \partial_t\varphi_t + \frac{1}{4}\|\nabla\varphi_t\|^2 = 0 \quad \text{on } \{\rho_t > 0\}. \tag{13.12}$$

*Equivalently, the optimal Eulerian velocity is $v_t = m_t/\rho_t = \nabla\varphi_t/2$.*

*Proof.* Let $(\rho, m)$ satisfy the continuity equation and let $\varphi$ be smooth. Multiplying the constraint by $\varphi$ and integrating by parts gives

$$\int_{\mathbb{R}^d} \varphi_1\,\mathrm{d}\alpha_1 - \int_{\mathbb{R}^d} \varphi_0\,\mathrm{d}\alpha_0 = \int_0^1 \int_{\mathbb{R}^d} (\rho_t\,\partial_t\varphi_t + \langle m_t, \nabla\varphi_t\rangle)\,\mathrm{d}x\mathrm{d}t.$$

If $\partial_t\varphi_t + \|\nabla\varphi_t\|^2/4 \leqslant 0$, Young's inequality yields, pointwise, $\rho\,\partial_t\varphi + \langle m, \nabla\varphi\rangle \leqslant -\frac{\rho}{4}\|\nabla\varphi\|^2 + \langle m, \nabla\varphi\rangle \leqslant \frac{\|m\|^2}{\rho}$, with the same perspective convention when $\rho = 0$. Hence the dual objective of every feasible $\varphi$ is no larger than the action of every feasible primal pair.

Conversely, introduce $\varphi$ as a Lagrange multiplier for $\partial_t\rho + \operatorname{div} m = 0$. After integration by parts, and up to the fixed endpoint contribution in (13.11), the pointwise minimization over $(\rho, m)$ contains $\frac{\|m\|^2}{\rho} - \rho\,\partial_t\varphi - \langle m, \nabla\varphi\rangle$. For fixed $\rho > 0$, its minimum over $m$ is attained at $m = \rho\nabla\varphi/2$ and equals $-\rho(\partial_t\varphi + \|\nabla\varphi\|^2/4)$. The infimum over $\rho \geqslant 0$ is finite exactly when the Hamilton–Jacobi inequality holds, in which case the pointwise infimum is 0. Fenchel–Rockafellar duality then gives no duality gap. Equality in the two pointwise inequalities gives (13.12). □

$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(\gamma(t)) = \partial_t\varphi_t(\gamma(t)) + \langle\nabla\varphi_t(\gamma(t)), \dot\gamma(t)\rangle \leqslant \|\dot\gamma(t)\|^2$, where the last inequality uses $\partial_t\varphi + \|\nabla\varphi\|^2/4 \leqslant 0$ and Young's inequality. Integrating and minimizing over curves gives
$\varphi_1(y) - \varphi_0(x) \leqslant \|x-y\|^2$. Thus $(\varphi_0, \varphi_1)$ is a feasible static Kantorovich dual pair for the quadratic cost. At optimality the inequality is saturated on the endpoint pairs connected by the primal characteristics, which is the Eulerian version of complementary slackness.

**Proximal splitting.**

$$J(\rho, m) = \begin{cases} \|m\|^2/\rho, & \rho > 0, \\ 0, & (\rho, m) = (0, 0), \\ +\infty, & \text{otherwise.} \end{cases}$$

$$\mathcal{F}(U) = \int_0^1 \int_{\mathbb{R}^d} J(\rho_t(x), m_t(x))\,\mathrm{d}x\,\mathrm{d}t, \qquad \mathcal{C} = \{(\rho, m)\ ;\ \partial_t\rho + \operatorname{div} m = 0,\ \rho_0\mathrm{d}x = \alpha_0,\ \rho_1\mathrm{d}x = \alpha_1\},$$

$\min_U\ \mathcal{F}(U) + \mathcal{G}(U)$.

$$\operatorname{prox}_{\tau\mathcal{F}}(\bar U) = \operatorname{argmin}_U \frac{1}{2}\|U - \bar U\|_{\mathsf{H}}^2 + \tau\mathcal{F}(U), \qquad \operatorname{prox}_{\tau\mathcal{G}}(\bar U) = \operatorname{argmin}_{U\in\mathcal{C}} \frac{1}{2}\|U - \bar U\|_{\mathsf{H}}^2.$$

For the standard product metric, the first map is local in $(t, x)$: it is the proximal operator of the convex perspective $J$. The second map is the orthogonal projection onto the affine set defined by the divergence equation and endpoint constraints. Douglas–Rachford then alternates these two simple operations:

$$\begin{aligned} U^{k+1} &= \operatorname{prox}_{\tau\mathcal{F}}(Z^k), \\ \widetilde U^{k+1} &= \operatorname{prox}_{\tau\mathcal{G}}(2U^{k+1} - Z^k), \\ Z^{k+1} &= Z^k + \widetilde U^{k+1} - U^{k+1}. \end{aligned}$$

**Path-space formulation.** Let $\mathcal{S} = C([0,1]; \mathbb{R}^d)$ be the space of continuous paths endowed with the uniform topology. For $t \in [0,1]$ define the evaluation map
$P_t : \mathcal{S} \to \mathbb{R}^d, \qquad P_t(\gamma) = \gamma(t)$.

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{M\in\mathcal{P}(\mathcal{S})} \left\{ \int_{\mathcal{S}}\int_0^1 \|\dot\gamma(t)\|^2\mathrm{d}t\,\mathrm{d}M(\gamma)\ ;\ (P_0)_\sharp M = \alpha_0,\ (P_1)_\sharp M = \alpha_1 \right\}.$$

If $\alpha_0$ has a density, the minimizer $M^*$ is unique. Its time marginals reproduce the optimal curve: $\alpha_t = (P_t)_\sharp M^*$ for all $t$. Furthermore, for a.e. $t$, denoting $Q_t(\gamma) = \dot\gamma(t)$ on absolutely continuous paths, the conditional law of the velocity is deterministic: $(P_t, Q_t)_\sharp M^*(\mathrm{d}x, \mathrm{d}q) = \alpha_t(\mathrm{d}x)\delta_{v_t^*(x)}(\mathrm{d}q)$, where $v_t^*$ is the optimal velocity field in the Benamou–Brenier formulation. Hence $M^*$ concentrates on straight-line geodesics and, for a.e. $t$, assigns exactly one direction at $\alpha_t$-a.e. spatial point.

### 13.3 Generalized Dynamic Wasserstein Distances

The quadratic Benamou–Brenier formula is only one instance of a broader fixed-mass dynamic language. By changing the local action, the mobility, the state graph, the velocity class or the allowed jumps, one obtains transport geometries adapted to constraints, Markov chains, normalized directions, kernelized particle motion or nonlocal motion.

**Local tangent actions.** The common infinitesimal object behind these dynamic distances is a local action on tangent directions. Let $T_\alpha$ denote the admissible tangent space at $\alpha$, for instance the $L^2(\alpha)$-closure of gradients for the usual $\mathcal{W}_2$ geometry.

$$q_\alpha(w) \coloneqq \lim_{t\to 0} \frac{d^2\big((\mathrm{Id}+tw)_\sharp\alpha, \alpha\big)}{t^2}, \tag{13.13}$$

whenever this limit exists. The notation $q_\alpha$ is intentionally broader than a quadratic form: it is the squared local norm, or more generally the squared local action, selected by $d$.

**Geodesic distances from quadratic tangent norms.** Here the action is quadratic and represented by an operator $Q_\alpha$ on $L^2(\alpha)$, or on a closed subspace in constrained applications.

$$\mathbb{A}(\alpha; w, z) = \langle Q_\alpha w, z\rangle_{L^2(\alpha)},$$

$$\mathsf{D}_Q^2(\alpha_0,\alpha_1) \coloneqq \inf_{\substack{\partial_t\alpha_t+\mathrm{div}(\alpha_t v_t)=0\\ \alpha_{t=0}=\alpha_0,\ \alpha_{t=1}=\alpha_1}} \int_0^1 \langle Q_{\alpha_t} v_t, v_t\rangle_{L^2(\alpha_t)}\,\mathrm{d}t. \tag{13.14}$$

$$\langle Q_\alpha v, v\rangle_{L^2(\alpha)} = \int \frac{m(x)^\top B_\rho(x) m(x)}{\rho(x)}\,\mathrm{d}x.$$

$A_Q(\rho,m,x) = \frac{m^\top B_\rho(x) m}{\rho}$.

**General mobilities and convex action densities.** The local action formalism keeps the continuity equation but replaces the quadratic cost $|m|^2/\rho$ by a convex momentum perspective. Starting from a velocity action $A(a,w)$, set

$$J_A(a,m) = \begin{cases} A(a, m/a), & a>0,\\ 0, & a=0,\ m=0,\\ \infty, & a=0,\ m\neq 0.\end{cases}$$

For a curve $\partial_t\alpha_t + \mathrm{div}\,\omega_t = 0$, with $\alpha_t = \rho_t\mathrm{d}x$ and $\omega_t = m_t\mathrm{d}x$, write $\mathbb{J}_A(\alpha_t,\omega_t) = \int J_A(\rho_t,m_t)\mathrm{d}x$. If $J_A$ is lower semicontinuous, convex, even in $m$, $r$-homogeneous in $m$, and vanishes only for $m=0$, define

$$\mathsf{D}_A(\alpha_0,\alpha_1) \coloneqq \inf_{\substack{\partial_t\alpha_t+\mathrm{div}\,\omega_t=0\\ \alpha_{t=0}=\alpha_0,\ \alpha_{t=1}=\alpha_1}} \left(\int_0^1 \mathbb{J}_A(\alpha_t,\omega_t)\mathrm{d}t\right)^{1/r}.$$

Equivalently, away from zero density, $A(a,-w) = A(a,w)$, $A(a,\xi w) = |\xi|^r A(a,w)$, and $A(a,w)=0$ iff $w=0$. For a smooth state $\alpha = \rho\mathrm{d}x$ and velocity $v$, the associated local squared action is

$$q_{A,\alpha}(v) \coloneqq \left(\int J_A(\rho(x), \rho(x)v(x))\,\mathrm{d}x\right)^{2/r}. \tag{13.15}$$

**Proposition 13.5** (Homogeneous dynamic actions define distances)**.** *Under the hypotheses above and after the usual lower-semicontinuous relaxation assuming that zero-action values are attained, $\mathsf{D}_A$ is an extended distance on each finite-action component: it is symmetric, satisfies the triangle inequality, and $\mathsf{D}_A(\alpha_0,\alpha_1)=0$ only when $\alpha_0=\alpha_1$.*

*Proof.* The constant curve gives zero self-distance. Conversely, zero action implies $\omega_t = 0$ a.e.; the continuity equation gives $\partial_t\alpha_t = 0$, hence equal endpoints.
Symmetry follows by time reversal and evenness in $m$. For the triangle inequality, concatenate two admissible curves with actions $a$ and $b$ using time fractions $\tau$ and $1-\tau$. Homogeneity gives the action $\tau^{1-r}a + (1-\tau)^{1-r}b$. Optimizing in $\tau$ yields $(a^{1/r}+b^{1/r})^r$, hence the triangle inequality after taking infima. □

**Dynamic spectral Wasserstein distances.** Let $\gamma$ be a monotone spectral gauge on $\mathbb{S}_+^d$. For a probability measure $\alpha$ and a velocity field $v\in L^2(\alpha;\mathbb{R}^d)$, define the spectral tangent norm

$$\mathcal{N}_{\gamma,\alpha}(v)^2 \coloneqq \gamma\left(\int v(x)v(x)^\top \mathrm{d}\alpha(x)\right). \tag{13.16}$$

$$\mathsf{W}^2_{\gamma,\mathrm{dyn}}(\alpha_0,\alpha_1) \coloneqq \inf_{\substack{\partial_t\alpha_t+\mathrm{div}(\alpha_t v_t)=0\\ \alpha_{t=0}=\alpha_0,\ \alpha_{t=1}=\alpha_1}} \int_0^1 \mathcal{N}_{\gamma,\alpha_t}(v_t)^2\,\mathrm{d}t. \tag{13.17}$$

$$(\rho,m)\longmapsto \gamma\left(\int \frac{m(x)m(x)^\top}{\rho(x)}\,\mathrm{d}x\right).$$

$A_{\mathrm{lin}}(\rho,m) = \frac{m^\top G m}{\rho}$,

**Proposition 13.6** (Static/dynamic equality for spectral Wasserstein)**.** *Let $\gamma$ be a monotone spectral gauge and let $\alpha_0,\alpha_1\in\mathcal{P}_2(\mathbb{R}^d)$. Then $\mathsf{W}^2_{\gamma,\mathrm{dyn}}(\alpha_0,\alpha_1) = \mathcal{W}^2_\gamma(\alpha_0,\alpha_1)$. Equivalently, the static spectral distance is the distance generated by the tangent norm $\mathcal{N}_{\gamma,\alpha}$.*

*Proof.* We first prove $\mathsf{W}^2_{\gamma,\mathrm{dyn}} \leqslant \mathcal{W}^2_\gamma$. Let $\pi \in \mathcal{U}(\alpha_0,\alpha_1)$ and let $(X,Y)\sim\pi$. Set $Z_t = (1-t)X + tY$ and $\alpha_t = (Z_t)_\sharp\pi$. Define the Eulerian velocity as the conditional mean $v_t(z) := \mathbb{E}[Y-X \mid Z_t = z]$. Then $(\alpha_t, v_t)$ solves the continuity equation. If $M_\pi := \int (x-y)(x-y)^\top \mathrm{d}\pi(x,y)$, $\quad C_t := \int v_t(z)v_t(z)^\top \mathrm{d}\alpha_t(z)$, conditional Jensen gives $C_t \preceq M_\pi$. Indeed, for every direction $u\in\mathbb{R}^d$,

$$u^\top C_t u = \mathbb{E}\big[\mathbb{E}[\langle u, Y-X\rangle \mid Z_t]^2\big] \leqslant \mathbb{E}[(\langle u, Y-X\rangle)^2] = u^\top M_\pi u.$$

By the Loewner monotonicity of $\gamma$, $\int_0^1 \gamma(C_t)\mathrm{d}t \leqslant \gamma(M_\pi)$. Taking the infimum over $\pi$ gives the first inequality.
Conversely, let $(\alpha_t, v_t)$ be an admissible finite-action competitor. Since $\gamma$ is a finite positive gauge on the finite-dimensional cone $\mathbb{S}^d_+$, it is equivalent to the trace on this cone; hence finite spectral action implies the usual finite kinetic energy needed for the superposition principle. Thus there exists a probability law $\eta$ on absolutely continuous paths $\omega.$ such that $(e_t)_\sharp\eta = \alpha_t$ and $\dot\omega_t = v_t(\omega_t)$ for a.e. $t$. Let $\pi = (e_0,e_1)_\sharp\eta$ be the induced endpoint coupling. With $C_t = \int v_t v_t^\top \mathrm{d}\alpha_t$, Jensen's inequality along each path gives

$$(\omega_1-\omega_0)(\omega_1-\omega_0)^\top = \left(\int_0^1 \dot\omega_t \mathrm{d}t\right)\left(\int_0^1 \dot\omega_t \mathrm{d}t\right)^\top \preceq \int_0^1 \dot\omega_t\dot\omega_t^\top \mathrm{d}t.$$

After integration over $\eta$, $M_\pi \preceq \int_0^1 C_t \mathrm{d}t$. Monotonicity of $\gamma$, followed by convexity, yields

$$\gamma(M_\pi) \leqslant \gamma\left(\int_0^1 C_t\mathrm{d}t\right) \leqslant \int_0^1 \gamma(C_t)\mathrm{d}t.$$

Since $\pi$ is admissible for the static problem, $\mathcal{W}^2_\gamma(\alpha_0,\alpha_1)$ is bounded by the action of the dynamic competitor. Taking the infimum over all competitors gives the reverse inequality.
The crucial assumption is the monotonicity of $\gamma$: both parts of the proof produce Loewner-order comparisons between covariance matrices, and these comparisons imply inequalities between actions only for monotone gauges. ☐

$\mathcal{W}_\gamma\big(\alpha, (\mathrm{Id}+\tau v)_\sharp\alpha\big)^2 = \tau^2\mathcal{N}_{\gamma,\alpha}(v)^2 + o(\tau^2)$.

**Nonlocal Wasserstein distances.** $\mathsf{J}(\mathrm{d}x,\mathrm{d}y) := K(x,\mathrm{d}y)\mathfrak{m}(\mathrm{d}x)$ $\bar\nabla\varphi(x,y) := \varphi(y)-\varphi(x)$, $\qquad \theta(a,b) := \frac{a-b}{\log a - \log b}$

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi\,\mathrm{d}\alpha_t = \frac{1}{2}\iint \bar\nabla\varphi(x,y)\,v_t(x,y)\,\theta(\rho_t(x),\rho_t(y))\,\mathsf{J}(\mathrm{d}x,\mathrm{d}y). \tag{13.18}$$

$\mathcal{A}_K(\rho,v) := \frac{1}{2}\iint |v(x,y)|^2\theta(\rho(x),\rho(y))\,\mathsf{J}(\mathrm{d}x,\mathrm{d}y)$,

$$\mathcal{W}^2_K(\alpha_0,\alpha_1) := \inf_{\rho_t,v_t}\int_0^1 \mathcal{A}_K(\rho_t,v_t)\,\mathrm{d}t, \tag{13.19}$$

where the infimum is over all curves solving (13.18) with endpoints $\alpha_0,\alpha_1$.
**Proposition 13.7** (Metric properties of nonlocal transport)**.** *Under standard regularity and irreducibility assumptions on the jump kernel, $\mathcal{W}_K$ is an extended distance on the set of probability measures absolutely continuous with respect to $\mathfrak{m}$, and finite-distance pairs are connected by constant-speed geodesics.*

*Proof.* We use the analytic compactness and lower-semicontinuity theorem for the logarithmic-mean action. In particular, action-bounded sequences of admissible curves are compact for the narrow topology, the weak nonlocal continuity equation is closed under this convergence, and the action is lower semicontinuous. These facts are the nonlocal analogues of the compactness theorem used in the Benamou–Brenier formulation.
Nonnegativity is immediate from the definition of $\mathcal{A}_K$. If $\alpha_0=\alpha_1$, the constant curve $\rho_t=\rho_0$, $v_t = 0$, is admissible and has zero action, hence $\mathcal{W}_K(\alpha_0,\alpha_0)=0$.
Symmetry follows by time reversal. If $(\rho_t,v_t)$ transports $\alpha_0$ to $\alpha_1$, set $\tilde\rho_t = \rho_{1-t}$, $\quad \tilde v_t = -v_{1-t}$. For every test function $\varphi$, differentiating $\int\varphi\tilde\rho_t\,\mathrm{d}\mathfrak{m}$ and using (13.18) shows that $(\tilde\rho_t,\tilde v_t)$ solves the same continuity equation from $\alpha_1$ to $\alpha_0$. Since the action is quadratic in $v$, $\mathcal{A}_K(\tilde\rho_t,\tilde v_t) = \mathcal{A}_K(\rho_{1-t},v_{1-t})$, so the action is unchanged.
For the triangle inequality, let $(\rho^0_t,v^0_t)$ connect $\alpha_0$ to $\alpha_1$ with action $A_0$, and let $(\rho^1_t,v^1_t)$ connect $\alpha_1$ to $\alpha_2$ with action $A_1$. For $0<\zeta<1$, concatenate the two curves after the rescaling

$$(\rho_t,v_t) = \begin{cases} \big(\rho^0_{t/\zeta},\,\zeta^{-1}v^0_{t/\zeta}\big), & 0\leqslant t\leqslant\zeta,\\ \big(\rho^1_{(t-\zeta)/(1-\zeta)},\,(1-\zeta)^{-1}v^1_{(t-\zeta)/(1-\zeta)}\big), & \zeta<t\leqslant 1.\end{cases}$$

The factors $\zeta^{-1}$ and $(1-\zeta)^{-1}$ are exactly those needed for the weak continuity equation after changing time. Since $\mathcal{A}_K$ is quadratic in the velocity, $\int_0^1\mathcal{A}_K(\rho_t,v_t)\,\mathrm{d}t = \frac{A_0}{\zeta}+\frac{A_1}{1-\zeta}$. Optimizing in $\zeta$, or taking $\zeta = \sqrt{A_0}/(\sqrt{A_0}+\sqrt{A_1})$ when both actions are positive, gives a concatenated action $(\sqrt{A_0}+\sqrt{A_1})^2$. Taking infima over the two curves yields $\mathcal{W}_K(\alpha_0,\alpha_2)\leqslant\mathcal{W}_K(\alpha_0,\alpha_1)+\mathcal{W}_K(\alpha_1,\alpha_2)$.
If $\mathcal{W}_K(\alpha_0,\alpha_1)=0$, choose admissible curves with actions tending to zero. By the compactness and lower-semicontinuity theorem quoted above, a subsequence converges to an admissible curve $(\rho_t,v_t)$ of zero action. Hence $v_t=0$ for $\theta(\rho_t(x),\rho_t(y))\mathsf{J}(\mathrm{d}x,\mathrm{d}y)\mathrm{d}t$-a.e. $(t,x,y)$. Substituting in the weak continuity equation gives $\frac{\mathrm{d}}{\mathrm{d}t}\int\varphi\,\mathrm{d}\alpha_t = 0$ for every admissible test function $\varphi$. The irreducibility/separation assumption in ensures that these test functions determine the measure, so $\alpha_t$ is constant and $\alpha_0=\alpha_1$.
Finally, if $\mathcal{W}_K(\alpha_0,\alpha_1)<+\infty$, the same direct-method compactness applied to a minimizing sequence gives a minimizer of (13.19). Reparametrizing this minimizing curve by metric arclength gives a constant-speed curve. Equivalently, for every $0\leqslant s<t\leqslant 1$, the restriction of the minimizer to $[s,t]$, rescaled to unit time, is admissible between $\alpha_s$ and $\alpha_t$; the triangle inequality and equality of the total minimal action force $\mathcal{W}_K(\alpha_s,\alpha_t) = (t-s)\mathcal{W}_K(\alpha_0,\alpha_1)$ after this constant-speed parametrization. Thus finite-distance pairs are joined by constant-speed geodesics. The nonlocal entropy-flow consequences and fractional PDE examples are developed in Section 14.5. ☐

**Kernelized Benamou–Brenier distances.** Let $k$ be a positive definite kernel on $\mathbb{R}^d$ with scalar RKHS $\mathcal{H}_k$. The vector-valued RKHS is
$\mathcal{H}_k^d := \mathcal{H}_k \times \cdots \times \mathcal{H}_k, \qquad \|v\|_{\mathcal{H}_k^d}^2 := \sum_{\ell=1}^d \|v_\ell\|_{\mathcal{H}_k}^2$. This is the vector-valued analogue of the scalar RKHS norm used for MMDs in Section 6.2. The kernelized dynamic action between two probabilities is defined by

$$\mathcal{W}_k^2(\alpha_0, \alpha_1) := \inf_{\alpha_t, v_t} \int_0^1 \|v_t\|_{\mathcal{H}_k^d}^2 \, \mathrm{d}t, \qquad \partial_t \alpha_t + \operatorname{div}(\alpha_t v_t) = 0, \tag{13.20}$$

where the endpoint constraints are $\alpha_{t=0} = \alpha_0$ and $\alpha_{t=1} = \alpha_1$. The corresponding tangent action is $q_{k,\alpha}(v) = \|v\|_{\mathcal{H}_k^d}^2$, with the important caveat that the admissible tangent space is the smooth RKHS class, not the whole Wasserstein tangent space.
**Discrete Wasserstein distances on Markov chains.**

$$\Sigma_n := \left\{ a \in \mathbb{R}_+^n : \sum_i a_i = 1 \right\}.$$

For $a \in \Sigma_n$, write the relative density $\rho_i(a) = a_i/\pi_i$. $\mathrm{Ent}_\pi(\rho) := \sum_i \pi_i \rho_i \log \rho_i$.

$$\theta(a, b) := \begin{cases} \dfrac{a-b}{\log a - \log b}, & a \neq b, \\ a, & a = b, \end{cases}$$

$\theta(a, b)(\log a - \log b) = a - b$,

$$(\mathcal{K}_\rho \psi)_i := \sum_j K_{ij} \theta(\rho_i, \rho_j)(\psi_i - \psi_j). \tag{13.21}$$

$(\mathcal{L}_a \psi)_i := \pi_i (\mathcal{K}_{\rho(a)} \psi)_i$.

$$\mathcal{A}(a, \psi) := \frac{1}{2} \sum_{i,j} \pi_i K_{ij} \theta(\rho_i(a), \rho_j(a))(\psi_i - \psi_j)^2. \tag{13.22}$$

$$\mathcal{W}_K^2(a_0, a_1) := \inf_{a_t, \psi_t} \int_0^1 \mathcal{A}(a_t, \psi_t) \, \mathrm{d}t, \qquad \dot{a}_t + \mathcal{L}_{a_t} \psi_t = 0, \tag{13.23}$$

with $a_{t=0} = a_0$ and $a_{t=1} = a_1$.

## 13.4 Dynamic Unbalanced Wasserstein Distances

This compact subsection records the dynamic distances that also allow mass variation.
**Balance equation and tangent variables.**

$$\partial_t \rho_t + \nabla \cdot m_t = s_t, \qquad \int_0^1 \int \left( \frac{|m_t|^2}{\rho_t} + \kappa^2 \frac{s_t^2}{\rho_t} \right) \mathrm{d}x \, \mathrm{d}t,$$

**Reaction–transport action.**

$$\mathcal{A}_\kappa(\rho, m, s) := \int \left( \frac{\|\dot{m}\|^2}{\dot{\rho}} + \kappa^2 \frac{\dot{s}^2}{\dot{\rho}} \right) \mathrm{d}\lambda, \qquad (\dot{\rho}, \dot{m}, \dot{s}) = \left( \frac{\mathrm{d}\rho}{\mathrm{d}\lambda}, \frac{\mathrm{d}m}{\mathrm{d}\lambda}, \frac{\mathrm{d}s}{\mathrm{d}\lambda} \right),$$

where $\lambda$ dominates $\rho$ and the total variations of $m$ and $s$, and the value is independent of this choice. The convention is $0/0 = 0$ and $a/0 = +\infty$ for $a > 0$, so finite action forces both the flux and source to be absolutely continuous with respect to the transported mass.
**Static and dynamic viewpoints.**
**Proposition 13.8** (Static/dynamic equivalence for unbalanced OT)**.** *Fix the action above and let $\mathrm{CW}_\kappa$ be the cone value of Theorem 10.4 with the cone metric normalized to the same growth scale $\kappa$. For nonnegative finite measures $\alpha_0, \alpha_1$ on $\mathbb{R}^d$, the dynamic value*

$$\mathrm{WFR}_\kappa^2(\alpha_0, \alpha_1) := \inf_{\substack{\partial_t \rho_t + \nabla \cdot m_t = s_t \\ \rho_0 = \alpha_0, \ \rho_1 = \alpha_1}} \int_0^1 \mathcal{A}_\kappa(\rho_t, m_t, s_t) \, \mathrm{d}t \tag{13.24}$$

*equals the static cone formulation $\mathrm{CW}_\kappa(\alpha_0, \alpha_1)$. Hence the static unbalanced problem and the balance-equation least-action problem define the same geodesic distance.*

*Proof.* The cone construction turns variation of mass into radial motion and spatial transport into angular motion on $\mathfrak{C}[\mathbb{R}^d]$. Applying the Benamou–Brenier theorem on the cone to the lifted endpoint measures gives a dynamic least-action problem on $\mathfrak{C}[\mathbb{R}^d]$ whose static value is the cone value $\mathrm{CW}_\kappa$ of Theorem 10.4. This is the standard static/dynamic identification for the Hellinger–Kantorovich and Wasserstein–Fisher–Rao metrics.
Projecting a cone curve back to the base space with the weight $r^2$ produces a measure curve $\rho_t$, a spatial flux $m_t$ and a source term $s_t$ satisfying the balance equation. With the matching normalization of the cone metric, the cone kinetic energy decomposes exactly into the perspective action $\mathcal{A}_\kappa$ in (13.24). Conversely, any finite-action triple $(\rho_t, m_t, s_t)$ can be lifted to a cone curve whose radial velocity realizes the growth term and whose spatial velocity realizes the transport term, with the same action after relaxation. The two infima are therefore equal; lower semicontinuity gives the general finite-measure statement from the smooth positive case. □

**Balanced versus unbalanced geodesics.**

## 14 Wasserstein Gradient Flows

### 14.1 Minimizing Movements and Wasserstein Gradients

$$\alpha_{t+\tau} := \arg\min_{\alpha} \frac{1}{2\tau}\mathcal{W}_2(\alpha_t,\alpha)^2 + f(\alpha). \tag{14.1}$$

**Euclidean gradient flows.** If we restrict (14.1) to finite dimensions and assume $\alpha_t = \delta_{x(t)}$ and $\alpha = \delta_x$ (single Dirac measures), this matches the implicit Euler scheme:
$x(t+\tau) := \arg\min_x \frac{1}{2\tau}\|x - x(t)\|^2 + h(x)$, where $h(x) = f(\delta_x)$. Its solution is formally given by the implicit Euler formula:
$x(t+\tau) = (\mathrm{Id} + \tau\nabla h)^{-1}(x(t))$. $x(t+\tau) = (\mathrm{Id} - \tau\nabla h)(x(t)) = x(t) - \tau\nabla h(x(t))$.

$$\dot{x}(t) = -\nabla h(x(t)). \tag{14.2}$$

**Wasserstein gradient formula.** $f((1-\tau)\alpha + \tau\beta) = f(\alpha + \tau\rho) = f(\alpha) + \tau\int[\delta f(\alpha)](x)\mathrm{d}\rho(x) + o(\tau)$.

**Definition 14.1** (Wasserstein gradient)**.** Assume that $f$ admits a smooth first variation $\delta f(\alpha)$. In the smooth formal calculus on $\mathcal{P}_2(\mathbb{R}^d)$, the Wasserstein gradient of $f$ at $\alpha$ is the gradient vector field $\nabla_{\mathcal{W}} f(\alpha) = \nabla_x \delta f(\alpha)$.

$$\frac{\partial\alpha_t}{\partial t} + \mathrm{div}(-\nabla_{\mathcal{W}} f(\alpha_t)\alpha_t) = 0. \tag{14.3}$$

**Proposition 14.2** (Formal Wasserstein gradient)**.** *Assume that $f$ admits a smooth first variation $\delta f(\alpha)$ and that $\alpha$ has a smooth positive density. For infinitesimal perturbations generated by a velocity field $v$ through $(\mathrm{Id}+\tau v)_\sharp\alpha$, the differential of $f$ is $\frac{\mathrm{d}}{\mathrm{d}\tau}_{|\tau=0} f\big((\mathrm{Id}+\tau v)_\sharp\alpha\big) = \int\langle\nabla\delta f(\alpha)(x),\, v(x)\rangle\mathrm{d}\alpha(x)$. Hence, for the Riemannian metric $\|v\|^2_{L^2(\alpha)} = \int\|v\|^2\mathrm{d}\alpha$, the Wasserstein gradient is the vector field $\nabla_{\mathcal{W}} f(\alpha) = \nabla\delta f(\alpha)$.*

*Proof.* The push-forward expansion gives, in the sense of distributions, $(\mathrm{Id}+\tau v)_\sharp\alpha = \alpha - \tau\,\mathrm{div}(\alpha v) + o(\tau)$. Using the definition of the first variation, $f\big((\mathrm{Id}+\tau v)_\sharp\alpha\big) = f(\alpha) - \tau\int\delta f(\alpha)\,\mathrm{div}(\alpha v)\mathrm{d}x + o(\tau)$. An integration by parts, with either compact support or vanishing boundary flux, gives $-\int\delta f(\alpha)\,\mathrm{div}(\alpha v)\mathrm{d}x = \int\langle\nabla\delta f(\alpha),\, v\rangle\mathrm{d}\alpha$. By definition of the Riesz representative for the $L^2(\alpha)$ metric, this representative is $\nabla\delta f(\alpha)$. □

**From the JKO step to the velocity field.** $\min_v \frac{1}{2\tau}\tau^2\|v\|^2_{L^2(\alpha_t)} + f((\mathrm{Id}+\tau v)_\sharp\alpha_t)$. $(\mathrm{Id}+\tau v)_\sharp\alpha_t = \alpha_t - \tau\,\mathrm{div}(v\alpha_t) + o(\tau)$ $f((\mathrm{Id}+\tau v)_\sharp\alpha_t) = f(\alpha_t) - \tau\int\delta f(\alpha_t)\,\mathrm{div}(v\alpha_t)\mathrm{d}x + o(\tau) = f(\alpha_t) + \tau\int\langle\nabla_x\delta f(\alpha_t)(x), v(x)\rangle\mathrm{d}\alpha_t(x) + o(\tau)$ $\min_v f(\alpha_t) + \tau\int\Big[\frac{1}{2}\|v(x)\|^2 + \langle\nabla_{\mathcal{W}} f(\alpha_t)(x), v(x)\rangle\Big]\mathrm{d}\alpha_t(x) + o(\tau)$.

**Metric derivative and curves of maximal slope.** The same descent principle admits a coordinate-free formulation, which is the right language for limits of JKO schemes. Let $(\mathcal{X}, d)$ be a metric space and let $x : (0,T) \to \mathcal{X}$ be an absolutely continuous curve.
$|\dot{x}_t| := \lim_{h\to 0}\frac{d(x_{t+h},x_t)}{|h|}$

$$|\partial f|(x) := \limsup_{\substack{y\to x\\ y\neq x}} \frac{(f(x)-f(y))_+}{d(x,y)}.$$

When this slope is a strong upper gradient for $f$, meaning that $|f(y_t) - f(y_s)| \leqslant \int_s^t|\partial f|(y_r)|\dot{y}_r|\mathrm{d}r$ along absolutely continuous curves $y_r$, an absolutely continuous curve $x_t$ is called a curve of maximal slope if it satisfies the energy-dissipation inequality
$f(x_t) + \frac{1}{2}\int_s^t|\dot{x}_r|^2\mathrm{d}r + \frac{1}{2}\int_s^t|\partial f|^2(x_r)\mathrm{d}r \leqslant f(x_s), \qquad 0 \leqslant s < t.$

**Discrete evolutions.** If $f(\alpha)$ can be evaluated on discrete distributions and $\nabla_{\mathcal{W}}$ is continuous in this case, the flow (14.3) maintains the number of Dirac masses, $\alpha_t = \frac{1}{n}\sum_i\delta_{x_i(t)}$. The particles $X(t) := (x_i(t))_i$ evolve according to a system of coupled ODEs:

$$\dot{x}_i(t) = -n\nabla_{x_i}F(X(t)), \tag{14.4}$$

where $F(X) := f\left(\frac{1}{n}\sum_i\delta_{x_i}\right)$ and the factor $n$ comes from the empirical Wasserstein metric $\frac{1}{n}\sum_i\|\dot{x}_i\|^2$.

**Linear Functionals.**

**Shannon Neg-Entropy.**

**Interaction Energies.**

$$f(\alpha) := \iint k(x,y)\mathrm{d}\alpha(x)\mathrm{d}\alpha(y). \tag{14.5}$$

For a symmetric kernel $k$:
$\delta f(\alpha)(x) = 2\int k(x,y)\mathrm{d}\alpha(y)$, $\quad\nabla_{\mathcal{W}} f(\alpha)(x) = 2\int\nabla_x k(x,y)\mathrm{d}\alpha(y)$. For $\alpha_0 = \frac{1}{n}\sum_i\delta_{x_i}$, the flow (14.3) implies particles $(x_i(t))_i$ obey:
$\dot{x}_i(t) = -\frac{2}{n}\sum_j\nabla k(x_i(t), x_j(t))$. If $k$ is positive definite, or more generally conditionally positive definite on signed measures of zero total mass as for the energy-distance kernel $k(x,y) = -\|x-y\|$, and one minimizes the squared kernel discrepancy to a teacher distribution $\beta$, then

$$\|\alpha-\beta\|_k^2 = \iint k\mathrm{d}\alpha\mathrm{d}\alpha - 2\int\left(\int k(x,y)\mathrm{d}\beta(y)\right)\mathrm{d}\alpha(x) + \text{constant}.$$

Thus MMD-type training energies are exactly an interaction energy plus a linear potential; the teacher distribution appears through the potential $x \mapsto -2\int k(x,y)\mathrm{d}\beta(y)$. The corresponding empirical Wasserstein gradient flow is
$\dot{x}_i(t) = -\frac{2}{n}\sum_j\nabla_x k(x_i(t), x_j(t)) + 2\int\nabla_x k(x_i(t), y)\mathrm{d}\beta(y)$.

**Stochastic particles and McKean–Vlasov limits.** $\mathrm{d}X_t = b(X_t)\mathrm{d}t + \sqrt{2}\sigma\mathrm{d}B_t$, and the one-particle law $\alpha_t = \rho_t \mathrm{d}x$ directly satisfies the linear Fokker–Planck equation
$\partial_t\rho_t = -\operatorname{div}(b\rho_t) + \sigma^2\Delta\rho_t$. $\mathrm{d}X_i^n(t) = b(X_i^n(t), \alpha_t^n)\mathrm{d}t + \sqrt{2}\sigma\mathrm{d}B_i(t)$, $\qquad \alpha_t^n = \frac{1}{n}\sum_{i=1}^n \delta_{X_i^n(t)}$. For finite $n$, the empirical law $\alpha_t^n$ is itself random. Under suitable Lipschitz, growth and chaotic-initialization assumptions, propagation of chaos states that finitely many particles become asymptotically independent as $n\to\infty$, all with the same deterministic law $\rho_t\mathrm{d}x$; equivalently, the empirical measure $\alpha_t^n$ converges in probability to this law.
$\partial_t\rho_t = -\operatorname{div}\big(b(x,\rho_t)\rho_t\big) + \sigma^2\Delta\rho_t$. $b(x,\rho) = -\nabla\frac{\delta\mathcal{E}}{\delta\rho}(x)$, $\mathcal{E}(\rho) + \sigma^2\int\rho\log\rho\,\mathrm{d}x$.

**Gradient flows under density constraints.**

$$F(\rho\mathrm{d}x) = \int_\Omega h(x)\rho(x)\mathrm{d}x + \iota_{\mathcal{K}_\kappa}(\rho\mathrm{d}x), \qquad \mathcal{K}_\kappa = \{\rho\mathrm{d}x \in \mathcal{P}_2(\Omega)\,;\, 0\leqslant\rho\leqslant\kappa \text{ a.e.}\}, \tag{14.6}$$

where $\Omega\subset\mathbb{R}^d$ is a convex domain and $\mathcal{K}_\kappa$ is assumed nonempty; if $\Omega$ has finite volume, this requires $\kappa|\Omega|\geqslant 1$. The indicator term has no classical first variation; it contributes the normal cone to $\mathcal{K}_\kappa$ in the Wasserstein first-order condition.

$$\alpha^{k+1} \in \operatorname*{argmin}_{\alpha\in\mathcal{K}_\kappa} \frac{1}{2\tau}\mathcal{W}_2^2(\alpha^k,\alpha) + \int_\Omega h\,\mathrm{d}\alpha. \tag{14.7}$$

$$\partial_t\rho_t = \operatorname{div}\left(\rho_t\nabla(h+p_t)\right), \qquad 0\leqslant\rho_t\leqslant\kappa, \qquad p_t\geqslant 0, \qquad p_t(\kappa-\rho_t) = 0. \tag{14.8}$$

The pressure vanishes in the unsaturated region and acts only where $\rho_t = \kappa$, pushing mass away from congested zones so that the density cap remains satisfied.

**Multi-species gradient flows.** Several applications involve several densities living on the same physical space: chemical species, color channels, cell populations or competing phases. Let $\Omega\subset\mathbb{R}^d$ be the physical domain and assume that the component masses $\mathbf{m} = (m_1,\dots,m_p)$ are positive and fixed.

$$\mathcal{P}_{2,\mathbf{m}}(\Omega;\mathbb{R}_+^p) = \left\{\alpha = (\alpha_1,\dots,\alpha_p)\,;\, \alpha_i\in\mathcal{M}_+(\Omega),\ \alpha_i(\Omega) = m_i,\ \int_\Omega \|x\|^2\mathrm{d}\alpha_i(x) < +\infty\right\}. \tag{14.9}$$

$$\mathcal{W}_{2,\oplus}^2(\alpha,\eta) = \sum_{i=1}^p m_i\,\mathcal{W}_2^2\left(\frac{\alpha_i}{m_i},\frac{\eta_i}{m_i}\right). \tag{14.10}$$

$$\alpha^{k+1} \in \operatorname*{argmin}_{\alpha\in\mathcal{P}_{2,\mathbf{m}}(\Omega;\mathbb{R}_+^p)} \frac{1}{2\tau}\mathcal{W}_{2,\oplus}^2(\alpha^k,\alpha) + F(\alpha). \tag{14.11}$$

For smooth densities and first variations $\varphi_i = \delta F/\delta\rho_i$, this yields

$$\partial_t\rho_i = \operatorname{div}\left(\rho_i\nabla\varphi_i\right), \qquad \varphi_i = \frac{\delta F}{\delta\rho_i}, \qquad i = 1,\dots,p. \tag{14.12}$$

$$\mathcal{C}_\beta = \left\{\alpha\in\mathcal{P}_{2,\mathbf{m}}(\Omega;\mathbb{R}_+^p)\,;\, \sum_{i=1}^p\alpha_i = \beta\right\}. \tag{14.13}$$

$$\alpha^{k+1} \in \operatorname*{argmin}_{\alpha\in\mathcal{C}_\beta} \frac{1}{2\tau}\mathcal{W}_{2,\oplus}^2(\alpha^k,\alpha) + F(\alpha). \tag{14.14}$$

$$\partial_t\rho_i = \operatorname{div}\left(\rho_i\nabla(\varphi_i-\lambda_t)\right), \qquad \operatorname{div}(b\nabla\lambda_t) = \operatorname{div}\left(\sum_{i=1}^p\rho_i\nabla\varphi_i\right). \tag{14.15}$$

For the separable Shannon entropy $F(\alpha) = \sum_i\int_\Omega\rho_i\log\rho_i\mathrm{d}x$, this becomes

$$\partial_t\rho_i = \Delta\rho_i - \operatorname{div}(\rho_i\nabla\lambda_t), \qquad \operatorname{div}(b\nabla\lambda_t) = \Delta b. \tag{14.16}$$

When $b$ is constant, the pressure can be chosen constant and the system reduces to independent heat equations which nevertheless preserve the pointwise sum $\sum_i\rho_i = b$. This product geometry should be contrasted with the vector-valued Benamou–Brenier distances of Section 12.1, where the metric itself can couple the components through a non-diagonal mobility.

## 14.2 Geodesic Convexity and Convergence

This compact subsection records the convexity mechanisms behind Wasserstein convergence.

**Geodesics and convexity.** $\alpha_t = ((1-t)P_0 + tP_1)_\sharp\pi^\star$, $\qquad t\in[0,1]$, where $P_0(x,y) = x$ and $P_1(x,y) = y$. If the optimal plan is induced by a Brenier map $T$, this reduces to $((1-t)\mathrm{Id} + tT)_\sharp\alpha_0$.

**Definition 14.3** (Geodesic convexity)**.** A functional $f$ on $\mathcal{P}_2(\mathbb{R}^d)$ is geodesically convex if for every $\mathcal{W}_2$ geodesic $(\alpha_t)_t$, $f(\alpha_t)\leqslant(1-t)f(\alpha_0) + tf(\alpha_1)$. It is $\lambda$-geodesically convex if the right-hand side is improved by $-\frac{\lambda}{2}t(1-t)\,\mathcal{W}_2^2(\alpha_0,\alpha_1)$.

**Proposition 14.4** (Geodesic convexity of linear and quadratic energies)**.** *1. If $h$ is convex, then $\alpha\mapsto\int h\mathrm{d}\alpha$ is geodesically convex; if $h$ is $\lambda$-strongly convex, it is $\lambda$-geodesically convex. Conversely, geodesic convexity for all Dirac endpoints forces $h$ to be convex.*

*2. More generally, if $H_k : (\mathbb{R}^d)^k\to\mathbb{R}\cup\{+\infty\}$ is convex, then $\alpha\mapsto\int H_k(x_1,\dots,x_k)\mathrm{d}\alpha(x_1)\cdots\mathrm{d}\alpha(x_k)$ is geodesically convex. In particular $\alpha\mapsto\frac{1}{2}\iint W(x,y)\mathrm{d}\alpha(x)\mathrm{d}\alpha(y)$ is geodesically convex when $W$ is convex on $\mathbb{R}^d\times\mathbb{R}^d$.*

*Proof.* Along a geodesic $X_t = (1-t)X_0 + tX_1$, convexity of $h$ gives the first claim after integration, and Dirac endpoints give necessity. For the polynomial claim, take $k$ independent copies of the same optimal coupling and apply convexity of $H_k$ to $(X_t^1,\dots,X_t^k) = (1-t)(X_0^1,\dots,X_0^k) + t(X_1^1,\dots,X_1^k)$. □

This criterion is restrictive for kernel losses. MMD kernels from Section 6.2 are chosen for positive definiteness, not convexity, so positive definiteness does not imply geodesic convexity of $\alpha \mapsto \mathrm{MMD}_k^2(\alpha,\beta)$. In one dimension, ordered quantiles give a useful exception for distance kernels.

**Proposition 14.5** (One-dimensional interactions and energy-distance MMD). *Let $\varphi : \mathbb{R}_+ \to \mathbb{R}$ be continuous with at most quadratic growth and set $\mathcal{I}_\varphi(\alpha) = \frac{1}{2}\iint \varphi(|x-y|)\mathrm{d}\alpha(x)\mathrm{d}\alpha(y)$. Then $\mathcal{I}_\varphi$ is geodesically convex on $\mathcal{P}_2(\mathbb{R})$ if and only if $\varphi$ is convex on $\mathbb{R}_+$. For $k(x,y) = -|x-y|$, the squared MMD, equivalently the squared energy distance,*

$$\mathcal{E}_\beta(\alpha) = \mathrm{MMD}_k^2(\alpha,\beta) = \iint -|x-y|\,\mathrm{d}(\alpha-\beta)(x)\mathrm{d}(\alpha-\beta)(y),$$

*is geodesically convex in $\alpha$.*

*Proof.* Let $Q_t = (1-t)Q_0 + tQ_1$ be the quantile geodesic. For $r > s$, $Q_t(r) - Q_t(s)$ is the convex combination of the nonnegative differences $Q_i(r) - Q_i(s)$, hence convexity of $\varphi$ on $\mathbb{R}_+$ gives convexity of $\mathcal{I}_\varphi$. Conversely, testing on $\frac{1}{2}(\delta_0 + \delta_a)$ recovers ordinary convexity of $\varphi$ after the $\varphi(0)$ terms cancel. For $k = -|\cdot|$, $\varphi(r) = -r$ is affine on $\mathbb{R}_+$, so the self-distance term is affine along quantile geodesics, while the cross term $2\int |Q_\alpha(r) - Q_\beta(s)|\mathrm{d}r\mathrm{d}s$ is convex. This proves the MMD claim. $\square$

This special one-dimensional mechanism explains the favorable energy-distance flows studied earlier; it does not extend to arbitrary positive definite kernels, whose MMD decomposition need not contain convex terms along Wasserstein geodesics.

Internal energies use a different Jacobian mechanism.

**Theorem 14.6** (McCann displacement convexity for internal energies). *Let $g : [0,+\infty) \to \mathbb{R} \cup \{+\infty\}$ be convex with $g(0) = 0$, and set $\psi(r) = r^d g(r^{-d})$. If $\psi$ is convex and nonincreasing on $(0,+\infty)$, then $\mathcal{U}_g(\rho\mathrm{d}x) = \int g(\rho)\mathrm{d}x$ is geodesically convex on $\mathcal{P}_2(\mathbb{R}^d)$.*

*Proof.* For smooth positive densities, let $T$ be the Brenier map, $T_t = (1-t)\mathrm{Id}+tT$, and $J_t = \det((1-t)I + t\nabla T)$. Since $\rho_t(T_t)J_t = \rho_0$ and $\det^{1/d}$ is concave, $s_t = (J_t/\rho_0)^{1/d} \geqslant (1-t)\rho_0^{-1/d} + t\rho_1(T)^{-1/d}$. Hence $\int g(\rho_t)\mathrm{d}x = \int \rho_0\psi(s_t)\mathrm{d}x \leqslant (1-t)\mathcal{U}_g(\alpha_0) + t\mathcal{U}_g(\alpha_1)$. The general case follows by approximation. $\square$

This includes Shannon entropy $g(s) = s\log s$, porous-medium powers $m > 1$, and more generally the fast-diffusion range $m \geqslant 1 - 1/d$ when the energy is well defined, with $m = 1$ understood as the entropy limit.

**Proposition 14.7** (Geodesic convexity of power divergences). *For $m > 0$, set $\varphi_m(r) = (r^m - mr + m - 1)/(m(m-1))$ if $m \neq 1$, and $\varphi_1(r) = r\log r - r + 1$. If $\beta = b\mathbb{1}_\Omega \mathrm{d}x$ on a convex domain, then $\mathcal{D}_{\varphi_m}(\cdot|\beta)$ is geodesically convex for $m \geqslant 1 - 1/d$. If $\beta = Z^{-1}e^{-V}\mathrm{d}x$ with $V$ convex, then $\mathcal{D}_{\varphi_m}(\cdot|\beta)$ is geodesically convex for $m \geqslant 1$. In the KL case $m = 1$, if $V$ is $\lambda$-strongly convex, then $\mathrm{KL}(\cdot|\beta)$ is $\lambda$-geodesically convex.*

*Proof.* For flat $\beta$, the non-affine part of $b\varphi_m(\rho/b)$ is $b^{1-m}\rho^m/(m(m-1))$, and McCann's criterion gives the threshold $m \geqslant 1 - 1/d$ for $m > 0$. For $m > 1$ and $V$ convex, $\mathcal{D}_{\varphi_m}$ differs by constants from $\int \rho^m e^{(m-1)V}\mathrm{d}x$; along a Brenier geodesic its integrand is $\rho_0^m \exp((m-1)(V(T_t) - \log J_t))$, which is convex in $t$ because $V(T_t)$ and $-\log J_t$ are convex. For $m = 1$, $\mathrm{KL}(\alpha|\beta) = \int \rho\log\rho\,\mathrm{d}x + \int V\mathrm{d}\alpha + \log Z$, and the second term is $\lambda$-geodesically convex when $V$ is $\lambda$-strongly convex; $\lambda = 0$ covers the log-concave case. $\square$

> **Remark 14.8 (Flat references and general $\varphi$-divergences).** For the Csiszár divergences of Definition 6.9, flat references reduce the problem to an internal energy. Proposition 14.7 gives the main nonflat extension for log-concave targets and $m \geqslant 1$, with a stronger $\lambda$-convex KL endpoint. This does not extend to arbitrary $\varphi$. Hellinger is the instructive borderline: $\varphi_H(r) = (\sqrt{r} - 1)^2 = \varphi_{1/2}(r)/2$, so it is covered in dimension 1 for flat references, but with $\beta = \mathcal{N}(0,1)$ and $\alpha_m = \mathcal{N}(m,1)$, the curve $m \mapsto \alpha_m$ is a $\mathcal{W}_2$-geodesic family and
>
> $$\mathcal{D}_{\varphi_H}(\alpha_m|\beta) = 2(1 - e^{-m^2/8}), \qquad \frac{\mathrm{d}^2}{\mathrm{d}m^2}\mathcal{D}_{\varphi_H}(\alpha_m|\beta) = \frac{1}{2}e^{-m^2/8}(1 - m^2/4),$$
>
> which is negative for $|m| > 2$. Thus Hellinger divergence to a Gaussian target is not even 0-geodesically convex. KL is exceptional because it splits as entropy plus a linear potential; Gaussian translations reduce a general $\varphi$ to the scalar convexity test $m \mapsto \mathbb{E}[\varphi(e^{mX - m^2/2})]$, which need not hold.

**Geodesically convex constraints.**

**Proposition 14.9** (Geodesic convexity of density caps). *Assume that $\Omega \subset \mathbb{R}^d$ is convex and let $\kappa > 0$. The set $\mathcal{K}_\kappa$ in (14.6) is geodesically convex in $\mathcal{P}_2(\Omega)$: if $\alpha_0, \alpha_1 \in \mathcal{K}_\kappa$, then the constant-speed $\mathcal{W}_2$ geodesic $(\alpha_t)_{t\in[0,1]}$ between them also belongs to $\mathcal{K}_\kappa$.*

*Proof.* Assume first that the endpoints have smooth positive densities $\alpha_i = \rho_i\mathrm{d}x$ with $\rho_i \leqslant \kappa$, and let $T = \nabla\varphi$ be the Brenier map from $\alpha_0$ to $\alpha_1$. Since $\Omega$ is convex, $T_t = (1-t)\mathrm{Id} + tT$ maps $\Omega$ into $\Omega$. The interpolation is $\alpha_t = (T_t)_\sharp\alpha_0$ and $\rho_t(T_t(x)) = \frac{\rho_0(x)}{\det((1-t)I + t\nabla T(x))}$. The concavity of $\det^{1/d}$ on positive semidefinite matrices and the identity $\rho_1(T(x))\det\nabla T(x) = \rho_0(x)$ give $\rho_t(T_t(x))^{-1/d} \geqslant (1-t)\rho_0(x)^{-1/d} + t\rho_1(T(x))^{-1/d} \geqslant \kappa^{-1/d}$. Hence $\rho_t \leqslant \kappa$ for all $t$. The general case follows by approximation and lower semicontinuity of the $L^\infty$ bound, as in McCann's displacement-convexity theory. $\square$

**Dissipation and finite metric length.** $\frac{\mathrm{d}}{\mathrm{d}t}h(x_t) = -\|\nabla h(x_t)\|^2 = -\|\dot{x}_t\|^2$. Thus, for $0 \leqslant s < t$, Cauchy's inequality gives

$$\int_s^t \|\dot{x}_r\|\,\mathrm{d}r \leqslant \sqrt{t-s}\left(\int_s^t \|\dot{x}_r\|^2\mathrm{d}r\right)^{1/2} = \sqrt{t-s}\,\big(h(x_s) - h(x_t)\big)^{1/2}. \tag{14.17}$$

**Proposition 14.10** (Finite metric length from dissipation). *Let $(\alpha_r)_{r\in[0,T]}$ be an absolutely continuous curve in $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$. Assume that, for all $0 \leqslant s < t \leqslant T$, it satisfies the exact energy-dissipation identity $f(\alpha_s) - f(\alpha_t) = \int_s^t |\dot{\alpha}_r|^2_{\mathcal{W}_2}\mathrm{d}r$. Then*

$$\mathrm{Length}_{\mathcal{W}_2}\big((\alpha_r)_{r\in[s,t]}\big) = \int_s^t |\dot{\alpha}_r|_{\mathcal{W}_2}\,\mathrm{d}r \leqslant \sqrt{t-s}\,\big(f(\alpha_s) - f(\alpha_t)\big)^{1/2}. \tag{14.18}$$

*In the smooth Wasserstein gradient-flow setting with velocity $v_r = -\nabla_{\mathcal{W}} f(\alpha_r)$, the dissipation identity reads equivalently $f(\alpha_s) - f(\alpha_t) = \int_s^t |\dot{\alpha}_r|^2_{\mathcal{W}_2}\mathrm{d}r = \int_s^t \int \|v_r(x)\|^2\mathrm{d}\alpha_r(x)\mathrm{d}r$. If one assumes only the metric energy-dissipation inequality for a curve of maximal slope, then the same length estimate holds with the right-hand side multiplied by $\sqrt{2}$.*

*Proof.* Cauchy's inequality gives
$$\int_s^t |\dot\alpha_r|_{\mathcal{W}_2}\,\mathrm{d}r \leqslant \sqrt{t-s}\left(\int_s^t |\dot\alpha_r|^2_{\mathcal{W}_2}\mathrm{d}r\right)^{1/2}.$$
The exact energy-dissipation identity substitutes the last integral by $f(\alpha_s)-f(\alpha_t)$, which proves (14.18). In the metric theory of curves of maximal slope, the energy-dissipation inequality gives instead $\frac{1}{2}\int_s^t |\dot\alpha_r|^2_{\mathcal{W}_2}\mathrm{d}r \leqslant f(\alpha_s)-f(\alpha_t)$, and the same argument yields the additional factor $\sqrt{2}$. □

$\mathcal{W}_2(\alpha_s,\alpha_t) \leqslant \mathrm{Length}_{\mathcal{W}_2}\big((\alpha_r)_{r\in[s,t]}\big) \leqslant \sqrt{f(\alpha_0)-m_T}\,\sqrt{t-s}$, with an additional factor $\sqrt{2}$ under the energy-dissipation inequality. Thus an energy-dissipating trajectory is 1/2-Hölder continuous as a curve in Wasserstein space on every time interval where the energy remains bounded from below.

**Convergence: the Wasserstein–PL viewpoint.** $|\partial f|^2(\alpha) := \int \|\nabla_{\mathcal{W}} f(\alpha)(x)\|^2\,\mathrm{d}\alpha(x)$

**Definition 14.11** (Wasserstein–Polyak–Łojasiewicz inequality)**.** Let $f_\star := \inf_{\alpha\in\mathcal{P}_2(\mathbb{R}^d)} f(\alpha) > -\infty$. The functional $f$ satisfies the Wasserstein–PL inequality with constant $\kappa>0$ if
$$|\partial f|^2(\alpha) \geqslant 2\kappa\big(f(\alpha)-f_\star\big) \tag{14.19}$$
for every $\alpha$ in the domain of $f$. In the smooth Otto notation this reads $\int \|\nabla_{\mathcal{W}} f(\alpha)\|^2\,\mathrm{d}\alpha \geqslant 2\kappa\big(f(\alpha)-f_\star\big)$.

**Proposition 14.12** (Energy decay for convex Wasserstein flows)**.** *Assume formally that $f$ is geodesically convex, admits a smooth first variation, and has a minimizer $\alpha^\star$. Let $(\alpha_t)_t$ be a smooth solution of the Wasserstein gradient flow $\partial_t\alpha_t + \mathrm{div}(\alpha_t v_t) = 0$, $\quad v_t = -\nabla_{\mathcal{W}} f(\alpha_t)$. Then $\frac{\mathrm{d}}{\mathrm{d}t}f(\alpha_t) = -\int \|\nabla_{\mathcal{W}} f(\alpha_t)(x)\|^2\,\mathrm{d}\alpha_t(x) \leqslant 0$. If $T_t$ is the optimal map from $\alpha_t$ to $\alpha^\star$, then $f(\alpha_t) - f(\alpha^\star) \leqslant -\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\mathcal{W}_2^2(\alpha_t,\alpha^\star)$, and consequently $f(\alpha_t)-f(\alpha^\star) \leqslant \frac{\mathcal{W}_2^2(\alpha_0,\alpha^\star)}{2t}$.*

*Proof.* The chain rule and Proposition 14.2 give $\frac{\mathrm{d}}{\mathrm{d}t}f(\alpha_t) = \int\langle\nabla_{\mathcal{W}} f(\alpha_t)(x),\, v_t(x)\rangle\,\mathrm{d}\alpha_t(x) = -\int\|\nabla_{\mathcal{W}} f(\alpha_t)(x)\|^2\,\mathrm{d}\alpha_t(x)$. Geodesic convexity along the geodesic $((1-s)\mathrm{Id}+sT_t)_\sharp\alpha_t$ gives $f(\alpha^\star)-f(\alpha_t) \geqslant \int\langle\nabla_{\mathcal{W}} f(\alpha_t)(x),\, T_t(x)-x\rangle\,\mathrm{d}\alpha_t(x)$. Since $v_t = -\nabla_{\mathcal{W}} f(\alpha_t)$, this reads $f(\alpha_t)-f(\alpha^\star) \leqslant \int\langle v_t(x),\, T_t(x)-x\rangle\,\mathrm{d}\alpha_t(x)$. $\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\mathcal{W}_2^2(\alpha_t,\alpha^\star) = \int\langle x-T_t(x),\, v_t(x)\rangle\,\mathrm{d}\alpha_t(x)$, which proves the differential inequality. Integrating it from 0 to $t$ and using the monotonicity of $s\mapsto f(\alpha_s)$ gives
$$t\big(f(\alpha_t)-f(\alpha^\star)\big) \leqslant \int_0^t \big(f(\alpha_s)-f(\alpha^\star)\big)\,\mathrm{d}s \leqslant \frac{1}{2}\mathcal{W}_2^2(\alpha_0,\alpha^\star).$$
□

**Theorem 14.13** (Wasserstein–PL convergence)**.** *Assume that $f_\star > -\infty$, that $f$ satisfies the Wasserstein–PL inequality (14.19) with constant $\kappa>0$, and that $(\alpha_t)_{t\geqslant 0}$ is a global Wasserstein gradient flow satisfying the full energy-dissipation identity*
$$\frac{\mathrm{d}}{\mathrm{d}t}f(\alpha_t) = -|\dot\alpha_t|^2 = -|\partial f|^2(\alpha_t) \qquad \textit{for a.e. } t>0. \tag{14.20}$$
*Set $E(t) := f(\alpha_t)-f_\star$. Then, for $0\leqslant s\leqslant t$, $E(t) \leqslant e^{-2\kappa(t-s)}E(s)$, and the length of the flow segment satisfies $\mathrm{Length}_{\mathcal{W}_2}\big((\alpha_r)_{r\in[s,t]}\big) \leqslant \sqrt{\frac{2}{\kappa}}\big(\sqrt{E(s)}-\sqrt{E(t)}\big)$. If, in addition, $f$ is lower semicontinuous for $\mathcal{W}_2$-convergence, then $\alpha_t$ converges in $\mathcal{W}_2$ to a minimizer $\alpha_\infty \in \mathrm{Argmin}\, f$, and $\mathcal{W}_2(\alpha_t,\alpha_\infty) \leqslant \sqrt{\frac{2}{\kappa}}\sqrt{E(t)} \leqslant \sqrt{\frac{2}{\kappa}}\sqrt{E(0)}\,e^{-\kappa t}$. In particular, $\mathrm{dist}_{\mathcal{W}_2}(\alpha_t, \mathrm{Argmin}\, f) \leqslant \sqrt{\frac{2}{\kappa}}\sqrt{E(0)}\,e^{-\kappa t}$.*

*Proof.* The energy estimate is immediate from (14.20) and (14.19): $E'(t) = -|\partial f|^2(\alpha_t) \leqslant -2\kappa E(t)$ for almost every $t$. Gronwall's lemma gives $E(t)\leqslant e^{-2\kappa(t-s)}E(s)$.
The distance estimate uses the same two identities in a slightly different way. On the set where $E(r)>0$, the PL inequality gives $|\partial f|(\alpha_r) \geqslant \sqrt{2\kappa E(r)}$. Using the full dissipation identity and $|\dot\alpha_r| = |\partial f|(\alpha_r)$,
$$\begin{aligned}\mathrm{Length}_{\mathcal{W}_2}\big((\alpha_r)_{r\in[s,t]}\big) &= \int_s^t |\dot\alpha_r|\,\mathrm{d}r = \int_s^t |\partial f|(\alpha_r)\,\mathrm{d}r\\ &= \int_s^t \frac{|\partial f|^2(\alpha_r)}{|\partial f|(\alpha_r)}\,\mathrm{d}r \leqslant \frac{1}{\sqrt{2\kappa}}\int_s^t \frac{|\partial f|^2(\alpha_r)}{\sqrt{E(r)}}\,\mathrm{d}r\\ &= \frac{1}{\sqrt{2\kappa}}\int_s^t \frac{-E'(r)}{\sqrt{E(r)}}\,\mathrm{d}r = \sqrt{\frac{2}{\kappa}}\big(\sqrt{E(s)}-\sqrt{E(t)}\big).\end{aligned}$$
If $E$ vanishes on a subinterval, then the flow is stationary there by (14.20), so the same estimate follows by applying the argument on the part where $E>0$. Since Wasserstein distance is bounded by metric length, the same upper bound controls $\mathcal{W}_2(\alpha_s,\alpha_t)$. Letting $t\to+\infty$, the energy estimate gives $E(t)\to 0$, and the length estimate shows that the tail of $(\alpha_t)_t$ is Cauchy. Since $(\mathcal{P}_2(\mathbb{R}^d),\mathcal{W}_2)$ is complete, there is a limit $\alpha_\infty$. Lower semicontinuity yields $f(\alpha_\infty) \leqslant \liminf_{t\to+\infty} f(\alpha_t) = f_\star$, hence $\alpha_\infty \in \mathrm{Argmin}\, f$. Sending the endpoint of the length estimate to $+\infty$ with $s=t$ fixed gives $\mathcal{W}_2(\alpha_t,\alpha_\infty) \leqslant \sqrt{\frac{2}{\kappa}}\sqrt{E(t)}$. The distance-to-the-set bound follows because $\alpha_\infty$ is one minimizer. □

**Remark 14.14 (Literature and attribution).** The exponential decay of the energy gap is the direct metric-gradient-flow analogue of the classical PL argument. Similar slope-based statements are standard for curves of maximal slope and for Wasserstein gradient flows under functional inequalities. The tail-length refinement leading to the explicit $\mathcal{W}_2$-distance-to-$\mathrm{Argmin}\, f$ estimate was communicated to us by Raphaël Barboni.

**Geodesic strong convexity.**

**Proposition 14.15** (Strong geodesic convexity implies Wasserstein–PL)**.** *Assume that $f$ is $\lambda$-geodesically convex with $\lambda > 0$, that $f_\star$ is attained at some $\alpha^\star$, and that the Wasserstein slope is finite at $\alpha$. Then $|\partial f|^2(\alpha) \geqslant 2\lambda\big(f(\alpha) - f_\star\big)$. Thus positive Wasserstein strong convexity implies the Wasserstein–PL inequality with the same constant.*

*Proof.* Fix $\alpha$ and a minimizer $\alpha^\star$. Set $g = f(\alpha) - f_\star, \quad r = \mathcal{W}_2(\alpha, \alpha^\star), \quad a = |\partial f|(\alpha)$. If $r = 0$, then $\alpha = \alpha^\star$ and there is nothing to prove. Otherwise let $(\alpha_s)_{s\in[0,1]}$ be a constant-speed geodesic from $\alpha$ to $\alpha^\star$. By $\lambda$-geodesic convexity, $f(\alpha_s) \leqslant (1-s)f(\alpha) + sf_\star - \frac{\lambda}{2}s(1-s)r^2$. Therefore $f(\alpha) - f(\alpha_s) \geqslant sg + \frac{\lambda}{2}s(1-s)r^2$. Dividing by $\mathcal{W}_2(\alpha, \alpha_s) = sr$ and letting $s \downarrow 0$ in the definition of the descending metric slope yields $a \geqslant \frac{g}{r} + \frac{\lambda}{2}r$. Equivalently, $g \leqslant ar - \lambda r^2/2$. Maximizing the right-hand side over $r \geqslant 0$ gives $g \leqslant a^2/(2\lambda)$, which is the desired PL inequality. ☐

> **Remark 14.16 (Convergence rates for classical examples).** The examples needed to use the PL implication are the earlier convexity criteria: Proposition 14.4 for linear, polynomial, and interaction energies, Theorem 14.6 for internal energies, and Proposition 14.7 for $\mathrm{KL}(\cdot|\beta)$ when $\mathrm{d}\beta = Z^{-1}e^{-V}\mathrm{d}x$ with $V$ strongly convex. When one of these criteria gives a positive constant, Proposition 14.15 yields Wasserstein–PL and Theorem 14.13 yields linear convergence.

**Convexity and curvature.** The same language is not restricted to subsets of $\mathbb{R}^d$. If $(\mathcal{X}, d, \mathfrak{m})$ is a geodesic metric-measure space, $\mathcal{W}_2$ geodesics can be defined by transporting each pair of endpoints along metric geodesics, or more intrinsically by dynamical optimal plans on path space, as discussed in Section 3.5.

$$\mathrm{Ent}_{\mathfrak{m}}(\alpha) \coloneqq \begin{cases} \displaystyle\int_{\mathcal{X}} \rho \log \rho \, \mathrm{d}\mathfrak{m}, & \text{if } \alpha = \rho\,\mathfrak{m}, \\ +\infty, & \text{otherwise.} \end{cases}$$

**Theorem 14.17** (Ricci curvature and entropy convexity)**.** *Let $(M, g)$ be a smooth compact connected Riemannian manifold without boundary, and let $\mathfrak{m} = \mathrm{vol}_g$. For $\lambda \in \mathbb{R}$, the lower Ricci bound $\mathrm{Ric}_g \geqslant \lambda g$ holds if and only if $\mathrm{Ent}_{\mathfrak{m}}$ is $\lambda$-geodesically convex on $(\mathcal{P}_2(M), \mathcal{W}_2)$.*

## 14.3 Functional Inequalities via Optimal Transport

This compact subsection records representative OT proofs of geometric and functional inequalities.

**Brunn–Minkowski and isoperimetry.**

**Proposition 14.18** (Brunn–Minkowski and Euclidean isoperimetry)**.** *Let $A, B \subset \mathbb{R}^d$ be bounded Borel sets with positive Lebesgue measure. For $t \in [0,1]$, set $(1-t)A + tB \coloneqq \{(1-t)x + ty \;:\; x \in A,\; y \in B\}$. Then $\mathrm{Vol}((1-t)A + tB)^{1/d} \geqslant (1-t)\mathrm{Vol}(A)^{1/d} + t\,\mathrm{Vol}(B)^{1/d}$. $\mathrm{Per}(A) \geqslant d\,\omega_d^{1/d}\mathrm{Vol}(A)^{(d-1)/d}$.*

*Proof.* We first give the argument for regular sets, the general Borel case following by approximation from inside and outside. Let $\alpha$ and $\beta$ be the uniform probability measures on $A$ and $B$, and let $T = \nabla\varphi$ be the Brenier map with $T_\sharp\alpha = \beta$. The interpolating map $T_t = (1-t)\mathrm{Id} + tT$ sends $\alpha$ to a measure $\alpha_t$ supported in $(1-t)A + tB$. At points where $T$ is differentiable, $\det DT_t = \det((1-t)\mathrm{Id} + tDT)$. Since $DT$ is symmetric positive semidefinite and $\det^{1/d}$ is concave on the positive semidefinite cone, $\det(DT_t)^{1/d} \geqslant (1-t) + t\det(DT)^{1/d}$. The change-of-variables identity for the uniform measures gives $\det(DT) = \mathrm{Vol}(B)/\mathrm{Vol}(A)$ almost everywhere. Hence the density $\rho_t$ of $\alpha_t$ satisfies $\rho_t^{-1/d} \geqslant (1-t)\mathrm{Vol}(A)^{1/d} + t\mathrm{Vol}(B)^{1/d}$. This means that $\rho_t \leqslant c_t^{-d}$ on its support, where $c_t = (1-t)\mathrm{Vol}(A)^{1/d} + t\mathrm{Vol}(B)^{1/d}$. Since $\alpha_t$ has total mass one and is supported in $(1-t)A + tB$, this gives $\mathrm{Vol}((1-t)A + tB) \geqslant c_t^d$, which is the displayed Brunn–Minkowski inequality.

For isoperimetry, apply Brunn–Minkowski with $B = B_1$ and write $A + \varepsilon B_1$ for the parallel set. Then $\mathrm{Vol}(A + \varepsilon B_1)^{1/d} \geqslant \mathrm{Vol}(A)^{1/d} + \varepsilon\omega_d^{1/d}$. Using $\mathrm{Vol}(A + \varepsilon B_1) = \mathrm{Vol}(A) + \varepsilon\,\mathrm{Per}(A) + o(\varepsilon)$ and differentiating at $\varepsilon = 0$ yields the claimed perimeter bound. ☐

**Prékopa–Leindler.**

**Proposition 14.19** (Prékopa–Leindler inequality)**.** *Let $u, v, w : \mathbb{R}^d \to [0, +\infty)$ be integrable, and let $t \in [0,1]$. Assume that $w((1-t)x + ty) \geqslant u(x)^{1-t}v(y)^t \qquad$ for all $x, y \in \mathbb{R}^d$. Then*

$$\int_{\mathbb{R}^d} w \geqslant \left(\int_{\mathbb{R}^d} u\right)^{1-t} \left(\int_{\mathbb{R}^d} v\right)^t.$$

*Proof.* Set $U = \int u$ and $V = \int v$, and assume $U, V > 0$. Let $\alpha = (u/U)\mathrm{d}x$ and $\beta = (v/V)\mathrm{d}x$, and let $T$ be the Brenier map from $\alpha$ to $\beta$. Put $T_t = (1-t)\mathrm{Id} + tT$. The change of variables gives $\int w \geqslant \int w(T_t(x))\det(DT_t(x))\mathrm{d}x$. By the assumption on $w$ and by $\det((1-t)\mathrm{Id} + tDT) \geqslant \det(DT)^t$ for positive semidefinite $DT$, $\int w \geqslant \int u(x)^{1-t}v(T(x))^t\det(DT(x))^t\mathrm{d}x$. Since $T_\sharp\alpha = \beta$, the Monge–Ampère identity gives $\det(DT(x)) = \frac{u(x)V}{v(T(x))U}$ almost everywhere. Substitution yields

$$\int w \geqslant \left(\frac{V}{U}\right)^t \int u(x)\mathrm{d}x = U^{1-t}V^t.$$

Approximation removes the smoothness and positivity assumptions. ☐

**Logarithmic Sobolev inequalities.** $\mathcal{I}_\beta(\alpha) \coloneqq \int \|\nabla \log h\|^2 \,\mathrm{d}\alpha = \int \frac{\|\nabla h\|^2}{h}\,\mathrm{d}\beta$,

$$\mathrm{KL}(\alpha|\beta) \leqslant \frac{1}{2\lambda}\mathcal{I}_\beta(\alpha) \qquad \text{for all probability measures } \alpha. \tag{14.21}$$

**Proposition 14.20** (Curvature implies logarithmic Sobolev)**.** *Let $\beta = Z^{-1}e^{-V}\mathrm{d}x$ be a smooth probability measure on $\mathbb{R}^d$. If $\nabla^2 V \succeq \lambda\mathrm{Id}$ for some $\lambda > 0$, then $\beta$ satisfies (14.21) with constant $\lambda$.*

*Proof.* Apply the HWI inequality of Proposition 14.25. Under the curvature assumption, for every absolutely continuous $\alpha$, $\mathrm{KL}(\alpha|\beta) \leqslant \mathcal{W}_2(\alpha,\beta)\sqrt{\mathcal{I}_\beta(\alpha)} - \frac{\lambda}{2}\mathcal{W}_2^2(\alpha,\beta)$. For fixed $s = \sqrt{\mathcal{I}_\beta(\alpha)}$, maximize the right-hand side over $r = \mathcal{W}_2(\alpha,\beta) \geqslant 0$. The elementary inequality $rs - \lambda r^2/2 \leqslant s^2/(2\lambda)$ gives (14.21). Approximation gives the usual nonsmooth statement. This is the Bakry–Emery mechanism expressed in the optimal-transport language; the direct $\Gamma_2$ proof gives the same criterion. ☐

**Proposition 14.21** (Logarithmic Sobolev inequality implies entropy decay). *Let $\beta = e^{-V}\mathrm{d}x/Z$ be a smooth probability measure on $\mathbb{R}^d$ satisfying the logarithmic Sobolev inequality (14.21) with constant $\lambda > 0$. Let $\alpha_t$ solve the Wasserstein gradient flow of $F(\alpha) = \mathrm{KL}(\alpha|\beta)$, namely $\partial_t\rho_t = \nabla\cdot(\rho_t\nabla V) + \Delta\rho_t$, $\quad \alpha_t = \rho_t\mathrm{d}x$. If $\mathrm{KL}(\alpha_0|\beta) < +\infty$, then $\mathrm{KL}(\alpha_t|\beta) \leqslant e^{-2\lambda t}\,\mathrm{KL}(\alpha_0|\beta)$. Moreover, if $\beta \in \mathcal{P}_2(\mathbb{R}^d)$, then $\mathcal{W}_2(\alpha_t,\beta) \leqslant \sqrt{\frac{2}{\lambda}\,\mathrm{KL}(\alpha_0|\beta)}\,e^{-\lambda t}$.*

*Proof.* The first variation of $F$ is $\log(\rho/\rho_\beta)$, up to an additive constant. Along the smooth Wasserstein gradient flow, $\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{KL}(\alpha_t|\beta) = -\int\|\nabla\log\frac{\mathrm{d}\alpha_t}{\mathrm{d}\beta}\|^2\,\mathrm{d}\alpha_t = -\mathcal{I}_\beta(\alpha_t)$. The logarithmic Sobolev inequality bounds the Fisher information below by $2\lambda\,\mathrm{KL}(\alpha_t|\beta)$. Hence $\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{KL}(\alpha_t|\beta) \leqslant -2\lambda\,\mathrm{KL}(\alpha_t|\beta)$, and Gronwall's lemma gives the entropy estimate. The Wasserstein estimate follows by applying Theorem 14.13 to $F = \mathrm{KL}(\cdot|\beta)$, whose unique minimizer is $\beta$, and using the logarithmic Sobolev inequality as the corresponding Wasserstein–PL inequality. ☐

### Gaussian logarithmic Sobolev inequality.

**Proposition 14.22** (Gaussian logarithmic Sobolev inequality). *Let $\gamma_d$ be the standard Gaussian measure on $\mathbb{R}^d$. If $f$ is smooth, positive, has sufficient decay, and $\int f\mathrm{d}\gamma_d = 1$, then $\mathrm{Ent}_{\gamma_d}(f) := \int f\log f\,\mathrm{d}\gamma_d \leqslant \frac{1}{2}\int\frac{\|\nabla f\|^2}{f}\,\mathrm{d}\gamma_d$.*

*Proof.* Let $\alpha = f\gamma_d$, and let $T = \nabla\varphi$ be the Brenier map with $T_\sharp\alpha = \gamma_d$. Write $T(x) = x + \theta(x)$. The Monge–Ampère equation reads $f(x)e^{-\|x\|^2/2} = e^{-\|T(x)\|^2/2}\det(DT(x))$. Thus $\log f(x) = -\langle x,\,\theta(x)\rangle - \frac{1}{2}\|\theta(x)\|^2 + \log\det(\mathrm{Id} + D\theta(x))$. Since $DT$ is positive semidefinite, $\log\det(\mathrm{Id}+D\theta) \leqslant \mathrm{tr}(D\theta) = \mathrm{div}\,\theta$. Multiplying by $f$ and integrating with respect to $\gamma_d$ gives $\mathrm{Ent}_{\gamma_d}(f) \leqslant \int f(\mathrm{div}\,\theta - \langle x,\,\theta\rangle)\,\mathrm{d}\gamma_d - \frac{1}{2}\int f\|\theta\|^2\,\mathrm{d}\gamma_d$. Gaussian integration by parts gives $\int f(\mathrm{div}\,\theta - \langle x,\,\theta\rangle)\,\mathrm{d}\gamma_d = -\int\langle\nabla f,\,\theta\rangle\,\mathrm{d}\gamma_d$. Completing the square pointwise, $-\langle\nabla f,\,\theta\rangle - \frac{1}{2}f\|\theta\|^2 \leqslant \frac{1}{2}\frac{\|\nabla f\|^2}{f}$. This proves the inequality. The general Sobolev-class statement follows by approximation. ☐

$$f_a(x) = \exp\left(\langle a,\,x\rangle - \frac{1}{2}\|a\|^2\right), \qquad a \in \mathbb{R}^d,$$

$\int f_a\log f_a\,\mathrm{d}\gamma_d = \frac{1}{2}\|a\|^2$, $\qquad \frac{1}{2}\int\frac{\|\nabla f_a\|^2}{f_a}\,\mathrm{d}\gamma_d = \frac{1}{2}\|a\|^2$. Thus equality is attained along translations, and the sharp logarithmic Sobolev constant of the standard Gaussian is $\lambda = 1$. Proposition 14.21 then gives the classical exponential convergence of the Ornstein–Uhlenbeck flow.

### Poincaré as a linearized Wasserstein–PL inequality.

$$\mathrm{KL}(h_\varepsilon\beta|\beta) = \frac{\varepsilon^2}{2}\int g^2\mathrm{d}\beta + O(\varepsilon^3), \qquad \int\|\nabla\log h_\varepsilon\|^2 h_\varepsilon\mathrm{d}\beta = \varepsilon^2\int\|\nabla g\|^2\mathrm{d}\beta + O(\varepsilon^3).$$

$\lambda\int g^2\mathrm{d}\beta \leqslant \int\|\nabla g\|^2\mathrm{d}\beta$, $\qquad \int g\mathrm{d}\beta = 0$. $|\partial E_2|^2_{\mathcal{W}_2}(h\beta) = \int h\|\nabla h\|^2\mathrm{d}\beta$.

**Proposition 14.23** (Poincaré as linearized Wasserstein–PL). *For a mean-zero perturbation $u$, define the frozen-mobility norm*

$$\|u\|^2_{-1,\beta} := \inf_{\substack{v\\ u=-\nabla_\beta\cdot v}}\int\|v\|^2\mathrm{d}\beta, \qquad \nabla_\beta\cdot v := \frac{1}{\rho_\beta}\nabla\cdot(\rho_\beta v).$$

*Equivalently, the associated dynamic metric on curves with fixed mass is obtained from $\partial_t h_t + \nabla_\beta\cdot v_t = 0$, $\quad \int_0^1\int\|v_t\|^2\mathrm{d}\beta\,\mathrm{d}t$. For a smooth density ratio $h$ with $\int h\mathrm{d}\beta = 1$, set $E_2(h) = \frac{1}{2}\int(h-1)^2\mathrm{d}\beta$. Its squared slope in this linearized Wasserstein metric is $|\partial E_2|^2_{-1,\beta}(h) = \int\|\nabla h\|^2\mathrm{d}\beta$. $2\lambda E_2(h) \leqslant |\partial E_2|^2_{-1,\beta}(h)$. The corresponding gradient flow is the linear Fokker–Planck equation $\partial_t h_t = Lh_t$, and it satisfies $\chi^2(h_t\beta|\beta) \leqslant e^{-2\lambda t}\chi^2(h_0\beta|\beta)$.*

*Proof.* A tangent vector for the frozen-mobility metric has the form $\dot h = -\nabla_\beta\cdot v$. Differentiating $E_2$ in this direction and integrating by parts with respect to $\beta$ gives $\mathrm{d}E_2(h)[\dot h] = \int(h-1)\dot h\,\mathrm{d}\beta = \int\langle\nabla h,\,v\rangle\,\mathrm{d}\beta$. The squared dual norm of this differential is at most $\int\|\nabla h\|^2\mathrm{d}\beta$ by Cauchy–Schwarz, and equality is attained, after normalization, by the tangent vector represented by $v = \nabla h$. This proves the slope identity. The Poincaré inequality is precisely $2\lambda E_2 \leqslant |\partial E_2|^2_{-1,\beta}$. The Riemannian gradient is $-Lh$, hence the gradient flow is $\partial_t h_t = Lh_t$. Along this flow, $\frac{\mathrm{d}}{\mathrm{d}t}E_2(h_t) = -\int\|\nabla h_t\|^2\mathrm{d}\beta$, and Gronwall's lemma gives the decay estimate. ☐

### Talagrand's transport-entropy inequality.

**Proposition 14.24** (Gaussian $T_2$ inequality). *Let $\alpha = f\gamma_d$ be a probability measure with finite second moment. Then $\frac{1}{2}\mathcal{W}_2^2(\alpha,\gamma_d) \leqslant \mathrm{KL}(\alpha|\gamma_d) = \int f\log f\,\mathrm{d}\gamma_d$.*

*Proof.* Assume first that $f$ is smooth and positive. Let $T = \nabla\varphi$ be the Brenier map with $T_\sharp\gamma_d = \alpha$, and write $T(x) = x+\theta(x)$. The change of variables gives $f(T(x))e^{-\|T(x)\|^2/2}\det(DT(x)) = e^{-\|x\|^2/2}$. Hence $\log f(T(x)) = \langle x,\,\theta(x)\rangle + \frac{1}{2}\|\theta(x)\|^2 - \log\det(\mathrm{Id}+D\theta(x))$. Using again $\log\det(\mathrm{Id}+D\theta) \leqslant \mathrm{div}\,\theta$ and integrating with respect to $\gamma_d$, $\mathrm{KL}(\alpha|\gamma_d) \geqslant \frac{1}{2}\int\|\theta\|^2\,\mathrm{d}\gamma_d + \int(\langle x,\,\theta\rangle - \mathrm{div}\,\theta)\,\mathrm{d}\gamma_d$. The last integral vanishes by Gaussian integration by parts. Since $T$ is optimal, $\int\|\theta\|^2\,\mathrm{d}\gamma_d = \mathcal{W}_2^2(\gamma_d,\alpha)$, which proves the claim. Approximation gives the general case. ☐

**The HWI bridge.**

**Proposition 14.25** (Otto–Villani HWI inequality)**.** *Let $\beta = \rho_\beta \mathrm{d}x = e^{-V}\mathrm{d}x/Z$ be a smooth probability measure in $\mathcal{P}_2(\mathbb{R}^d)$, and assume that $\nabla^2 V \succeq \lambda \mathrm{Id}$ for some $\lambda \in \mathbb{R}$. For $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$, define the relative Fisher information by $\mathcal{I}_\beta(\alpha) \coloneqq \int \|\nabla \log \frac{\rho_\alpha}{\rho_\beta}\|^2\,\mathrm{d}\alpha$, when $\alpha = \rho_\alpha \mathrm{d}x$ and the expression is finite, and set $\mathcal{I}_\beta(\alpha) = +\infty$ otherwise. Then $\mathrm{KL}(\alpha|\beta) \leqslant \mathcal{W}_2(\alpha,\beta)\sqrt{\mathcal{I}_\beta(\alpha)} - \frac{\lambda}{2}\mathcal{W}_2^2(\alpha,\beta)$.*

*Proof.* If $\mathcal{I}_\beta(\alpha) = +\infty$, there is nothing to prove. Assume first that $\rho_\alpha$ is smooth and positive. Let $T$ be the Brenier map from $\alpha$ to $\beta$, set $v = T - \mathrm{Id}$, and let $\alpha_t = ((1-t)\mathrm{Id} + tT)_\sharp \alpha$. By Theorem 14.6 and Proposition 14.4, $F(\eta) = \mathrm{KL}(\eta|\beta)$ is $\lambda$-geodesically convex. Thus the one-dimensional function $f(t) = F(\alpha_t)$ satisfies $f(1) \geqslant f(0) + f'(0) + \frac{\lambda}{2}\mathcal{W}_2^2(\alpha,\beta)$. Since $f(1) = F(\beta) = 0$, it remains to compute the initial derivative. The continuity equation for the geodesic gives $\partial_t \alpha_t + \nabla\cdot(\alpha_t v_t) = 0$ and $v_0 = v$. Hence $f'(0) = \int \langle \nabla \log \frac{\rho_\alpha}{\rho_\beta},\, v\rangle\,\mathrm{d}\alpha$. Therefore $\mathrm{KL}(\alpha|\beta) \leqslant -\int \langle \nabla \log \frac{\rho_\alpha}{\rho_\beta},\, T - \mathrm{Id}\rangle\,\mathrm{d}\alpha - \frac{\lambda}{2}\mathcal{W}_2^2(\alpha,\beta)$. Cauchy's inequality and $\int \|T - \mathrm{Id}\|^2 \mathrm{d}\alpha = \mathcal{W}_2^2(\alpha,\beta)$ give the announced bound. The general case follows by the standard approximation and lower-semicontinuity of entropy, Fisher information, and $\mathcal{W}_2$. □

When $\lambda > 0$, the HWI inequality is a bridge from curvature to PL. It is not itself a PL inequality because it still contains the distance term $\mathcal{W}_2(\alpha,\beta)$. Optimizing the right-hand side, or simply using Young's inequality $rs - \lambda r^2/2 \leqslant s^2/(2\lambda)$, gives the logarithmic Sobolev estimate
$\mathrm{KL}(\alpha|\beta) \leqslant \frac{1}{2\lambda}\mathcal{I}_\beta(\alpha)$. Since $\mathcal{I}_\beta = |\partial\,\mathrm{KL}(\cdot|\beta)|^2$, this is the Wasserstein–PL inequality with constant $\lambda$. Thus HWI recovers Proposition 14.20 and explains why the same curvature assumption controls both the static inequality and the exponential decay of the Fokker–Planck flow in Proposition 14.21.

### 14.4 Training Two-Layer MLPs as Wasserstein Flows

This compact subsection records the mean-field description of two-layer training.

**Wasserstein training of two-layer MLPs.** $x = (u,v) \in \mathbb{R}^d \times \mathbb{R}^{d'}$, where $u$ is the inner weight and $v$ is the outer vector weight. For a scalar nonlinearity $\sigma$, define the vector-valued feature
$\psi(x,z) = v\,\sigma(\langle u,\, z\rangle) \in \mathbb{R}^{d'}$. $G_X(z) = \frac{1}{n}\sum_{i=1}^n \psi(x_i,z)$, $\qquad G_\alpha(z) = \int \psi(x,z)\mathrm{d}\alpha(x)$, $\qquad \alpha = \frac{1}{n}\sum_i \delta_{x_i}$. Let $\rho$ be a probability distribution on data-label pairs $(z,y) \in \mathbb{R}^d \times \mathbb{R}^{d'}$. The population risk is
$f(\alpha) = \int \ell(G_\alpha(z),y)\mathrm{d}\rho(z,y)$,

$$\dot{x}_i = -n\nabla_{x_i} F(X), \qquad F(X) = f\left(\frac{1}{n}\sum_i \delta_{x_i}\right),$$

which is the gradient flow of $F(X) = f(\alpha_X)$ for this Wasserstein particle metric, equivalently Euclidean gradient descent with the time scale multiplied by $n$. It gives a particle discretization of (14.3).
Assume that $\ell$ is differentiable in its first variable. The first variation is

$$\delta f(\alpha)(x) = \int \langle \nabla_1 \ell(G_\alpha(z),y),\, \psi(x,z)\rangle \mathrm{d}\rho(z,y), \tag{14.22}$$

$\nabla_{\mathcal{W}} f(\alpha)(x) = \nabla_x \delta f(\alpha)(x) = \int [D_x\psi(x,z)]^\top \nabla_1 \ell(G_\alpha(z),y)\mathrm{d}\rho(z,y)$. For the squared Euclidean loss $\ell(s,y) = \frac{1}{2}\|s-y\|^2$, the energy is the sum of a quadratic interaction and a linear potential:

$$f(\alpha) = \frac{1}{2}\iint k(x,x')\mathrm{d}\alpha(x)\mathrm{d}\alpha(x') + \int g(x)\mathrm{d}\alpha(x) + \frac{1}{2}\int \|y\|^2 \mathrm{d}\rho(z,y), \tag{14.23}$$

$$k(x,x') = \int \langle \psi(x,z),\, \psi(x',z)\rangle \mathrm{d}\rho(z,y), \qquad g(x) = -\int \langle y,\, \psi(x,z)\rangle \mathrm{d}\rho(z,y). \tag{14.24}$$

$\delta f(\alpha)(x) = \int k(x,x')\mathrm{d}\alpha(x') + g(x)$, $\qquad \nabla_{\mathcal{W}} f(\alpha)(x) = \int \nabla_x k(x,x')\mathrm{d}\alpha(x') + \nabla_x g(x)$.

**Classical convexity and stationarity.** $F(\alpha) = \frac{1}{2}\iint k(x,x')\mathrm{d}\alpha(x)\mathrm{d}\alpha(x') + \int V(x)\mathrm{d}\alpha(x) + C$, $Q((1-s)\alpha + s\beta) \leqslant (1-s)Q(\alpha) + sQ(\beta)$, $\qquad Q(\alpha) = \frac{1}{2}\iint k\mathrm{d}\alpha\mathrm{d}\alpha$.

**Proposition 14.26** (Affine convexity and stationary densities)**.** *Let $F = Q + \int V\,\mathrm{d}\alpha + C$ be as above, and assume that $Q$ is classically convex. Suppose that a Wasserstein gradient flow for $F$ converges to a measure $\alpha_\infty = \rho_\infty \mathrm{d}x$. Assume also the standard regularity needed to pass to the limit in the first variation, and assume that the support and positivity of $\rho_\infty$ allow the stationary condition to be tested against all admissible zero-mass density perturbations. In the form needed here, assume that this stationarity yields, for every competitor $\beta$, the variational inequality $\int \delta F(\alpha_\infty)(x)\mathrm{d}(\beta - \alpha_\infty)(x) \geqslant 0$. Then $\alpha_\infty$ is a global minimizer of $F$.*

*Proof sketch.* The dissipation identity for the gradient flow gives stationarity of the limit: formally, after passing to the limit, $\int \|\nabla \delta F(\alpha_\infty)\|^2 \mathrm{d}\alpha_\infty = 0$. Without such a support and positivity assumption, this identity only controls the first variation on the region explored by the limit. The density hypothesis permits testing against sufficiently many signed density perturbations of total mass zero. By approximation and the assumed regularity, this yields the displayed first-order variational inequality for arbitrary competitors $\beta$. Classical convexity of $F$ in the affine variable $\alpha$ then gives the usual subgradient inequality $F(\beta) \geqslant F(\alpha_\infty) + \int \delta F(\alpha_\infty)\mathrm{d}(\beta - \alpha_\infty) \geqslant F(\alpha_\infty)$. Thus no competitor has smaller energy. For square-loss two-layer mean-field models, (14.23) is exactly of this quadratic-plus-linear form, and positive semidefiniteness of the induced kernel $k$ is the classical convexity assumption. □

**Proposition 14.27** (Formal global optimality for two-homogeneous mean-field flows)**.** *Assume that the feature is positively two-homogeneous in the neuron variable, $\psi(\lambda x, z) = \lambda^2 \psi(x,z)$ $\quad (\lambda > 0)$, and that $f(\alpha) = J(G_\alpha)$ with $J$ convex and differentiable as a functional of the predictor. Let $\alpha$ be a smooth stationary point of the Wasserstein flow, so that $\nabla_x \delta f(\alpha)(x) = 0$ on $\mathrm{supp}(\alpha)$. Assume also full directional support: for every unit direction $\omega$, the support of $\alpha$ intersects the ray $\{\lambda\omega : \lambda > 0\}$. Then $\alpha$ is a global minimizer of $f$ over the mean-field model class.*

*Proof.* Choose the first-variation representative

$$h_\alpha(x) = \delta f(\alpha)(x) = \langle \nabla J(G_\alpha), \psi(x,\cdot)\rangle_\rho\,.$$

Additive constants in first variations do not affect the Wasserstein gradient or first-order inequalities between probability measures; this normalization is useful because it inherits the homogeneity of $\psi$. By two-homogeneity of $\psi$, one has $h_\alpha(\lambda x) = \lambda^2 h_\alpha(x), \qquad \lambda > 0$. Taking $x = 0$ and any $\lambda \neq 1$ in this identity gives $h_\alpha(0) = 0$. We first record that stationarity forces $h_\alpha$ to vanish on $\mathrm{supp}(\alpha)$. Indeed, if $x = r\omega \in \mathrm{supp}(\alpha)$ with $r > 0$ and $\|\omega\| = 1$, then $0 = \langle \nabla h_\alpha(r\omega), \omega\rangle = \frac{\mathrm{d}}{\mathrm{d}s} h_\alpha(s\omega)\Big|_{s=r} = 2r h_\alpha(\omega)$, so $h_\alpha(\omega) = 0$, and hence $h_\alpha(x) = r^2 h_\alpha(\omega) = 0$. The case $x = 0$ has already been covered, so $h_\alpha = 0$ on $\mathrm{supp}(\alpha)$ and $\int h_\alpha \,\mathrm{d}\alpha = 0$.

Suppose that a competitor $\beta$ satisfies $f(\beta) < f(\alpha)$. Since $G_\beta - G_\alpha = \int \psi(x, \cdot)\mathrm{d}(\beta - \alpha)(x)$, convexity of $J$ gives $f(\beta) - f(\alpha) \geqslant \int h_\alpha(x)\mathrm{d}(\beta - \alpha)(x)$. The left-hand side is negative, while $\int h_\alpha \,\mathrm{d}\alpha = 0$, so $\int h_\alpha \,\mathrm{d}\beta < 0$. Since $h_\alpha(0) = 0$, this strict negativity must occur away from the origin: there exists a nonzero point $x = r\omega$ with $r > 0$, $\|\omega\| = 1$, and $h_\alpha(x) < 0$. Equivalently $h_\alpha(\omega) = h_\alpha(x)/r^2 < 0$. By full directional support, there is $r_\omega > 0$ such that $r_\omega\omega \in \mathrm{supp}(\alpha)$. At this support point, $\langle \nabla h_\alpha(r_\omega\omega), \omega\rangle = 2r_\omega h_\alpha(\omega) < 0$, which contradicts the stationarity condition $\nabla h_\alpha = 0$ on $\mathrm{supp}(\alpha)$. Thus no competitor has smaller risk.

The rigorous Chizat–Bach theorem replaces the full directional support assumption by propagation and overparameterization hypotheses ensuring that any negative first-variation direction is represented by the evolving support. The same radial-derivative contradiction, applied after this support-propagation step, then rules out non-optimal stationary limits. □

### 14.5 Generalized Dynamic Wasserstein Flows

This compact subsection records how changing the dynamic geometry changes gradient flows.

**Generalized Wasserstein flows.**

$$\alpha_{t+\tau} \in \operatorname*{argmin}_{\alpha} \frac{1}{2\tau} d^2(\alpha_t, \alpha) + f(\alpha). \tag{14.25}$$

$\partial_t \alpha_t + \mathrm{div}(\alpha_t v_{\alpha_t}) = 0$, The metric ingredient needed below is the local squared action $q_\alpha$ introduced in Section 13.3. It is the second-order tangent cost in (13.13).

Given a first-order direction $g \in T_\alpha$, with $T_\alpha$ the admissible tangent space used in (13.13), and usually $g = \nabla_{\mathcal{W}} f(\alpha)$, the local steepest-descent velocity associated with the action $q_\alpha$ is formally

$$\mathcal{J}_\alpha(g) \in \operatorname*{argmin}_{w \in T_\alpha} \left\{ \langle g, w\rangle_{L^2(\alpha)} + \frac{1}{2} q_\alpha(w) \right\}. \tag{14.26}$$

$$\partial_t \alpha_t + \mathrm{div}(\alpha_t \,\mathcal{J}_{\alpha_t}(\nabla_{\mathcal{W}} f(\alpha_t))) = 0. \tag{14.27}$$

$$\mathrm{LMO}_\alpha(g) \in \operatorname*{argmin}_{w \in T_\alpha} \left\{ \langle g, w\rangle_{L^2(\alpha)} \;;\; q_\alpha(w) \leqslant 1 \right\}. \tag{14.28}$$

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\alpha_t) = \langle \nabla_{\mathcal{W}} f(\alpha_t), v_t\rangle_{L^2(\alpha_t)} = -q_{\alpha_t}(v_t), \qquad v_t = \mathcal{J}_{\alpha_t}(\nabla_{\mathcal{W}} f(\alpha_t)).$$

$$v_\alpha = -\,Q_\alpha^{-1} \nabla_{\mathcal{W}} f(\alpha), \tag{14.29}$$

The examples below are organized by the same dictionary. Concave-mobility actions use $q_{\theta,\alpha}$ from Section 13.3; spectral flows use $q_{\gamma,\alpha}(v) = \mathcal{N}_{\gamma,\alpha}(v)^2$ from (13.16); kernelized Benamou–Brenier flows use $q_{k,\alpha}(v) = \|v\|^2_{\mathcal{H}_k^d}$ with a restricted RKHS tangent space.

**General mobility flows.**

$$\partial_t \rho_t = \nabla \cdot \left( \theta(\rho_t) \nabla \frac{\delta F}{\delta \rho}(\rho_t) \right). \tag{14.30}$$

$F(\rho) = \int U(\rho(x))\,\mathrm{d}x + \int V(x)\rho(x)\,\mathrm{d}x$,

$$\partial_t \rho = \nabla \cdot \big( \theta(\rho) \big( U''(\rho) \nabla \rho + \nabla V \big) \big).$$

**Dynamic spectral Wasserstein flows.**

**Definition 14.28** (Spectral steepest-descent direction)**.** For $g \in L^2(\alpha; \mathbb{R}^d)$, the spectral steepest-descent direction is any minimizer

$$\mathcal{J}_\gamma^\alpha(g) \in \operatorname*{argmin}_{v \in L^2(\alpha;\mathbb{R}^d)} \left\{ \int \langle g(x),\, v(x)\rangle \mathrm{d}\alpha(x) + \frac{1}{2} \mathcal{N}_{\gamma,\alpha}(v)^2 \right\}. \tag{14.31}$$

$q_{\gamma,\alpha}(v) = \mathcal{N}_{\gamma,\alpha}(v)^2$. The field $g$ should be read as the Wasserstein gradient $g = \nabla_{\mathcal{W}} f(\alpha) = \nabla \delta f(\alpha)$. Since $\mathcal{N}_{\gamma,\alpha}$ is 1-homogeneous in $v$, (14.31) is the penalized steepest-descent step associated with the spectral tangent norm.

**Proposition 14.29** (Polar formula for the spectral direction)**.** *Let $S_\alpha(g) \coloneqq \int g(x)g(x)^\top \mathrm{d}\alpha(x)$. Assume that the inverse-trace problem admits a positive definite minimizer*

$$A_\alpha^\star \in \operatorname*{argmin}_{A \in \mathcal{B}_\gamma \cap \mathbb{S}^d_{++}} \mathrm{tr}\big(A^{-1} S_\alpha(g)\big).$$

*Then*

$$\mathcal{J}_\gamma^\alpha(g)(x) = -(A_\alpha^\star)^{-1} g(x) \tag{14.32}$$

*is a spectral steepest-descent direction. If the minimizer lies on the boundary of $\mathcal{B}_\gamma$, the same formula is understood by a limiting argument along positive definite minimizers, or equivalently by replacing the inverse by the corresponding inverse on the range of $S_\alpha(g)$.*

*Proof.* Using the polar formula $\gamma(M) = \sup_{A \in \mathcal{B}_\gamma} \mathrm{tr}(AM)$, the objective in (14.31) can be written as

$$\sup_{A \in \mathcal{B}_\gamma} \left\{ \int \langle g,\, v\rangle \mathrm{d}\alpha + \frac{1}{2} \int v^\top A v \mathrm{d}\alpha \right\}.$$

For a fixed $A \succ 0$, the quadratic lower bound $\int \langle g,\, v\rangle \mathrm{d}\alpha + \frac{1}{2} \int v^\top A v \mathrm{d}\alpha$ is minimized by $v_A = -A^{-1} g$, with value $-\frac{1}{2}\,\mathrm{tr}(A^{-1} S_\alpha(g))$. Let $v_\star = -(A_\alpha^\star)^{-1} g$ and $M_\star = \int v_\star(x) v_\star(x)^\top \mathrm{d}\alpha(x) = (A_\alpha^\star)^{-1} S_\alpha(g) (A_\alpha^\star)^{-1}$. Since $A_\alpha^\star$ minimizes $A \mapsto \mathrm{tr}(A^{-1} S_\alpha(g))$ over the convex polar set, its first-order optimality condition gives $\operatorname{tr}(AM_\star) \leqslant \operatorname{tr}(A^\star_\alpha M_\star)$ for all $A \in \mathcal{B}_\gamma$. Thus $A^\star_\alpha$ is active in the polar representation of $\gamma(M_\star)$, and $\gamma(M_\star) = \operatorname{tr}(A^\star_\alpha M_\star) = \operatorname{tr}((A^\star_\alpha)^{-1} S_\alpha(g))$. For any $v$, the polar formula gives $\int \langle g, v\rangle \mathrm{d}\alpha + \frac{1}{2}\mathcal{N}_{\gamma,\alpha}(v)^2 \geqslant \int \langle g, v\rangle \mathrm{d}\alpha + \frac{1}{2}\int v^\top A^\star_\alpha v \mathrm{d}\alpha$, and the right-hand side is minimized at $v_\star$. The activity identity above shows that equality holds at $v_\star$, hence $v_\star$ minimizes (14.31). The boundary case follows by approximation with positive definite matrices in the polar set. ☐

For the trace gauge, $\mathcal{B}_\gamma = \{A\,;\, 0 \preceq A \preceq \mathrm{Id}\}$ and the minimizer is $A^\star_\alpha = \mathrm{Id}$. For the operator gauge $\gamma(M) = \lambda_{\max}(M)$, one has $\mathcal{B}_\gamma = \{A \succeq 0\,;\, \operatorname{tr}(A) \leqslant 1\}$; if $S_\alpha(g) \succ 0$, then

$$A^\star_\alpha = \frac{S_\alpha(g)^{1/2}}{\operatorname{tr}(S_\alpha(g)^{1/2})}, \qquad \mathcal{J}^\alpha_\gamma(g)(x) = -\operatorname{tr}(S_\alpha(g)^{1/2})\, S_\alpha(g)^{-1/2} g(x).$$

The static/dynamic equality of Proposition 13.6 implies that the same local action $q_{\gamma,\alpha}$ is obtained either from the dynamic path-length geometry (13.17) or from the small-step expansion of the static distance $\mathcal{W}_\gamma$. Thus the normalized dynamics can be obtained formally from the JKO scheme

$$\alpha^{k+1} \in \operatorname*{argmin}_{\alpha\in\mathcal{P}_2(\mathbb{R}^d)} \frac{1}{2\tau}\mathcal{W}_\gamma(\alpha^k,\alpha)^2 + f(\alpha). \tag{14.33}$$

For a small displacement $\alpha = (\mathrm{Id} + \tau v)_\sharp \alpha^k$, the tangent expansion is
$\mathcal{W}_\gamma(\alpha^k, (\mathrm{Id}+\tau v)_\sharp \alpha^k)^2 = \tau^2 \mathcal{N}_{\gamma,\alpha^k}(v)^2 + o(\tau^2)$.

**Proposition 14.30** (Formal normalized spectral gradient flow)**.** *Assume that $f$ has a smooth first variation and that the tangent expansion of $\mathcal{W}_\gamma$ above is valid along smooth perturbations. Then the formal small-step limit of (14.33), equivalently (14.27) with $q_\alpha = q_{\gamma,\alpha}$, solves*

$$\partial_t \alpha_t + \operatorname{div}\big(\alpha_t\, \mathcal{J}^{\alpha_t}_\gamma(\nabla_{\mathcal{W}} f(\alpha_t))\big) = 0. \tag{14.34}$$

*Equivalently, if $A^\star_t$ solves the inverse-trace problem for $S_t = \int \nabla_{\mathcal{W}} f(\alpha_t)(x)\nabla_{\mathcal{W}} f(\alpha_t)(x)^\top \mathrm{d}\alpha_t(x)$, then the velocity is $v_t(x) = -(A^\star_t)^{-1}\nabla_{\mathcal{W}} f(\alpha_t)(x)$. Moreover, along smooth solutions, $\frac{\mathrm{d}}{\mathrm{d}t} f(\alpha_t) = -\mathcal{N}_{\gamma,\alpha_t}(v_t)^2$.*

*Proof.* Write $\alpha = (\mathrm{Id}+\tau v)_\sharp \alpha_t$. The spectral JKO objective equals, up to terms independent of $v$ and $o(\tau)$,

$$\tau\left[\frac{1}{2}\mathcal{N}_{\gamma,\alpha_t}(v)^2 + \int \langle \nabla_{\mathcal{W}} f(\alpha_t)(x),\, v(x)\rangle \mathrm{d}\alpha_t(x)\right].$$

Minimizing the bracket gives $v = \mathcal{J}^{\alpha_t}_\gamma(\nabla_{\mathcal{W}} f(\alpha_t))$. Substituting this velocity into the transport expansion $(\mathrm{Id}+\tau v)_\sharp\alpha_t = \alpha_t - \tau \operatorname{div}(\alpha_t v) + o(\tau)$ gives (14.34). Finally, the variation formula gives $\frac{\mathrm{d}}{\mathrm{d}t} f(\alpha_t) = \int \langle \nabla_{\mathcal{W}} f(\alpha_t),\, v_t\rangle \mathrm{d}\alpha_t$. Since $v_t$ minimizes a 2-homogeneous penalized objective, the one-dimensional rescaling $s \mapsto s v_t$ has derivative zero at $s = 1$, hence $\int \langle \nabla_{\mathcal{W}} f(\alpha_t),\, v_t\rangle \mathrm{d}\alpha_t + \mathcal{N}_{\gamma,\alpha_t}(v_t)^2 = 0$, which proves the dissipation identity. ☐

### Discrete Wasserstein flows on Markov chains.

$$\rho^{k+1} \in \operatorname{argmin}_\rho \frac{1}{2\tau}\mathcal{W}_K^2(\rho,\rho^k) + \mathrm{Ent}_\pi(\rho). \tag{14.35}$$

$\frac{\rho^{k+1}-\rho^k}{\tau} = -\mathcal{K}_{\rho^k}\log\rho^k + O(\tau) = K\rho^k + O(\tau)$,

**Proposition 14.31** (Entropy gradient flow of a reversible Markov chain)**.** *Let $K$ be reversible with invariant law $\pi$. The gradient flow of $\mathrm{Ent}_\pi$ for the metric $\mathcal{W}_K$ is the forward equation of the Markov chain,*

$$\dot\rho_i(t) = \sum_j K_{ij}\big(\rho_j(t) - \rho_i(t)\big). \tag{14.36}$$

*Equivalently, for the masses $a_i(t) = \pi_i\rho_i(t)$, this is $\dot a_i(t) = \sum_j (a_j(t)K_{ji} - a_i(t)K_{ij})$.*

*Proof.* The first variation of the entropy, with respect to the weighted pairing $\sum_i \pi_i\xi_i\varphi_i$, is $\log\rho_i + 1$. Constants do not contribute to $\mathcal{K}_\rho$, hence the metric gradient-flow equation is $\dot\rho = -\mathcal{K}_\rho\log\rho$. Using the identity $\theta(a,b)(\log a - \log b) = a - b$, one obtains, componentwise, $\dot\rho_i = -\sum_j K_{ij}\theta(\rho_i,\rho_j)(\log\rho_i - \log\rho_j) = \sum_j K_{ij}(\rho_j - \rho_i)$, which is the density form of the Kolmogorov forward equation. Multiplying by $\pi_i$ and using detailed balance gives the equation for the probability masses. ☐

### Nonlocal Wasserstein flows and fractional PDEs.

**Proposition 14.32** (Entropy gradient flow for reversible jump kernels)**.** *Under the regularity and irreducibility assumptions used in Proposition 13.7, the Markov semigroup generated by the closure of $L\psi(x) = \int \big(\psi(y)-\psi(x)\big)K(x,\mathrm{d}y)$ is the gradient flow, for the distance $\mathcal{W}_K$ defined in (13.19), of the relative entropy $\mathrm{Ent}_{\mathfrak{m}}(\alpha) = \int \rho\log\rho\,\mathrm{d}\mathfrak{m}$, $\alpha = \rho\mathfrak{m}$. Equivalently, in weak form, the entropy flow is*

$$\partial_t\rho_t = L\rho_t. \tag{14.37}$$

*When $K$ is singular, the integral defining $L$ is understood in the principal-value sense.*

*Proof idea.* The key identity is the logarithmic-mean relation $\theta(a,b)(\log a - \log b) = a - b$. The first variation of $\mathrm{Ent}_{\mathfrak{m}}$ is $\log\rho + 1$. With the sign convention of (13.18), the steepest descent velocity is $v(x,y) = -\bar\nabla(\log\rho+1)(x,y) = \log\rho(x) - \log\rho(y)$. Hence $\theta(\rho(x),\rho(y))v(x,y) = \rho(x) - \rho(y)$. Substituting this in (13.18) and using the symmetry of $\mathsf{J}$ gives $\frac{\mathrm{d}}{\mathrm{d}t}\int\varphi\rho\,\mathrm{d}\mathfrak{m} = \int\varphi(x)\int(\rho(y)-\rho(x))K(x,\mathrm{d}y)\,\mathfrak{m}(\mathrm{d}x)$, which is the weak form of $\partial_t\rho = L\rho$. The metric properties needed to justify this computation as a gradient-flow identity are the distance and geodesic results recalled in Proposition 13.7. ☐

$K_s(x,\mathrm{d}y) = c_{d,s}\frac{\mathrm{d}y}{|x-y|^{d+s}}$, $\quad 0 < s < 2$. $L_s\psi(x) = \mathrm{p.v.}\int_{\mathbb{R}^d}\big(\psi(y)-\psi(x)\big)\frac{c_{d,s}}{|x-y|^{d+s}}\,\mathrm{d}y = -(-\Delta)^{s/2}\psi(x)$,

$$\partial_t\rho_t = -(-\Delta)^{s/2}\rho_t. \tag{14.38}$$

More generally, when a target-dependent jump kernel $K_\beta$ satisfies detailed balance with a desired invariant measure $\beta = \rho_\beta\mathfrak{m}$, the same construction can be applied to $r_t = \mathrm{d}\alpha_t/\mathrm{d}\beta$. The gradient flow of $\mathrm{KL}(\alpha|\beta)$ is then $\partial_t r_t = L_\beta r_t$.

### 14.6 Dynamic Unbalanced OT and WFR Flows

The generalized dynamic Wasserstein flows of Section 14.5 are primarily mass-preserving: their tangent vectors are generated by continuity equations and are therefore velocity fields advecting the measure. Unbalanced transport changes the state space from probabilities to positive finite measures, where descent may both move and create or remove mass.

**Definition 14.33** (WFR minimizing movement)**.** Let $F : \mathcal{M}_+(\mathbb{R}^d) \to \mathbb{R} \cup \{+\infty\}$ be an energy on positive finite measures and let $\mathrm{WFR}_\kappa$ be the Wasserstein–Fisher–Rao distance with growth scale $\kappa > 0$, defined by the dynamic balance-equation formulation (13.24) and its static equivalence in Proposition 13.8. The unbalanced JKO step is

$$\alpha^{k+1} \in \underset{\alpha \in \mathcal{M}_+(\mathbb{R}^d)}{\operatorname{argmin}} \ \frac{1}{2\tau} \mathrm{WFR}_\kappa^2(\alpha^k, \alpha) + F(\alpha). \tag{14.39}$$

The continuous-time limit, when it exists, is called the $\mathrm{WFR}_\kappa$ gradient flow of $F$.

$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = g_t \rho_t$,

**Proposition 14.34** (Formal WFR gradient-flow PDE)**.** *Assume that $F$ has a smooth first variation $\varphi_t = \delta F(\alpha_t)$ and that $\alpha_t = \rho_t \mathrm{d}x$ is smooth and positive. The formal $\mathrm{WFR}_\kappa$ gradient flow of $F$ is*

$$\partial_t \rho_t = \operatorname{div}(\rho_t \nabla \varphi_t) - \frac{1}{\kappa^2} \varphi_t \rho_t. \tag{14.40}$$

*Along such a smooth solution,*

$$\frac{\mathrm{d}}{\mathrm{d}t} F(\alpha_t) = -\int \left( \|\nabla \varphi_t\|^2 + \frac{1}{\kappa^2} \varphi_t^2 \right) \rho_t \, \mathrm{d}x \leqslant 0. \tag{14.41}$$

*Proof.* For an infinitesimal tangent pair $(v, g)$, the density variation is $\zeta = -\operatorname{div}(\rho v) + g\rho$. The first variation of $F$ is $\int \varphi \zeta = \int \langle \nabla \varphi, v \rangle \rho \, \mathrm{d}x + \int \varphi g \rho \, \mathrm{d}x$, after integration by parts. For the inner product $\langle (v, g), (\tilde{v}, \tilde{g}) \rangle_\rho = \int \langle v, \tilde{v} \rangle \rho \, \mathrm{d}x + \kappa^2 \int g \tilde{g} \rho \, \mathrm{d}x$, the Riesz representative is $(\nabla \varphi, \varphi/\kappa^2)$. Steepest descent therefore uses $(v, g) = (-\nabla \varphi, -\varphi/\kappa^2)$, which gives (14.40). Substituting this pair in the first-variation formula gives

$$\frac{\mathrm{d}}{\mathrm{d}t} F(\alpha_t) = -\int \left( \|\nabla \varphi_t\|^2 + \frac{1}{\kappa^2} \varphi_t^2 \right) \rho_t \, \mathrm{d}x,$$

which is (14.41). □

### 14.7 Conditional Wasserstein Training of Infinite ResNets

This compact subsection records the conditional-measure geometry of infinite ResNets.

**Finite-depth finite-width ResNets.**

$$z_{r+1} = z_r + \tau G_{r,n_r}(z_r), \qquad G_{r,n_r}(z) := \frac{1}{n_r} \sum_{i=1}^{n_r} \psi(\theta_{r,i}, z), \qquad z_0 = z, \qquad r = 0, \dots, L-1. \tag{14.42}$$

$f_{L,n}((\theta_{r,i})_{r,i}) = \int \ell(z_L^\theta(z), y) \mathrm{d}\rho(z, y)$. $\alpha_r = \frac{1}{n_r} \sum_{i=1}^{n_r} \delta_{\theta_{r,i}}$, $G_{r,n_r}(z) = \int_\Theta \psi(\theta, z) \mathrm{d}\alpha_r(\theta)$. $\alpha^{L,n}(\mathrm{d}s, \mathrm{d}\theta) = \frac{1}{L} \sum_{r=0}^{L-1} \delta_{s_r}(\mathrm{d}s) \alpha_r(\mathrm{d}\theta)$, $s_r = r/L$,

**Finite-depth mean-field limit.** $z_{r+1} = z_r + \tau G_{\alpha_r}(z_r)$, $z_0 = z$, $G_{\alpha_r}(z) := \int_\Theta \psi(\theta, z) \mathrm{d}\alpha_r(\theta)$. $f_L(\alpha_0, \dots, \alpha_{L-1}) = \int \ell(z_L^\alpha(z), y) \mathrm{d}\rho(z, y)$.

**Continuous-depth conditional geometry.** $\alpha(\mathrm{d}s, \mathrm{d}\theta) = \alpha_s(\mathrm{d}\theta)\mathrm{d}s$, or, equivalently, a probability measure on $[0, 1] \times \Theta$ whose first marginal is Lebesgue measure. The general conditional Wasserstein distance is defined in Section 10.6.

$$\mathcal{W}_{2,\lambda}^2(\alpha, \beta) = \int_0^1 \mathcal{W}_2^2(\alpha_s, \beta_s) \mathrm{d}s. \tag{14.43}$$

**Infinite-depth mean-field ResNet.**

$$\partial_s z_s = G_{\alpha_s}(z_s) := \int_\Theta \psi(\theta, z_s) \mathrm{d}\alpha_s(\theta), \qquad z_0 = z. \tag{14.44}$$

$f_{\mathrm{Res}}(\alpha) = \int \ell(z_1^\alpha(z), y) \mathrm{d}\rho(z, y)$.

**Conditional Wasserstein gradient flow.**

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} F(\alpha + \varepsilon \xi) \bigg|_{\varepsilon = 0} = \int_S \int_\Theta \frac{\delta F}{\delta \alpha_s}(\alpha)(\theta) \mathrm{d}\xi_s(\theta) \mathrm{d}\lambda(s).$$

$$\partial_t \alpha_{t,s} + \operatorname{div}_\theta(\alpha_{t,s} v_{t,s}) = 0, \qquad v_{t,s}(\theta) = -\nabla_\theta \frac{\delta F}{\delta \alpha_s}(\alpha_t)(\theta), \tag{14.45}$$

for $\lambda$-a.e. condition $s$, whenever the first variation is differentiable in the parameter variable. In the ResNet case one takes $F = f_{\mathrm{Res}}$.

**Particle discretization and layerwise matching.** $\alpha_r = \frac{1}{n_r} \sum_{i=1}^{n_r} \delta_{\theta_{r,i}}$, $\beta_r = \frac{1}{n_r} \sum_{i=1}^{n_r} \delta_{\theta'_{r,i}}$, $\mathcal{W}_{2,\lambda_L}^2(\alpha^{L,n}, \beta^{L,n}) = \frac{1}{L} \sum_{r=0}^{L-1} \mathcal{W}_2^2(\alpha_r, \beta_r)$. $\frac{1}{L} \sum_{r=0}^{L-1} \frac{1}{n_r} \sum_{i=1}^{n_r} \|\theta_{r,i} - \theta'_{r,i}\|^2$,

### 14.8 Second-Order Momentum Flows

This compact subsection records inertial extensions of Wasserstein particle flows.

**Finite-dimensional momentum.** Let $F:\mathbb{R}^N\to\mathbb{R}$ be a smooth finite-dimensional objective. With step size $h>0$ and damping $\gamma>0$, one convenient velocity form is

$$S^{k+1}=(1-\gamma h)S^k-h\nabla F(X^k),\qquad X^{k+1}=X^k+hS^{k+1}. \tag{14.46}$$

If $S^k$ approximates $\dot X(kh)$, this is the explicit Euler discretization of the first-order system

$$\dot X(t)=S(t),\qquad \dot S(t)=-\gamma S(t)-\nabla F(X(t)), \tag{14.47}$$

$$\ddot X(t)+\gamma\dot X(t)+\nabla F(X(t))=0. \tag{14.48}$$

$$Y^k=X^k+\theta_k(X^k-X^{k-1}),\qquad X^{k+1}=Y^k-h\nabla F(Y^k), \tag{14.49}$$

$$\ddot X(t)+\frac{r}{t}\dot X(t)+\nabla F(X(t))=0. \tag{14.50}$$

**Empirical Wasserstein lift.**

$$\alpha_X\coloneqq\frac1n\sum_{i=1}^n\delta_{x_i},\qquad F(X)\coloneqq f(\alpha_X). \tag{14.51}$$

$\|\dot X\|_n^2\coloneqq\frac1n\sum_{i=1}^n\|\dot x_i\|^2$,

$$\frac{\mathrm d}{\mathrm d\varepsilon}\Big|_{\varepsilon=0}F(x_1,\dots,x_i+\varepsilon\xi,\dots,x_n)=\frac1n\langle\nabla\delta f(\alpha_X)(x_i),\,\xi\rangle.$$

Thus, if $\nabla^{(n)}F$ denotes the gradient for the metric $\|\cdot\|_n$, then

$$(\nabla^{(n)}F(X))_i=n\nabla_{x_i}F(X)=\nabla\delta f(\alpha_X)(x_i)=\nabla_{\mathcal W}f(\alpha_X)(x_i). \tag{14.52}$$

$$v_\alpha(x)\coloneqq-\nabla_{\mathcal W}f(\alpha)(x)=-\nabla\delta f(\alpha)(x). \tag{14.53}$$

$$\dot x_i(t)=s_i(t),\qquad \dot s_i(t)=-\gamma s_i(t)+v_{\alpha_t}(x_i(t)),\qquad \alpha_t=\frac1n\sum_{i=1}^n\delta_{x_i(t)}. \tag{14.54}$$

$\ddot x_i(t)+\gamma\dot x_i(t)=v_{\alpha_t}(x_i(t))=-\nabla_{\mathcal W}f(\alpha_t)(x_i(t))$.

**Proposition 14.35** (Discrete mechanical energy dissipation)**.** *Assume that $f$ and the particle solution are smooth enough to justify differentiating the identities above along (14.54). Define $\mathcal H_n(X,S)\coloneqq F(X)+\frac{1}{2n}\sum_{i=1}^n\|s_i\|^2$, $\quad S=(s_1,\dots,s_n)$. Then along (14.54),*

$$\frac{\mathrm d}{\mathrm dt}\mathcal H_n(X(t),S(t))=-\frac{\gamma}{n}\sum_{i=1}^n\|s_i(t)\|^2\leqslant 0. \tag{14.55}$$

*Proof.* By (14.52) and the definition $v_\alpha=-\nabla_{\mathcal W}f(\alpha)$, one has $\nabla_{x_i}F(X)=-v_{\alpha_X}(x_i)/n$. Hence $\frac{\mathrm d}{\mathrm dt}F(X(t))=\sum_{i=1}^n\langle\nabla_{x_i}F(X(t)),\,s_i(t)\rangle=-\frac1n\sum_{i=1}^n\langle v_{\alpha_t}(x_i(t)),\,s_i(t)\rangle$. On the other hand, $\frac{\mathrm d}{\mathrm dt}\frac{1}{2n}\sum_i\|s_i(t)\|^2=\frac1n\sum_i\langle s_i(t),\,-\gamma s_i(t)+v_{\alpha_t}(x_i(t))\rangle$. The force terms cancel, leaving (14.55). □

**Phase-space formulation.**

**Proposition 14.36** (Phase-space Liouville formulation)**.** *Let $v_\alpha$ be smooth enough that (14.54) has a classical solution. Define the empirical phase-space measure $\eta_t^n\coloneqq\frac1n\sum_{i=1}^n\delta_{(x_i(t),s_i(t))}\in\mathcal P(\mathbb R_x^d\times\mathbb R_s^d)$, $\quad\alpha_t^n=(\pi_x)_\sharp\eta_t^n$. Then $\eta_t^n$ solves, in the sense of distributions,*

$$\partial_t\eta_t^n+\nabla_x\cdot(s\,\eta_t^n)+\nabla_s\cdot\big((-\gamma s+v_{\alpha_t^n}(x))\,\eta_t^n\big)=0. \tag{14.56}$$

*If, as $n\to\infty$, the empirical phase-space laws converge to a smooth density $\eta_t(x,s)$ and the nonlinear fields converge accordingly, the limiting kinetic equation is*

$$\partial_t\eta_t+\nabla_x\cdot(s\,\eta_t)+\nabla_s\cdot\big((-\gamma s+v_{\alpha_t}(x))\,\eta_t\big)=0,\qquad\alpha_t=(\pi_x)_\sharp\eta_t. \tag{14.57}$$

*Proof.* Let $\varphi\in C_c^\infty(\mathbb R_x^d\times\mathbb R_s^d)$. Along the particle system,

$$\frac{\mathrm d}{\mathrm dt}\int\varphi\,\mathrm d\eta_t^n=\frac1n\sum_{i=1}^n\big[\langle\nabla_x\varphi(x_i,s_i),\,s_i\rangle+\langle\nabla_s\varphi(x_i,s_i),\,-\gamma s_i+v_{\alpha_t^n}(x_i)\rangle\big].$$

Rewriting the right-hand side as an integral against $\eta_t^n$ and integrating by parts gives (14.56). Passing formally to the mean-field limit gives (14.57). □

$k(x,y)=-\|x-y\|,\qquad f(\alpha)=\mathrm{MMD}_k^2(\alpha,\beta)=\iint k(x,y)\,\mathrm d(\alpha-\beta)(x)\mathrm d(\alpha-\beta)(y)$. $v_\alpha(x)=2\int\frac{x-y}{\|x-y\|}\,\mathrm d(\alpha-\beta)(y)$.

**Entropy without an empirical energy.** For the numerical illustration, we add the quadratic confinement $V(x)=\|x\|^2/2$. In one dimension the overdamped equation becomes the Ornstein–Uhlenbeck Fokker–Planck equation
$\partial_t\rho_t=\partial_{xx}\rho_t+\partial_x(x\rho_t)$, whose stationary law is the centered Gaussian $\mathcal N(0,1)$. The Newton lift is discretized directly in phase space: for a density $\eta_t(x,s)$ with spatial marginal $\rho_t(x)=\int\eta_t(x,s)\,\mathrm ds$, it reads
$\partial_t\eta_t+\partial_x(s\eta_t)+\partial_s\big((-\partial_x\log\rho_t(x)-x)\eta_t\big)=0$.

# 15 Generative Models via Transportation

## 15.1 Generative Models via Flow Matching

This compact subsection records transport-based interpolants for generative modeling.

**Stochastic interpolant.** The word "stochastic" can hide two different levels of randomness. We first use the simpler one: after drawing a latent variable $U \sim \pi$, the path $t \mapsto P_t(U)$ is deterministic and differentiable.
We assume first that $\alpha_t$ is obtained by pushing a latent distribution $\pi \in \mathcal{P}(\mathbb{R}^{d'})$ through a time-dependent map $P_t : \mathbb{R}^{d'} \to \mathbb{R}^d$; the latent dimension $d'$ may be larger than the data dimension $d$, i.e.

$$\forall t \in [0,1], \quad \alpha_t := (P_t)_\sharp \pi. \tag{15.1}$$

If $\pi = \alpha \otimes \beta$ and $\alpha = \frac{1}{n}\sum_i \delta_{x_i}$, $\beta = \frac{1}{m}\sum_j \delta_{y_j}$, then $\alpha_t$ consists of $n \times m$ Dirac masses $\alpha_t = \frac{1}{nm}\sum_{i,j} \delta_{P_t(x_i,y_j)}$. If $\pi = (\mathrm{Id}, T)_\sharp \alpha$ is a Brenier-type coupling, then $\alpha_t = ((1-t)\mathrm{Id} + tT)_\sharp \alpha$ is the so-called McCann OT interpolation.

**Flow matching formula.**

**Proposition 15.1** (Flow matching vector field)**.** *For each fixed $t$, assume $\partial_t P_t \in L^2(\pi; \mathbb{R}^d)$. Consider the flow-matching problem over measurable fields $v_t : \mathbb{R}^d \to \mathbb{R}^d$*

$$\min_{v_t} \int_{\mathbb{R}^{d'}} \|v_t(P_t(u)) - [\partial_t P_t](u)\|^2 \, \mathrm{d}\pi(u). \tag{15.2}$$

*Its minimizer is characterized $\alpha_t$-almost everywhere by the conditional expectation*

$$v_t(z) = \mathbb{E}_{u\sim\pi}\big([\partial_t P_t](u) \,\big|\, z = P_t(u)\big). \tag{15.3}$$

*Then the pair $(\alpha_t, v_t)$ satisfies the continuity equation (13.2).*

*Proof.* We first recall the two equivalent ways of writing the interpolated measure. Formally, one may write $\alpha_t(z) = \int_{\mathbb{R}^{d'}} \delta(z - P_t(u))\,\mathrm{d}\pi(u)$, while the rigorous meaning is that, for every smooth test function $\varphi$,

$$\int_{\mathbb{R}^d} \varphi(z)\,\mathrm{d}\alpha_t(z) = \int_{\mathbb{R}^{d'}} \varphi(P_t(u))\,\mathrm{d}\pi(u). \tag{15.4}$$

The minimizer in (15.2) is the orthogonal projection in $L^2(\pi; \mathbb{R}^d)$ of the latent velocity $\partial_t P_t(u)$ onto the closed subspace of functions that depend on $u$ only through $P_t(u)$. This projection is the conditional expectation (15.3). Formally, this can be read as $v_t(z) = \frac{1}{\alpha_t(z)} \int_{\mathbb{R}^{d'}} \delta(z - P_t(u))[\partial_t P_t](u)\,\mathrm{d}\pi(u)$, and rigorously it means that, for every smooth test vector field $m$,

$$\int \langle m(z),\, v_t(z)\rangle\,\mathrm{d}\alpha_t(z) = \int \langle m(P_t(u)),\, [\partial_t P_t](u)\rangle\,\mathrm{d}\pi(u). \tag{15.5}$$

The weak form of $\partial_t \alpha_t + \mathrm{div}(\alpha_t v_t) = 0$ is that, for every smooth scalar test function $\varphi$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(z)\,\mathrm{d}\alpha_t(z) - \int \langle v_t(z),\, \nabla\varphi(z)\rangle\,\mathrm{d}\alpha_t(z) = 0. \tag{15.6}$$

Using (15.4) and differentiating under the integral sign gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(z)\mathrm{d}\alpha_t(z) = \int \langle \nabla\varphi(P_t(u)),\, [\partial_t P_t](u)\rangle \mathrm{d}\pi(u). \tag{15.7}$$

On the other hand, applying (15.5) with $m = \nabla\varphi$ gives

$$\int \langle v_t(z),\, \nabla\varphi(z)\rangle\,\mathrm{d}\alpha_t(z) = \int \langle \nabla\varphi(P_t(u)),\, [\partial_t P_t](u)\rangle \mathrm{d}\pi(u). \tag{15.8}$$

Comparing (15.7) and (15.8) yields (15.6), which is the desired continuity equation. ◻

The conditional expectation in (15.3) has a simple measure-theoretic meaning. Let $\alpha_t = (P_t)_\sharp \pi$ and define the vector-valued measure $m_t$ on $\mathbb{R}^d$ by
$\int_{\mathbb{R}^d} \langle \psi(z),\, \mathrm{d}m_t(z)\rangle := \int_{\mathbb{R}^{d'}} \langle \psi(P_t(u)),\, [\partial_t P_t](u)\rangle \mathrm{d}\pi(u)$ $\mathrm{d}m_t(z) = v_t(z)\mathrm{d}\alpha_t(z)$, $\quad v_t = \frac{\mathrm{d}m_t}{\mathrm{d}\alpha_t}$. In the language of Lebesgue decomposition, the flux measure $m_t$ has only an absolutely continuous part with respect to $\alpha_t$ and no singular part; the conditional expectation is precisely this density. Equivalently, disintegrating $\pi$ with respect to the map $P_t$ gives $\pi(\mathrm{d}u) = \pi_{t,z}(\mathrm{d}u)\alpha_t(\mathrm{d}z)$, where $\pi_{t,z}$ is supported on the fiber $\{u : P_t(u) = z\}$, and
$v_t(z) = \int_{\{P_t(u)=z\}} [\partial_t P_t](u)\mathrm{d}\pi_{t,z}(u)$. Thus the solution of (15.2) is the conditional expectation of the velocities $\partial_t P_t$: intuitively, $v_t(z)$ is the average velocity of all trajectories passing through $z$.
For the exact field $v_t$, integrating the ODE $\dot{x} = v_t(x)$ defines a transport map $T_t$. If $v_t$ is regular enough, or more generally if the continuity equation has a unique solution for this velocity, then $(T_t)_\sharp \alpha_0 = \alpha_t$.

**Connection with diffusion models.** In the special case where $P_t(x,y) = (1-t)x + ty$ is a linear interpolation and $\pi = \alpha \otimes \beta$, the curve $\alpha_t$ is a convolution of rescaled versions of $\alpha_0$ and $\alpha_1$. The flow-matching problem (15.2) becomes
$\min_{(v_t)_t} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|v_t((1-t)x + ty) - (y - x)\|^2 \,\mathrm{d}\alpha_0(x)\mathrm{d}\alpha_1(y)$.

**Proposition 15.2** (Tweedie identity)**.** *Let $W$ be a random vector in $\mathbb{R}^d$ with density $\beta$. For $\sigma > 0$, observe $Z = W + \sigma\,\varepsilon$, where $\varepsilon \sim \mathcal{N}(0, I_d)$ is independent of $W$. Denote by $\beta_\sigma = \beta * \mathcal{N}(0, \sigma^2 I_d)$ the density of $Z$. Then $\mathbb{E}\big[W \mid Z = z\big] = z + \sigma^2 \nabla \log \beta_\sigma(z)$ for all $z \in \mathbb{R}^d$.*

*Proof.* Bayes' rule gives the conditional density $p_{W|Z}(w \mid z) = \dfrac{\beta(w)\,\varphi_\sigma(z-w)}{\beta_\sigma(z)}$ with $\varphi_\sigma$ the $\mathcal{N}(0, \sigma^2 I_d)$ density. Hence $\mathbb{E}[W \mid Z = z] = \frac{1}{\beta_\sigma(z)} \int_{\mathbb{R}^d} w\,\beta(w)\,\varphi_\sigma(z-w)\,\mathrm{d}w$. Differentiating the Gaussian convolution under the integral sign and using $\nabla_z \varphi_\sigma(z-w) = -\sigma^{-2}(z-w)\,\varphi_\sigma(z-w)$ yields $\nabla_z \beta_\sigma(z) = \int \beta(w)\,\nabla_z\varphi_\sigma(z-w)\,\mathrm{d}w = -\sigma^{-2}\Big(z - \mathbb{E}[W \mid Z = z]\Big)\,\beta_\sigma(z)$. Rearranging finishes the proof. ◻

**Proposition 15.3** (Gaussian-endpoint flow-matching field). *Let $X \sim \alpha$ and $Y \sim \mathcal{N}(0, I_d)$ be independent. For $t \in (0,1)$ set* $Z_t = (1-t)\,X + t\,Y, \qquad \alpha_t = Law(Z_t)$. *The regression minimizer $v^\star : \mathbb{R}^d \times (0,1) \to \mathbb{R}^d$ of* $\min_v \int_0^1 \iint_{\mathbb{R}^d\times\mathbb{R}^d} \big|y - x - v\big((1-t)x + ty, t\big)\big|^2 \,\mathrm{d}\alpha(x)\,\mathrm{d}\mathcal{N}(y)\,\mathrm{d}t$ *is* $v^\star(x,t) = -\frac{1}{1-t}\,x \;-\; \frac{t}{1-t}\,\nabla \log \alpha_t(x) \qquad (x \in \mathbb{R}^d,\ t \in (0,1))$. *In particular, for each $t \in (0,1)$ this field is a gradient field,*

$$v^\star(\cdot, t) = -\nabla\left(\frac{\|\cdot\|^2}{2(1-t)} + \frac{t}{1-t}\log \alpha_t\right).$$

*Proof.* Fix $t \in (0,1)$ and write $W = (1-t)X$, $\sigma = t$, so that $Z_t = W + \sigma Y$ matches the setting of Proposition 15.2. Conditional expectations satisfy $v^\star(z,t) = \mathbb{E}[Y - X \mid Z_t = z] = \frac{1}{t}\,\mathbb{E}[Z_t - W \mid Z_t = z] - \frac{1}{1-t}\,\mathbb{E}[W \mid Z_t = z]$. Applying Proposition 15.2 to $\mathbb{E}[W \mid Z_t = z]$ and noting $\mathbb{E}[Y \mid Z_t = z] = -\,t\,\nabla \log \alpha_t(z)$ gives the claimed formula. □

**When is the induced map optimal?** Integrating the learned velocity gives a deterministic map from $\alpha_0$ to $\alpha_1$, but this map is not automatically the Brenier optimal map. It is optimal only in special cases where the accumulated flow remains the gradient of a convex potential.

**Proposition 15.4** (Gaussian flow matching and optimality). *Let $\Sigma_0, \Sigma_1 \succ 0$ and let $X_0 \sim \mathcal{N}(0, \Sigma_0)$ and $X_1 \sim \mathcal{N}(0, \Sigma_1)$ be independent. Consider the linear flow-matching interpolation* $Z_t = (1-t)X_0 + tX_1, \qquad \alpha_t = \mathrm{Law}(Z_t) = \mathcal{N}(0, \Sigma_t)$, *where*

$$\Sigma_t = (1-t)^2\Sigma_0 + t^2\Sigma_1. \tag{15.9}$$

*Then the exact flow-matching velocity is affine, $v_t(z) = A_t z$, with*

$$A_t = \big(t\Sigma_1 - (1-t)\Sigma_0\big)\Sigma_t^{-1}. \tag{15.10}$$

*The induced flow map $T_t^{\mathrm{FM}}$ from $\alpha_0$ to $\alpha_t$ is*

$$T_t^{\mathrm{FM}} = \Sigma_0^{1/2}\Big((1-t)^2\mathrm{Id} + t^2\Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2}\Big)^{1/2}\Sigma_0^{-1/2}. \tag{15.11}$$

*In particular,*

$$T_1^{\mathrm{FM}} = \Sigma_0^{1/2}\big(\Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2}\big)^{1/2}\Sigma_0^{-1/2}. \tag{15.12}$$

*This terminal map coincides with the quadratic optimal transport map*

$$T^{\mathrm{OT}} = \Sigma_0^{-1/2}\big(\Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2}\big)^{1/2}\Sigma_0^{-1/2} \tag{15.13}$$

*if and only if $\Sigma_0\Sigma_1 = \Sigma_1\Sigma_0$.*

*Proof.* The conditional-expectation formula gives $v_t(z) = \mathbb{E}[X_1 - X_0 \mid Z_t = z]$. Since all variables are jointly Gaussian, this conditional expectation is linear and $v_t(z) = \mathrm{Cov}(X_1 - X_0, Z_t)\,\mathrm{Cov}(Z_t)^{-1}z = \big(t\Sigma_1 - (1-t)\Sigma_0\big)\Sigma_t^{-1}z$, which proves (15.10). To solve the characteristic equation, whiten the source by setting $C = \Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2}, \qquad \widetilde{Z}_t = \Sigma_0^{-1/2}Z_t$. In these coordinates the source covariance is Id and $\widetilde{\Sigma}_t = (1-t)^2\mathrm{Id} + t^2C$. Because Id and $C$ commute, the affine flow map in whitened coordinates is simply $\widetilde{T}_t = \widetilde{\Sigma}_t^{1/2}$. Indeed, $\frac{\mathrm{d}}{\mathrm{d}t}\widetilde{\Sigma}_t^{1/2} = \big(tC - (1-t)\mathrm{Id}\big)\widetilde{\Sigma}_t^{-1/2}$, which is exactly the equation $\dot{\widetilde{T}}_t = \widetilde{A}_t\widetilde{T}_t$ with $\widetilde{T}_0 = \mathrm{Id}$. Returning to the original coordinates gives (15.11), and $t = 1$ gives (15.12).

Both $T_1^{\mathrm{FM}}$ and $T^{\mathrm{OT}}$ push $\mathcal{N}(0, \Sigma_0)$ to $\mathcal{N}(0, \Sigma_1)$. The Brenier map between nondegenerate Gaussians is the unique symmetric positive definite linear map with this property. Hence $T_1^{\mathrm{FM}} = T^{\mathrm{OT}}$ if and only if $T_1^{\mathrm{FM}}$ is symmetric positive definite. The map $T_1^{\mathrm{FM}}$ is similar to $C^{1/2}$, so if it is symmetric then it is automatically positive definite. Since $C^{1/2}$ is symmetric positive definite, $(T_1^{\mathrm{FM}})^\top = \Sigma_0^{-1/2}C^{1/2}\Sigma_0^{1/2}$. Thus symmetry of $T_1^{\mathrm{FM}}$ is equivalent to $\Sigma_0C^{1/2} = C^{1/2}\Sigma_0$, hence to $\Sigma_0C = C\Sigma_0$ by functional calculus. Multiplying this identity on the left and right by $\Sigma_0^{1/2}$ gives $\Sigma_0\Sigma_1 = \Sigma_1\Sigma_0$. Conversely, if $\Sigma_0$ and $\Sigma_1$ commute, they are orthogonally co-diagonalizable, and both (15.12) and (15.13) reduce in that basis to the diagonal map with entries $\sqrt{\lambda_{1,k}/\lambda_{0,k}}$. This proves the equivalence. □

**Variations on the interpolant.** $v_t(z) = \lambda'(t)\,v^{\mathrm{lin}}_{\lambda(t)}(z), \qquad v^{\mathrm{lin}}_r(z) = \mathbb{E}[Y - X \mid (1-r)X + rY = z]$. $Z_t = a_tX + b_tY, \qquad Y \sim \mathcal{N}(0, \sigma^2\mathrm{Id})$, then both the mixture centers and the component variances are changed. Writing $p_t$ for the density of $Z_t$ and $s_t = \nabla \log p_t$, Tweedie's formula gives, away from times where $a_t = 0$,

$$v_t(z) = a_t'\,\mathbb{E}[X \mid Z_t = z] + b_t'\,\mathbb{E}[Y \mid Z_t = z] = \frac{a_t'}{a_t}z + \left(\frac{a_t'b_t^2}{a_t} - b_t'b_t\right)\sigma^2 s_t(z).$$

For the linear bridge, $a_t = 1-t$ and $b_t = t$, this recovers the formula above. For the variance-preserving Ornstein–Uhlenbeck noising used in diffusion models,
$a_\tau = e^{-\tau}, \qquad b_\tau = \sqrt{1 - e^{-2\tau}}$, one obtains the forward probability-flow velocity $v_\tau(z) = -z - \sigma^2\nabla \log p_\tau(z)$. Sampling follows the reverse field $z + \sigma^2\nabla \log p_\tau(z)$ as $\tau$ decreases.
$a_t = (1-t)(1-2t), \qquad b_t = t,$

## 15.2 One-Step Generative Models

This compact subsection records one-step generators trained by distributional discrepancies.

**One-step models via parameter-domain discrepancy flows.** The most direct one-step strategy is to train the generator parameters themselves by descending a discrepancy between generated and data distributions. Let $\zeta$ be a simple latent distribution, let $f_\theta : \mathcal{Z} \to \mathcal{X}$ be a neural generator, and let $\beta$ be the data distribution.
$(f_\theta)_\sharp\zeta = \beta$, or, in practice, such that the generated law is close to $\beta$. Given a discrepancy $\mathcal{D}$ on probability measures, define

$$\mathcal{E}_\beta(\gamma) \coloneqq \mathcal{D}(\gamma, \beta), \qquad H_\beta(\theta) \coloneqq \mathcal{E}_\beta\big((f_\theta)_\sharp\zeta\big) = \mathcal{D}\big((f_\theta)_\sharp\zeta, \beta\big). \tag{15.14}$$

$$\dot\theta_t = -\nabla_\theta H_\beta(\theta_t). \tag{15.15}$$

$\alpha_t \coloneqq (f_{\theta_t})_\sharp\zeta, \qquad X_t = f_{\theta_t}(Z), \qquad Z \sim \zeta$. If $t \mapsto \theta_t$ is smooth and $f_\theta$ is differentiable with respect to $\theta$, then $\dot X_t = \partial_\theta f_{\theta_t}(Z)\dot\theta_t$. For $\alpha_t$-a.e. $x$, let $\eta_{t,x}$ be the disintegration of the latent law $\zeta$ with respect to the map $f_{\theta_t}$. Thus $\eta_{t,x}$ is supported on the fiber $\{z \,;\, f_{\theta_t}(z) = x\}$, and for every bounded measurable $\psi$,

$$\int \psi(z)\,\mathrm{d}\zeta(z) = \int_{\mathcal{X}} \left( \int \psi(z)\,\mathrm{d}\eta_{t,x}(z) \right) \mathrm{d}\alpha_t(x).$$

$$v_t(x) = \int \partial_\theta f_{\theta_t}(z)\dot\theta_t\,\mathrm{d}\eta_{t,x}(z) = -\int \partial_\theta f_{\theta_t}(z)\nabla_\theta H_\beta(\theta_t)\,\mathrm{d}\eta_{t,x}(z). \tag{15.16}$$

$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(x)\,\mathrm{d}\alpha_t(x) = \int \langle \nabla\varphi(x),\, v_t(x)\rangle\,\mathrm{d}\alpha_t(x),$

$$\partial_t\alpha_t + \mathrm{div}(\alpha_t v_t) = 0. \tag{15.17}$$

The velocity (15.16) depends on the parameter value $\theta_t$ and on the latent disintegration, not only on the measure $\alpha_t$. If the parametrization is non-identifiable, two different parameters can generate the same measure while inducing different velocities.
$-\nabla\delta_\gamma\mathcal{E}_\beta(\gamma)\big|_{\gamma=\alpha_t}$. $\hat\beta_n = \frac{1}{n}\sum_{i=1}^n \delta_{Y_i}, \qquad Y_i \overset{\text{i.i.d.}}{\sim} \beta,$

$$\mathbb{E}\left[\nabla_\theta H_{\hat\beta_n}(\theta)\right] \neq \nabla_\theta H_\beta(\theta).$$

**One-step model using Wasserstein flow of discrepancy.** Let $\zeta$ be a simple latent distribution and let $\alpha_\theta = (G_\theta)_\sharp\zeta$ be the model distribution. Assume that the target data distribution is $\beta$. The Wasserstein-flow construction chooses a discrepancy $\mathcal{E}_\beta(\alpha)$,

$$\partial_t\alpha_t + \mathrm{div}(\alpha_t w_t) = 0, \qquad w_t(x) = -\nabla\delta_\alpha\mathcal{E}_\beta(\alpha_t)(x). \tag{15.18}$$

$$\min_\eta \int_0^1 \int \|U_\eta(t,x) - w_t(x)\|^2\,\mathrm{d}\alpha_t(x)\,\mathrm{d}t. \tag{15.19}$$

$$\alpha_\theta^+ = (\mathrm{Id} + \tau U_\eta)_\sharp\alpha_\theta, \qquad \text{or equivalently} \qquad G_\theta^+(z) = G_\theta(z) + \tau U_\eta(G_\theta(z)).$$

**Sliced-Wasserstein flow.** $\mathcal{E}_\beta(\alpha) = \frac{1}{2}\mathrm{SW}_2^2(\alpha,\beta)$, with $\mathrm{SW}_2$ defined in Section 10.2. With the continuity-equation convention $\partial_t\alpha_t + \mathrm{div}(\alpha_t w_t) = 0$, the descent velocity points from the current projected law toward the target projected law.

$$w_{\mathrm{SW}}[\alpha,\beta](x) = \int_{\mathbb{S}^{d-1}} \big(T_\theta(P_\theta x) - P_\theta x\big)\,\theta\,\mathrm{d}\sigma(\theta). \tag{15.20}$$

The sign follows from the one-dimensional identity: the first variation of $\frac{1}{2}\mathcal{W}_2^2(\rho,\nu)$ has spatial derivative $s - T(s)$, so the descent velocity is $T(s) - s$, and composing with $P_\theta$ lifts it in the direction $\theta$. Thus empirical implementations only require one-dimensional sorting along sampled directions, followed by averaging the lifted projected displacements.
**Stein variational gradient descent.** For the formal density-level calculation, assume $\alpha = \rho_\alpha\,\mathrm{d}x$ and $\beta = \rho_\beta\,\mathrm{d}x$ have smooth positive densities, and let $v$ be a smooth compactly supported vector field. For the perturbation $\alpha_\varepsilon = (\mathrm{Id} + \varepsilon v)_\sharp\alpha$, integration by parts gives the Stein first variation

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\mathrm{KL}(\alpha_\varepsilon|\beta)\bigg|_{\varepsilon=0} = -\int \big(\langle\nabla\log\rho_\beta(x),\, v(x)\rangle + \mathrm{div}\,v(x)\big)\mathrm{d}\alpha(x).$$

$$v_\alpha^{\mathrm{SVGD}}(x) = \int \Big(k(y,x)\nabla\log\rho_\beta(y) + \nabla_y k(y,x)\Big)\mathrm{d}\alpha(y). \tag{15.21}$$

$$\partial_t\alpha_t + \mathrm{div}\big(\alpha_t v_{\alpha_t}^{\mathrm{SVGD}}\big) = 0. \tag{15.22}$$

For particles $\alpha_n^\ell = n^{-1}\sum_i \delta_{X_i^\ell}$, this becomes the explicit update

$$X_i^{\ell+1} = X_i^\ell + \tau\,\frac{1}{n}\sum_{j=1}^n \Big(k(X_j^\ell, X_i^\ell)\nabla\log\rho_\beta(X_j^\ell) + \nabla_{X_j} k(X_j^\ell, X_i^\ell)\Big). \tag{15.23}$$

**Self-corrected drifting fields.**

$$B_\varepsilon[\nu](x) \coloneqq \frac{\int (y-x)K_\varepsilon(x,y)\,\mathrm{d}\nu(y)}{\int K_\varepsilon(x,y)\,\mathrm{d}\nu(y)}. \tag{15.24}$$

For the Gaussian kernel $K_\varepsilon(x,y) = \exp(-\|x-y\|^2/(2\varepsilon))$, this normalized field is a score of a smoothed density:

$$B_\varepsilon[\nu](x) = \varepsilon\nabla\log\left(\int K_\varepsilon(x,y)\,\mathrm{d}\nu(y)\right). \tag{15.25}$$

$$u_t(x) = B_\varepsilon[\beta](x) - B_\varepsilon[\alpha_t](x) = \varepsilon\nabla\log\frac{\int K_\varepsilon(x,y)\,\mathrm{d}\beta(y)}{\int K_\varepsilon(x,y)\,\mathrm{d}\alpha_t(y)}. \tag{15.26}$$

The first term pulls samples toward data, while the second term corrects self-attraction and prevents all particles from collapsing onto the same high-density region. For a fixed reference measure, $B_\varepsilon[\nu]$ is precisely the Gaussian mean-shift displacement in (15.33): it moves $x$ toward the local kernel barycenter of $\nu$.

**Proposition 15.5** (Drifting as a time-dependent Wasserstein gradient)**.** *Let $\alpha_t$ be a smooth curve of positive densities and let $u_t = \nabla\varphi_t$ be a smooth time-dependent gradient field. Define the semi-relaxed functional*

$$\mathcal{R}_t(\alpha|\alpha_t) \coloneqq -\int \varphi_t(x)\,\mathrm{d}\alpha(x) + \int \varphi_t(x)\,\mathrm{d}\alpha_t(x). \tag{15.27}$$

*Here $\alpha_t$ and $\varphi_t$ are frozen when taking the first variation with respect to the first argument $\alpha$. Then the continuity equation $\partial_t\alpha_t + \operatorname{div}(\alpha_t u_t) = 0$ is the formal Wasserstein gradient descent of the time-dependent functional $\alpha \mapsto \mathcal{R}_t(\alpha|\alpha_t)$.*

*Proof.* Since $\alpha_t$ and $\varphi_t$ are fixed in the variation with respect to $\alpha$, the first variation is $\delta_\alpha\mathcal{R}_t(\alpha|\alpha_t)(x) = -\varphi_t(x)$. By Proposition 14.2, $\nabla_{\mathcal{W}}\mathcal{R}_t(\alpha|\alpha_t) = \nabla\delta_\alpha\mathcal{R}_t(\alpha|\alpha_t) = -\nabla\varphi_t = -u_t$. The Wasserstein gradient-descent velocity is the negative of this gradient, namely $u_t$. Substituting this velocity in the continuity equation gives the claimed flow. □

### 15.3 Moment Measures

**Definition 15.6** (Moment measure of a convex function)**.** Let $u : \mathbb{R}^d \to \mathbb{R}\cup\{+\infty\}$ be a proper lower-semicontinuous convex function such that $Z_u \coloneqq \int_{\mathbb{R}^d} e^{-u(x)}\,\mathrm{d}x < +\infty$. The normalized log-concave measure associated with $u$ is $\rho_u \coloneqq Z_u^{-1}e^{-u(x)}\,\mathrm{d}x$. Whenever $\nabla u$ is defined $\rho_u$-almost everywhere, the moment measure of $u$ is $\mathfrak{M}(u) \coloneqq (\nabla u)_\sharp\rho_u$.

$\int y\,\mathrm{d}\mathfrak{M}(u)(y) = Z_u^{-1}\int \nabla u(x)e^{-u(x)}\,\mathrm{d}x = -Z_u^{-1}\int \nabla(e^{-u(x)})\,\mathrm{d}x = 0$. Thus moment measures are necessarily centered. The non-smooth theory uses essentially continuous convex functions, meaning convex functions whose restriction to the closure of their effective domain is continuous except possibly on a boundary set of zero $\mathcal{H}^{d-1}$ measure.

**Theorem 15.7** (Cordero–Erausquin–Klartag)**.** *Let $\alpha \in \mathcal{P}_1(\mathbb{R}^d)$ be a probability measure. There exists an essentially continuous convex function $u$ such that $\alpha = \mathfrak{M}(u)$ if and only if $\int y\,\mathrm{d}\alpha(y) = 0 \quad$ and $\quad \operatorname{supp}(\alpha)$ is not contained in a hyperplane. Under these conditions, $u$ is unique up to translations of the argument and additive constants.*

**Optimal-transport variational formulation.**

$$\mathcal{C}_\alpha(\rho) \coloneqq \sup_{\pi\in\mathcal{U}(\rho,\alpha)} \int_{\mathbb{R}^d\times\mathbb{R}^d} x\cdot y\,\mathrm{d}\pi(x,y). \tag{15.28}$$

$$\mathcal{C}_\alpha(\rho) = \inf_{v\text{ convex}} \left\{\int v(x)\,\mathrm{d}\rho(x) + \int v^*(y)\,\mathrm{d}\alpha(y)\right\}, \tag{15.29}$$

where $v^*$ is the Legendre transform. If $\rho, \alpha \in \mathcal{P}_2(\mathbb{R}^d)$, then

$$\mathcal{C}_\alpha(\rho) = \frac{1}{2}\int \|x\|^2\,\mathrm{d}\rho(x) + \frac{1}{2}\int \|y\|^2\,\mathrm{d}\alpha(y) - \frac{1}{2}\mathcal{W}_2^2(\rho,\alpha). \tag{15.30}$$

$$\min_{\rho\in\mathcal{P}_1(\mathbb{R}^d)} \{\mathcal{H}(\rho) + \mathcal{C}_\alpha(\rho)\}, \qquad \mathcal{H}(r\,\mathrm{d}x) \coloneqq \int r(x)\log r(x)\,\mathrm{d}x, \tag{15.31}$$

**Proposition 15.8** (Variational characterization of moment measures)**.** *Let $\alpha \in \mathcal{P}_1(\mathbb{R}^d)$ be centered and not supported on a hyperplane. Then the minimization problem (15.31) admits a solution, unique up to translations. Every minimizer is a log-concave density of the form $\rho = Z_u^{-1}e^{-u}\,\mathrm{d}x$, where $u$ is convex, essentially continuous, and satisfies the moment-measure equation $\alpha = (\nabla u)_\sharp\rho$. Conversely, any essentially continuous convex $u$ satisfying this equation yields a global minimizer. In the finite-second-moment case $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$, the objective $\rho \mapsto \mathcal{H}(\rho) + \mathcal{C}_\alpha(\rho)$ is displacement convex along $\mathcal{W}_2$ geodesics on $\mathcal{P}_2(\mathbb{R}^d)$ whenever the terms are finite. For general $\alpha \in \mathcal{P}_1(\mathbb{R}^d)$, the same statement is obtained by approximation and lower semicontinuity.*

*Proof.* The existence proof has two ingredients. First, since $\alpha$ is centered, translating $\rho$ does not change either $\mathcal{H}(\rho)$ or $\mathcal{C}_\alpha(\rho)$, so one can center a minimizing sequence. Second, the assumption that $\alpha$ is not supported on a hyperplane gives a coercive lower bound of the form $\mathcal{C}_\alpha(\rho) \geqslant c_\alpha\int\|x\|\,\mathrm{d}\rho(x)$ for centered absolutely continuous $\rho$, with $c_\alpha > 0$. Together with the lower semicontinuity estimates for entropy and maximal correlation, this yields a minimizer.

Let $\rho = r\,\mathrm{d}x$ be a minimizer and let $u$ be a convex optimizer in the dual formula (15.29). Keeping $u$ fixed and varying $\rho$ in (15.31) gives the Euler equation $\log r(x) + 1 + u(x) = \text{constant} \quad$ on $\{r > 0\}$, so $\rho = Z_u^{-1}e^{-u}\mathrm{d}x$. The optimality condition for the scalar-product transport problem says that an optimal coupling is supported on $\{(x,y) : y \in \partial u(x)\}$. Since $\rho$ is absolutely continuous and $u$ is convex, $\partial u(x) = \{\nabla u(x)\}$ for $\rho$-almost every $x$, hence $\alpha = (\nabla u)_\sharp\rho$.

Conversely, assume $\rho = Z_u^{-1}e^{-u}\mathrm{d}x$ and $\alpha = (\nabla u)_\sharp\rho$. Let $\nu$ be a smooth compactly supported competitor, and let $T$ be the Brenier map from $\rho$ to $\nu$. Along the geodesic $\rho_t = ((1-t)\mathrm{Id}+tT)_\sharp\rho$, the right derivative of the entropy at $t = 0$ is $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{H}(\rho_t)\Big|_{t=0^+} = -\int\langle T(x)-x,\,\nabla u(x)\rangle\,\mathrm{d}\rho(x)$, where the identity follows by differentiating the Jacobian formula and integrating by parts. Santambrogio's derivative estimate for the maximal-correlation functional gives $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{C}_\alpha(\rho_t)\Big|_{t=0^+} \geqslant \int\langle T(x)-x,\,\nabla u(x)\rangle\,\mathrm{d}\rho(x)$, because $(\mathrm{Id},\nabla u)_\sharp\rho$ is optimal for the scalar-product problem. The first-order terms cancel, hence the one-sided derivative of $\mathcal{H}+\mathcal{C}_\alpha$ at $\rho$ in every such direction is nonnegative. Displacement convexity then implies global minimality, and approximation removes the smooth compact-support restriction. Strict displacement convexity of entropy gives uniqueness, except for translations; translations do not change $\mathcal{C}_\alpha$ because $\alpha$ is centered.

The entropy term is displacement convex by McCann's theorem, recalled in Theorem 14.6. If $\alpha$ has finite second moment, identity (15.30) writes $\mathcal{C}_\alpha$ as the sum of the 1-convex moment term $\rho \mapsto \frac{1}{2}\int\|x\|^2\,\mathrm{d}\rho$ and the $(-1)$-convex term $\rho \mapsto -\frac{1}{2}\mathcal{W}_2^2(\rho,\alpha)$, hence $\mathcal{C}_\alpha$ is displacement convex. For merely finite first moment $\alpha$, the same convexity and the variational characterization follow by the approximation and semicontinuity arguments of Santambrogio. □

**Conjugate moment measures for generation.**

$$\beta = (\nabla w^*)_\sharp\left(Z_w^{-1}e^{-w(z)}\,\mathrm{d}z\right). \tag{15.32}$$

### 15.4 Evolution in Depth of Transformers

This compact subsection records the continuous-depth view of attention dynamics.

**Attention as a context-dependent velocity.** After tokenization, embedding, and positional encoding, each input (from a set of tokens) is represented as a point cloud $(x_i)_{i=1}^n$ of $n$ points in the space of vectorized tokens. An attention layer with skip connection and rescaling by $1/T$ (where $T$ is the depth) defines a transformation of the tokens:
$x_i \mapsto x_i + \frac{1}{T}\sum_j \frac{e^{\langle Qx_i, Kx_j\rangle} Vx_j}{\sum_\ell e^{\langle Qx_i, Kx_\ell\rangle}}$, where $\theta = (K, Q, V)$ are the parameters of the attention layer, represented by three matrices.

**Token measure evolution.** To handle an arbitrary number of tokens, we define $\alpha = \frac{1}{n}\sum_i \delta_{x_i}$ as the empirical measure of tokens and rewrite the transformer mapping as:
$x_i \mapsto x_i + \frac{1}{T}\Gamma_\theta[\alpha](x_i)$, $\Gamma_\theta[\alpha](x) := \frac{\int e^{\langle Qx, Ky\rangle} Vy \,\mathrm{d}\alpha(y)}{\int e^{\langle Qx, Kz\rangle}\,\mathrm{d}\alpha(z)}$. $\alpha_{t+\tau} = (\mathrm{Id} + \tau\Gamma_{\theta_t}[\alpha_t])_\sharp \alpha_t$. $\partial_t \alpha_t + \mathrm{div}(\alpha_t \Gamma_{\theta_t}[\alpha_t]) = 0$.

**$L^2$ attention and mean shift.**

$$K_\varepsilon(x, y) = \exp(-\|x - y\|^2/(2\varepsilon)), \qquad \rho_\varepsilon[\alpha](x) = \int K_\varepsilon(x, y)\mathrm{d}\alpha(y), \qquad m_\varepsilon[\alpha](x) = \frac{\int yK_\varepsilon(x, y)\mathrm{d}\alpha(y)}{\rho_\varepsilon[\alpha](x)}.$$

$$M_\varepsilon[\alpha](x) := m_\varepsilon[\alpha](x) - x = \frac{\int (y - x)K_\varepsilon(x, y)\mathrm{d}\alpha(y)}{\rho_\varepsilon[\alpha](x)} = \varepsilon\nabla \log \rho_\varepsilon[\alpha](x) \tag{15.33}$$

$x_i^{k+1} = (1 - \tau)x_i^k + \tau m_\varepsilon[\alpha_k](x_i^k) = x_i^k + \tau M_\varepsilon[\alpha_k](x_i^k)$ $\partial_t \alpha_t + \mathrm{div}\big(\alpha_t M_\varepsilon[\alpha_t]\big) = 0$.

**Gradient structure and limitations.** When the token space has dimension $d$ and the query/key space has dimension $r$, take $Q, K \in \mathbb{R}^{r\times d}$ and $V \in \mathbb{R}^{d\times d}$. If $V = Q^\top K$, the field $\Gamma_\theta[\alpha]$ is a gradient vector field in the token variable. Indeed, define the log-partition potential
$\Phi_\alpha(x) = \int \exp(\langle Qx, Ky\rangle)\mathrm{d}\alpha(y)$, $\qquad U_\alpha(x) = \log \Phi_\alpha(x)$. $\nabla_x U_\alpha(x) = \frac{\int Q^\top Ky \exp(\langle Qx, Ky\rangle)\mathrm{d}\alpha(y)}{\int \exp(\langle Qx, Kz\rangle)\mathrm{d}\alpha(z)} = \Gamma_\theta[\alpha](x)$. This is an instantaneous gradient in $x$. It is not, however, the gradient of the first variation of a fixed functional of $\alpha$, because the potential $U_\alpha$ itself depends on the current measure through the same attention normalization.

### 15.5 Flows over the Gaussian Manifold

This compact subsection records Gaussian closures for transport-driven flows.

**Gaussianity preservation.**

**Proposition 15.9** (Affine velocities preserve Gaussianity)**.** *Let $\alpha_t = \mathcal{N}(\mathbf{m}_t, \boldsymbol{\Sigma}_t)$, with $\boldsymbol{\Sigma}_t$ positive definite, solve the continuity equation with an affine velocity $v_t(x) = b_t + A_t(x - \mathbf{m}_t)$. Then $\alpha_t$ remains Gaussian and its moments solve $\dot{\mathbf{m}}_t = b_t$, $\quad \dot{\boldsymbol{\Sigma}}_t = A_t\boldsymbol{\Sigma}_t + \boldsymbol{\Sigma}_t A_t^\top$. Conversely, any smooth Gaussian curve with positive definite covariance can be generated by such an affine velocity. If one wants the velocity to be a Wasserstein tangent gradient, one chooses the unique symmetric solution of the Lyapunov equation $A_t\boldsymbol{\Sigma}_t + \boldsymbol{\Sigma}_t A_t = \dot{\boldsymbol{\Sigma}}_t$.*

*Proof.* Let $X_t$ follow the characteristic ODE $\dot{X}_t = b_t + A_t(X_t - \mathbf{m}_t)$. This linear ODE maps Gaussian random variables to Gaussian random variables. Taking expectation gives $\dot{\mathbf{m}}_t = b_t$. Writing $\tilde{X}_t = X_t - \mathbf{m}_t$, one has $\dot{\tilde{X}}_t = A_t\tilde{X}_t$, hence $\dot{\boldsymbol{\Sigma}}_t = \frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}(\tilde{X}_t\tilde{X}_t^\top) = A_t\boldsymbol{\Sigma}_t + \boldsymbol{\Sigma}_t A_t^\top$. For the converse, set $b_t = \dot{\mathbf{m}}_t$ and choose any matrix $A_t$ satisfying the covariance equation. Since $\boldsymbol{\Sigma}_t$ is positive definite, the Lyapunov map $A \mapsto A\boldsymbol{\Sigma}_t + \boldsymbol{\Sigma}_t A$ is invertible on symmetric matrices, which gives the unique symmetric choice when a gradient velocity is required. In that case $v_t$ is the gradient of the quadratic potential $x \mapsto \langle b_t, x\rangle + \langle A_t(x - \mathbf{m}_t), x - \mathbf{m}_t\rangle/2$. ☐

**Gaussian-preserving gradient flows.** The following centered catalogue tracks representative cases where the Wasserstein gradient is linear on Gaussian inputs. Thus Gaussianity is a conclusion of the flow, not an ansatz imposed in advance.

**Proposition 15.10** (Centered Gaussian closure catalogue)**.** *Let $\gamma = \mathcal{N}(0, \mathrm{Id})$ and let $\alpha_0 = \mathcal{N}(0, C_0)$ with $C_0 \succ 0$. For each row below, let $\alpha_t$ be the corresponding Wasserstein gradient flow initialized at $\alpha_0$. Then the centered Gaussian family is invariant: as long as $C_t \succ 0$,*

$$\alpha_t = \mathcal{N}(0, C_t), \qquad \dot{C}_t = H(C_t),$$

*with*

$$\begin{aligned}
\mathrm{KL}(\alpha|\gamma) \quad &: \quad H(C) = 2(\mathrm{Id} - C),\\
\mathcal{I}(\alpha|\gamma) \quad &: \quad H(C) = 4(C^{-1} - C),\\
\mathcal{W}_2^2(\alpha, \gamma) \quad &: \quad H(C) = 4(C^{1/2} - C),\\
\mathrm{MMD}_k^2(\alpha, \gamma), \quad k(x, y) = \langle x, y\rangle^2 \quad &: \quad H(C) = 8(C - C^2),\\
\bar{\mathcal{L}}^\varepsilon_{\|\cdot - \cdot\|^2}(\alpha, \gamma) \quad &: \quad H(C) = 4\left(C + \tfrac{\varepsilon^2}{16}\mathrm{Id}\right)^{1/2} - 4\left(C^2 + \tfrac{\varepsilon^2}{16}\mathrm{Id}\right)^{1/2},\\
\mathrm{SW}_2^2(\alpha, \gamma) \quad &: \quad H(C) = V(C)C + CV(C),
\end{aligned}$$

*where $\bar{\mathcal{L}}^\varepsilon_{\|\cdot - \cdot\|^2}$ is the debiased Sinkhorn divergence for the quadratic cost $\|x - y\|^2$ and KL regularization strength $\varepsilon$, and*

$$V(C) = 2\int_{\mathbb{S}^{d-1}} \left(\frac{1}{\sqrt{\theta^\top C\theta}} - 1\right)\theta\theta^\top\,\mathrm{d}\sigma(\theta)$$

*for the normalized spherical measure $\sigma$. Here*

$$\mathcal{I}(\alpha|\gamma) = \int |\nabla \log \rho(x) + x|^2\,\rho(x)\,\mathrm{d}x \qquad (\alpha = \rho\,\mathrm{d}x).$$

*Multiplying any of these energies by a constant simply rescales the corresponding right-hand side.*

*Proof.* Each row is obtained by computing the ambient Wasserstein first variation on centered Gaussian inputs and identifying the affine descent velocity $v(x) = M_C x$. Proposition 15.9 then gives $\dot{C} = M_C C + C M_C^\top$. For $\mathrm{KL}(\cdot|\gamma)$, the Fokker–Planck velocity is $(C^{-1} - \mathrm{Id})x$, hence $\dot{C} = 2(\mathrm{Id} - C)$. For the Fisher row, the ambient first variation is quadratic on centered Gaussian inputs; equivalently, the restriction of $\frac{1}{2}\mathcal{I}$ to centered Gaussians is

$$\frac{1}{2}\left(\mathrm{tr}(C) + \mathrm{tr}(C^{-1}) - 2d\right).$$

Multiplying by two for $\mathcal{I}$ gives the descent velocity $2(C^{-2} - \mathrm{Id})x$, hence $\dot{C} = 4(C^{-1} - C)$. Although the ambient Fisher flow is a fourth-order PDE for a generic density, its velocity is linear on centered Gaussian inputs, so the centered Gaussian family is closed.
For $\mathcal{W}_2^2(\cdot, \gamma)$, the Brenier map from $\mathcal{N}(0, C)$ to $\gamma$ is $C^{-1/2}x$, so the descent velocity for the unhalved squared distance is $2(C^{-1/2} - \mathrm{Id})x$, giving $4(C^{1/2} - C)$. For the polynomial MMD row, centered Gaussians satisfy $\mathrm{MMD}_k^2(\alpha, \gamma) = \|C - \mathrm{Id}\|_{\mathrm{F}}^2$; the first variation is quadratic and its descent velocity is $4(\mathrm{Id} - C)x$, giving $8(C - C^2)$.
Gaussian Sinkhorn dual potentials are quadratic, so the velocity is again linear; differentiating the closed Gaussian formula yields the displayed spectral expression. The square roots are spectral functions of $C$, hence commute with $C$, which is why the covariance ODE closes as a matrix function of $C$ alone. For sliced Wasserstein, each one-dimensional projection is a Gaussian transport with velocity $2((\theta^\top C\theta)^{-1/2} - 1)\langle\theta, x\rangle\theta$; averaging these velocities over $\mathbb{S}^{d-1}$ gives $v(x) = V(C)x$ and thus $\dot{C} = V(C)C + CV(C)$. □

As a simple exact family, if a functional depends on a measure only through its first two moments, $f(\alpha) = g(\mathbf{m}_\alpha, \mathbf{\Sigma}_\alpha)$, then

$$\delta f(\alpha)(x) = \langle\nabla_{\mathbf{m}} g,\, x\rangle + \langle\nabla_{\mathbf{\Sigma}} g,\, (x - \mathbf{m}_\alpha)(x - \mathbf{m}_\alpha)^\top\rangle, \qquad \nabla_x \delta f(\alpha)(x) = \nabla_{\mathbf{m}} g + 2\nabla_{\mathbf{\Sigma}} g(x - \mathbf{m}_\alpha),$$

where $\nabla_{\mathbf{\Sigma}} g$ is the symmetric Frobenius gradient. Thus Gaussian laws are invariant under the full Wasserstein gradient flow and

$$\dot{\mathbf{m}}_t = -\nabla_{\mathbf{m}} g, \qquad \dot{\mathbf{\Sigma}}_t = -2(\mathbf{\Sigma}_t \nabla_{\mathbf{\Sigma}} g + \nabla_{\mathbf{\Sigma}} g\, \mathbf{\Sigma}_t).$$

**Constrained evolution on the Gaussian manifold.** The affine-gradient cases above are exceptional: a generic Wasserstein gradient flow does not remain Gaussian. The constrained substitute projects the evolution onto $\mathcal{G} = \{\mathcal{N}(\mathbf{m}, \mathbf{\Sigma}) : \mathbf{m} \in \mathbb{R}^d,\ \mathbf{\Sigma} \succ 0\}$, the Gaussian submanifold of $\mathcal{P}_2(\mathbb{R}^d)$. The Wasserstein gradient of a functional constrained to a smooth submanifold $\mathcal{M} \subset \mathcal{P}_2$ is defined as the Riesz representative of the differential restricted to tangent velocities of $\mathcal{M}$.
$\alpha^{k+1} \in \underset{\alpha \in \mathcal{M}}{\mathrm{argmin}}\ \frac{1}{2\tau}\mathcal{W}_2^2(\alpha, \alpha^k) + f(\alpha)$. For $\mathcal{M} = \mathcal{G}$, tangent velocities are affine gradient fields $v(x) = b + A(x - \mathbf{m})$ with $A = A^\top$.
The constrained gradient is therefore the $L^2(\mathcal{N}(\mathbf{m}, \mathbf{\Sigma}))$ projection of the ambient Wasserstein gradient onto this finite-dimensional affine space, whenever the ambient gradient exists.
**Proposition 15.11** (Gaussian-constrained Wasserstein gradients)**.** *Let $f$ be a smooth functional and assume that its restriction to nondegenerate Gaussian measures can be written as $f(\mathcal{N}(\mathbf{m}, \mathbf{\Sigma})) = F(\mathbf{m}, \mathbf{\Sigma})$. Then the Wasserstein gradient constrained to the Gaussian family is the affine vector field $v_F(x) = \nabla_{\mathbf{m}} F(\mathbf{m}, \mathbf{\Sigma}) + 2\nabla_{\mathbf{\Sigma}} F(\mathbf{m}, \mathbf{\Sigma})(x - \mathbf{m})$, where $\nabla_{\mathbf{\Sigma}} F$ denotes the symmetric matrix derivative. Equivalently, $v_F$ is the $L^2(\mathcal{N}(\mathbf{m}, \mathbf{\Sigma}))$ projection of the ambient Wasserstein gradient onto affine gradient fields, whenever the ambient gradient exists. Hence the gradient descent flow constrained to Gaussian measures satisfies*

$$\dot{\mathbf{m}}_t = -\nabla_{\mathbf{m}} F(\mathbf{m}_t, \mathbf{\Sigma}_t), \qquad \dot{\mathbf{\Sigma}}_t = -2\big(\mathbf{\Sigma}_t \nabla_{\mathbf{\Sigma}} F(\mathbf{m}_t, \mathbf{\Sigma}_t) + \nabla_{\mathbf{\Sigma}} F(\mathbf{m}_t, \mathbf{\Sigma}_t)\mathbf{\Sigma}_t\big), \tag{15.34}$$

*and the descent velocity is affine.*

*Proof.* Test the functional along a Gaussian tangent vector, represented by an affine gradient field $v(x) = b + A(x - \mathbf{m})$ with $A$ symmetric. The induced first-order variations are $\dot{\mathbf{m}} = b$ and $\dot{\mathbf{\Sigma}} = A\mathbf{\Sigma} + \mathbf{\Sigma}A$. Therefore

$$\mathrm{d}F(\mathbf{m}, \mathbf{\Sigma})[b, A\mathbf{\Sigma} + \mathbf{\Sigma}A] = \langle\nabla_{\mathbf{m}} F,\, b\rangle + \mathrm{tr}(\nabla_{\mathbf{\Sigma}} F(A\mathbf{\Sigma} + \mathbf{\Sigma}A)).$$

Since $A$, $\mathbf{\Sigma}$ and $\nabla_{\mathbf{\Sigma}} F$ are symmetric, the second term equals $2\,\mathrm{tr}(\nabla_{\mathbf{\Sigma}} F\, A\mathbf{\Sigma}) = \int\langle 2\nabla_{\mathbf{\Sigma}} F(x - \mathbf{m}),\, A(x - \mathbf{m})\rangle \mathrm{d}\mathcal{N}(\mathbf{m}, \mathbf{\Sigma})(x)$. Together with the mean term, this gives $\mathrm{d}F(\mathbf{m}, \mathbf{\Sigma})[\dot{\mathbf{m}}, \dot{\mathbf{\Sigma}}] = \int\langle v_F(x),\, v(x)\rangle \mathrm{d}\mathcal{N}(\mathbf{m}, \mathbf{\Sigma})(x)$ for all affine gradient fields $v$. This identifies the constrained Wasserstein gradient in the induced $L^2(\alpha)$ metric, or equivalently the projection of the ambient gradient when it exists. Substituting the descent velocity $-v_F$ in Proposition 15.9 gives (15.34). □

**Non-variational Gaussian-preserving flows.**
**Contractive Gaussian projection.**
**Theorem 15.12** (Gelbrich theorem)**.** *For $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$, let $\mathcal{R}\alpha \coloneqq \mathcal{N}(\mathbf{m}_\alpha, \mathbf{\Sigma}_\alpha)$ be the Gaussian with the same mean and covariance as $\alpha$. Then $\mathcal{W}_2^2(\mathcal{R}\alpha, \mathcal{R}\nu) = \|\mathbf{m}_\alpha - \mathbf{m}_\nu\|^2 + \mathcal{B}^2(\mathbf{\Sigma}_\alpha, \mathbf{\Sigma}_\nu) \leqslant \mathcal{W}_2^2(\alpha, \nu)$.*

*Proof.* Take any coupling $(X, Y)$ of $\alpha$ and $\nu$, center the variables, and write $C = \mathbb{E}[(X - \mathbf{m}_\alpha)(Y - \mathbf{m}_\nu)^\top]$. In the positive definite case, positivity of the block covariance matrix implies the factorization $C = \mathbf{\Sigma}_\alpha^{1/2} K \mathbf{\Sigma}_\nu^{1/2}$ with $\|K\|_{\mathrm{op}} \leqslant 1$, and therefore, by operator/nuclear norm duality,

$$\mathrm{tr}\, C \leqslant \mathrm{tr}\left((\mathbf{\Sigma}_\alpha^{1/2}\mathbf{\Sigma}_\nu\mathbf{\Sigma}_\alpha^{1/2})^{1/2}\right).$$

The semidefinite case follows by adding $\eta\mathrm{Id}$ to both covariance matrices and letting $\eta \downarrow 0$. Expanding $\mathbb{E}\|X - Y\|^2$ gives the lower bound $\mathbb{E}\|X - Y\|^2 \geqslant \|\mathbf{m}_\alpha - \mathbf{m}_\nu\|^2 + \mathcal{B}^2(\mathbf{\Sigma}_\alpha, \mathbf{\Sigma}_\nu)$. Taking the infimum over couplings proves the inequality, while equality for Gaussian laws is Proposition 2.35. □

**Theorem 15.13** (Hugo Lavenant Gaussian-preservation criterion)**.** *Let $F : \mathcal{P}_2(\mathbb{R}^d) \to (-\infty, +\infty]$ satisfy $F(\mathcal{R}\alpha) \leqslant F(\alpha) \quad \forall\alpha \in \mathcal{P}_2(\mathbb{R}^d)$, with $\mathcal{R}$ defined in Theorem 15.12. If $\gamma$ is Gaussian and $\nu$ minimizes the JKO step $\eta \mapsto F(\eta) + \frac{1}{2\tau}\mathcal{W}_2^2(\gamma, \eta)$, then $\mathcal{R}\nu$ is also a minimizer. If the JKO minimizer is unique, it is Gaussian. Thus any unique Wasserstein gradient flow obtained as the limit of this JKO scheme preserves Gaussian initial data.*

*Proof.* For the JKO claim, $\mathcal{R}\gamma = \gamma$ because $\gamma$ is Gaussian. Hence, for any competitor $\eta$, $F(\mathcal{R}\eta) + \frac{1}{2\tau}\mathcal{W}_2^2(\gamma, \mathcal{R}\eta) \leqslant F(\eta) + \frac{1}{2\tau}\mathcal{W}_2^2(\gamma, \eta)$. Applying this to a minimizer $\eta = \nu$ shows that $\mathcal{R}\nu$ is again a minimizer. Uniqueness forces $\nu = \mathcal{R}\nu$. □